

Guidance on the Drinking Water Directive

Volume I: Methodologies for testing starting substances, compositions and constituents for use in the manufacture of materials or products in contact with water intended for human consumption

Version 1.0



Legal notice

This document aims to assist users in complying with their obligations under Directive (EU) 2020/2184 of the European Parliament and of the Council of 16 December 2020 on the quality of water intended for human consumption (OJ L 435, 23.12.2020, <http://data.europa.eu/eli/dir/2020/2184/oj>), hereafter referred to as the drinking water directive (DWD). However, users are reminded that the text of the DWD (Article 11(2)(a)) and Commission Implementing Decision (EU) 2024/365 are the only authentic legal references and that the information in this document does not constitute legal advice. Usage of the information remains the sole responsibility of the user. The European Chemicals Agency does not accept any liability with regard to the use that may be made of the information contained in this document.

Guidance on the Drinking Water Directive – Volume I: Methodologies for testing starting substances, compositions and constituents for use in the manufacture of materials or products in contact with water intended for human consumption

Reference: ECHA-24-G-11-EN
Cat. number: ED-01-24-022-EN-N
ISBN: 978-92-9468-423-3
DOI: 10.2823/3635089
Publ. date: January 2025
Language: EN

© European Chemicals Agency, 2025

Luxembourg: Publications Office of the European Union, 2025

If you have questions or comments in relation to this document, please send them (quoting the reference and issue date) using the information request form. The information request form can be accessed via the Contact ECHA page at <http://echa.europa.eu/contact>.

European Chemicals Agency

Mailing address: P.O. Box 400, FI-00121 Helsinki, FINLAND
Visiting address: Telakkakatu 6, FI-00150 Helsinki, FINLAND

Preface

Guidance on the Drinking Water Directive – Volume I (testing methodology) is to be used in preparing applications for starting substances, compositions and constituents for approval for inclusion in the European positive list (EUPL).

This document describes the applicants' obligations according to Article 11(2)(a) of the drinking water directive (DWD) and how to fulfil them.

This guidance provides technical scientific advice on how to comply with the methodologies for testing and how to perform the exposure assessment and the assessment of human health risks set out in Commission Implementing Decision (EU) 2024/365.



Notes for the reader

This guidance must be read in conjunction with the relevant DWD implementing legislation, particularly:

- Commission Implementing Decision (EU) 2024/365 of 23 January 2024 laying down rules for the application of Directive (EU) 2020/2184 of the European Parliament and of the Council as regards methodologies for testing and accepting starting substances, compositions and constituents to be included in the European positive lists;
- Commission Implementing Decision (EU) 2024/367 of 23 January 2024 laying down rules for the application of Directive (EU) 2020/2184 of the European Parliament and of the Council by establishing the European positive lists of starting substances, compositions and constituents authorised for use in the manufacture of materials or products that come into contact with water intended for human consumption;
- Commission Delegated Regulation (EU) 2024/369 of 23 January 2024 supplementing Directive (EU) 2020/2184 of the European Parliament and of the Council by laying down the procedure regarding inclusion in or removal from the European positive lists of starting substances, compositions and constituents.

In addition, this guidance document is part of wider guidance on DWD applications comprising a total of five volumes. The remaining volumes are:

- DWD Guidance Volume II on methodologies for accepting starting substances, compositions and constituents for use in the manufacture of materials or products in contact with water intended for human consumption;
- DWD Guidance Volume III on the scope of DWD applications;
- DWD Guidance Volume IV on the contents of a notification of intention;
- DWD Guidance Volume V on calibrants and representative samples of accepted compositions in connection to applications submitted under Commission Delegated Regulation (EU) 2024/369 (this guidance was prepared by the European Commission Joint Research Centre).

Contents

Legal notice	2
Preface	3
Contents	4
List of figures	7
List of tables	8
Abbreviations	9
Glossary of key terms	11
1. Introduction to the guidance on testing methodologies for the drinking water directive	17
1.1. Aim of this guidance	17
1.2. How to use this guidance document	17
1.3. General structure of this guidance document	31
1.4. Use of additional guidance documents	31
1.5. Intellectual property rights and data sharing	32
1.6. Alignment between a DWD notification of intention and a DWD application	32
2. Identity of the applied-for entry	33
2.1. Name or any other identifier (point 1.1 in Annex I to Commission Implementing Decision (EU) 2024/365)	34
2.2. Information related to molecular and structural formula or crystal structure (point 1.2 in Annex I to Commission Implementing Decision (EU) 2024/365)	38
2.3. Chemical characterisation (point 1.3 in Annex I to Commission Implementing Decision (EU) 2024/365)	39
2.4. Additional requirements for polymers and pre-polymers (point 1.5 in Annex I to Commission Implementing Decision (EU) 2024/365)	43
2.5. Individual versus group entries	44
3. Intended use of the applied-for entry	46
3.1. Main information on intended use	47
3.2. Conditions of the intended use	53
3.3. Processing and internal structure of the material, final material and products (points 2.5.1–2.5.3 in Annex II to Commission Implementing Decision (EU) 2024/365)	55
3.4. EU- and national-level assessments and authorisations (points 2.6 and 2.7 in Annex II to Commission Implementing Decision (EU) 2024/365)	61
3.5. Potentially relevant chemical species other than the applied-for entry	62
3.6. Other information	64
4. Physico-chemical properties	65
4.1. Appearance/state of the substance (point 4.1 in Annex III to Commission Implementing	

Decision (EU) 2024/365)	65
4.2. Melting/freezing point (point 4.2 in Annex III to Commission Implementing Decision (EU) 2024/365)	66
4.3. Thermal stability (IUCLID Section 3.4).....	66
4.4. Boiling point (point 4.3 in Annex III to Commission Implementing Decision (EU) 2024/365)	66
4.5. Density (point 4.4 in Annex III to Commission Implementing Decision (EU) 2024/365)	67
4.6. Vapour pressure (point 4.5 in Annex III to Commission Implementing Decision (EU) 2024/365)	67
4.7. Surface tension of an aqueous solution (point 4.6 in Annex III to Commission Implementing Decision (EU) 2024/365)	68
4.8. Water solubility (point 4.7 in Annex III to Commission Implementing Decision (EU) 2024/365)	68
4.9. Partition coefficient (<i>n</i> -octanol–water) and its pH dependency (point 4.8 in Annex III to Commission Implementing Decision (EU) 2024/365)	69
4.10. Granulometry (point 4.9 in Annex III to Commission Implementing Decision (EU) 2024/365)	69
4.11. Dissociation constant (point 4.10 in Annex III to Commission Implementing Decision (EU) 2024/365)	70
4.12. Hydrolysis as a function of pH.....	70
5. Data on migration of the applied-for entry and identification of relevant chemical species	71
5.1. Physical testing of migration (points 5.1–5.3 in Annex IV to Commission Implementing Decision (EU) 2024/365)	71
5.2. Calculation of the migration rate and C_{tap} (point 5.2 in Annex IV to Commission Implementing Decision (EU) 2024/365)	92
5.3. Alternative methods (adaptations of migration testing).....	94
5.4. Identification of relevant chemical species.....	98
6. Toxicological data for the relevant chemical species	100
6.1. Considerations before initiating testing	100
6.2. General considerations for reporting animal data	101
6.3. Approach to toxicity testing – principles.....	102
6.4. Genotoxicity/mutagenicity (point 6.1 of Table 1 of Annex V to Commission Implementing Decision (EU) 2024/365)	103
6.5. Toxicokinetic and metabolism studies in mammals (points 6.2, 7.1 and 8.1 of Tables 1, 2 and 3 of Annex V to Commission Implementing Decision (EU) 2024/365)	109
6.6. Repeated dose toxicity (points 7.2 and 8.2 in Tables 2 and 3 of Annex V to Commission Implementing Decision (EU) 2024/365)	113
6.7. Reproductive toxicity (points 7.3 and 8.3 of Tables 2 and 3 of Annex V to Commission Implementing Decision (EU) 2024/365)	117
6.8. Carcinogenicity (point 8.4 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365)	126

6.9. Appropriate additional studies, referred to under point 6.2 of Table 1 and point 7.4 of Table 2 of Annex V to Commission Implementing Decision (EU) 2024/365	127
6.10. Additional toxicity properties (point 8.5 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365)	128
6.11. Appropriate mechanistic data or studies (point 8.5.4 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365)	130
7. Testing for nanoforms	131
7.1. Overview	131
7.2. Substance identification and characterisation of nanoforms (point 1.4 in Annex I to Commission Implementing Decision (EU) 2024/365)	134
7.3. Physico-chemical properties beyond granulometry	135
7.4. Migration	136
7.5. Toxicological data	138
8. References and background documents	147
8.1. General sources	147
8.2. Specific sources	153
8.3. Test methods and standards	154
9. Annex I – Material categories	158
9.1. Organics	158
9.2. Metallics	160
9.3. Cementitious materials	161
9.4. Enamels, ceramics and other inorganics	162
10. Annex II – Product groups and their conversion factors	163
10.1. Metallic materials	163
10.2. Organic, cementitious, enamel, ceramic and other inorganic materials	164
11. Annex III – Technical functions for starting substances and organic cementitious constituents	167
11.1. Organic materials	167
11.2. Cementitious materials	170
12. Annex IV – Existing metallic composition categories (as at January 2024)	177

List of figures

Figure 1. Schematic representation of stepwise approach to fulfilling the testing methodology for human health hazard for starting substances and organic cementitious constituents for the purposes of the DWD	21
Figure 2. Schematic representation of stepwise approach to fulfilling the testing methodology for human health hazard for metallic, enamel, ceramic and other inorganic compositions for the purposes of the DWD.....	22
Figure 3. General approach to identifying relevant chemical species for starting substances of organic materials – step 1	23
Figure 4. General approach to identifying relevant chemical species for starting substances of organic materials – step 2	24
Figure 5. General approach to identifying relevant chemical species for starting substances of organic materials – step 3	25
Figure 6. General approach to identifying relevant chemical species for metallic compositions – step 1	26
Figure 7. General approach to identifying relevant chemical species for metallic compositions – step 2	27
Figure 8. General approach to identifying relevant chemical species for organic constituents of cementitious materials – step 1.....	27
Figure 9. General approach to identifying relevant chemical species for organic constituents of cementitious materials – step 2.....	28
Figure 10. General approach to identify relevant chemical species for organic constituents of cementitious materials – step 3.....	29
Figure 11. General approach to identify relevant chemical species for enamel, ceramic and other inorganic compositions	30
Figure 12. Plastic and rubber processing methods.....	55
Figure 13. Basic metal processing methods.....	57
Figure 14. Plating processes	58
Figure 15. Steps of a plating process	59
Figure 16. Basic ceramic processing methods.....	61
Figure 17. Glass and ceramic treatments.....	61
Figure 18. Decision tree for deciding on the testing strategy for nanoforms (step 1) ..	133
Figure 19. Decision tree for deciding on the testing strategy for nanoforms (step 2) ..	134

List of tables

Table 1. Examples of restrictions on the conditions of use for EUPL entries	54
Table 2. Selecting the worst foreseeable conditions for migration testing	77
Table 3. Relevant chemical species for toxicological assessment of different material types	99
Table 4. Stepwise approach to toxicity testing	102
Table 5. Material categories and subcategories for organic materials	158
Table 6. Material categories and subcategories for metallic materials	160
Table 7. Material categories and subcategories for enamel, ceramic and other inorganic materials	162
Table 8. Product groups for metallic compositions	163
Table 9. Product groups for organic, cementitious, enamel, ceramic and other inorganic materials	165
Table 10. Risk groups for organic, cementitious, enamel, ceramic and other inorganic materials	166
Table 11. Technical functions of starting materials applied for inclusion in the organics EUPL	167
Table 12. List of generic constituents of cementitious materials	172
Table 13. Technical functions of constituents applied for inclusion in the cementitious EUPL	175
Table 14. Existing metallic composition categories	177

Abbreviations

AAS	atomic absorption spectrometry
ADME	absorption, distribution, metabolism and excretion
AF	assessment factor
AUC	area under the curve
BPR	Regulation (EU) No 528/2012 (biocidal products regulation)
CF	conversion factor
CLP regulation	Regulation (EC) No 1272/2008 on classification, labelling and packaging
C _{max}	maximum serum concentration
C _{tap}	concentration at the tap
DNEL	derived no-effect level
DWCM	drinking water contact material
DWD	Directive (EU) 2184/2020 (drinking water directive)
EATS	oestrogen, androgen, thyroid and steroidogenic
ED	endocrine disruptor
EFSA	European Food Safety Authority
EOGRTS	extended one-generation reproductive toxicity study
EUPL	European positive list
FCM	food contact material
GC	gas chromatography
GC-MS	gas chromatography–mass spectrometry
HPLC-MS	high-performance liquid chromatography–mass spectrometry
ICP	inductively coupled plasma
ICP-MS	inductively coupled plasma–mass spectrometry
ICP-OES	inductively coupled plasma–optical emission spectroscopy
ID	internal diameter
ISO	International Organization for Standardization
IUCLID	International Uniform Chemical Information Database
IUPAC	International Union of Pure and Applied Chemistry
JRC	Joint Research Centre

K_{ow}	<i>n</i> -octanol–water partition coefficient
LC-MS	liquid chromatography–mass spectrometry
LOD	limit of detection
LOQ	limit of quantification
MEP _n (T)	mean of the equivalent pipe concentration
M_n	number average molecular weight (total weight of polymer divided by the total number of molecules)
MS	mass spectrometry
MTC _{tap}	maximum tolerable concentration at the tap
MTD	maximal tolerated dose
NIAS	non-intentionally added species
NOAEL	no observed adverse effect level
OECD	Organisation for Economic Co-operation and Development
REACH regulation	Regulation (EC) No 1907/2006 on the registration, evaluation, authorisation and restriction of chemicals
SCCS	Scientific Committee on Consumer Safety
SIP	substance identity profile
S-OES	spark optical emission spectrometry
spICP-MS	single particle inductively coupled plasma mass spectrometry
STOT RE	specific target organ toxicity – repeated exposure
S/V	surface to volume ratio (in this guidance, the S/V ratio refers to the amount of water contact surface area of a material or a product relative to the volume of water it is in contact with; to calculate the S/V ratio, simply divide the surface area by the water volume)
TG	Test Guideline
T _{max}	time to maximum plasma concentration
UVCB	substance of unknown or variable composition, complex reaction product or biological material
w/w	weight by weight

Glossary of key terms

Additive – a starting substance that is intentionally added to an organic material to achieve a physical or chemical effect during processing of the plastic or in the final material or product; it is intended to be present in the final material or product. Additives are not bound to the polymeric chain.

Adhesive – according to European standard EN 923, a non-metallic substance capable of joining materials by surface bonding (adhesion) with a bond of adequate internal strength (cohesion). In the context of the DWD, adhesives are a category of organic materials including plastics and polymers, rubber and elastomers. In practice, adhesives used in drinking water contact materials are typically formulations of several substances including polymeric organic substances of medium or high molecular weight (also mixtures of more than one polymer and usually of lower molecular weight than self-supporting plastics) but also various additives of lower molecular weight. Adhesives usually contain substances of molecular weight below 1 000 Da.

Admixture – material added during the mixing process of a cementitious material in a quantity not more than 5 % by mass of the inorganic binder content of the cementitious material in order to modify the properties of the mix in the fresh and/or hardened state. Admixtures are one of the generic constituent categories of cementitious materials. An admixture contains organic cementitious constituents that are to be listed in the EUPL for cementitious materials.

Agency – the European Chemicals Agency set up pursuant to Article 75(1) of Regulation (EC) No 1907/2006.

Alloy – a metallic material consisting of two or more constituents.

Assessment factor – a factor that reflects the degree of uncertainty in extrapolating from experimental test data (e.g. obtained in a limited number of subjects from a limited number of animal species) to the situation in the human (sub)population for which the risk characterisation is performed. Sources of uncertainty typically considered by using assessment factors include inter- and intraspecies variability in terms of toxicodynamics and/or toxicokinetics, differences in route, frequency or duration of exposure between the experimental data and the scenario considered for risk characterisation, a particular severity of effect, or a poor database. A non-exhaustive list of expressions that have been used as synonyms or for specific types of assessment factors would include any of the following terms: uncertainty factor, extrapolation factor, modifying factor, adjustment factor or safety factor.

Aid to polymerisation – a starting substance that helps initiate polymerisation and/or control the formation of the polymer structure but does not end up as a building block of the polymer structure.

Applied-for entry – unless explicitly described otherwise, this term is used to refer to any starting substance, composition or organic cementitious constituent that may be tested by an applicant with the intention of submitting an application to the agency to request its inclusion in, updating or removal from the EUPL.

Cement – a finely ground inorganic material that, when mixed with water, forms a paste that sets and hardens by means of hydration reactions and processes and which, after hardening, retains its strength and stability even under water.

Cementitious material – a material that contains a hydraulic cement in sufficient proportion to act as the main binder by forming a hydrate structure that governs the performance of the material.

Ceramic material – an inorganic poly- or single crystalline, non-metallic solid material subjected to a high temperature during manufacture.

Coating – a non-self-supporting layer composed of substances applied to an already existing substrate in order to impart special properties or improve the technical performance of the product.

Composition – in the context of the DWD, the chemical composition of a metal, enamel, ceramic or other inorganic material.

Constituent of an enamel, ceramic or other inorganic composition – a constituting substance of an enamel, ceramic or other inorganic composition, for example oxides, carbides, nitrides, silicides, borides, phosphides, sulphides, tellurides, selenides, carbon and silicon.

Constituent of a metallic composition – an intentionally added constituting element that produces a certain desired effect on the service properties of the metallic composition.

Corrosion – the degradation of (typically) metals to lower energy-containing forms, such as oxides. Corrosion requires suitable thermodynamic and kinetic conditions. The interaction of metals with (drinking) water typically results in oxidation of metals and consumption of oxygen. This type of corrosion requires sites for anodic and cathodic reactions and their linkage via a conducting electrolyte. There are several known corrosion types. The most relevant types for materials in contact with water are uniform and localised corrosion.

Curing compound – a liquid substance that is added as a surface coating on freshly installed concrete. Curing compounds are used to reduce the loss of water or heat in order to create conditions that are favourable for concrete formation.

Elastomer – a low shear modulus material, either natural or synthetic, made up of carbonaceous macromolecules, and characterised by long polymer chains arranged in a three-dimensional flexible network held together by chemical covalent cross-links. Elastomers have, at service temperatures and until their decomposition, elastic physical properties that allow the material to be substantially deformed under stress and recover almost its original shape when the stress is removed. **Rubber** is a type of elastomer, typically a natural material. Thermoplastic elastomers may occur as plastics or elastomers, depending on the mode of cross-linking and the method of their manufacture.

Enamel – a vitreous material obtained by melting at temperatures higher than 1 200 °C and fritting of a mixture of inorganic substances.

European positive list (EUPL) – one of the lists of starting substances, compositions or organic cementitious constituents established in Commission Implementing Decision (EU) 2024/367.

Final material – a material that is subject to testing and acceptance in accordance with the testing requirements and acceptance criteria set out in the annexes to Commission Implementing Decision (EU) 2024/368.

Formulation – the list of all substances or constituents used in the preparation of an organic material or of a cementitious material and their relative quantities.

Formwork release agent – coating applied to the formwork surface, prior to concreting, to facilitate easy removal the formwork or shuttering.

Generic constituent category – a type of inorganic or organic ingredient of a cementitious mixture. Generic constituent categories include cement, inorganic additions, aggregates, mixing water, organic additions, admixtures, fibres, formwork release agents and curing compounds, and they may encompass inorganic and organic constituents that may need to be listed in one of the EUPLs (cementitious, organic or metallic EUPLs).

Grinding aid – a substance that, when mixed into the mill contents, improves the efficiency and effectiveness of the grinding process by increasing the rate of reduction in particle size of cement and/or inorganic additions.

Impurity of a metallic composition – a constituent (element) that is unintentionally present in a metallic composition following its manufacturing and subsequent processing and which does not produce any (intentional) desired effect on the overall service characteristics of the metallic composition. Impurities might pose health risks and must therefore be banned or their maximum content controlled.

International Uniform Chemical Information Database (IUCLID) – the software that applicants must use for the preparation and submission of their DWD applications to ECHA.

Localised corrosion – occurs when only certain regions of a material surface undergo corrosion, such as pitting of passivated materials (see below) or crevice corrosion. Pitting corrosion affects mainly the mechanical stability of a material with negligible substance migration and can occur in a passivated material in contact with water. The tendency to passivate and the stability of the passive film can be determined using an electrochemical test in accordance with relevant standards (e.g. EN 16056 for steels).

Lubricant – material used to reduce friction between bearing surfaces.

Material or drinking water contact material (DWCM) – a solid, semi-solid or liquid that is used for the manufacturing of a product and that is:

- an organic material prepared from one or more starting substances;
- a cementitious material prepared from one or more constituents;
- a metallic, enamel, ceramic or other inorganic composition.

‘Material’ here does not refer to interim or final products.

Material type (level 1) – the top-level distinction between organics, metallics, cementitious materials, enamels, ceramics and other inorganic materials that is made in the DWD legal text.

Material category (level 2) – the level below level 1 when, for instance, referring to plastics, rubbers, etc., within the organic material type – see Annex I to this guidance document.

Material subcategory (level 3) – the level below level 2 when, for instance, referring to polyvinyl chloride plastic, polyethylene plastic, polyethylene terephthalate plastic, etc. – see Annex I to this guidance document.

Metallic composition category – a group of metallic compositions with the same metallic composition constituents and the same behaviour when they come into contact with water intended for human consumption.

Metallic material – a material that is a metal or metal alloy and that is used either in bulk form or as metallic plating.

Migration – the net mass transfer of substances incorporated within solid matrices into water intended for human consumption.

Migration water or contact water – the test water that has been in contact with the test piece under specified conditions.

Monomer – a substance that is capable of forming covalent bonds with a sequence of additional like or unlike molecules under the conditions of the relevant polymer-forming reaction used for the particular process.

Nanoform – a form of a natural or manufactured substance containing particles in an unbound state, as an aggregate or as an agglomerate and where, for 50 % or more of the particles in the number size distribution, one or more external dimensions is in the size range 1–100 nm, including by derogation fullerenes, graphene flakes and single-walled carbon nanotubes with one or more external dimensions below 1 nm. In that context:

- ‘particle’ means a minute piece of matter with defined physical boundaries;
- ‘agglomerate’ means a collection of weakly bound particles or aggregates in which the resulting external surface area is similar to the sum of the surface areas of the individual components;
- ‘aggregate’ means a particle comprising strongly bound or fused particles.

Non-intentionally added species (NIAS) – one of the following:

- an impurity of a starting substance or organic cementitious constituent or composition;
- a reaction product or a degradation product of a starting substance or organic cementitious constituent that forms during the processing or use of the material;
- a reaction product or a degradation product of a starting substance or organic cementitious constituent that forms in contact with water during the use of the material.

NIAS are chemical species that have not been added for a technical reason during the manufacturing process. Beyond impurities, NIAS can include breakdown and reaction (e.g. hydrolysis) products, degradation products, unreacted initiators and end groups of polymers, unreacted starting materials of substances of unknown or variable composition, complex reaction products or biological materials (UVCBs) and substance constituents in the case of multi-constituent and UVCBs.

Organic cementitious constituent – an organic substance intentionally used in the manufacture of cementitious materials and listed in the EUPL for cementitious materials. Each organic cementitious constituent falls under a generic constituent category. Exceptionally, polymers used as components of admixtures are considered organic cementitious constituents, but their monomers are allowed to be listed in the EUPL and polymers are considered to be covered by the listed monomers. Additional guidance is provided in DWD Guidance Volume III on the scope of DWD applications.

Organic material – a material that consists of mainly carbon-based substances. An organic material may fall under material categories such as plastics, rubbers, coatings, adhesives, lubricants or silicones.

Other reactant – a starting substance other than a monomer that is incorporated into the polymer structure. Examples include hardeners, surface treatment agents and certain cross-linking agents.

Passivity – refers to a slow, undetectable, rate of corrosion as a result of a stable thin barrier layer (typically an oxide) on the surface of a material. This passive layer practically stops any further corrosion and passivates the material. Examples of passive metals are aluminium and stainless steel. Passive materials are subject to localised rather than uniform corrosion. In the presence of a stable passive film under constant conditions, a material is hygienic and only undetectable metal release should occur.

Polymer – a substance consisting of molecules characterised by the sequence of one or more types of monomer units. Such molecules must be distributed over a range of molecular weights, and differences in the molecular weight are primarily attributable to differences in the number of monomer units. A polymer comprises the following:

- a simple weight majority of molecules containing at least three monomer units that are covalently bound to at least one other monomer unit or other reactant;
- less than a simple weight majority of molecules of the same molecular weight.

In the context of this definition a 'monomer unit' means the reacted form of a monomer substance in a polymer.

Polymer production aid – a starting substance used to provide a medium suitable for polymer or plastic manufacturing; it may be present but is neither intended to be present in the final materials or products nor has a physical or chemical effect in the final material or product.

Polymeric additive – any polymer that is used as an additive that has a physical or chemical effect on the plastic and which cannot be used in the absence of other polymers as a main structural component of final materials and products.

Polymerised part – the part of a substance composition consisting of molecules characterised by the sequence of one or more types of monomer units. Molecules such as dimers and trimers also contribute to the polymerised part. However, the term 'polymerised part' does not cover unreacted monomers or unreacted other reactants.

Pre-polymer – a substance resulting from a polymerisation-type of reaction that is further reacted into a final polymer in a material or product.

Product – an object that comes into contact with water intended for human consumption that is made of final materials and with the intention of being placed on the market.

Relevant chemical species – for the purpose of toxicological testing and risk assessment and acceptance, a relevant chemical species is a substance identified in accordance with Section 3 of Annex IV to Commission Implementing Decision (EU) 2024/365. A starting substance, an organic cementitious constituent, a NIAS, a constituent or impurity of a metallic, enamel, ceramic or other inorganic composition may potentially be a relevant chemical species depending on the applied-for entry, the intended use and the results of migration testing or modelling.

Rubber and elastomers – a material category of organic materials. According to standard ISO 1382, rubber is a family of polymeric materials that are flexible and elastic, and elastomers are macromolecular materials that return rapidly to approximately their initial dimensions and shape after substantial deformation by a weak stress and release of the stress. While rubbers can be from either natural or synthetic sources (e.g. ethylene-propylene-non-conjugated diene

rubbers), the term 'rubber' is typically used to refer to natural rubbers as opposed to synthetic elastomers.

'Thermoplastic elastomer' means a polymer or blend of polymers that does not require vulcanisation or cross-linking during processing, yet, at its service temperature, has properties similar to those of vulcanised rubber. These properties disappear at processing temperature, so that further processing is possible, but return when the material is returned to its service temperature. Rubbers, however, while cross-linked, have no thermoplastic properties, that is, they cannot be molten or thermoplastically formed and regain their elasticity.

This differentiation is misleading (see also comment on rubbers): thermoplastic elastomers are elastomers because of their properties. In food contact materials, thermoplastic elastomers fall under the regulations on plastics, but rubber is not covered in this way. Synthetic elastomers with rubber properties are not thermoplastic elastomers but (synthetic) rubber.

Silicone – macromolecular substances or materials based on organopolysiloxanes that are cross-linked forming a three-dimensional network with elastomeric or rubber-like properties.

Specific migration limit – the maximum permitted amount of a given substance released from a material or article into food or food simulants.

Starting substance – a substance that has been intentionally added during the production of organic materials or of admixtures for cementitious materials.

Substance – a chemical element and its compounds in the natural state or obtained by any manufacturing process, including any additive necessary to preserve its stability and any impurity deriving from the process used, but excluding any solvent that may be separated without affecting the stability of the substance or changing its composition. Important note: substance constituents are relevant only to starting substances and organic cementitious constituents.

Substance constituent – any single chemical species present in a substance that can be characterised by its unique chemical identity. The term is particularly relevant to mono-constituent, multi-constituent substances and UVCBs as defined under the REACH regulation (Regulation (EC) No 1907/2006). A mono-constituent substance is a substance where one main constituent makes up at least 80 % of the substance. A mono-constituent substance is named after the main constituent, using its International Union of Pure and Applied Chemistry name. A multi-constituent substance is a substance that consists of several main constituents. Each of these main constituents is present in the substance at a concentration of between 10 % and 80 %. A multi-constituent substance is named as a 'reaction mass of' the main constituents present in the substance. Constituents that are not the main constituents of mono-constituent and multi-constituent substances are considered in practice to be impurities and are not mentioned in the substance name.

Test piece – a representative object composed of the final material that is used for carrying out migration testing in accordance with the procedures and methods for testing set out in Annex IV to Commission Implementing Decision (EU) 2024/365. Note that this term is also relevant to migration testing under Annexes I, II, III and IV of Commission Implementing Decision (EU) 2024/368.

Uniform corrosion – in uniform corrosion the whole surface area of a (non-passive) material corrodes releasing metals into (drinking) water, compromising its hygienic status. This type of corrosion can be tested, for example by the **standard rig test EN 15664-1**.

1. Introduction to the guidance on testing methodologies for the drinking water directive

1.1. Aim of this guidance

Directive (EU) 2020/2184 of the European Parliament and of the Council of 16 December 2020 on the quality of water intended for human consumption (the drinking water directive (DWD)) sets specific minimum hygiene requirements for the authorisation of starting substances, compositions and organic cementitious constituents for use in the manufacture of materials or products in contact with water intended for human consumption on the basis of the principles set out in its Annex V. These requirements ensure the uniform application of paragraph 1 of Article 11 of the DWD.

This document describes the methodologies for testing starting substances, compositions and organic cementitious constituents for inclusion in the European positive lists (EURLs), including specific migration limits and scientific pre-conditions related to starting substances, compositions or organic cementitious constituents according to Article 11(2)(a) of the DWD. Methodologies for testing and accepting starting substances, compositions and organic cementitious constituents into the EURLs are set out in Commission Implementing Decision (EU) 2024/365.

This guidance needs to be considered together with DWD Guidance Volume II (see Section 1.2.2)

This guidance is primarily addressed to applicants seeking approval for a starting substance, composition or organic cementitious constituent to be included in the EURL and also those seeking the removal or updating of an existing entry in the EURL.

In this document, unless explicitly described otherwise, the term 'applied-for entry' is used to refer to starting substances, compositions and organic cementitious constituents that may be applied for at ECHA (the agency) for inclusion in, renewal or removal from an EURL. This is how the starting substance, composition or organic cementitious constituent would be presented if included in the EURL. In contrast, for the purpose of assessment of toxicological properties and risk assessment and acceptance, the term 'relevant chemical species', as defined according to Section 3 of Annex IV to Commission Implementing Decision (EU) 2024/365, is used.

1.2. How to use this guidance document



Notes for the reader

The section headings in Chapters 2–6 include references to the relevant points in Annexes I–V to Commission Implementing Decision (EU) 2024/365 (http://data.europa.eu/eli/dec_impl/2024/365/oj).

The term 'chapter' is used to refer to the highest level parts of the document (Chapters 1, 2, 3, etc.). Lower level headings (e.g. 1.1) are referred to as 'sections'.

1.2.1. Guiding principles for fulfilling data requirements

The following guiding principles reflect the general guidance on testing methodologies.

- **The standard information** specified in the tables of Annexes I–V to Commission Implementing Decision (EU) 2024/365 forms the basis of the information requirements. It is the minimum set required for all applied-for entries and relevant chemical species. Any other relevant information that is available should also be identified and considered.
- **The additional information** includes any other relevant information, including on physico-chemical and toxicological properties and on methods of detection and identification.
- Information on the intrinsic properties of relevant chemical species may be generated with **alternative test methods**, as described in Part 2 of Section 2 of Annex V to Commission Implementing Decision (EU) 2024/365 (General rules for adaptation), or can be omitted, provided that the conditions of the specific rules for adaptation set out in column 2 of Tables 1–3 of Annex V to Commission Implementing Decision (EU) 2024/365 are met.
- No physico-chemical property information or testing is required for starting substances or constituents if one of the criteria set in Section 1 of Annex III of Commission Implementing Decision (EU) 2024/365 is met. Physico-chemical property information supports the assessment of toxicological data and the interpretation of migration data. Where a robust maximum tolerable concentration at the tap (MTC_{tap}) value is already available or can be readily calculated from an existing parametric value in Annex I to the DWD or a specific migration limit (SML) value for food contact materials (FCMs) adopted by the European Food Safety Authority (EFSA) less than 15 years before the submission of the DWD application, there is no need for information or testing of the physico-chemical properties of the starting substance or organic cementitious constituent.
- No toxicological information or testing is required for relevant chemical species if one of the criteria set in Section 1 of Annex V to Commission Implementing Decision (EU) 2024/365 is met. The criteria explain that, where a robust MTC_{tap} for the relevant chemical species is available or can be readily calculated, or where the hazards of the relevant chemical species are known and severe thus leading to the adoption of a default MTC_{tap} value, toxicological information and testing will not be necessary. This waiving of toxicological information does not apply if new or updated information has become available since the adoption of the current MTC_{tap} value.
- The application system prescribed by the implementing legislation of the DWD does not allow for the type of approach set out in Regulation (EC) No 1907/2006 on the regulation, evaluation, authorisation and restriction of chemicals (REACH regulation) to testing proposals being submitted and examined by ECHA.
- Toxicological and other relevant testing should be carried out in compliance with the principles of good laboratory practice provided for in Directive 2004/10/EC, or international standards recognised as equivalent by the Commission or ECHA, and with the provisions of Directive 2010/63/EU of the European Parliament and of the Council ⁽¹⁾, if applicable. Migration testing must be carried out in compliance with the principles of good laboratory practice provided for in standard EN ISO/IEC 17025 or other international standards recognised as equivalent by the Commission or ECHA as stipulated in Commission Implementing Decision (EU) 2024/365.

⁽¹⁾ Directive 2010/63/EU of the European Parliament and of the Council of 22 September 2010 on the protection of animals used for scientific purposes (OJ L 276, 20.10.2010, p. 33).

- Test results must be reported as described in the guideline used. The robust study summaries of all key studies must be included in the application submitted to ECHA. The full study report should be provided on request by ECHA.
- For toxicological studies, where a test method offers flexibility in the determination/choice of the study design, for example in relation to dose levels, **the study design should ensure that the data generated are adequate for hazard identification and risk assessment.** To this end, testing is to be performed at appropriately high dose levels. Of these doses, the highest dose needs to induce minimal toxicity, such as characterised by an approximately 10 % reduction in body weight (maximal tolerated dose (MTD)) ⁽²⁾. If dose (concentration) selection is limited by the physico-chemical properties or biological effects of the test substance, a scientifically valid justification should be provided.
- All toxicological data included in the application must be provided in the form of a robust study summary in the International Uniform Chemical Information Database (IUCLID) dossier. The information submitted on the applied-for entry, or any relevant chemical species related to the applied-for entry, must be sufficient for conducting the risk assessment, submitted as part of the application. Further guidance on risk assessment and acceptance is provided in DWD Guidance Volume II.
- There must be at least **one key study** or an acceptable general or specific rule for adaptation for each standard information requirement listed in Part 2, Section 2, of Annex V to Commission Implementing Decision (EU) 2024/365. A key study is the study required for a certain end point. It must be reliable and adequate for use in the hazard identification and risk assessment. For criteria on the selection of key studies, see ECHA guidance on the evaluation of available information (Chapter R.4) ⁽³⁾.
- If there is more than one adequate study available for a certain information requirement, a study summary should be provided for each of the studies to enable an independent evaluation of the information provided. If the studies show a difference in potency or divergent results, the worst-case value should be used, unless a valid scientific justification is provided for not doing so.
- The information provided must make it possible to assess the risks under the relevant conditions of use. The following information should be taken into account when preparing an application:
 - the final material in which the substance is present (e.g. plastic, cementitious material);
 - the product in which the final material is present and specific conditions affecting migration from such product into the drinking water.
- It is **the responsibility of the applicant to secure any necessary rights to use any third-party intellectual property in their application.** As set out in the terms and conditions of the ECHA submission tool ⁽⁴⁾, when submitting the application, the applicant is required to warrant that they have obtained all the necessary permission, rights or licences to use third-party intellectual property for the purpose of their application. The fact that a study or any other third-party material is available on ECHA's website does not mean that

⁽²⁾ The MTD is the highest dose of the test agent during the bioassay that can be predicted not to alter the animal's normal longevity from effects other than carcinogenicity. Data obtained from a subchronic or other repeated dose toxicity study are used as the basis for determining the MTD.

⁽³⁾ https://echa.europa.eu/documents/10162/17235/information_requirements_r4_en.pdf/d6395ad2-1596-4708-ba86-0136686d205e?t=1323782558175.

⁽⁴⁾ The tool will be accessible from ECHA's website.

this material is not subject to the third party's intellectual property rights and may be freely used without its permission.

1.2.2. General principles on generating new information

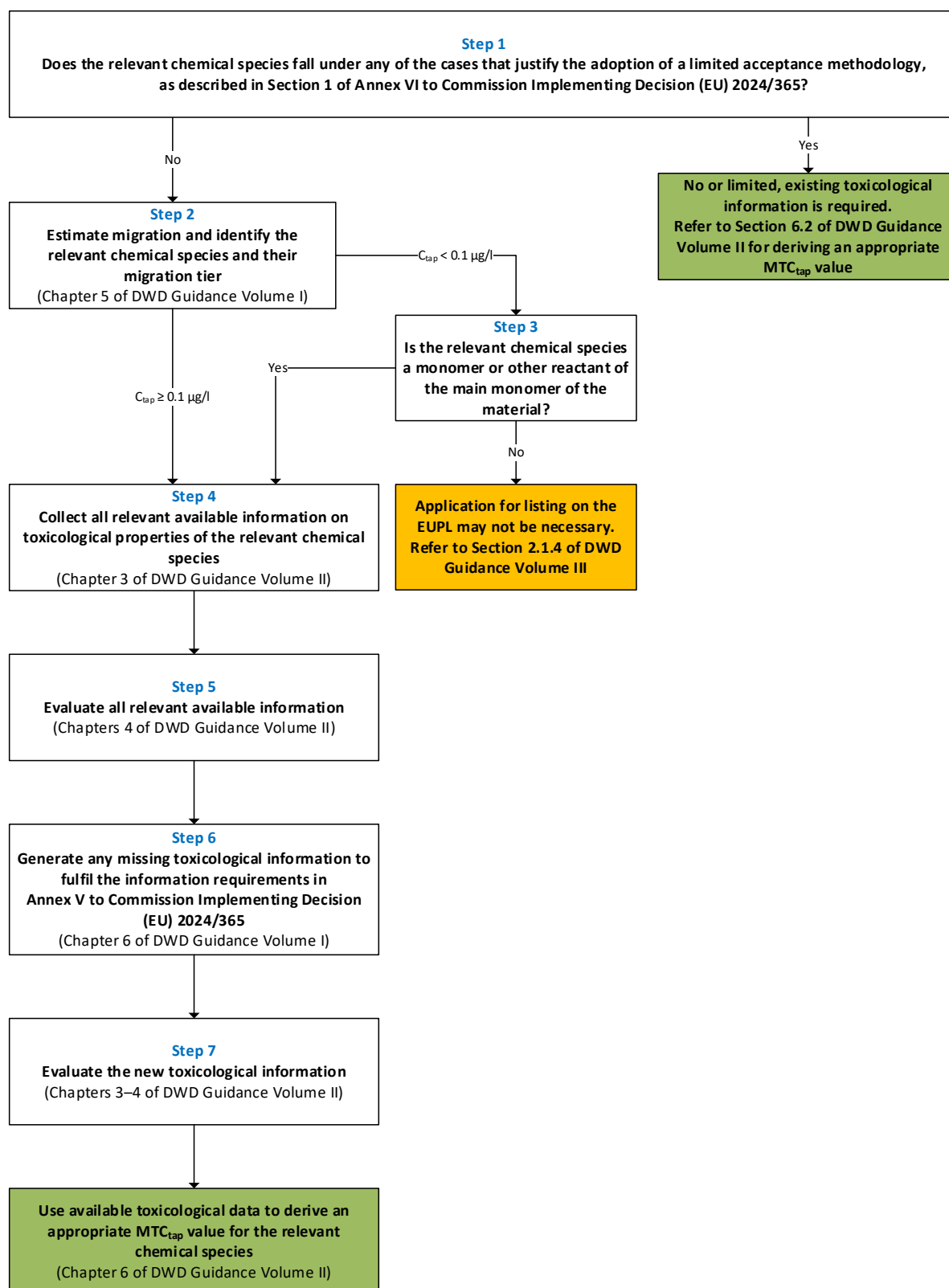
Before generating any new information, in particular new *in vivo* toxicological studies according to the testing methodologies explained in this guidance, the applicants need to collect and evaluate all relevant available information (e.g. experimental data, human data, grouping and read-across, quantitative structure–activity relationships, *in vitro* data). Consideration should also be given to tests already performed/submitted for the purposes of other regulatory programmes (e.g. REACH regulation, Regulation (EU) No 528/2012 on biocidal products (BPR), Regulation (EC) No 1935/2004 on food contact materials (FCM regulation)).

The steps to be followed in evaluating and assessing the information available are provided in DWD Guidance Volume II. If the assessment of the available information concludes that further information needs to be generated to fulfil the information requirements, this should be generated taking into account the principles described in this guidance document.

The relationship between the two guidance documents is presented in Figure 1 and Figure 2.

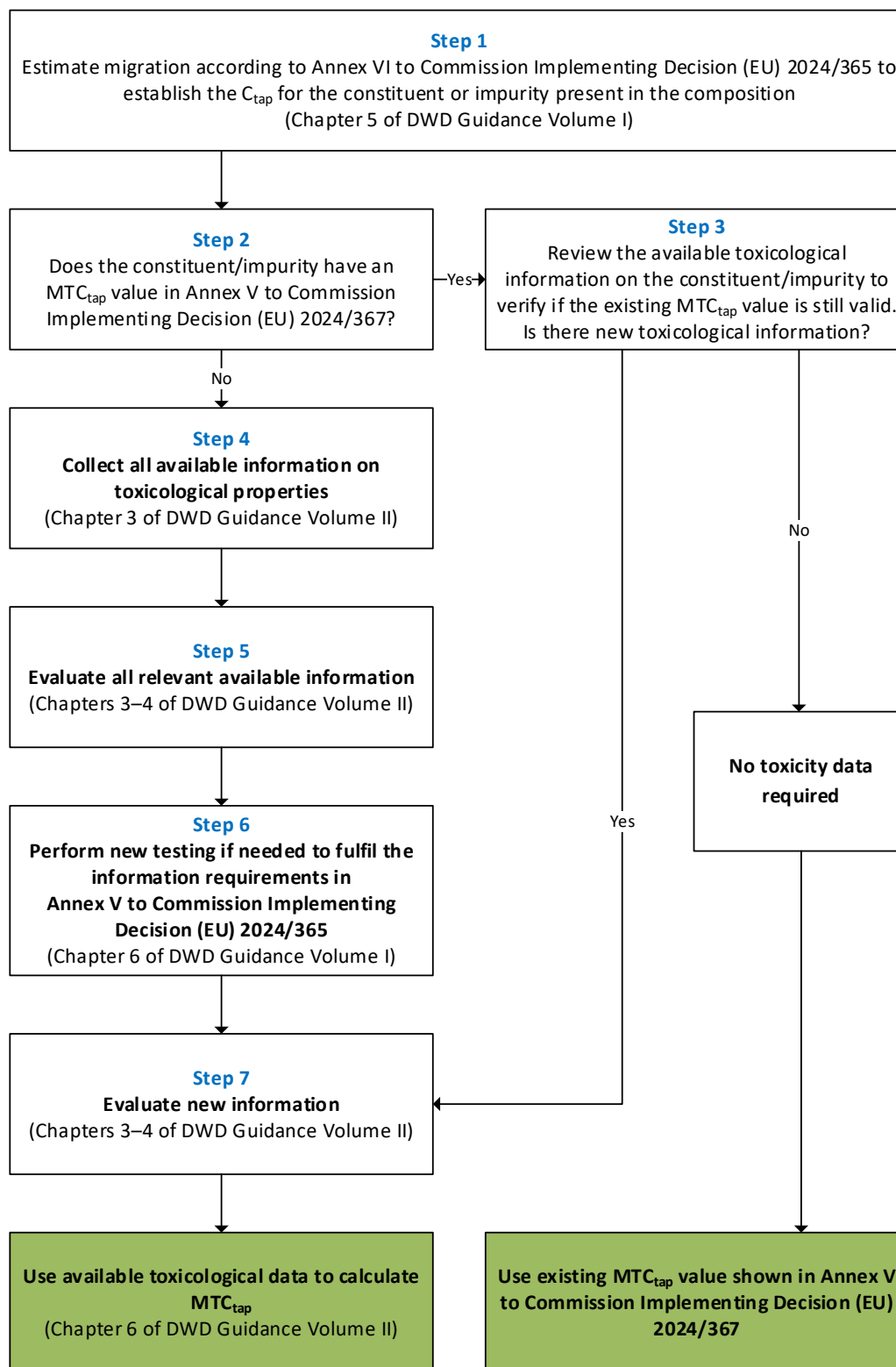
Figure 1 summarises the stepwise approach to be followed for relevant chemical species that are associated with the use of starting substances and organic cementitious constituents, and Figure 2 shows the approach that should be followed for relevant chemical species that are associated with the use of metallic, enamel, ceramic and other inorganic compositions.

Figure 1. Schematic representation of stepwise approach to fulfilling the testing methodology for human health hazard for starting substances and organic cementitious constituents for the purposes of the DWD



NB: C_{tap} , concentration at the tap; MTC_{tap} , maximum tolerable concentration at the tap.

Figure 2. Schematic representation of stepwise approach to fulfilling the testing methodology for human health hazard for metallic, enamel, ceramic and other inorganic compositions for the purposes of the DWD



NB: C_{tap} , concentration at the tap.

1.2.3. General principles of identifying the relevant chemical species

A key aspect in assessing the risks to human health arising from the intentional use of a starting substance, composition or constituent in the manufacture of materials in contact with drinking water is the identification and assessment of the relevant chemical species. Their identification is normally a multistep process that considers:

- the chemical composition of the applied-for entry, including any impurities;
- the way the applied-for entry is used and the reactions it is involved in during manufacture and use of the contact material;
- the migration rate of potentially relevant chemical species into water.

The figures that follow provide an overview of the steps involved for each material type. For ease of presentation, for organic and cementitious materials, the process has been split into three steps; for metallic materials, the process has been split into two steps.

Figure 3. General approach to identifying relevant chemical species for starting substances of organic materials – step 1

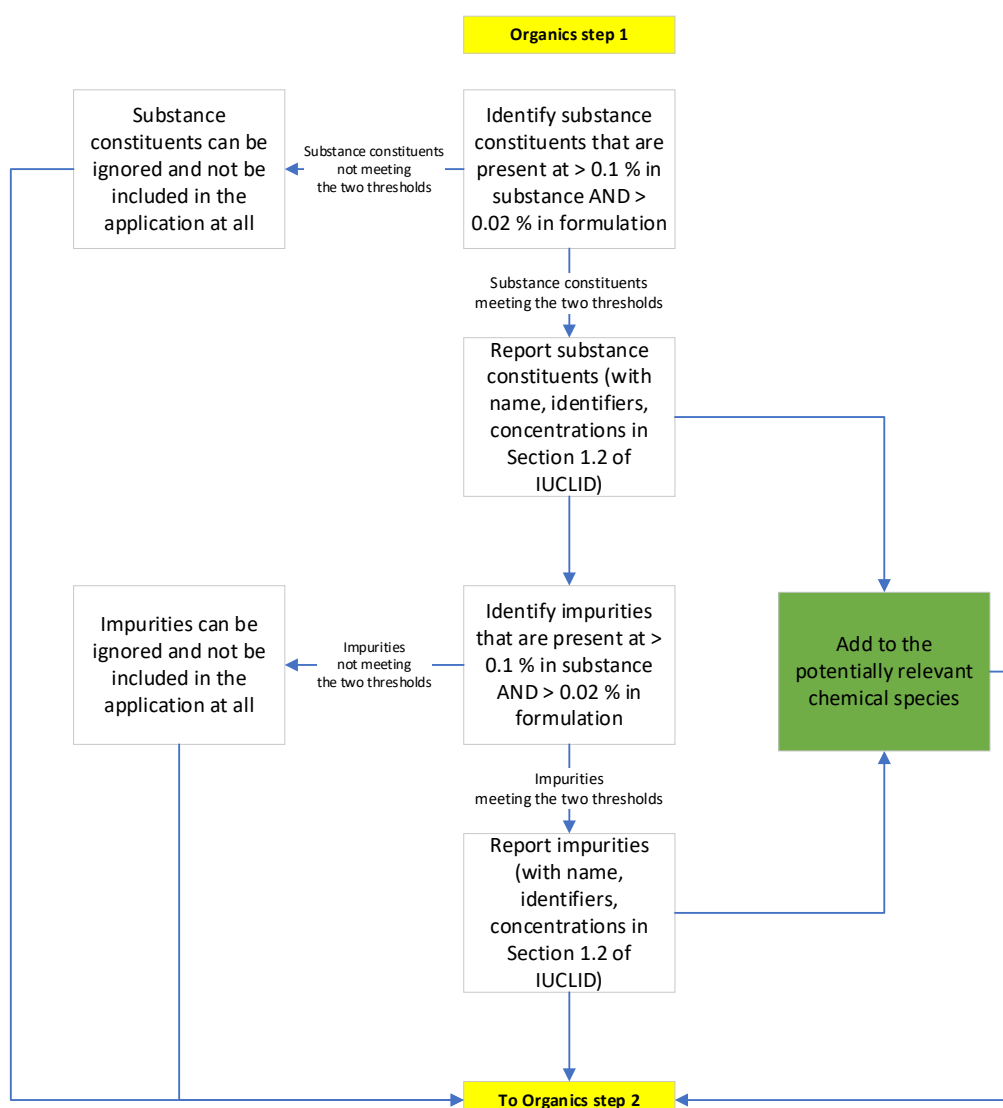
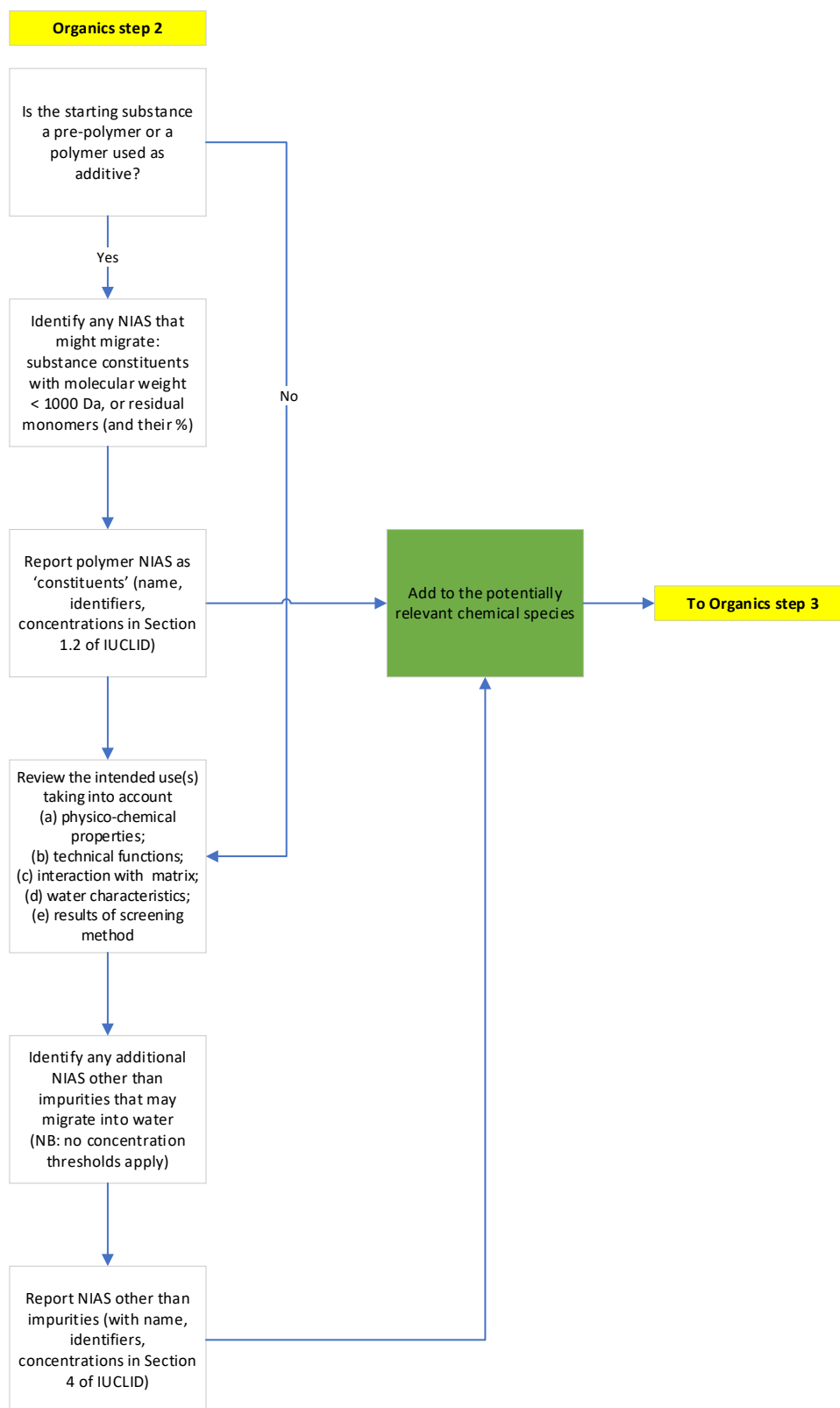
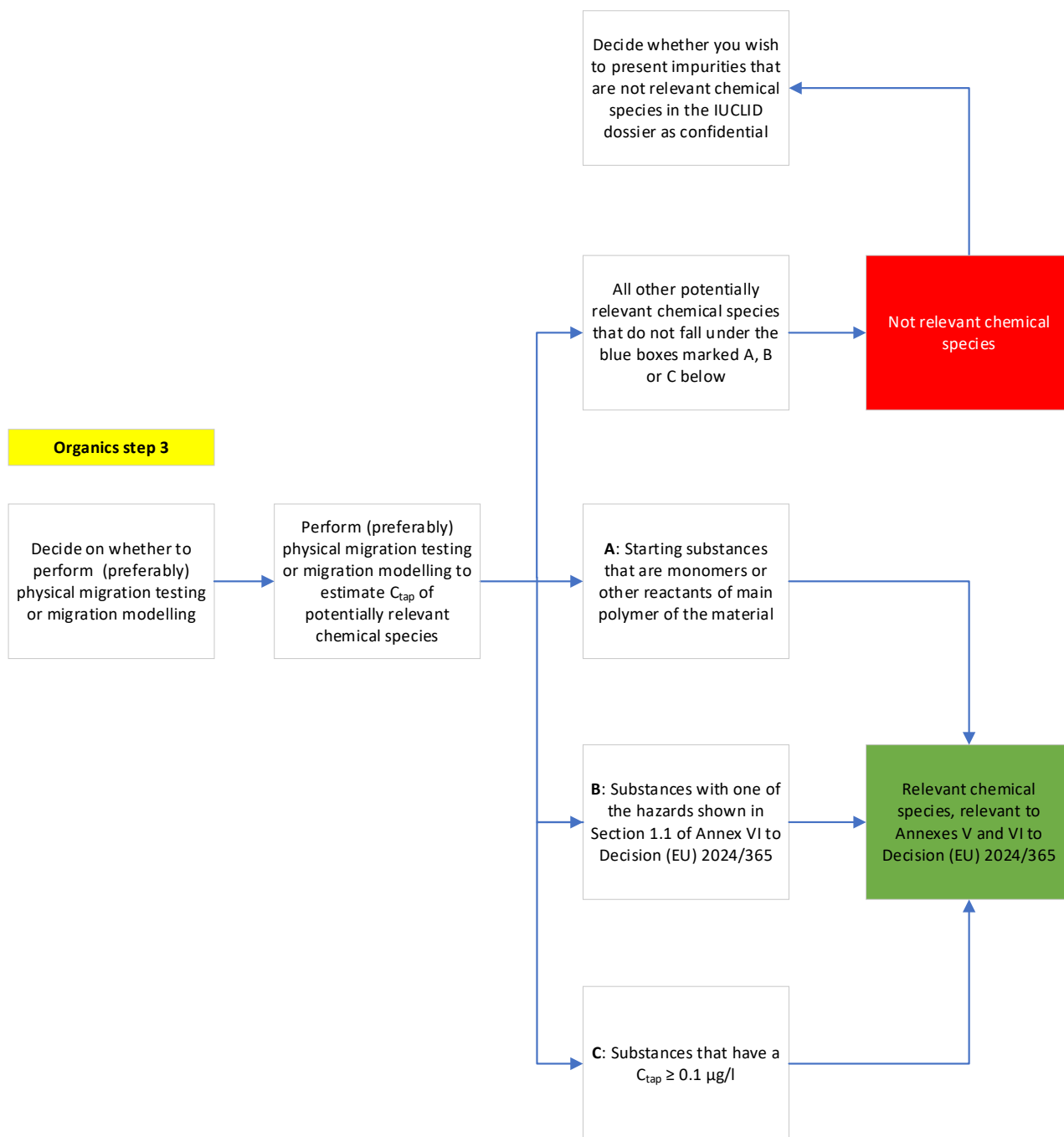


Figure 4. General approach to identifying relevant chemical species for starting substances of organic materials – step 2

NB: NIAS, non-intentionally added species.

Figure 5. General approach to identifying relevant chemical species for starting substances of organic materials – step 3



NB: C_{tap} , concentration at the tap.

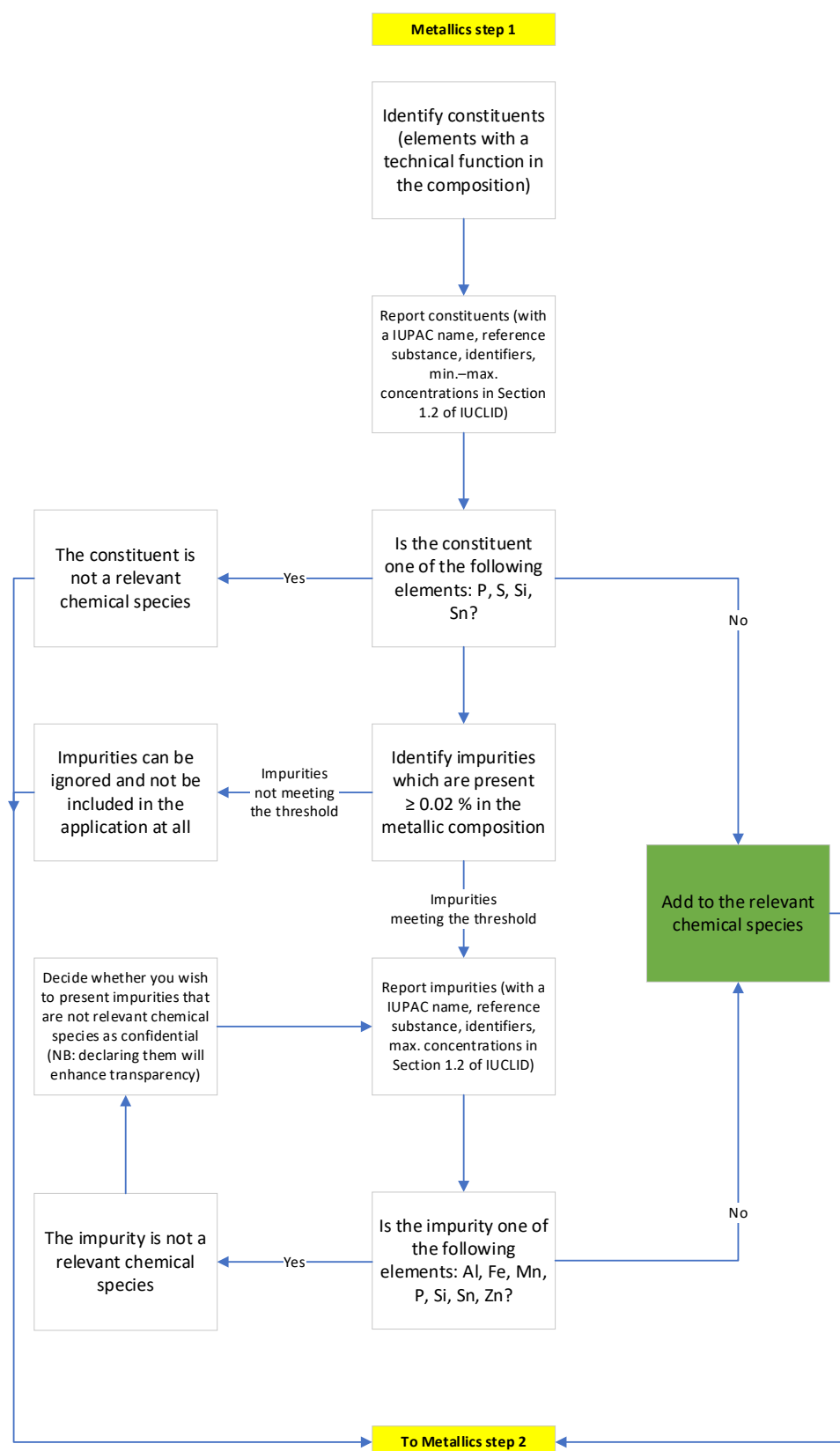
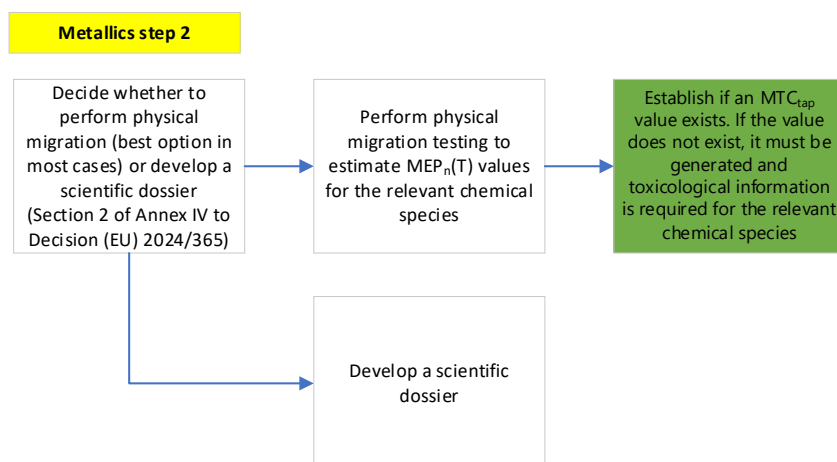
Figure 6. General approach to identifying relevant chemical species for metallic compositions – step 1

Figure 7. General approach to identifying relevant chemical species for metallic compositions – step 2



NB: MEP_n(T), mean of the equivalent pipe concentrations.

Figure 8. General approach to identifying relevant chemical species for organic constituents of cementitious materials – step 1

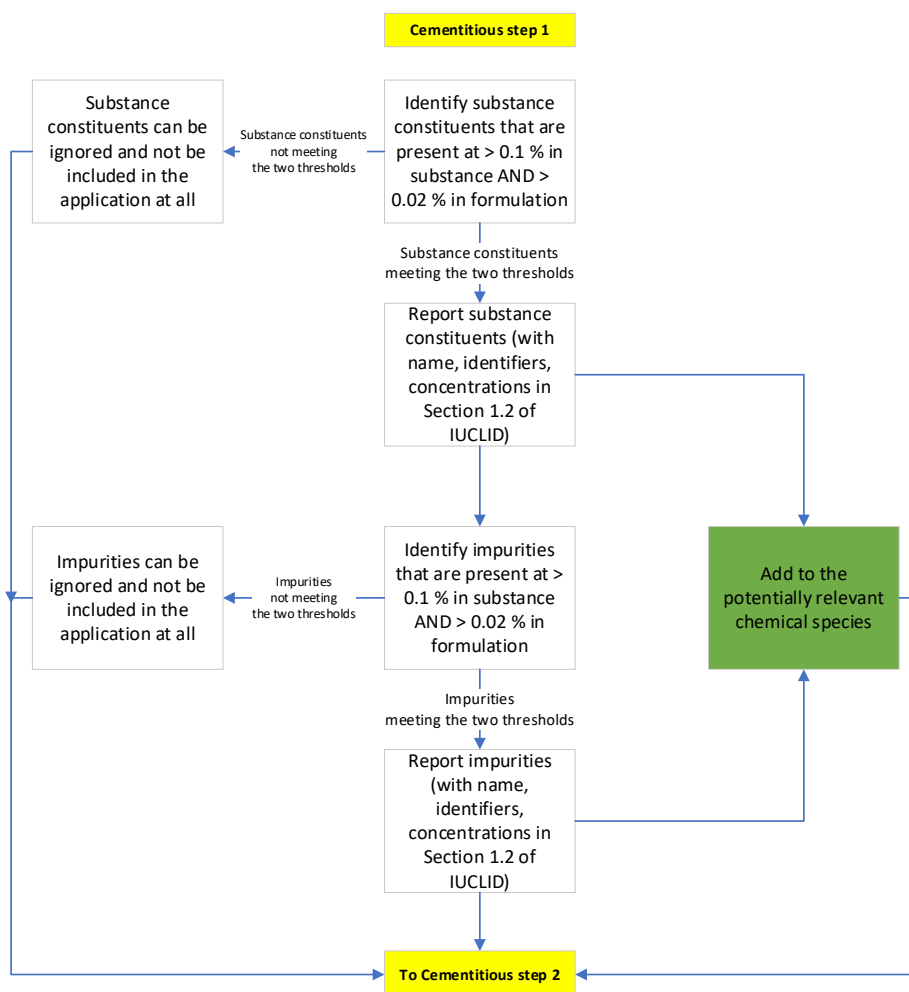
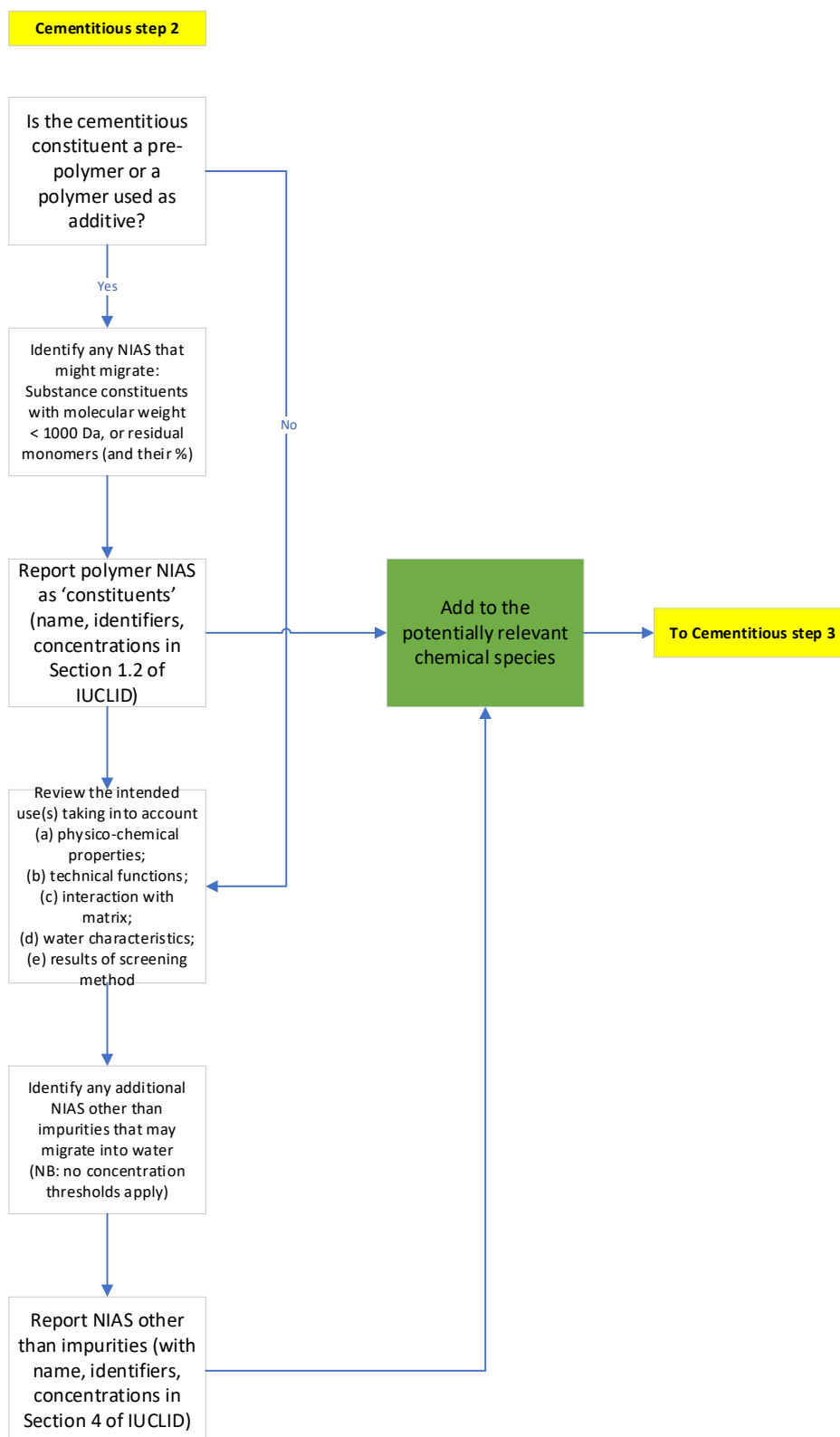
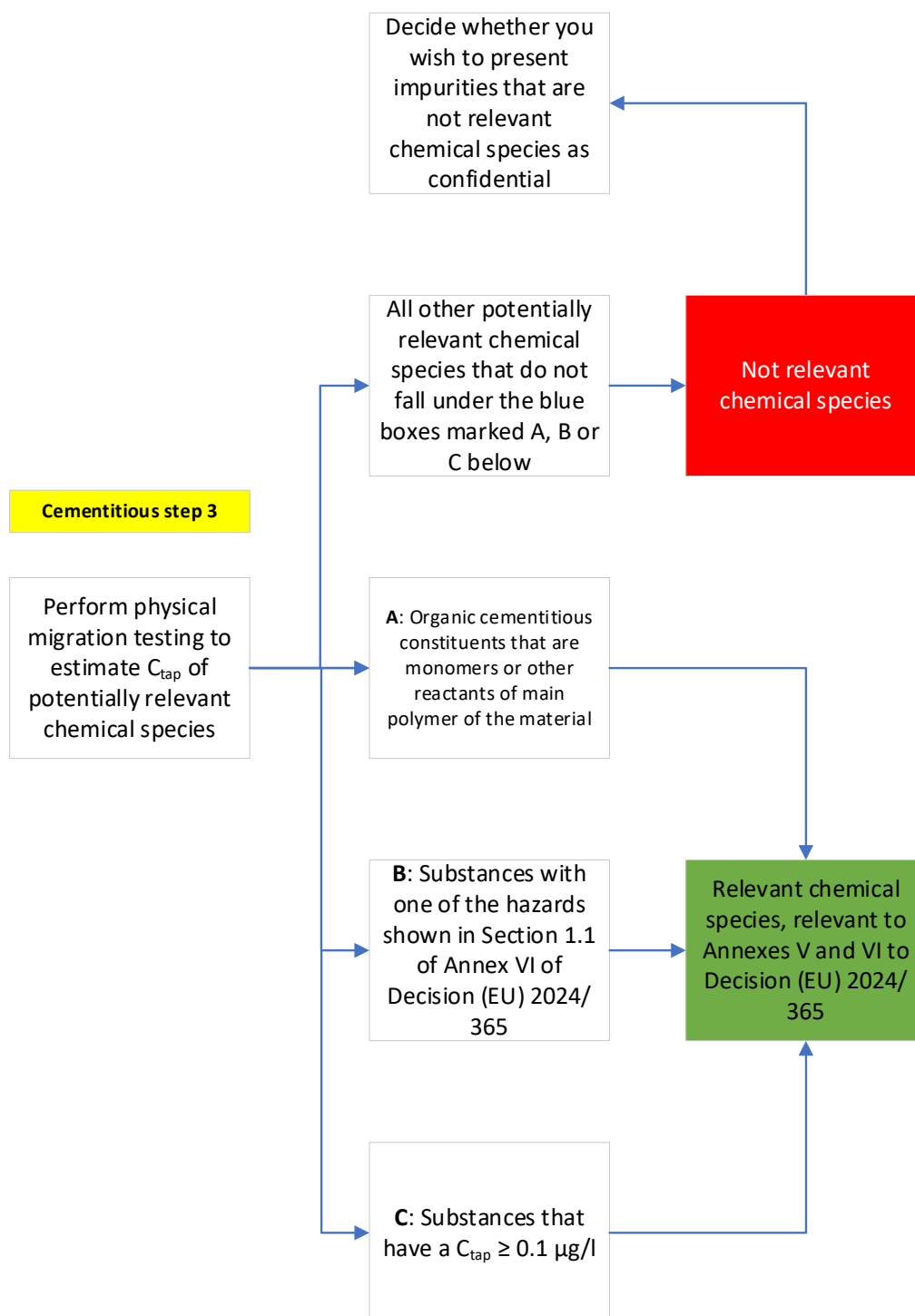


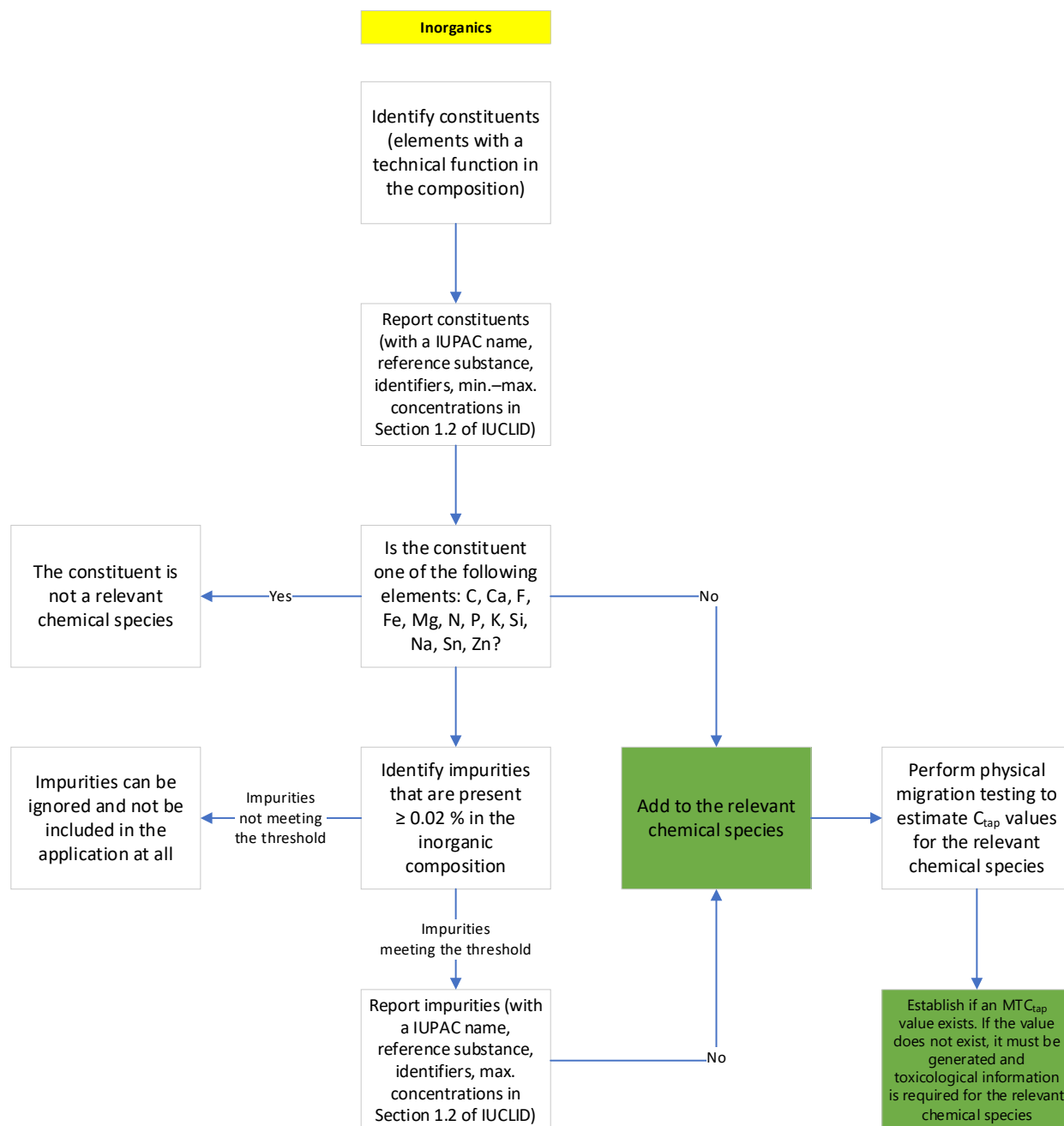
Figure 9. General approach to identifying relevant chemical species for organic constituents of cementitious materials – step 2

NB: NIAS, non-intentionally added species.

Figure 10. General approach to identify relevant chemical species for organic constituents of cementitious materials – step 3



NB: C_{tap} , concentration at the tap.

Figure 11. General approach to identify relevant chemical species for enamel, ceramic and other inorganic compositions

NB: C_{tap} , concentration at the tap.

1.3. General structure of this guidance document

Data requirements are divided into six parts as follows.

- **Identity of the applied-for entry** (Chapter 2): the identity of the applied-for entry and its compositional information together with the purity and relevant impurities for which an application is submitted to ECHA.
- **Intended use of the applied-for entry** (Chapter 3): instructions on how to describe the intended use of the applied-for entry in the materials and products that come into contact with drinking water. In particular, this gives advice on how to define the material type and products in which the starting substance, composition or organic cementitious constituent is or is intended to be present on the EU market and on how to define the technical function and concentration of the substance in the material. The description of the intended use should include information on the processing and internal structure of the material, final material and product, the reactions that occur during processing and the use of the final material and products that may result in the formation of non-intentionally added species (NIAS) other than impurities and details of existing EU- and national-level authorisations of the applied-for entry.
- **Physico-chemical properties** (Chapter 4): different basic physico-chemical properties are required to be reported for different material types. In the case of organic and cementitious materials, the basic physical and chemical properties of a starting substance or organic cementitious constituent, such as physical state, vapour pressure, solubility in water and the *n*-octanol–water partition coefficient. Thermal stability and chemical reactivity are also of relevance but are expected to be presented in the context of the forming of NIAS. For metallic, enamel, ceramic and other inorganic compositions, only colour, melting/freezing point and density are relevant. For relevant chemical species other than a starting substance or an organic cementitious constituent (in order words, for NIAS) water solubility and partition coefficient are required.
- **Data on migration of the applied-for entry and other potentially relevant chemical species** (Chapter 5): the tests used to estimate the migration of a substance from the relevant material types (organic, cementitious, metallic, enamel, ceramic or other inorganic). Alternative methods such as modelling are also described together with the boundaries of their applicability and specific requirements such as diffusion and partition coefficient (see also physico-chemical properties above), material relative density (specific gravity), etc.
- **Toxicological data for the relevant chemical species** (Chapter 6): the toxicological testing methodologies to be used according to the level of migration of the relevant chemical species into drinking water.
- **Testing for nanoforms** (Chapter 7): additional information required for applied-for entries in nanoform, intentionally used to manufacture materials coming into contact with drinking water.

1.4. Use of additional guidance documents

In addition to this guidance, REACH guidance and BPR guidance provided by ECHA represent major sources of guidance. Where relevant, those guidance sources should also be taken into account for the evaluation of substances falling under the scope of Article 11 of the DWD. The use of BPR guidance is recommended for some testing methodologies (see Sections 6.10.1, 6.10.2 and 6.10.3 of this guidance) with the intention of facilitating a harmonised approach. ECHA guidance can be obtained from the ECHA website:

- REACH guidance (<https://echa.europa.eu/guidance-documents/guidance-on-reach>);
- BPR guidance (<https://echa.europa.eu/regulations/biocidal-products-regulation/understanding-bpr>);
- ECHA guidance on the application of Regulation (EC) No 1272/2008 on classification, labelling and packaging (CLP regulation) criteria (<https://echa.europa.eu/guidance-documents/guidance-on-clp>).

1.5. Intellectual property rights and data sharing

Users of this guidance are also reminded that information that is relevant to the testing required under the DWD may be available in relevant registrations under the REACH regulation. The issue of access rights to information originating from REACH registration is considered appropriate. It is the responsibility of the applicant to secure any necessary rights to use any third-party intellectual property in their application. As set out in the terms and conditions in the ECHA submission tool, when submitting the application, the applicant is required to warrant that they have obtained all the necessary permission, rights or licences to use the third-party intellectual property for the purpose of their application. The fact that a study or any other third-party material is available on ECHA's website does not mean that this material is not subject to the third party's intellectual property rights or that it may be freely used without their permission.

The DWD neither makes provision for data sharing among potential applicants nor does it mandate joint submissions. Collaboration among economic operators with an interest in the same starting substance, composition or constituent is strongly encouraged, in particular with the aim of avoiding unnecessary animal and other testing; however, the DWD imposes no obligation on economic operators to collaborate in such a way, and any interactions among them are their responsibility. ECHA cannot intervene in the event of disputes over data sharing.

For advice on data sharing, applicants may want to consider the provisions established by ECHA (in accordance with the REACH regulation) ⁽⁵⁾, ECHA's REACH guidance on data sharing ⁽⁶⁾ and the explanatory note clarifying which chapters are of relevance to applicants under the BPR ⁽⁷⁾.

1.6. Alignment between a DWD notification of intention and a DWD application

As described in DWD Guidance Volume IV on the contents of a notification or intention, certain essential information must be identical in both notification of intention and application, specifically:

- the type of intended application (review, new, removal);
- the material type;
- the EUPL number, in the case of applications for review or removal;
- the substance identifiers (e.g. International Union of Pure and Applied Chemistry (IUPAC) name, European Community (EC) number, Chemical Abstracts Service (CAS) number, etc.).

⁽⁵⁾ <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A02006R1907-20220501>.

⁽⁶⁾ https://echa.europa.eu/documents/10162/2324906/guidance_on_data_sharing_en.pdf.

⁽⁷⁾ <https://echa.europa.eu/web/guest/guidance-documents/guidance-on-biocides-legislation>.

2. Identity of the applied-for entry

The applied-for entry needs to be identified by a combination of the required identification parameters:

- the EUPL number (if available);
- the EUPL name (if available and appropriate) and/or the IUPAC name and/or any other names and other available and appropriate identifiers, for example CAS number, EC number;
- information related to molecular and structural formula or crystal structure;
- the compositional information, that is, chemical composition (this is a meaning of the term 'composition' different from that used in the DWD to refer to metallic, enamel, ceramic and other inorganic materials).

Testing methodologies for applied-for entry identity vary among substances and depend on whether an application is for a starting substance for organic materials, an organic constituent of cementitious materials, a composition of a metallic material or a composition of an enamel, ceramic or other inorganic material. Material types may be divided into two groups within which similar testing methodologies apply:

- starting substances and organic cementitious constituents;
- metallic, enamel, ceramic or other inorganic compositions.

Starting substances of organic materials and organic cementitious constituents can be divided into two main groups.

- **Well-defined substances.** These are starting substances or organic cementitious constituents with a defined qualitative and quantitative composition that can be sufficiently identified based on the abovementioned identification parameters in this chapter. As with the REACH regulation, the '≥ 80 %' rule for mono-constituent substances as well as the '80-10 %' rule (≥ 10 % and < 80 % weight by weight (w/w)) for multi-constituent substances should be applied to identify and name a well-defined substance according to its main constituent(s) (see the *Guidance for identification and naming of substances under REACH and CLP* referenced below). While some variability of the reported composition for a well-defined substance is expected, the concentration range of the constituents should be neither large nor be poorly predictable.
- **Substances of unknown or variable composition, complex reaction products or biological materials (UVCBs).** These starting substances or organic cementitious constituents cannot be sufficiently identified by the required identification parameters because of one or more of the following:
 - the number of constituents is relatively large;
 - the composition is, in significant part, unknown;
 - the variability of composition in UVCBs is relatively large and/or poorly predictable.

Polymers ⁽⁸⁾ fall within the definition of UVCBs. The information required for the identification of UVCBs is therefore also relevant for polymers. However, owing to their chemical nature, in particular the molecular weight distribution and the nature of the backbone of their constituents, additional polymer-specific information is also required for their identification. Specific considerations are therefore provided in this document for substances obtained from polymerisation-type reactions. Further information on the definition of polymer is available in Chapter 2 of the *Guidance for monomers and polymers* under the REACH and CLP regulations ⁽⁹⁾.

Metallic, enamel, ceramic and other inorganic compositions are characterised by a series of composition constituents (elements in the case of metallic compositions; substances in the case of enamel, ceramic and other inorganic compositions), typically with a minimum and maximum concentration. Metallic compositions may also have specified impurities (elements) accompanied by a maximum concentration.

Applications made by economic operators must cover a single starting substance, composition, organic cementitious constituent, nanoform or entry in the EUPL. However, an application from a relevant authority may cover several starting substances, compositions, organic cementitious constituents, nanoforms or entries, according to Article 3(6) of Commission Delegated Regulation (EU) 2024/369. More detailed guidance on individual versus group entries is provided in Section 2.5 of this guidance.

Independently of the material type, information submitted to ECHA must be sufficient to identify the applied-for entry depending on the requirements of the specific material type and to characterise it accordingly. An applied-for entry must be fully identified by its chemical composition, that is, the chemical identity and the content of each of its chemical constituents ⁽¹⁰⁾.

Further details on the identification of substances, in particular on starting substances and organic cementitious constituents is available in Chapter 4 of the *Guidance for identification and naming of substances under REACH and CLP* ⁽¹¹⁾. The principles in that guidance should be applied for the purpose of identification of starting substances and organic cementitious constituents.

2.1. Name or any other identifier (point 1.1 in Annex I to Commission Implementing Decision (EU) 2024/365)

2.1.1. EUPL number (point 1.1.5 in Annex I to Commission Implementing Decision (EU) 2024/365)

For an applied-for entry already listed in the EUPL, its EUPL number must be provided in the application for its renewal, updating or removal from the EUPL. For members of an existing group entry in the EUPL, the EUPL number of that group entry must be provided.

⁽⁸⁾ See Article 2(5) of Commission Implementing Decision (EU) 2024/367. The definition of polymer in the DWD is the same as under the REACH regulation. This definition is also provided in the 'Glossary of key terms' in this guidance.

⁽⁹⁾ <https://echa.europa.eu/-/guidance-for-monomers-and-polymers>.

⁽¹⁰⁾ 'Chemical constituent' as used here does not refer to a starting substance for admixtures for cementitious materials.

⁽¹¹⁾ <https://echa.europa.eu/-/guidance-for-identification-and-naming-of-substances-under-reach-and-clp>

2.1.2. Chemical name of entries (point 1.1.1 in Annex I to Commission Implementing Decision (EU) 2024/365)

For starting substances and organic cementitious constituents (organic and cementitious materials). If an EUPL entry specifically corresponds to the substance, the EUPL name should be used as the chemical name and be provided in the 'IUPAC name' field of the 'Reference substance' for the applied-for entry. If such an EUPL entry is not available but an EUPL group entry covering the substance exists, the EUPL name should not be used as the chemical name of the substance. Instead, the chemical name of the substance according to the naming conventions in the *Guidance for identification and naming of substances under REACH and CLP*⁽¹²⁾ should be provided. When applying these conventions, attention should be paid to establishing the correct substance type (i.e. mono-constituent substance, multi-constituent substance or UVCB).

For applied-for starting substances and organic cementitious constituents (organic and cementitious materials) of polymeric nature. Concerning the reporting of the EUPL name, the abovementioned rules for starting substances and organic cementitious constituents (organic and cementitious materials) also apply to substances of polymeric nature. However, it should be noted that EUPL entries for substances of polymeric nature typically do not differentiate between substances meeting the polymer definition and their non-polymer equivalents. Given that polymers and their non-polymer equivalents are considered different substances, EUPL entries for substances of polymeric nature generally refer to group entries. In this situation, the EUPL name should not be used to define the applied-for substance. Instead, a name following the conventions specified below should be provided.

UVCBs obtained from a polymerisation reaction but not meeting the 'polymer' definition⁽¹³⁾, including certain pre-polymers, should be named following the same conventions as for other UVCBs, adapted as necessary. These substances are typically named as 'oligomerisation products of' the reactants used for their manufacture. Further details of the naming conventions can be found in the introductory text of the European Commission document *No-Longer Polymer List – Version 3*⁽¹⁴⁾.

Substances meeting the polymer definition, irrespective of whether they are considered pre-polymers or polymers in the context of the DWD, should generally be named following the convention 'polymerisation products of' the reactants used for their manufacture, including the monomers and any other reactant ending up in the polymer structure. For polymers obtained from more than one monomer, the kind of arrangement in which the different monomeric units are related to each other should be qualified by one of the following terms: alternating, periodic, random, statistical, block or graft. If it is not possible to establish the arrangement, the polymerisation type should be qualified as copolymerisation. For non-linear polymers, the backbone type, such as 'branched', should furthermore be qualified. Further information on the qualifiers is available in Chapter 14.2.1 of the *IUPAC Compendium of Polymer Terminology and Nomenclature – IUPAC Recommendations 2008*⁽¹⁵⁾. If reactions other than polymerisations need to be mentioned in the name, as it may be the case for post-modified polymers, the naming conventions for UVCBs should also be considered to refer to these reactions.

⁽¹²⁾ <https://echa.europa.eu/-/guidance-for-identification-and-naming-of-substances-under-reach-and-clp>

⁽¹³⁾ The definition of 'polymer' is specified in Article 2(5) of Commission Implementing Decision (EU) 2024/367. This definition is also provided in the 'Glossary of key terms' in this Guidance document.

⁽¹⁴⁾ See JRC document number JRC36863 (<https://publications.jrc.ec.europa.eu/repository/>).

⁽¹⁵⁾ The compendium (also known as the 'Purple book') is available online (<https://iupac.org/what-we-do/books/purplebook/>).

Examples of polymer names:

- polymerisation products of ethanol and ethylene oxide;
- alternating polymerisation products of maleic anhydride and styrene;
- branched random polymerisation products of divinylbenzene and styrene;
- copolymerisation products of but-1-ene and pent-1-ene;
- branched polymerisation products of ethylene;
- block polymerisation products of 1-(chloromethyl)-4-vinylbenzene and 2-methylpropene, graft polymerisation products with methyl methacrylate.

The name of a substance of polymeric nature should be representative of its composition. Additional specifications, such as the specific polymerisation reaction type, may for instance occasionally be needed at the level of the name to avoid any ambiguity about the substance actually applied for.

For metallic, enamel, ceramic or other inorganic compositions. Point 1.1.1 in Annex I to Commission Implementing Decision (EU) 2024/365 is not applicable to compositions. However, applicants must be aware of the following requirements when completing their application in the IUCLID software (and specifically Section 1.1 'Identification'), which are for technical reasons:

- the applied-for entry must be assigned a 'reference substance';
- the name of the applied-for entry (e.g. the name of an alloy or the name of an enamel or ceramic composition) should be provided in the 'IUPAC name' field of the 'reference substance'.

2.1.3. Other names and identifiers (points 1.1.2–1.1.4 in Annex I to Commission Implementing Decision (EU) 2024/365)

The following points apply to all applied-for entries.

- The EC number and EC name from the following inventories must be provided for the applied-for entry if available and appropriate:
 - the *European Inventory of Existing Commercial Chemical Substances*;
 - the *European List of New or Notified Chemical Substances*;
 - the *No-Longer Polymer list*.

If the applied-for entry is not listed in any of these inventories, the number (currently referenced as 'list number' ⁽¹⁶⁾) and associated name assigned by ECHA in the context of the REACH regulation must be provided if available and appropriate.

⁽¹⁶⁾ Further information on list numbers is available in Chapter 3.2.2 of the ECHA *Guidance for identification and naming of substances under REACH and CLP* that is available at <https://echa.europa.eu/-/guidance-for-identification-and-naming-of-substances-under-reach-and-clp>.

- For those substances that are the subject of harmonised classification, the name specified in Part 3 of Annex VI to the CLP regulation should be specified.
- The International Organization for Standardization (ISO) common name of the substance, if available, should be included. ECHA's inventory database may be used as a source for (common) names. Generally known names, trade names, abbreviations, etc., should be provided.
- The CAS number and CAS name must be provided if available and appropriate. For substances of polymeric nature, it should be noted that the CAS Registry does not take the DWD polymer definition into account. Therefore, the CAS entries for this type of substance do normally not refer to one specific substance and are to be treated as secondary identifiers.

2.1.4. Identifying compositions (points 1.1.6–1.1.12 in Annex I to Commission Implementing Decision (EU) 2024/365)

For metallic compositions, the following must be provided.

- The name of the metallic composition that is applied for. The name could take several forms, depending on the characteristics, complexity and standardisation of the metallic composition and should be unique in clearly distinguishing the composition from any other metallic composition in the EUPL. This may require the inclusion of references to specific grades, standards, etc. Examples of names, as shown in the first EUPL, are:
 - the name of a specific grade of an alloy, for instance 'Ti1 according to ISO 23515:2022';
 - the common name of metallic composition together with any relevant composition standards that it adheres to, for instance 'carbon steel according to EN 10025/EN 10213/EN 10222';
 - the normative designation of a metallic composition (alloys) where this exists, for instance 'CuSi4Zn9MnP';
 - the normative and standardised designations of a metallic composition where both exist, for instance 'CW702R (CuZn20Al2As)'.
- Any available reference to a relevant alloy designation according to material number and/or symbol (EN 1412, ISO 1190-1, CEN/TS 13388) and correspondence with European standards (EN) must be provided, where applicable.
- Reference to any other systematic notation must be provided if it exists.
- The identity of the relevant metallic composition category must be provided.

Where the applied-for metallic composition belongs to an existing category ⁽¹⁷⁾, applicants need to identify only the corresponding metallic composition category. Exceptionally, if the concentration range of any of the constituents or the maximum concentration of any of the impurities of the metallic composition exceed the corresponding ranges or maximum values previously accepted for that existing category, the applicant must also propose and justify new concentration ranges and maximum values for the constituents and the impurities of the metallic composition category as necessary.

⁽¹⁷⁾ As identified in Section 9.2 of Annex I to this guidance document; to be published on the ECHA website.

If the inclusion of the metallic composition in the EUPL would establish a new metallic composition category, the applicant must:

- describe the new metallic composition category by providing:
 - the identities of category constituents and their respective concentration ranges (expressed in % w/w);
 - the identities of category impurities present above 0.02 % w/w and their respective maximum concentrations (expressed in % w/w);
- define the reference material for the new metallic composition category by providing (see also Section 5.1.4.3):
 - the identities of constituents of the category's reference material and their respective concentration ranges (expressed in % w/w)
 - the identities of impurities of the category's reference material that are present above 0.02 % w/w and their respective concentration ranges (expressed in % by weight).

For enamel, ceramic or other inorganic compositions, the material category (e.g. ceramics) and material subcategory (e.g. hard ferrite ceramics) and the description of the composition must be provided. If the applied-for entry is not an enamel or a ceramic material, then it should be described as 'other inorganic'. Glasses fall under the 'enamels' category.

2.2. Information related to molecular and structural formula or crystal structure (point 1.2 in Annex I to Commission Implementing Decision (EU) 2024/365)

2.2.1. Molecular and structural formula; description of crystal structures (points 1.2.1, 1.2.2 and 1.5.4 in Annex I to Commission Implementing Decision (EU) 2024/365)

For starting substances and organic cementitious constituents (organic and cementitious materials). The molecular formula must be provided in accordance with the traditional Hill system and, in addition, in accordance with the CAS system if it differs from the Hill system formula. A structural formula for the substance, as well as the international chemical identifier (InChI) and simplified molecular input line entry system (SMILES) notation must be provided if available and appropriate. Representative molecular and structural formulae must also be determined for UVCBs as far as possible. For substances of polymeric nature, a structural representation of the polymer, including the structure of the monomer unit(s) and any other reactant and their relative order in the polymer backbone, must be provided.

Information on optical activity and typical ratio of (stereo)isomers, if available, must also be provided.

For compositions (metallic, enamel, ceramic and other inorganic materials). A description of crystal structure(s) including crystalline phases, if available, should be provided.

2.2.2. Molecular weight or molecular weight range (points 1.2.3 and 1.5.5–1.5.7 in Annex I to Commission Implementing Decision (EU) 2024/365)

For starting substances and organic cementitious constituents (organic and cementitious materials). The molar mass (g/mol) must be provided. For applied-for (pre-)polymers, the number average molecular mass (M_n) and the molecular mass range are required. Furthermore, the molecular weight distribution of the polymerised part must be documented (see also Section 2.3.3 of this guidance).

For compositions (metallic, enamel, ceramic and other inorganic materials). Not applicable.

2.3. Chemical characterisation (point 1.3 in Annex I to Commission Implementing Decision (EU) 2024/365)

2.3.1. Composition, purity and identity of any impurities (points 1.3.1–1.3.3 in Annex I to Commission Implementing Decision (EU) 2024/365)

For starting substances and organic cementitious constituents (organic and cementitious materials). The degree of purity (point 1.3.1) as well as the identity and concentration levels of the substance constituents (point 1.3.2) and impurities (point 1.3.3) are needed for identification. Therefore, the following information must be provided for each substance constituent (including any additive used to preserve the stability of the starting substance or organic cementitious constituent) and impurity present both at a concentration above 0.02 % w/w in the formulation of the final organic material or related to the cement content for cementitious materials and at concentration ≥ 0.1 % w/w in the substance:

- common name and chemical name in conformity with Section 2.1.1 of this guidance;
- CAS and EC numbers, if available, in conformity with Section 2.1.3 of this guidance;
- molecular and structural formula;
- molar mass;
- concentration values (minimum, maximum and typical concentrations expressed in % w/w or %/cement content for cementitious materials).

It should be noted that any difference in the structural formula is indicative of the existence of different constituents. Accordingly, different isomers refer to different constituents and should normally be reported separately in the composition.

It is recognised that, under the specific process conditions used to manufacture a given substance, some variations in the concentrations of the constituents/impurities/additives are unavoidable from one batch to another or, in the case of a continuous manufacturing process, over time. The concentration values to be reported should be solely representative of the variations observed under the process conditions used. The range should therefore not be overly wide.

As explained in the *Guidance for identification and naming of substances under REACH and CLP*⁽¹⁸⁾, for UVCBs no differentiation is made between constituents and impurities. This principle has some implications for the formatting of the information to be provided as part of an application. It does, however, not affect the level of detail to which the composition of an applied-

⁽¹⁸⁾ <https://echa.europa.eu/-/guidance-for-identification-and-naming-of-substances-under-reach-and-clp>

for UVCB should be resolved in comparison with well-defined substances. In practice, 'impurities' in UVCBs will be reported as constituents. The composition of UVCBs will be presented as consisting of constituents and any additive(s) eventually used to preserve the stability of the starting substance or organic cementitious constituent. All these UVCB constituents and additives present both at a concentration above 0.02 % w/w in the formulation of the final organic material or related to the cement content for cementitious materials and at a concentration of 0.1 % w/w or above in the substance must be identified and quantified.

For compositions (metallic, enamel, ceramic and other inorganic materials). The following information must be provided:

For each individual constituent:

- chemical name in conformity with Section 2.1.2 of this guidance;
- CAS and EC number, if available, in conformity with Section 2.1.3 of this guidance;
- concentration ranges (minimum and maximum concentrations expressed in % w/w).

For each impurity present at a concentration above 0.02 % w/w:

- chemical name in conformity with Section 2.1.12 of this guidance;
- CAS and EC numbers, if available, in conformity with Section 2.1.3 of this guidance;
- maximum concentrations in the composition (expressed in % w/w).
- in addition, **for enamel, ceramic and other inorganic compositions only**, maximum content of cadmium and lead (expressed in % w/w).

The applicant may give any other information about the composition they consider relevant, such as an explanation of the chemical composition and ratios between specific alloying elements that affect the composition's behaviour.

For applied-for entries whose compositions cannot be technically determined as precisely as required (e.g. certain UVCBs). The applicant must provide a robust scientific justification why the composition cannot be determined precisely. However, the chemical composition and the identity of the substance constituents should still be given as far as known. Substance constituents which cannot be identified individually should be described as constituent groups based on chemical nature. In this case, each group should be identified by at least a chemical name describing the chemical nature of the constituents as well as the typical concentration and concentration range. In addition, if available, molecular and structural information should be provided. The fraction of unknown substance constituents should be as low as possible and the unknown substance constituents should be identified as far as possible by a generic description of their chemical nature.

For all UVCBs. The manufacturing process must always be described. This information is necessary to circumvent the limitations of identifying UVCBs by their composition only. As indicated in Chapter 4 of the *Guidance for identification and naming of substances under REACH and CLP*, the manufacturing process description is a main identifier for UVCBs. The description generally consists of the following:

- the identity of the starting materials and their ratios;
- a description of the relevant manufacturing steps in the order they occur; if chemical reactions take place in a step, the reaction type and the chemical structures obtained from the reaction should be specified;
- for each step, the relevant operating parameters applied to control the composition (e.g. temperature, pressure, solvent, catalysis type);
- the methods of extraction and purification (if relevant).

The reader is reminded that Article 10(2)(b) of Commission Delegated Regulation (EU) 2024/369 prescribes that information on the manufacturing process of a starting substance, composition or organic cementitious constituent or the manufacturing process in which it is involved should be considered confidential and may not be made publicly available.

For applied-for starting substances and organic cementitious constituents (organic and cementitious materials) that are polymers or pre-polymers. The abovementioned requirements for starting substances and organic cementitious constituents (organic and cementitious materials) and for UVCBs also apply to (pre-)polymers. In addition, the following information must be provided.

- The minimum, maximum and typical concentrations of each residual monomer used to manufacture the polymer, irrespective of their concentration level in the polymer/pre-polymer. For each monomer to be reported in the composition, a remark should be made to flag the role of the constituent as a monomer.
- The constituents contributing to the polymerised part of molecular weight below 1 000 Da must always be specified and be reported separately from any other reported constituents. For those constituents below 1 000 Da, the minimum, maximum and typical concentrations should also be provided.

2.3.2. Composition of an applied-for entry that is supported by multiple economic operators

Applicants are reminded that, if an application is submitted on behalf of several economic operators (e.g. more than one manufacturer of a starting substance), the chemical characterisation that is declared in the application must adequately cover and be representative of the requirements of all economic operators covered ⁽¹⁹⁾. The application may be submitted by, for example, a trade association, a consortium or a consultant in order to cover the same starting substance, composition or constituent from these economic operators.

If several economic operators have an interest in applying to ECHA for the same starting substance, composition or constituent, and the applied-for entry is regarded as the same in the context of DWD, an approach analogous to the 'one substance, one registration' principle under the REACH regulation can be implemented by defining the substance identity profile (SIP). The SIP is expected to specify the boundaries of the applied-for entry that the applicants agreed to cover with the data jointly submitted. In the case of an application for a composition, the identities and concentration ranges of the constituents of the applied-for composition together

⁽¹⁹⁾ The approach to be followed resembles that for REACH regulation registration in which the boundary composition for the registered substance, including constituents or groups of constituents, impurities and additives, as well as their concentrations, is made on behalf of all member registrants. If a large variety of compositions exists, or if compositions have different classifications, you should report more than one boundary composition. The *Guidance for identification and naming of substances under REACH and CLP* is available at <https://echa.europa.eu/-/guidance-for-identification-and-naming-of-substances-under-reach-and-clp>.

with the identities and maximum concentrations of the impurities of the composition must be such that they cover all economic operators involved in the application.

The agreement on the scope of the identity of the applied-for entry covered by the application is a prerequisite to submitting a single application. Transparency on the scope and on the data to which it refers is central to implementation. Consequently, the scope of the applied-for entry or SIP should be reported in clear terms on behalf of all economic operators that wish to be covered by that application.

The scope of the SIP can be described in a generic manner in terms of its compositional information (including any relevant parameters such as morphology, e.g. physical form, shape), its name and other identifiers for which the classification and hazard data jointly submitted will be relevant. The definition of the SIP should not take an overly conservative approach to avoid excluding competitors from the submission. The SIP establishes the inherent link between the substance identity and the hazard data to be jointly submitted. If established early enough, it can facilitate the stage of information generation/collection during the process of fulfilling application obligations.

When covering different chemical characterisations of the same starting substance, composition or constituent, economic operators remain responsible for ensuring that other parts of the dossier, in particular substance identification, migration data, toxicological data and risk acceptance, adequately cover the whole SIP. In respect of migration, economic operators may, for example, consider using a worst-case approach or more than one migration study for that purpose.

The *Guidance for identification and naming of substances under REACH and CLP*⁽²⁰⁾, and its Appendix III in particular, provides information on the SIP concept, its relevance for joint applications and how it should be defined and reported.

2.3.3. Analytical information (points 1.3.4, 1.3.5 and 1.5.5 in Annex I to Commission Implementing Decision (EU) 2024/365)

For starting substances and organic constituents (organic and cementitious materials). If spectral data elucidate the composition of the applied-for entry, these should be given. Several spectroscopic methods can be suitable, for example ultraviolet and visible absorption spectroscopy, infrared spectroscopy, nuclear magnetic resonance spectroscopy and mass spectrometry. For inorganic substances and minerals, X-ray diffraction, X-ray fluorescence, atomic absorption spectroscopy (AAS) or elementary analysis may be more suitable.

The use of chromatographic methods, such as gas chromatography (GC), high-performance liquid chromatography (HPLC) and gel permeation chromatography, coupled with a detection technique, is necessary to confirm the composition of the applied-for entry. If appropriate, other valid constituent separation techniques may be used.

Spectroscopic and analytical methods are continuously improving, so the applicant should choose the most appropriate methods.

The applicant should describe the analytical methods used for the identification of the applied-for entry or give their bibliographical references, including the identification and quantification of the applied-for entry, its constituents and, where appropriate, its impurities and additives. The description should consist of the standard operating procedure followed and the relevant

⁽²⁰⁾ <https://echa.europa.eu/-/guidance-for-identification-and-naming-of-substances-under-reach-and-clp>

interpretation of the results. This information should be sufficient to allow the methods to be reproduced. When applicable, bibliographical references should be given.

For applied-for starting substances and organic cementitious constituents (organic and cementitious materials) that are polymers or pre-polymers. In addition to the analytical information requirements set out above in this chapter for starting substances and organic cementitious constituents, the analytical methods used to determine the number average molecular mass and molecular mass distribution should be described. Whenever applicable, size-exclusion chromatography may be most convenient in current practice for this purpose. When this chromatographic analytical technique is used, attention should be paid to the experimental conditions, in particular the selection of the column, the calibration standards and the solvent used. Care should in particular be taken in selecting conditions where the content in the polymerised part below 1 000 Da is not underestimated. Other analytical methods may also be appropriate for this purpose.

As for any other analytical method, the description should consist of the standard operating procedure followed and the relevant interpretation of the results. The analysis reports should include chromatograms along with key results for defined percentiles related to, for example, 1 000 Da as a typical cut-off parameter. This information should be sufficient to allow the methods to be reproduced. When applicable, bibliographical references should be given.

For compositions (metallic, enamel, ceramic and other inorganic materials). The above applies likewise, with the understanding that some techniques may be particularly suitable for certain compositions (e.g. inductively coupled plasma mass spectrometry (ICP-MS) for enamel, ceramics and other inorganic compositions) while others will not be suitable (e.g. chromatography).

It should be noted that this section refers only to analytical information concerning the applied-for entry. Analysis of substances in migration or contact water is described in Section 5.1.7 of this guidance. As further detailed in Section 5.1.7, the application needs to contain data on the most appropriate analytical method.

2.4. Additional requirements for polymers and pre-polymers (point 1.5 in Annex I to Commission Implementing Decision (EU) 2024/365)

For guidance on reporting the substance identity information (name, structure, etc.) of the monomers and reactants, see the corresponding sections in this chapter. Additives should be reported in an analogous way to substance constituents. Residual monomers and substance constituents with molecular weight below 1 000 Da should be identified and quantified because they are particularly relevant for risk assessment.

2.4.1. Viscosity (point 1.5.10 in Annex I to Commission Implementing Decision (EU) 2024/365)

The viscosity should be determined for liquid substances at 20 and 40 °C. Test according to Organisation for Economic Co-operation and Development (OECD) Test Guideline (TG) 114 (Viscosity of liquids), which recommends the following determination methods:

- capillary viscometer
- flow cup
- rotational viscometer

- rolling ball viscometer
- drawing ball viscometer.

Further guidance. ECHA *Guidance on Information Requirements and Chemical Safety Assessment – Chapter R.7a: Endpoint specific guidance*, R.7.1.18 Viscosity⁽²¹⁾.

2.4.2. Melt flow index (point 1.5.11 in Annex I to Commission Implementing Decision (EU) 2024/365)

Melt flow index, melt flow rate or melt index are the same test under different test conditions and are used to measure melt viscosity under a constant load and low shear rates. The melt flow index is the amount measured in grams that exits the die in 10 minutes. A higher melt flow index correlates with a lower viscosity resin. As the resin viscosity decreases, the flow per unit time increases. A lower melt flow index or fewer grams passing through the die in 10 minutes relates to higher viscosity (Giles et al., 2005).

Melt flow supports the characterisation of a polymer, notably when data on its diffusivity may be lacking. It may inform also on its crystallisation, which can provide information on its migration potential. This is usually something well known and characterised (basic measurement and info on polymers). The melt flow index for thermoplastics may be determined using standard EN ISO 1133 (Determination of the melt mass-flow rate (MFR) and melt volume-flow rate (MVR) of thermoplastics) ⁽²²⁾.

2.5. Individual versus group entries

2.5.1. Applications for members of group entries in the first European positive lists

The first EUPLs for starting substances, organic cementitious constituents and metallic compositions contain entries that have been characterised as 'group entries'. A group entry is the one that may encompass more than one distinct starting substance, organic cementitious constituent or metallic composition and originate from national positive lists or the plastic FCM Union list of Regulation (EU) 10/2011. Group entries are presented in Table 2 of Annex I, Table 2 of Annex II and Table 3 of Annex III to Commission Implementing Decision (EU) 2024/367.

According to Article 3(6) of Commission Delegated Regulation (EU) 2024/369, 'an application from a person other than a relevant authority shall cover only one starting substance, composition, organic cementitious constituent, or, in the case of an application for a nanoform, one nanoform.' Applicants who are not such authorities should apply only for specific, individual members in a group entry rather than the whole group.

If a starting substance, metallic composition or constituent falls under a group entry, the applicant should justify in their application how and why it belongs to that group entry. If the applied-for entry encompasses a UVCB, this is a single substance and not considered a group entry.

If review applications are made for one or more specified members of a group entry, on approval of the authorisation of such members of the group, these members of the group will replace the previous group entry in the EUPL. Thereafter, only those specific members of the original group

⁽²¹⁾ Chapter R.7a of the guidance and the endpoint specific guidance R.7.1.18 therein are available at <https://echa.europa.eu/guidance-documents/guidance-on-information-requirements-and-chemical-safety-assessment>.

⁽²²⁾ Details on the testing conditions for determining the melt flow index may be found online (<https://omnexus.specialchem.com/polymer-properties/properties/melt-flow-index#MFI-conditions>).

will be considered authorised for use in the manufacture of materials in contact with drinking water.

Exceptionally, if the applicant can demonstrate and provide evidence that an existing group entry had been incorrectly identified as a group entry in the EUPL, and rather comprises an individual starting substance, composition or organic cementitious constituent, they may submit an application for that individual entry that has been incorrectly labelled as a group. When the applicant is an authority, applications for removal or updating of a group entry may cover the entire group entry.

2.5.2. Applications for the review of entries incorrectly identified as individual entries

Exceptionally, it is possible that an entry in the EUPL has been incorrectly identified as an individual one by it being listed in Table 1 of Annex I or Table 1 of Annex II or Table 2 of Annex III or Table 1 of Annex IV to Commission Implementing Decision (EU) 2024/367. In such a case, an entry listed as an individual entry may actually be a group entry, that is, it encompasses two or more individual starting substances, compositions or constituents.

In the case of reviewing such an EUPL entry, the applicant should select and apply for one individual entry that belongs to that group. The applied-for entry may have an IUPAC name or identifiers that may differ from those currently shown in the EUPL. The applicant must provide sufficient evidence and explanation in their application to justify why the applied-for entry falls within the boundaries or is otherwise covered by the existing EUPL entry.

As described in the previous section, if review applications are made for one or more specified members of what was previously incorrectly identified as an individual entry, on approval of the authorisation of such members of the group, these members of the group will replace the previous group entry in the EUPL. Thereafter, only those specific members of the original group will be considered authorised for use in the manufacture of materials in contact with drinking water.

3. Intended use of the applied-for entry

This section focuses on (1) the standard information requirements related to intended use of the applied-for entry and (2) the identification of NIAS other than impurities.

Because of the differences in the nature of the applied-for entries for the different EUPLs, and the way in which the EUPLs are structured, the required use information varies according to the EUPL.

The EUPLs represent four distinct material types: organic, metallic, cementitious, and enamel and ceramic or other inorganic materials. Applied-for entries for the organic and cementitious EUPLs are individual substances that can be accepted for use in different materials such as plastics or rubber. In contrast, the applied-for entries for the metallic, enamel, ceramic and other inorganic EUPLs are compositions which, for practical purposes, are materials and used as such. Nonetheless, in all cases, any given application may encompass several relevant chemical species because of, for example, the presence of impurities and reactions occurring during product manufacture and use.

It should be noted that a chemical substance may be used both as a starting substance for organic materials and as an organic constituent for cementitious materials. In such a case, its approval for inclusion in the two EUPLs would require separate applications submitted to the agency, as stated in Article 3(6) of Commission Delegated Regulation (EU) 2024/369: 'An application from a person other than a relevant authority shall cover only one starting substance, composition, organic cementitious constituent or nanoform.' This means that an application by an economic operator may cover only one starting substance or organic cementitious constituent irrespective of whether it is the same substance. Scientifically it is necessary to test a substance in the different materials for reliable migration estimation, as organic and cementitious materials are very different.

For a case in which two applications are submitted concerning the same substance, but for different EUPLs, this could mean the following.

- If both applications are submitted close to each other, there is a possibility that they may be assessed together.
- If one application is submitted after the approval of the substance as a result of a prior application, the MTC_{tap} that has been approved by the European Commission can be reused, as long as new information has not become available. The MTC_{tap} is based on the intrinsic toxicological properties of the substance, which are the same irrespective of the material(s) for which the substance is used.

Relevant physico-chemical properties of the starting substances for organic materials and organic constituents for cementitious materials are covered in Chapter 4, although some of them, for example properties needed to perform modelling, are also briefly described in this chapter.

The applicant should provide information on the intended use of the applied-for entry, covering the following aspects:

- **contact material(s)** for which the applied-for entry will be used, that is, organics, metallics, cementitious materials or enamels, ceramics and other inorganics, followed by further subcategorisation, as applicable;

- **type of product or component** (for complex products) and product group the applied-for entry is intended to be used in, for example pipes and fittings, including whether the area of use is a domestic ⁽²³⁾ or non-domestic installation;
- **technical function** of a starting substance or organic constituent of cementitious materials in the manufacture of the product;
- **conditions of use** of the applied-for entry, final material and product, including concentrations in the formulation (organic/cementitious) or concentrations of constituents (metallics/enamels, ceramics and other inorganics) and potential restrictions on its use;
- **processing and structure of material, final material and product**, including processing methods and temperature, as these are relevant to the migration of relevant chemical species into drinking water;
- **EU- and national-level assessments and authorisations** in case they exist, including **EU authorisation of biocidal active substances**, in the case of starting substances for organic substances and organic constituents of cementitious materials;
- **identification of NIAS (other than impurities) from processing, use or water contact of starting substances for organic materials or organic constituents of cementitious materials that may migrate into drinking water**; such NIAS could be reaction products or degradation products of a starting substance or organic cementitious constituent that form during the processing or use of the material or in contact with water during the use of the material;
- **identification of other substances (polymerised part) that may migrate into the drinking water when using starting substances and organic cementitious constituents that are monomers or other reactants** (point 3.4 in Annex II to Commission Implementing Decision (EU) 2024/365) considering identity, physico-chemical properties, analytical data, etc.).

It should be noted that the following phrase in Annex II to Commission Implementing Decision (EU) 2024/365 refers to any information the applicant holds on the intended use: 'Sufficient information on the intended use of starting substances, compositions, constituent as well as of final materials and products shall be generated, including the information set out in Table 1.' refers to any information the applicant holds on the intended use. The information does not need to be exhaustive. It is expected that applicants may have different levels of knowledge on the intended use, depending on their position in the overall supply chain, for example whether they are a manufacturer of starting substances or a manufacturer of pipes, valves, taps, etc.

More details on these topics are provided in the sections below.

3.1. Main information on intended use

3.1.1. Material type, category and subcategory (point 2.1 in Annex II to Commission Implementing Decision (EU) 2024/365)

The material into which the applied-for entry is incorporated is important, as it affects its potential to migrate into drinking water. The identification of a material must include the following.

⁽²³⁾ Domestic here means as defined in Article 2(2) of the DWD: 'pipework, fittings and appliances which are installed between the taps that are normally used for water intended for human consumption in both public and private premises and the distribution network'. Everything else is considered non-domestic.

- **The material type.** Each type of material covers materials with similar physical/chemical characteristics. In the context of the DWD there are four types of materials: (1) organics, (2) metallics, (3) cementitious materials and (d) enamel, ceramic and other inorganic materials.
- **The material category and subcategory.** The applicant must indicate one or more categories and subcategories falling under the material type. This information is relevant for defining the appropriate migration test(s) required and/or the input parameters for modelling the migration. For example, the organic materials can cover plastics and polymers, rubbers and elastomers, coatings, adhesives, silicones and lubricants (these can be considered categories). If the substance is used in different material subcategories, for example in different plastics (e.g. polyolefins or polyvinyl chloride), such information must also be provided. Metallics can be classified, for example into ferrous (carbon steels, stainless steels, etc.) or non-ferrous (e.g. copper, copper alloys, etc.) alloys. Nevertheless, for the purposes of the DWD the metallic alloys are categorised based on their corrosion behaviour in drinking water. A metallic material must belong to one of the DWD-related metallic composition categories (see glossary of key terms). If an application is made for a new metallic alloy that does not belong to an existing category, a new category must be created for it.

A very restricted or a very broad field of application may influence the final approval and the conditions on the use of the applied-for entry. For example, if the applied-for entry is an additive used in both plastics and rubber and the migration test has been run on the most conservative matrix (typically, rubber), the use might be approved in both. Following that rationale, if migration tests are carried out on a worst-case polymeric material, the use might be approved in all plastics. In this context, the selection of the worst-case material (matrix) with the highest potential migration for the applied-for entry and any NIAS is important for correctly setting up the migration testing.

Different lists of material categories and subcategories exist in the context of other legislation or regulatory obligations ⁽²⁴⁾. They have been used as references, when suitable for the purposes of this document. These lists are presented in Annex I to this guidance, separately for each of the four material types.

The sections below provide more details on the information that needs to be provided for each material type in a DWD application submitted to ECHA.

3.1.1.1. Organic materials

In the scope of this guidance, organic materials comprise plastics and polymers, rubbers and elastomers, organic coatings, adhesives, silicones and lubricants. These materials may contain both organic and inorganic starting substances. Simple plastics consist of a main polymer, that is, the polymer with the highest content in the material, and additives, polymer production aids and aids to polymerisation (see their definitions in the glossary and details of the different types in Annex III to this guidance), which are used to modify structural, surface or physico-chemical properties. More complex plastics may consist of many different polymers, potentially in different layers. Elastomers are materials with rubbery properties (e.g. viscoelastic polymers) resulting from a structure consisting of a tangled mass of long-chain, flexible polymer molecules. Rubbers are vulcanised (cured) elastomers. Silicones are macromolecular substances or materials based on organopolysiloxanes and which are cross-linked forming a three-dimensional network with elastomeric or rubber-like properties.

⁽²⁴⁾ For example, substances of concern in articles as such or in complex objects (products).

As broad subcategories, the following organic materials can be selected by the applicant when submitting an application:

- plastics
- rubber and elastomers
- coatings
- adhesives
- silicones
- lubricants.

Annex I to this guidance lists several polymer-based materials within the material categories 'plastics and polymers' and 'rubbers and elastomers' that can be selected by the applicant to qualify the matrices in which the applied-for entry is used, when completing their IUCLID dossier.

3.1.1.2. Metallic materials

Metallic materials consist of metallic elements such as iron and copper and often also non-metallic elements such as carbon and nitrogen. They can be broadly categorised into ferrous and non-ferrous according to the main alloying element. They can be used as almost pure, one-constituent materials but are often alloyed to enhance their properties. A metallic material consisting of two or more constituents can be called an **alloy**. Metallic materials contain intentionally added **constituents** (elements) and **impurities** (elements). Impurities arise from raw materials and the production process.

For the purposes of the DWD, alloys listed in the EUPL have been assigned to a metallic composition category based on their similar constituents and/or corrosion behaviour among alloys in that category. The metallic compositions of the EUPL that belong to such a category are shown in Annex IV to this guidance⁽²⁵⁾. Metallic composition categories containing alloys exhibiting uniform corrosion and requiring migration testing, that is, in accordance with standard EN 15664-1, are accompanied by a reference material that represents the worst reproducible corrosion within the category. The composition of the reference material depends on the composition ranges of the category. In contrast, alloys exhibiting passive behaviour and accepted on to the EUPL via an electrochemical test (e.g. in accordance with standard EN 16056) do not have reference materials in their categories, since 'passive' compositions vary greatly and are practically different materials.

For metallic materials, a category, old or new, must be indicated as well as the identity of the applied-for composition. No other categorisation is required: for metallic materials, the levels of classification are material type (metallic) → metallic composition category → applied-for composition.

In principle, for each metallic composition and, where applicable, its corresponding metallic composition category (see Annex IV to this guidance for a list), the following information must be provided:

⁽²⁵⁾ The current list of metallic composition categories will become available on the ECHA website.

- **applied-for metallic composition:** standardised alloy notation (if available), constituents, impurities;
- **metallic composition category, reference material:** constituents, impurities.

This information should already be reported in the IUCLID section related to the 'identity of the substance'.

3.1.1.3. Cementitious materials

For the purposes of the DWD, according to Commission Implementing Decision (EU) 2024/368, a cementitious material is a material that contains a hydraulic cement in sufficient proportion to act as the main binder by forming a hydrate structure that governs the performance of the material.

Cementitious materials, such as concrete and mortar, can be also considered composite materials. They consist of an inorganic, hydraulic cement that reacts with water. The hydraulic reaction is characterised by setting and hardening after mixing with water leading to a three-dimensional network of cement hydrate phases (and unreacted cement grains and/or aggregates) that can bind together a solid aggregate. Apart from cement, aggregate and mixing water, they may contain the following ingredients: inorganic additions, organic additions, admixtures, fibres and curing compounds. In addition, formwork release agents may be applied on the formwork surface during manufacture of the cementitious material, potentially leading to residues on the cementitious material surface. Some of these ingredients represent generic categories (i.e. groupings) of organic cementitious constituents that must be listed in the EUPL for cementitious materials, as shown in Annex I to this guidance. Cementitious materials are used in drinking water installations, for example in water storage tanks, and the lining of water distribution installations.

While cementitious materials contain both organic and inorganic constituents, only organic constituents can be listed in the EUPL, according to point 3 of Annex V to the DWD which allows the use of inorganic constituents without their being listed in the EUPL. As a result, the EUPL for cementitious materials contains only organic substances, and applications for cementitious constituents will be accepted only for organic substances.

The tables in Annex I to this guidance list the generic constituent categories for cementitious materials. The applicant must report which generic category the individual organic cementitious constituent falls under. Once a generic category is selected, a technical function needs to be specified, as shown in Annex III to this guidance.

3.1.1.4. Enamels, ceramics and other inorganic materials

Enamels and ceramics are inorganic materials (oxides, carbides, nitrides, etc.) defined in the EUPL by their chemical composition. In the scope of the DWD, ceramics are crystalline bulk materials, while enamel is a glassy coating formed by melting and quenching the enamel composition. For simplicity, hereafter, this group of materials will be called 'inorganics'. Glasses and mixed metal oxide coatings also fall within this material type.

In principle, the composition to be included in the EUPL (defined by the concentration of each constituent) of inorganic materials should be already reported in the IUCLID section related to the identity of the substance.

Table 7 in Annex I to this guidance lists the categories and subcategories of inorganic materials.

3.1.2. Products and relevant product groups in contact with drinking water (points 2.2.1 and 2.2.2 in Annex II to Commission Implementing Decision (EU) 2024/365)

The **size of the water contact surface** of the material/product containing the substances (in organic, metallic, ceramic or cementitious material) affects the level of exposure, with larger contact surfaces leading to higher concentrations of substances in water within a similar time. Another factor is the **volume of water the surface is in contact with**, since higher volumes decrease the concentrations of migrated substances by dilution. Consequently, product group and the related ratio of the surface of a product to the water volume is an important parameter in assessing the risk of the applied-for entry. The higher this ratio, the higher the level of migration in a same material composition. The surface to volume (S/V) ratio of a product is used also to estimate a conversion factor (CF) – explained further in Chapter 5 for organic, cementitious and inorganic materials. See also Annex II to this guidance.

Concerning product groups, the applicant must provide the following information:

- for organic, cementitious and inorganic materials: product group, as specified in Table 9 of Annex II to this guidance (and in Commission Implementing Decision (EU) 2024/368), from which the S/V ratio (in dm^2/dm^3) and the CF can be inferred;
- for metallic materials: relative water contact surface area of the metallic composition exposed to direct contact with drinking water as a percentage of the total surface area, as shown in the product group in Table 8 of Annex II to this guidance (and in Commission Implementing Decision (EU) 2024/365).

Annex II to this guidance includes tables listing the existing **product groups** (e.g. pipes, fittings) for each material type. These product groups are distinguished based on the size of their water contact surface, which is relevant to quantifying the level of migration of the substance. Products contained in these product groups can be used as a reference by the applicant when identifying typical products and/or components in which the applied-for entry is used, in case it is not clear to which product group a given product would belong.

For instance, a metallic composition in the EUPL will be linked to certain products – the ‘relevant product groups’. Product groups are designated A, B, C and D, in order of decreasing surface area. If a composition has been accepted for a higher product group, it can also be used in a ‘minor’ product group, but not the other way round, for example. a metallic composition approved for B and C can also be used for D but not for A.

In addition to product group, the product (as defined in Commission Implementing Decision (EU) 2024/368 ⁽²⁶⁾) that comes in contact with drinking water, in its finished form (e.g. pipes, tanks, valves, taps), should also be identified, as far as possible, by the applicant. In addition, the area of uses of the product should be identified, if known. Area of use in this context means domestic or non-domestic use ⁽²³⁾. Information on the product and its area of uses is relevant for the determination of the parameters that drive migration and subsequent risk assessment.

Concerning different products, at least the following information should be provided:

- pipe with internal diameter (ID);
- ancillary/fitting for piping system with ID and type of ancillary/fitting;

⁽²⁶⁾ ‘Product’ means an object intended to come into contact with water intended for human consumption, which is made of final materials and with the intention to be placed on the market.

- components (relative surface area compared with the water contact surface area of the whole product: > 10 %, 1–10 %, < 1 %) of ancillary for piping system with ID;
- tank (range of water volume) and kind of tank.

Products can be classified as homogeneous or non-homogeneous.

- **Homogeneous product.** The water contact surface area and the remainder of the product/component of an assembled product are made of the same material (e.g. a solid wall pipe, frame of a pump made from a metallic alloy). These are 'single material products'.
- **Non-homogeneous product.** The water contact surface area and the remainder of the product are made of different materials (e.g. multilayered pipes, coated products). These can be 'assembled products' (comprising two or more components, possibly of different materials) or 'multilayer products' (comprising two or more layers, bonded together to form a single item). It should be noted that, for the assessment of starting substances, components of assembled products are considered homogeneous products.

3.1.2.1. Platings and coatings

Specifically, for platings (metallic) and coatings (organic), the product to which they are applied, as well as the extent and nature of the surface to which they are applied, must be described (complete component /product, intended water contact surface, plated/coated surface (inner surface, outer surface with scattering on the inner surface, etc.)). In the case of an inner surface plating, information on the durability and completeness of the coating/plating must be given.

3.1.3. Water characteristics (point 2.2.3 in Annex II to Commission Implementing Decision (EU) 2024/365)

The applicant must identify the temperature of the water that is relevant to the product(s) associated with the applied-for entry. Typically, water intended for human consumption is classified as cold ($\leq 23 \pm 2$ °C), warm ($23\text{--}60 \pm 2$ °C) and hot ($60\text{--}85 \pm 2$ °C). These classifications may be used to identify the water temperature to fulfil this information requirement for the application.

The applicant may identify, as far as possible, any other relevant characteristics of the water(s) (e.g. chlorinated) that will be in contact with the product(s) associated with the applied-for entry, including pH, specific electrical conductivity, dissolved oxygen, total alkalinity, total organic carbon, free CO₂ and concentrations of elements (e.g. calcium, magnesium, potassium, sodium, chloride, nitrate, sulphate, phosphorus, silicon dioxide, copper, iron, manganese, zinc, other metals).

Identifying water characteristics is useful in the identification and evaluation of the presence of NIAS, for example potential transformation products originating from the interaction between migrating substance and water.

3.1.4. Technical function of the applied-for entry in the product (for organic and cementitious materials) (point 2.3 in Annex II to Commission Implementing Decision (EU) 2024/365)

The technical function in the scope of the DWD is required for all substances used in organic and cementitious materials. It describes the effect a substance produces in a manufacturing process, material, mixture or product during its lifetime. Some examples of technical functions are monomer, antioxidant, antistatic agent, preservative and biocide.

For metallics, identification of technical functions for the alloying elements (e.g. hardener, stabiliser, corrosion inhibitor) is required only in certain cases described elsewhere in the guidance (see Section 5.1.5.1).

Knowledge of the technical function for starting substances is relevant for the migration and exposure considerations, as well as for the identification of the likelihood of formation of by-products (e.g. reaction during processing and use, inclusion into matrix material, processing aid).

For organics, technical function must be given, as a minimum, at a generic level and may be further defined by specifying the additive type of the substance (see Annex III to this guidance). For cementitious materials, the technical function is sometimes defined by the constituent category selected, but in other cases it may need to be defined further (e.g. admixtures → an auxiliary to admixtures).

Annex III to this guidance includes a list of technical functions relevant in the context of the DWD. The applicant must provide information on the generic technical function and on the technical function specific level, depending on the material type. It should be noted that the technical specifications for cementitious constituents listing European standards are meant only as guidelines and are not a legal requirement, as they may not cover every mortar and grout additive used.

3.2. Conditions of the intended use

3.2.1. Maximum dosage (concentration) in formulation and/or material (organic and cementitious) (points 2.4.1 and 2.4.2 in Annex II to Commission Implementing Decision (EU) 2024/365)

The dosage (concentration) is relevant only for applications for starting substances on the organics EUPL and organic constituents on the cementitious EUPL. The maximum dosage (concentration) of the starting substance in the formulation of organic materials (or related to the cement content for cementitious materials) must be indicated in the application.

In the case of polymers used in admixtures for cementitious materials, what needs to be provided is:

- the maximum dosages (concentrations) of the polymer used in the formulation to produce the admixture;
- the maximum dosages (concentrations) of the monomers used in the formulation of the polymer.

The maximum dosage values should not exceed the typical maximum indicated for the substance to exert its technological property.

For an application for a composition of metallic or inorganic material, the concentrations of the different composition constituents are defined at the level of material composition (see also Section 2.1.1 of this guidance).

The concentration at which the starting substance or organic cementitious constituent is added to the manufactured material(s) (or its residual concentration in the material) will influence its migration into drinking water⁽²⁷⁾, with higher concentrations typically leading to higher

⁽²⁷⁾ Note for guidance for the preparation of an application for the safety assessment of a substance to be used in plastic food contact materials (<https://efsa.onlinelibrary.wiley.com/doi/epdf/10.2903/j.efsa.2008.21r>).

migration potential. Consequently, the typical maximum concentration of the applied-for entry necessary to demonstrate the desired technological property should be given, for both testing and modelling of migration. This is also one indicator of the worst foreseeable conditions for the substance / technical function / material combination. In addition to the maximum, an applicant should also indicate the minimum and typical concentrations of the starting substance or organic cementitious constituent, as this may help when assessing migration and risk acceptance.

3.2.2. Proposed conditions of use (point 2.4.3 in Annex II to Commission Implementing Decision (EU) 2024/365)

Testing of migration for the applied-for entry should be conducted under the worst foreseeable conditions of use, and consequently any additional restrictions on its use should typically be unnecessary. However, the applicant may propose specific conditions of use and advise against other uses, when necessary, to ensure that the uses of the substance do not result in unacceptable risk.

In principle, any conditions of use can be proposed, and examples are shown in Table 1.

Table 1. Examples of restrictions on the conditions of use for EUPL entries

Restriction type	Examples
Properties, structure	Particle size, purity, viscosity, molecular weight, molecular unit
Processing	Production process, production temperature
Use environment	Water characteristics, water temperature
Products	Restricted contact surface size, only in multilayer material with indirect water contact
Materials	Allowed only in plastics, maximum concentration limit, only specific technical function allowed, restricted film thickness
Other factors	Link to other regulations or specifications that restrict the use, potential deteriorating effect on organoleptic properties

A typical example for specifying restricted conditions of use would be for metallic pipes – product group A. Despite the acceptance procedure using three different test waters for pipes, a restriction on the waters allowed may still be needed to ensure long-term safety. In setting restrictions, it is considered that a material's potential service life may extend up to 50 years for products in buildings and 100–150 years in distribution networks.

3.3. Processing and internal structure of the material, final material and products (points 2.5.1–2.5.3 in Annex II to Commission Implementing Decision (EU) 2024/365)

3.3.1. General information

For all material types, the applicant must describe the processing of the material, final material and product as well as the internal structure. This is particularly important for metallic materials (see Section 5.1.3) ⁽²⁸⁾.

The internal structure of materials can be described using microstructural information such as micrographs, X-ray diffractograms with phase distribution analyses, level of crystallinity and/or amorphousness.

In the following sections, guidance is given on how to describe the processing for the different material types. In practice, the processing can be described on a granular level indicating the main processing types. If these are insufficient, additional clarifying information can be given.

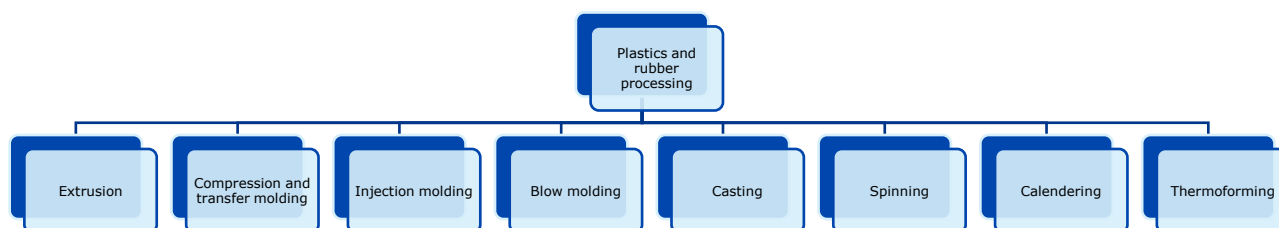
3.3.2. Organic materials

Processing options for organic materials depend on the type of material (e.g. thermoplastic or thermosetting resin) but also on the type of product (e.g. pipe, film). In the DWD, the main organic material categories are plastics and polymers, rubbers and elastomers, silicones, coatings, adhesives and lubricants. Some common plastic and rubber processing methods to produce solid forms are presented in Figure 12.

In addition to information on the processing and shaping into intermediate materials, final materials and/or final articles (see Figure 12), any other information the applicant considers relevant, such as polymerisation methods and additional processing steps, for example vulcanisation or reinforcement, should be given.

For any organic material, the processing temperature of the final material must be given.

Figure 12. Plastic and rubber processing methods



⁽²⁸⁾ The reader is reminded that Article 10(2)(b) of Commission Delegated Regulation (EU) 2024/369 prescribes that information on the manufacturing process of a starting substance, composition or organic cementitious constituent or the manufacturing process in which it is involved shall be considered confidential and may not be made publicly available.

3.3.2.1. Coatings

In the context of the DWD, 'coating' refers specifically to **organic coatings**. A coating is a distinct, homogeneous layer on the surface of a solid material separating the bulk material from the neighbouring phase, i.e. air or liquid.

Organic coatings can be further divided, for example into thermoplastic and thermosetting according to their resin type, and they vary in formulation and application. Typical organic coatings include paints and plastic coatings and also include organic polymers similar to those in plastics. The ingredients of coatings include binders (resin, polymer), pigments, solvents and additives. Based on the ingredients used, coatings vary in structure and properties and consequently in migration behaviour. Despite the various coating types, coating processes generally include the phases pretreatment, coating and post-treatment.

The following information on an organic coating should be given:

- coating formulation and main ingredients;
- coating process;
- coating properties (thickness and structure).

3.3.2.2. Adhesives

Adhesives are used to join materials together via surface attachment by applying them to the surfaces of the materials to be joined. Adhesive materials include epoxies, acrylates, urethanes and phenolics. They can be classified in various ways based on composition, uses, etc. In the context of this guidance, they are classified into solvent-based adhesives and adhesives obtained by polymerisation.

To fulfil the legal requirements, the intended use of an applied-for adhesive should be described similarly to coatings (see Section 3.3.2.1), including adhesive formulation, application process and properties.

3.3.2.3. Lubricants

Lubricants are materials that are used to reduce wear between surfaces in contact. They may be in the form of solid, liquid, gas or grease and made from organic or inorganic substances. Solid lubricants may be based, for example, on inorganic compounds or organic polymers, while liquid lubricants may be based, for example, on mineral oils. In the context of the DWD, lubricants that are applied to the surface of a solid material are similar to coatings. Thus, the processing information for them should be similar to that for coatings, covering formulation, application and properties.

Currently, there are no subcategories for lubricants. Consequently, applicants need to describe their product in writing in the free text fields in IUCLID provided for this purpose.

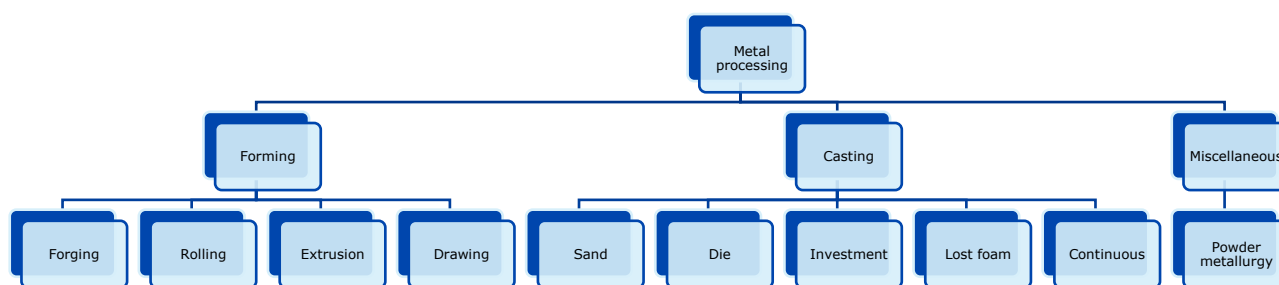
3.3.3. Metallic materials

For metallic compositions, the processing steps used to produce the final material must be described. In practice, this means the basic processing steps (see Figure 13) used to produce the desired shape and thermal treatments and/or surface treatment or plating. It must be indicated, for example, whether an alloy is cast or formed and the method of casting or forming. Most alloys are either cast or formed, while powder metallurgy may be less used in the DWD

context. It should be noted that some of the information on processing may already be communicated via standard notation. In addition to the methods shown in Figure 13, it is possible to specify other methods, such as welding, if relevant.

The corrosion resistance of a metal or a metallic alloy in various environments depends on its chemical composition and also on its microstructure (crystallography, grain size and shape, grain heterogeneity, second phases, impurity inclusions), residual stress and surface quality. Therefore, the applicant should describe the alloy's processing history and its surface condition (e.g. as-cast, ground) and roughness value, including any treatments affecting crystallography, grain morphologies (size and shape), phase structure, impurities and their distribution, residual stresses, microstructure and/or surface condition. These can include forming processes (casting, drawing, etc.), heat treatments (annealing, etc.) or surface treatments (work-hardening, sandblasting, grinding, etc.). The description of the alloy processing should contain a representative micrograph of its final structure as in the final product for contact with drinking water.

Figure 13. Basic metal processing methods



In addition to the basic processing, any thermal processing must be indicated and whether it affects the bulk material or the surface. This may mean treatments such as annealing, stress relief or hardening.

3.3.3.1. Platings

If the applied-for entry is used in a surface layer such as plating, according to Commission Implementing Decision (EU) 2024/365: *'For specifically produced surface layers, platings, the process type and main processing conditions should be described.'* Process type and process conditions are important, as they affect the properties of the resulting plating (e.g. composition, structure, thickness) and thus the amount of metal potentially migrating from the plating or from the substrate material through the plating.

The term 'plating' usually refers to a surface layer produced either with electroplating or electroless plating. However, in the context of the DWD, it includes all metallic coatings on a metallic substrate. Nonetheless, the most commonly used plating processes in the DWD are electroplating, galvanising and electroless plating.

To fulfil the legal requirements, the following information on a plating should be given:

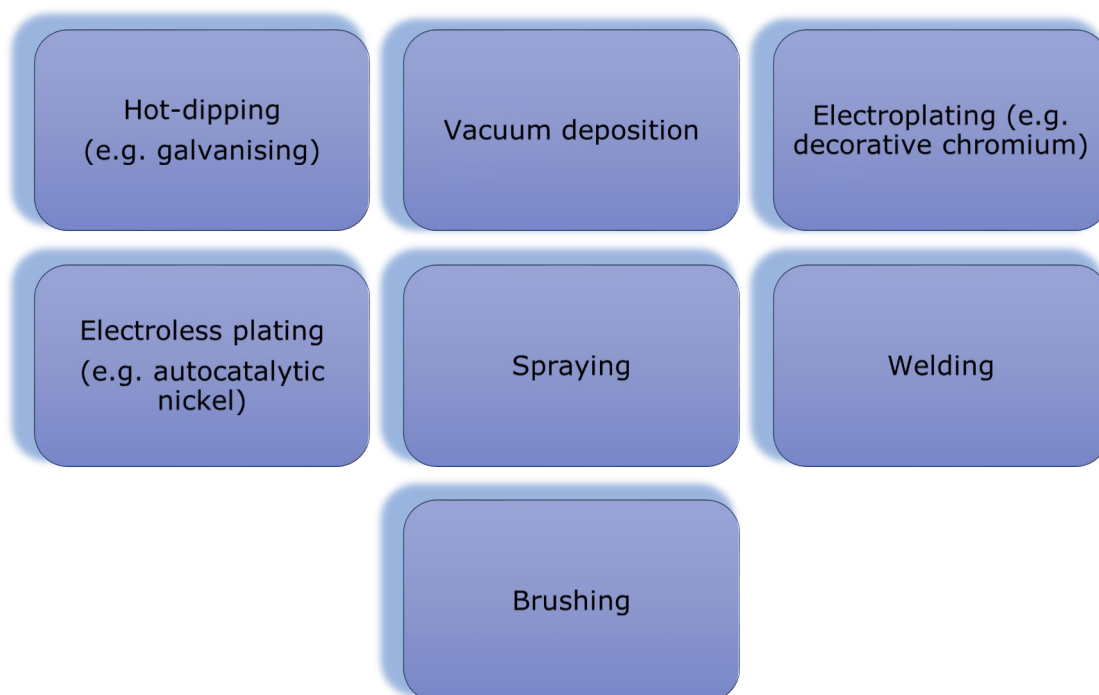
- applied surface of the plating:
 - complete component/product, inner surface, outer surface with scattering on the inner surface, etc.;
 - if applied on inner surface, information on the completeness of plating;

- plating process:
 - single- or multilayer plating and if multilayer the number of layers;
 - process type, for example electroplating;
 - process steps (pretreatments, plating, post-treatments);
 - substrate material;
 - plating bath composition, including organic substances present above 0.02 % concentration in formulation;
 - metal source and its purity (anodes or chemical);
- plating properties:
 - thickness, number of layers, composition of each layer, density, porosity, adhesion, ductility, hardness, hydrogen embrittlement, fatigue, internal stress.

Further information on how to describe platings is given below.

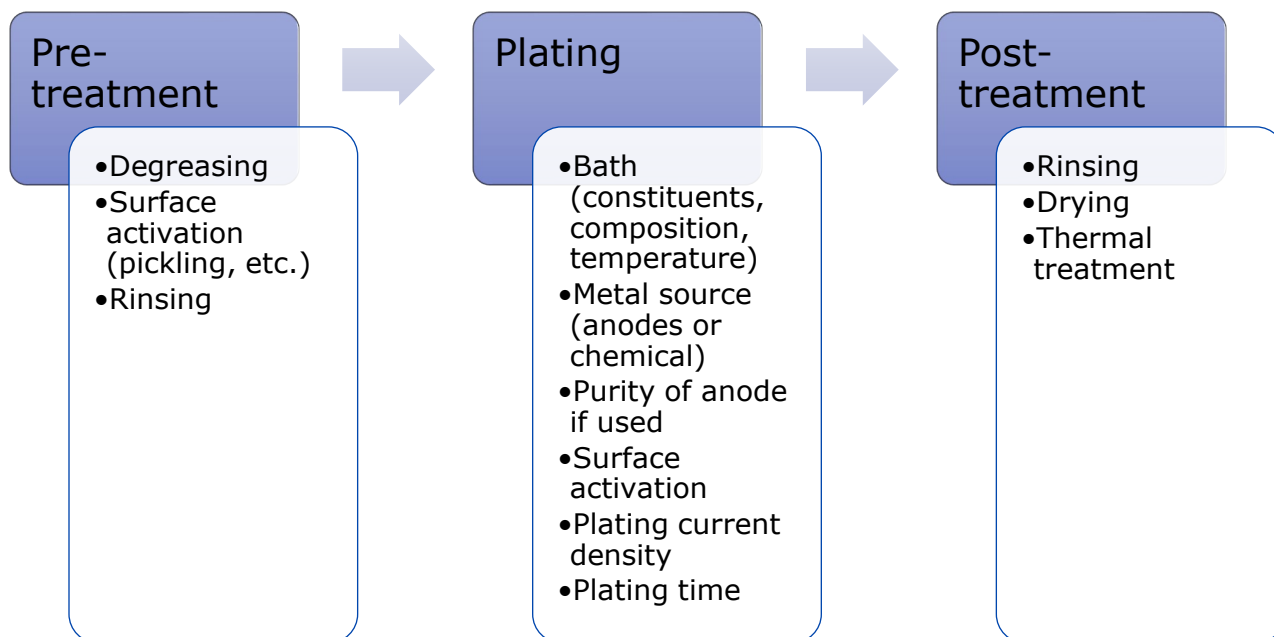
Platings may be produced using a variety of techniques such as hot-dipping (e.g. galvanising), vacuum deposition, electroplating, electroless plating, (thermal) spraying, welding or brushing. Figure 14 shows some common techniques used to produce platings. The list shown is not exhaustive, and any relevant methods missing from the figure can (and should) be described by the applicant.

Figure 14. Plating processes



If an applied-for entry is a metallic composition used in a surface layer (and indicated thus in IUCLID), to fulfil the legal obligations, the process type can be described by choosing the type from a picklist (Figure 15). In describing the process conditions, sufficient information on the different process steps must be given. These include pretreatments (e.g. degreasing, activation such as pickling, rinsing, etc.), plating process (e.g. bath composition, chemicals used to generate the bath, metal source, i.e. anode or chemical in bath, metal source purities, bath temperature, surface activation current density, plating time).

Figure 15. Steps of a plating process



The processing steps in Figure 15 describe steps in 'liquid-phase metallizing' techniques, for example electroplating, electroless plating and hot-dipping. Nevertheless, the main processing steps of pretreatment, plating and post-treatment may be applicable to any plating process and must be described with the relevant parameters.

3.3.4. Cementitious materials

Processing of cementitious materials consists of processing of the raw materials and the final material by mixing the ingredients (cement, aggregate, water, admixtures, etc.) in appropriate proportions.

For cementitious materials, the applied-for entries in the EUPL are the organic components of grinding aids, organic additions, admixtures, organic fibres, formwork release agents and curing compounds for which processing details should be given at the substance identity stage (see Chapter 2). Importantly, in the case of polymers used as components of admixtures, the applied-for entry should be the polymer, but the EUPL will list the monomers, for reasons of regulatory clarity and practicality (see further information in DWD Guidance Volume III).

For the other ingredients apart from the applied-for entry, a high-level description of their processing may be sufficient. Processing of the most common binder, cement such as Portland cement, typically includes mixing of the raw materials (silicates, etc.), firing and grinding.

The constituents and curing affect the chemical reactions (hydration) and consequently the properties, microstructure, porosity and potentially interactions of the cementitious material with drinking water. Thus, sufficient information on the processing of the cementitious material, that is, its constituents, curing and final structure, should be given. This should include the type of cement, water and aggregate used and also curing conditions and porosity. The processing (forming, setting and hardening) temperature of the final material must be given. This could typically be in the range of 5–40 °C.

If the applied-for entry is used in a cementitious lining, the same requirements as for organic coatings (see Section 3.3.2.1) apply.

Lastly, in describing the processing of the cementitious material, it should be ensured that it matches the information given for the test piece for migration, so that the representativeness of the test piece for migration can be confirmed.

3.3.5. Enamel, ceramic or other inorganic materials

In the context of the DWD, enamel, ceramic and other inorganic compositions are similar, consisting typically of oxides, nitrides and carbides. The main differences between ceramics and glassy materials (e.g. enamels) are microstructure and whether processing takes place above or below melting temperature. Glass forming usually entails heating of the composition to above its melting temperature and then cooling, which results in an amorphous microstructure. In contrast, ceramics often have extremely high melting temperatures. Thus, they are processed below melting temperature, which results in microstructures containing crystalline phases. Certain ceramic processing methods, such as hydroplastic forming and slip casting, initially result in porous, structurally weak water-containing materials. Therefore, the materials need drying and firing to reach the desired service properties. Similarly, for glassy materials cooling may result in unwanted internal stresses that weaken the material. In contrast, compressive residual surface stresses may be intentionally induced to improve glass strength. Elimination of internal stresses or enhancement of glass strength via introduction of surface stresses can be obtained by heat treatment such as annealing or thermal tempering, respectively.

For enamels or other coatings, that is, surface layers made from ceramic or inorganic raw materials, their processing steps and properties should be described in similar detail to those for organic coatings, see Section 3.3.2.1.

Processing information concerning ceramics or other inorganics can be given using, for example, the terms shown in Figure 16 and Figure 17. In addition, if necessary, it is possible to provide additional information in the application to better describe the processing.

In addition to the processing methods, the processing temperature range of the final material produced from an enamel, ceramic or other inorganic composition must be given.

Figure 16. Basic ceramic processing methods

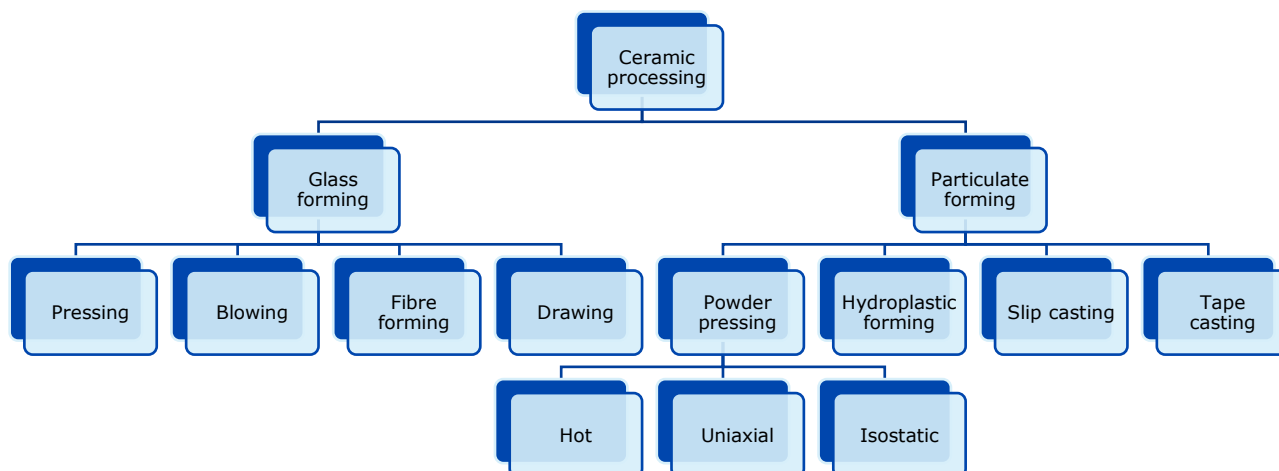
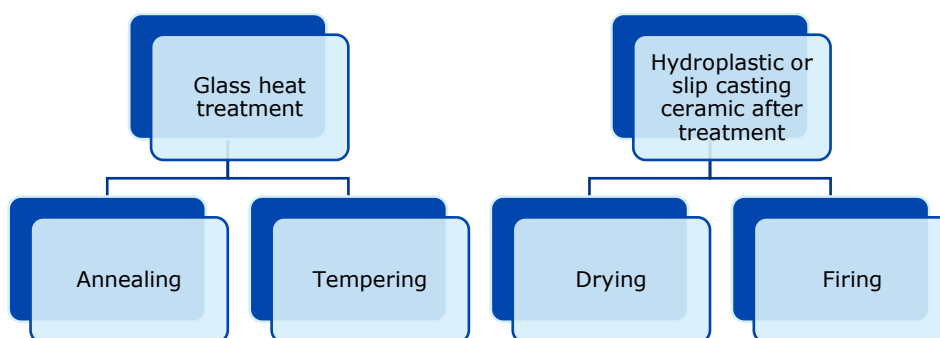


Figure 17. Glass and ceramic treatments



3.4. EU- and national-level assessments and authorisations (points 2.6 and 2.7 in Annex II to Commission Implementing Decision (EU) 2024/365)

In the case of starting substances for organic substances and organic constituents of cementitious materials, relevant authorisations that an applicant should provide details of may extend beyond those relevant to drinking water contact materials (DWCs). Food contact material authorisations are relevant and should be indicated, providing the corresponding scientific opinion, where available. Biocidal active substance authorisations are also relevant and should be provided if the applied-for entry is a biocidal active substance. For biocides, applicants are reminded that, according to Commission Implementing Decision (EU) 2024/367, only in-can preservatives (BPR product type 6) are permitted to be included in the EUPLs.

Information that the applicant is required to provide includes:

- the jurisdiction in which a previous application was submitted and an authorisation was granted and the corresponding evaluating competent authority;
- the EU or national legislation under which authorisation was granted and any associated approval identifiers (e.g. the BAS number of a biocidal active substance), including the period of validity of the authorisation, where applicable;

- a description of the intended use of the applied-for entry that has been previously authorised;
- a description of any conditions attached to the authorisation, for example migration limits and restrictions on use. Particularly in the case of an existing migration limit, and where the relevant information is available, the applicant should provide the derived no-effect level (DNEL) value on which that migration limit was based and details of the study from which the DNEL value originated;
- a copy of migration test results or migration modelling descriptions, where authorisations are relevant to a use in DWCMs, if available to the applicant;
- a copy of the authorisation opinion by the corresponding evaluating competent authority.

3.5. Potentially relevant chemical species other than the applied-for entry

3.5.1. Non-intentionally added species associated with the use of a starting substance or organic cementitious constituent other than a monomer or other reactant (points 3.1–3.3 in Annex II to Commission Implementing Decision (EU) 2024/365)

An applicant needs to assess the potential presence and migration of any NIAS when applying for a starting substance or organic cementitious constituent. The aim is to identify the potentially relevant chemical species that should be confirmed as relevant or not from the migration tests. In the DWD, NIAS may consist of:

- impurities (see Section 2.3 of this guidance);
- reaction products or degradation products that may result from the use of the intended substance either (1) during the processing or (2) as a result of contact with water during the use of the material.

NIAS could be any degradation/reaction products envisaged, hydrolysis products, etc. The assessment of NIAS does not include the formation of unexpected substances ⁽²⁹⁾ not linked to the starting substance, which are dealt with at the final material/product acceptance stage ⁽³⁰⁾.

Impurities are relevant to all material types and should have been captured already at the chemical characterisation stage (see Section 2.3). Reaction and degradation products are relevant only to starting substances and organic cementitious constituents.

In the case of starting substances and organic cementitious constituents, an applicant must identify NIAS other than impurities that may migrate into drinking water based on:

- the physico-chemical properties of the applied-for entry;
- the technical function(s) of the applied-for entry;
- any relevant interaction with the matrix;
- water characteristics relevant to the intended use;

⁽²⁹⁾ See definition under final material acceptance.

⁽³⁰⁾ See also Commission Implementing Decision (EU) 2024/368.

- the results of analysis of migration testing waters by applying an appropriate screening method (see Section 5.1.7).

To ensure the completeness and robustness of the identification of all relevant NIAS, the applicant must identify any reactions the applied-for entry may undergo; identification of such reactions may be supported by appropriate tests such as thermal stability and/or hydrolysis testing (see Section 4.2 and Regulation (EC) No 440/2008, methods A.1 and C.7). In addition, the results of an appropriate screening test from the migration water, for example based on mass spectrometric techniques (see Section 5.1.7 of this guidance and Commission Implementing Decision (EU) 2024/368) or another relevant screening test must be presented. It should be noted that the migration water used for the screening originates from the migration testing carried out as instructed in Chapter 5 and a test piece specified in 5.1.3.1. As it may sometimes be easier to detect NIAS from the material than from water, the representative test piece should be analysed before and after testing – all the more so if from the water analysis it is impossible to reliably distinguish the presence of NIAS.

Further guidance. ECHA guidance on the application of the CLP criteria, Section 2.8.4.3.1 Thermal stability tests and temperature control; ECHA *Guidance on Information Requirements and Chemical Safety Assessment – Chapter R.7b: Endpoint specific guidance*, R.7.9 Degradation/biodegradation.

3.5.2. Non-intentionally added species associated with the use of a starting substance or organic cementitious constituent as a monomer or other reactant (point 3.4 in Annex II to Commission Implementing Decision (EU) 2024/365)

When a starting substance or organic cementitious constituent is used as a monomer or other reactant, the polymerisation or cross-linking during the material manufacturing process may generate NIAS that could be relevant to the migration water. In such cases, the applicant must provide the following analysis:

- an evaluation of the presence of polymerised parts with molecular weight below 1 000 Da that are relevant to the use of the applied-for entry;
- a description of the process that leads to the formation of the polymerised parts identified;
- the molecular weight distribution for the polymerised parts identified on the basis of an appropriate test report and the number average molecular weight of the polymerised parts;
- the molecular mass range (min. and max.) of polymerised parts;
- evaluation of the residual concentrations of polymerised parts (% w/w) in the material.

This information may be based, for example, on appropriate test data for physico-chemical properties, analytical data, etc.

3.5.3. Substance constituents

Substance constituents present at a concentration above 0.02 % w/w in the formulation (or related to the cement content for cementitious materials) and at concentration of 0.1 % w/w or above in the substance should have already been identified under the chemical characterisation of the applied-for entry (see Section 2.3) and should also be considered in the migration testing.

For complex substances (e.g. UVCBs) for which the determination of composition has proven difficult (see Section 2.3), a gas chromatography–mass spectrometry (GC-MS) screening or other relevant tests (high-performance liquid chromatography–mass spectrometry (HPLC-MS),

etc.) of migration test water should always be used to identify the main structures, potentially even individual constituents, and be used as a starting point in identifying relevant chemical species.

If the applied-for entry is a complex substance, only if an applicant can otherwise describe the composition, potentially migrating compounds and relevant chemical species should GC-MS screening or another relevant test be omitted.

3.5.4. Identities of potentially relevant chemical species (point 3.5 in Annex II to Commission Implementing Decision (EU) 2024/365)

For the NIAS and substance constituents identified following the approach described above, the names and other identifiers (CAS and EC numbers) should be provided (see also Section 2.1).

3.6. Other information

3.6.1. Treatment of water contact material prior to use

The applicant may give information on any treatment of the DWCM prior to contact with water, for example sterilisation, cleaning with pressurised steam, rinsing, de-greasing, pre-conditioning, irradiation, e-beam or UV light treatment, etc., that might affect the potential for migration of substances into water.

4. Physico-chemical properties

When reporting physico-chemical properties, first-hand experimental results (primary data) take priority if the methods are suitable for the applied-for entry and are within their validity range. In exceptional cases, it is acceptable to use reference book data, but such data should be of sufficient quality, that is, they must be reliable. Instead of qualitative descriptions, an actual quantitative (numeric) value or a range should be reported, avoiding qualitative terms such as 'high' or 'low' as far as possible.

For UVCBs, some tests may be scientifically not reasonable. In such cases, a range of values, a representative value or values of the individual components should be given instead of a single value depending on the substance. Alternatively, a justification for not providing an experimental result should be given.

The required information below applies primarily to starting substances and organic cementitious constituents. For metallic and inorganic compositions, only colour, melting/freezing point and density apply. For relevant chemical species other than a starting substance or organic cementitious constituent, water solubility and the partition coefficient are required.

In meeting the requirements of Annex III to Commission Implementing Decision (EU) 2024/365, the general rules for adaptations set out in Sections 1 and 2 of Annex XI to the REACH Regulation (EC) 1907/2006 apply *mutatis mutandis*.

Report information on the test materials used to generate the data by including the identities of the species present and their concentration values, following the same rules for reporting the applied-for entry.

Note. Under points 3.2 and 3.3 of Annex II to Commission Implementing Decision (EU) 2024/365, applicants are required to consider thermal stability and hydrolysis tests in the context of the identification of NIAS. While these are undoubtedly physico-chemical properties, they are not included in Annex III nor are they discussed in this section of this guidance document. Thermal stability and hydrolysis are addressed in Sections 3.5.1–3.5.2 of this document in relation to the identification of NIAS other than impurities. Notably, however, thermal stability and hydrolysis testing information must be presented in the physico-chemical properties section of the IUCLID dossier.

Further guidance. ECHA *Guidance on Information Requirements and Chemical Safety Assessment*. See *Chapter R.4: Evaluation of available information*; *Chapter R.7a: Endpoint specific guidance*, R.7.1 Physicochemical properties; *Appendix R7-1 for nanomaterials*⁽³¹⁾.

4.1. Appearance/state of the substance (point 4.1 in Annex III to Commission Implementing Decision (EU) 2024/365)

The information required in this section is that observed at an ambient temperature of 20 °C (293 K) and an ambient atmospheric pressure of 101.3 kPa (1 bar).

Physical state. The physical state may be solid, liquid or gaseous.

Form (aggregate state). The information required is the aggregate state, namely a description of the form or structure: for example aerosol, compact, crystalline, colloid, dispersion (except colloids), fibre, filament, flakes, particulates, paste, pellets, powder, suspension, viscous. A

⁽³¹⁾ Further guidance referenced in this chapter 4 is accessible from <https://echa.europa.eu/guidance-documents/guidance-on-information-requirements-and-chemical-safety-assessment>

starting substance or organic cementitious constituent may be a colloid, that is, it is microscopically dispersed evenly throughout another substance in a system consisting of two separate phases.

Colour. The colour must be reported, including for metallic, enamel, ceramic or other inorganic compositions.

Odour. The report must include both the odour associated with the starting substance or organic cementitious constituent as manufactured and that of a purified substance as noted during the handling of the materials in laboratories or production plants if they are different. This can be, for example, ammonia-like, biting, characteristic of sulphur-containing compounds, characteristic of aromatic compounds, faint, garlic-like, odourless, pungent, slight or sweetish.

Odour should not be investigated for substances that are hazardous by inhalation.

Further guidance. ECHA guidance on the application of the CLP criteria, Chapter 2.0.4 Physical state.

4.2. Melting/freezing point (point 4.2 in Annex III to Commission Implementing Decision (EU) 2024/365)

The measurement of the melting/freezing point should be taken up to 360 °C. Usually the freezing point of liquid substances should be determined if above – 20 °C. An indication that no freezing has occurred during preliminary tests is also acceptable. For viscous liquids, the pour point is an acceptable alternative.

Test according to Regulation (EC) No 440/2008 method A.1 (Melting/freezing temperature). It is advisable to use the differential scanning calorimetry or differential thermo-analysis, since they give additional information about the thermal stability of the applied-for entry such as decomposition onset and energy.

If the melting/freezing point cannot be determined, the sublimation or decomposition temperature should be provided.

Further guidance. ECHA *Guidance on Information Requirements and Chemical Safety Assessment*. See *Chapter R.7a: Endpoint specific guidance*, R.7.1.2 Melting point/freezing point.

4.3. Thermal stability (IUCLID Section 3.4)

For starting substances and organic cementitious constituents, apart from melting/freezing point, specific information concerning thermal stability is required to identify reactions that the applied-for entry may undergo during processing; identification of such reactions may be supported by appropriate thermal stability tests (see Section 4.2, Regulation (EC) No 440/2008 A.1. and Section 3.5).

4.4. Boiling point (point 4.3 in Annex III to Commission Implementing Decision (EU) 2024/365)

The boiling point should be measured up to 360 °C. It should be measured at the normal atmospheric pressure of 101.3 kPa (1 bar) unless decomposition occurs, in which case reduced pressure can be used. The sublimation or decomposition temperature should be provided if the boiling point cannot be determined.

Test according to EU method A.2 (Boiling temperature). Differential scanning calorimetry allows the determination of the melting and boiling point in a single test.

Further guidance. ECHA *Guidance on Information Requirements and Chemical Safety Assessment – Chapter R.7a: Endpoint specific guidance*, R.7.1.3 Boiling point.

4.5. Density (point 4.4 in Annex III to Commission Implementing Decision (EU) 2024/365)

The density of gas starting substances or organic cementitious constituents can be calculated from their molecular weight and the ideal gas law. Polymer density should be determined by buoyancy methods, where appropriate.

For liquids and solids, test according to OECD TG 109 (Density of liquids and solids).

Further guidance. ECHA *Guidance on Information Requirements and Chemical Safety Assessment – Chapter R.7a: Endpoint specific guidance*, R.7.1.4 Relative density.

4.6. Vapour pressure (point 4.5 in Annex III to Commission Implementing Decision (EU) 2024/365)

Test according to OECD TG 104 (Vapour pressure); the reporting requirements are set out in paragraph 50 of the test guideline. Where the vapour pressure is less than 10^{-5} Pa, the vapour pressure at 20 or 25 °C may be estimated by a vapour pressure curve. The vapour pressure does not need to be measured if calculations indicate that the value is significantly less than 10^{-5} Pa.

According to column 2 of Table 1 in Annex III to Commission Implementing Decision (EU) 2024/365, there is 'no need to provide information if the melting point is above 300 °C. If the melting point is between 200 and 300 °C, a limit value based on measurement or a recognised calculation method is sufficient'.

Further guidance. ECHA *Guidance on Information Requirements and Chemical Safety Assessment – Chapter R.7a: Endpoint specific guidance*, R.7.1.5 Vapour pressure.

4.6.1. Henry's law constant (point 4.5.1 in Annex III to Commission Implementing Decision (EU) 2024/365)

According to the above point in Annex III to Commission Implementing Decision (EU) 2024/365, 'Henry's law constant must always be stated for solids and liquids if it can be calculated.' It depends on the water solubility, vapour pressure and molecular weight of a starting substance or organic cementitious constituent and expresses the tendency of a substance to evaporate from aqueous solutions. The unit should be stated as $\text{Pa} \times \text{m}^3 \times \text{mol}^{-1}$. The water solubility and vapour pressure values used to calculate the Henry's law constant need to be measured and given at the same temperature.

Further guidance. ECHA *Guidance on Information Requirements and Chemical Safety Assessment – Chapter R.7a: Endpoint specific guidance*, Appendix R.7.1-1 Henry's law constant and evaporation rate.

4.7. Surface tension of an aqueous solution (point 4.6 in Annex III to Commission Implementing Decision (EU) 2024/365)

The surface tension should be measured using an aqueous solution of sufficient concentration that any surface activity potential is expressed, that is, at 90 % saturation (the concentration must be quoted) to a maximum concentration of 1 g/l (where viscosity permits).

Test according to EU method A.5 (Surface tension), based on OECD TG 115 (Surface tension of aqueous solutions). If the data demonstrate that the starting substance or organic cementitious constituent is surface active, the critical micelle concentration needs to be determined.

Further guidance: ECHA *Guidance on Information Requirements and Chemical Safety Assessment – Chapter R.7a: Endpoint specific guidance*, R.7.1.6 Surface tension.

4.8. Water solubility (point 4.7 in Annex III to Commission Implementing Decision (EU) 2024/365)

The studies must include the effect of pH (5–9) and temperature on solubility. Water solubility should be determined at or near 20 °C. If temperature dependence is suspected, temperature-dependent solubility at 10 and 30 °C should be reported.

Water solubility should be determined unless the starting substance, organic cementitious constituent or other relevant chemical species (as defined in Section 3 of Annex IV to Commission Implementing Decision (EU) 2024/365) is hydrolytically unstable. Phrases such as 'insoluble in water' are insufficient; instead, a limit test should be performed so that a positive statement can be made (e.g. down to the analytical limit). For complex mixtures, a mass balance may be the only practical method. However, the extract should be compared (e.g. by HPLC, GC, liquid chromatography–mass spectrometry (LC-MS), GC-MS)) with the mixture to check for differential solubility values of the constituents.

Where the stability of the substance in aqueous media is such that the water solubility cannot be determined, a justification based on test data must be submitted.

Colloid and micelle formation and other possible observations must be reported.

EU method A.6 (Water solubility), based on OECD TG 105, includes two test methods that cover the whole range of solubility values but are not applicable to volatile substances.

Column 2 of Table 1 in Annex III to Commission Implementing Decision (EU) 2024/365 states that 'for metals and sparingly soluble metal compounds, information on transformation/dissolution in aqueous media shall be provided.' A specific water solubility approach (OECD Guidance Document 29 on transformation/dissolution of metals and metal compounds in aqueous media) has been designed for measuring transformation to the dissolved fraction under standard conditions.

Further guidance. ECHA *Guidance on Information Requirements and Chemical Safety Assessment – Chapter R.7a: Endpoint specific guidance*, R.7.1.7 Water solubility.

4.9. Partition coefficient (*n*-octanol–water) and its pH dependency (point 4.8 in Annex III to Commission Implementing Decision (EU) 2024/365)

For substances that dissociate within an environmentally relevant pH range (i.e. pK_a values from 5 to 9), values for the partition coefficient (K_{ow}) must be derived both for the neutral form and for the dissociated form.

Where the stability of the substances in aqueous media is such that the partition coefficient cannot be determined, a justification based on test data must be submitted.

For substances that are extremely soluble in one of the phases, a limit value should be provided. If necessary, it can be based on the individual solubility values in *n*-octanol and water.

Column 2 of Table 1 in Annex III to Commission Implementing Decision (EU) 2024/365 states that there is 'no need to provide information if the substance is inorganic. Further: If the test cannot be performed (e.g. the substance decomposes, has a high surface activity, reacts violently during the performance of the test or does not dissolve in water or in octanol, or it is not possible to obtain a sufficiently pure substance), a calculated value for the partition coefficient as well as details of the calculation method shall be provided'.

Test according to OECD TG 107 (Shake flask method) and OECD TG 117 (HPLC method; EU method A.24). In addition, OECD TG 123 (Slow-stirring method; EU method A.23) can be used for this end point.

Further guidance. ECHA) *Guidance on Information Requirements and Chemical Safety Assessment – Chapter R.7a: Endpoint specific guidance*, R.7.1.8 Partition coefficient.

4.10. Granulometry (point 4.9 in Annex III to Commission Implementing Decision (EU) 2024/365)

This must be determined and reported for substances such as powders or granules that are nanomaterials ⁽³²⁾. Additional guidance on nanoforms used intentionally in the manufacture of material coming into contact with drinking water is provided in Chapter 7.

Granulometry determines the particle size distribution. A presentation of the particle size distribution is necessary to interpret the data (e.g. in the form of a histogram of particle size vs mass, particle size vs number of particles).

The percentage of particles in mass with an aerodynamic diameter of less than 50 μm must be determined. Many methods are available for particle size measurements, but none of them are applicable to the entire size range. The most appropriate method for the substance must be used.

Further guidance. ECHA *Guidance on Information Requirements and Chemical Safety Assessment – Chapter R.7a: Endpoint specific guidance*, R.7.1.14 Granulometry.

⁽³²⁾ Commission Recommendation of 10 June 2022 on the definition of nanomaterial (2022/C 229/01) (OJ C 229, 14.6.2022, p. 1), [https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32022H0614\(01\)](https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32022H0614(01)).

4.11. Dissociation constant (point 4.10 in Annex III to Commission Implementing Decision (EU) 2024/365)

Column 2 of Table 1 in Annex III to Commission Implementing Decision (EU) 2024/365 states that there is:

no need to provide information in the following cases:

- (a) the substance is hydrolytically unstable (half-life less than 12 hours) or is readily oxidisable in water;
- (b) it is scientifically not possible to perform the test, for instance if the analytical method is not sensitive enough;
- (c) based on the structure, the substance does not have any chemical group that can dissociate.

Test according to EU method A.25 (Dissociation constant), equivalent to OECD TG 112.

Further guidance. ECHA *Guidance on Information Requirements and Chemical Safety Assessment – Chapter R.7a: Endpoint specific guidance*, R.7.1.17 Dissociation constant.

4.12. Hydrolysis as a function of pH

To ensure the completeness and robustness of the identification of all relevant NIAS, the applicant must identify any reactions the applied-for entry may undergo when in contact with water; identification of such reactions may be supported by appropriate hydrolysis tests (see Section 4.2 and Regulation (EC) No 440/2008 C.7).

5. Data on migration of the applied-for entry and identification of relevant chemical species

This chapter develops concepts related to the determination of the migration of a **substance** or **constituent** from its **matrix** (DWCM). To include/renew starting substances, compositions or constituents in the EUPLs, accurate assessment of the migration of relevant chemical species is essential. The amount migrated determines the extent of the hazard data requirements for the relevant chemical species (see Chapter 6) and the setting of the MTC_{tap} (see DWD Guidance Volume II).

The determination of migration consists of three separate steps.

- Migration contact with repeated periods of specified length. For organic, cementitious, enamel, ceramic and other inorganic materials, the contact periods are static. For metallic materials, the contact is dynamic with alternating static and flow periods.
- Sampling and analysis of the migration test water(s) for specified periods or time intervals to identify and quantify the concentration of relevant chemical species identified in the test water (see Section 5.4).
- Screening analysis of the test water to detect NIAS (for organic and cementitious materials) and additional analyses to quantify these substances.

The test results enable either direct calculation of an average concentration in water (metallics) or an average migration rate (organic, cementitious materials, enamels, ceramics and other inorganics). These can be used to derive a value representative of the concentration of the substance at the tap (C_{tap}) under the worst foreseeable conditions of use or known intended use with predetermined CFs (worst-case $CF = 20$ (see Commission Implementing Decision (EU) 2024/368) for organic, cementitious, enamel, ceramic and other inorganic materials) or relative surface area and the 'a'-factor (metallic compositions). The CFs depend on (1) contact surface to water volume ratio and (2) contact time of the product. Higher CF value are correlated with higher surface to volume ratio, longer contact time and consequently higher estimated C_{tap} values and greater risk from using the product.

While the general principle in migration testing is the same for all material types, because of the different nature of entries in different EUPLs (the organic and cementitious materials EUPLs contain chemical substances, while the other EUPLs contain compositions), and the differences in their interaction with their environment (drinking water), test methods and their operating parameters differ among material types.

5.1. Physical testing of migration (points 5.1–5.3 in Annex IV to Commission Implementing Decision (EU) 2024/365)

This section describes the methodologies related to the physical migration testing for accepting starting substances, compositions and constituents on to the EUPLs.

The aims of physical migration testing are to:

- quantify the long-term release of organic and inorganic substances from the contact layer, for example bulk material or coating;
- define the subsequent human health hazard data requirements for relevant chemical species according to their migration tier.

The physical testing of substance migration from DWCMs aims to simulate the real interaction between drinking water and the DWCM taking place in water abstraction, treatment, transport, distribution and storage installations. While the basic concept of migration testing is simple, there are certain principles that should be noted to ensure that tests are comparable and produce meaningful results.

The terms (and definitions) used to describe the spontaneous transport of substances from a solid matrix into the liquid matrix in scientific and regulatory language may differ (leaching, migration, release, etc.). For clarity, in this guidance the spontaneous transfer of substances from a solid matrix into a liquid matrix (i.e. drinking water) is from now on referred to as **migration** and defined as the net mass transfer of substance A incorporated within solid matrix B into water intended for human consumption. This aligns with the definition of migration given, for example, in standard EN 12873-1. Scientific concepts relevant to migration include thermodynamics (e.g. chemical potential for organic materials and electrochemical for metals) (Mäkinen, 2008) and kinetics (e.g. diffusion). Thus, the migration of a substance depends on the **properties of the substance** (molecular mass, etc.), the **properties of the matrix** (density, etc.) within which it is contained, the **characteristics of water** (dissolved oxygen, pH, etc.) and the **system** (temperature, pressure, etc.). Consequently, careful reporting of the test conditions, intended use and test water composition is important.

When testing migration for the purposes of the DWD, it is mandatory to use the most recent versions of material- and application-specific European standard testing and analytical methods, where available. These European standards are referred to in this guidance and proposed in, for example, a Joint Research Centre (JRC) technical report (Senaldi et al., 2021). The test and analytical protocols given in the standards should enable reliable substance migration assessments.

This chapter describes the requirements for and scope of physical migration testing necessary to have substances accepted on to the EUPLs. It is divided into important test parameters (Section 5.1.1), migration testing for each material type (Sections 5.1.2–5.1.5), considerations for acceptance criteria (Section 5.1.6) and analysis of relevant chemical species in the test water (Section 5.1.7). In addition, the sections on migration testing for different materials are divided into sections concerning test pieces and migration testing.

As indicated in Section 1.1 of Annex IV to Commission Implementing Decision (EU) 2024/365, 'any testing under this Section shall be carried out in compliance with the principles of good laboratory practice provided for in standard EN ISO/IEC 17025 or other international standards recognised as being equivalent by the Commission or ECHA.'

For any migration test, a full study report adhering to the relevant European standard must be included in an application to ECHA and such adherence must be clearly stated by the person or organisation that has performed the migration testing (see Article 3(10) of Commission Delegated Regulation (EU) 2024/369).

5.1.1. Important test parameters

The migration test for non-metallic materials resembles a kinetic test with the purpose of quantifying the rate of change in concentration, or migration rate, of the migrant in the chosen liquid (water) as a function of a specified duration. It consists of repeated stagnated contact periods and exchanges of migration water between the contact periods as defined in the test procedure. It is assumed that during contact equilibrium (saturation) concentration is not reached. In contrast, for metallic materials, the migration test aims to investigate the rate of corrosion under steady-state conditions. It should be noted that a change in a migration rate is related to a change in the properties of the migration system. With knowledge of the average migration rate, the concentration of the migrating substance at the tap at a given time can be

estimated. The testing methods used to fulfil legal requirements assume that the migration rate is constant.

To properly conduct a migration test and to assess its results, specific key test parameters are important and depend on the material type. In reporting migration test results, key test parameters are always included.

5.1.1.1. Key test parameters

- **Surface to volume ratio.** The contact surface area of material to volume of water it is in contact with affects the observed migration rate. It is generally preferable that the shape of the test subject(s) resembles the actual product in use when possible and that the surface area has been determined with a relative accuracy of 10 %. **It is recommended that the S/V ratio ranges presented in standards EN 12873-1 and EN 14944-3 (for organic and cementitious materials) are adhered to.** In addition, for metallic materials, the geometry of and correction factor for the surface area of test pieces in the rig test, as presented in EN 15664-1, should be respected to ensure the use of an appropriate S/V ratio in the interpretation of test results to derive realistic concentrations. In testing in accordance with standards EN 12873-1 and 14944-3, too high or too low an S/V ratio can result in a saturated liquid or in concentrations below the limit of detection (LOD) and/or limit of quantification (LOQ) of the analytical methods.
- **Time.** To accurately compare individual migration test results, the order and duration of the respective migration period(s) must be equal. A migration period is:
 - for organic, cementitious, enamel, ceramic and other inorganic materials, the duration of contact between the test piece and stagnant test water before sampling, analysis and subsequent water change;
 - for metallic materials, the order number of the week of sampling.

In most cases, the first test period may show concentrations higher than the later test periods.

- **Temperature.** The temperature of the migration test water is categorised as cold (23 ± 2 °C), warm (60 ± 2 °C) or hot (85 ± 2 °C). An increase in temperature usually increases the rate, diffusion and mass transfer of migration and can lead to significant changes in migration. Organic materials are particularly susceptible to increases in migration caused by increases in temperature. In addition, some metallic compositions also show increasing corrosion rates with increasing contact water temperature.
- **Water type.** The water type affects migration, particularly in the case of metallic materials. For the purposes of initial acceptance on to or renewal of an entry on the EUPL, the test waters specified in the relevant migration testing standards must be used. For organic, enamel, ceramic and other inorganic materials, testing can use demineralised chlorinated or non-chlorinated water, as set out in standard EN 12873-1. For metallic materials, the test waters specified in standard EN 15664-2 are required. Lastly, for cementitious materials, the test waters defined in standard EN 14944 are required.

It should be noted, that according to the DWD, the water should not be aggressive or corrosive. While these terms are not defined in the directive, limit values, as set out in part C of Annex I to the directive, for chloride content, conductivity, hydrogen ion concentration (pH) and sulphate define non-aggressiveness and non-corrosiveness of water.

For metals, the waters in standard EN 15664-2 are specified in terms of pH, alkalinity, sum of $[\text{Cl}^-]$ and $[\text{SO}_4^{2-}]$, and oxygen and total organic carbon concentrations. In the scientific literature, other parameters, such as free CO_2 concentration, have been used to determine the corrosivity of water for metals. In the test waters specified in standard EN 15664-2, the CO_2 concentration is determined via pH and alkalinity, which determine CO_2 levels in equilibrium. The amount of CO_2 in the test water should be reported in the migration test report for metallic materials, and this is required in accordance with standard EN 15664-1.

In addition to key test parameters, there are physico-chemical properties indicative of migration that can complement physical test data. The indicative physico-chemical properties water solubility and partition coefficient (*n*-octanol–water) are always required for relevant chemical species in organic and cementitious materials and in addition the partition coefficient (organic substance–water) if modelling is used. The diffusion coefficient is also required when modelling is used (see Section 5.3.1).

5.1.1.2. Indicative physico-chemical parameters

- **Partition coefficients *n*-octanol–water and organic substance–water** (see also Sections 4.9 and 5.3.1). These concern organic substances and represent the thermodynamic tendency of a substance to dissolve in water, thus reflecting its tendency to migrate. The ***n*-octanol–water** partition coefficient is a legal requirement for organic substances. The organic substance–water partition coefficient is a mandatory requirement for diffusion-based modelling. It is required for each individual substance modelled whether it is the applied-for entry or an NIAS. It can be estimated using various methods, as described in standard CEN/TR 16364, for example by direct measurement, using existing values or deriving it from the *n*-octanol–water partition coefficient (see Section 5.3.1 for more detail).
- **Diffusion coefficients.** These represent kinetic aspects of the release rate. A small diffusion coefficient indicates that atoms and molecules are propagating slowly in the matrix, suggesting slow migration. This parameter is generally estimated from the basic physico-chemical properties of a substance and the material matrices. The diffusion coefficient is mandatory if diffusion-based migration modelling is used (see Section 5.3.1).
- **Water solubility** (see Section 4.8). This can be used to derive partition coefficients.

5.1.2. Worst foreseeable conditions

According to Section 1.3 of Annex IV to Commission Implementing Decision (EU) 2024/365, testing of **migration should take place under conditions representative of the worst foreseeable conditions of use**. In practice, the worst foreseeable conditions means conditions in which the migration rate would be highest. This section gives general guidelines on what to consider when determining the worst foreseeable conditions.

For applied-for entries where an MTC_{tap} value needs to be derived for one or more constituents, impurities or other NIAS, migration testing under the worst foreseeable conditions of use ensures an appropriate volume of toxicological information for determination of the MTC_{tap} value and a more realistic representation of the long-term migration behaviour in real use conditions. Inaccurate testing might result in an inaccurate (even too low) MTC_{tap} value (see also Chapter 4 of DWD Guidance Volume II on MTC_{tap} setting). Inaccurate (excessively low) MTC_{tap} values can then create problems at the level of testing the final material for product certification (see criteria in Commission Implementing Decision (EU) 2024/368), leading to the rejection of the product.

Parameters that may affect migration can be roughly divided into the following categories.

- **Parameters related to the substance** (i.e. the individual constituent migrating into water). These parameters depend on the substance itself and may vary according to the nature of the substance, for example whether it is organic or inorganic. They include parameters such as molecular weight, polarity, charge, concentration and solubility.
- **Parameters related to the material** (i.e. the matrix from which substances migrate). These parameters depend on the host material in which the substance is incorporated and vary according to material type. They are descriptors of the **diffusivity** (i.e. rate of diffusion) of the material and include parameters such as molecular weight, crystallinity, viscosity, phases present, density, polarity, tacticity, presence of copolymers or additives, state of degradation, molecular structure and degree of cross-linking.
- **Parameters related to the environment, or 'system'**. These properties depend on the surroundings of the test material such as temperature, water type and S/V ratio.

The choice of worst foreseeable conditions may depend on interactions between multiple parameters of the substance, material and environment. Therefore, in selecting the worst foreseeable conditions, the intended use conditions should be scrutinised as a whole, taking into account all substance-, material- and environment-related parameters together.

It should be noted that the parameters and their importance may vary according to the substance and material types. While applications for organic and cementitious materials concern single substances and those for metallics, enamels, ceramics and other inorganics concern a material consisting of one or more constituents, the relevant chemical species migrate from each material type as individuals.

Selection of the worst foreseeable conditions is likely to become progressively difficult as the complexity of the substance, matrix or environmental conditions increases. For example, for a pure substance in a simple one- or two-component, homogeneous matrix in cold water, selection of the worst-case test piece could be much easier than for a UVCB in a complex multipolymer matrix.

For complex, UVCB-like substances, support can be taken from a recent EFSA opinion where the lowest average carbon number was used to determine the worst-case composition of a non-defined substance (EFSA Panel on Food Contact Materials, Enzymes and Processing Aids et al., 2023).

Table 2 lists indications for selecting the worst foreseeable conditions for migration testing depending on substance-, material- and environment-related parameters. These indications are general and do not always apply directly. Nonetheless, these are parameters that should be considered when manufacturing test pieces and designing testing conditions for migration testing.

The terms 'high', 'low' and 'varies' in Table 2 can be interpreted as follows.

- 'High' indicates that a higher value is typically (but not always) leads to higher migration and is indicative of the worst-case conditions.
- 'Low' indicates that a lower value is typically (but not always) leads to higher migration and is indicative of the worst-case conditions.

- 'Varies' indicates that the value of the parameter indicative of the worst-case conditions can be high or low depending on the case. Particularly for water type, there is considerable variance in the effect of water on migration from the solid it is in contact with.

Selecting the worst foreseeable conditions for some of the parameters, for example temperature, is typically straightforward for all the materials except metallics, where higher temperatures may lead to decreased migration. Thus, care should be taken in selecting and justifying the worst foreseeable conditions.

Justification of the worst foreseeable conditions, as described in the application, is the responsibility of the applicant.

When the worst-case conditions in actual use are unknown or uncertain, the following provides general guidance on **default worst-case testing conditions** that comply with the hygienic requirements for DWCMs.

- organic, cementitious and inorganic materials:
 - water type:
 - cold water test: both non-chlorinated (and demineralised) and chlorinated (as in standards EN 12873-1 or -2 and EN 14944-3);
 - warm and hot water test: non-chlorinated water (as in standards EN 12873-1 or -2 and EN 14944-3);
 - water temperature: hot (85 ± 2 °C) and only in non-chlorinated water;
 - test duration:
 - single-layer organic materials, cementitious and inorganic materials: minimum three migration periods at 72 hours each in cold water or seven migration periods at 24 hours each in warm or hot water if relevant;
 - multilayer organic materials in which the applied-for entry or the relevant chemical species studied is not on the surface layer and directly in contact with water: minimum 3 migration periods at 72 hours each in cold water and 22 migration periods at 24 hours each in warm or hot water;
 - CF = 20.
- metallic materials:
 - water type: as specified in standard EN 15664-2;
 - water temperature: cold (23 ± 2 °C);
 - test duration: minimum 26 weeks;
 - sampling: as specified in standard EN 15664-1.

Table 2. Selecting the worst foreseeable conditions for migration testing

Category	Parameter	Additional description and remarks	Material type (EUPL)			
			Organic	Cementitious	Metallic	Enamels, etc.
Substance related	Content/ concentration of a substance/ constituent in a material	Higher content or concentration generally leads to higher chemical activity and higher migration potential.	High	High	High for non-corrosion-inhibiting constituents Low (for corrosion-inhibiting constituents)	High
	Molecular weight	Weight of the molecule in Daltons. Molecular weight has considerable influence on migration. Light molecules migrate faster than large ones.	Low	Low	N.R.	N.R.
	Molecular size	Physical dimensions of the molecule, e.g. diameter. Small molecules migrate faster than large ones.	Low	Low	N.R.	N.R.
	Average carbon number	Indicator for weight and size.	Low	Low	N.R.	N.R.
	Polarity	Polar or non-polar. Because of the polarity of water, polar molecules may dissolve in water, leading to migration, while non-polar molecules may be water insoluble, leading to lower migration.	High or variable	High or variable	N.R.	N.R.
	Charge		Varies	Varies	N.R.	N.R.

Category	Parameter	Additional description and remarks	Material type (EUPL)			
			Organic	Cementitious	Metallic	Enamels, etc.
	Solubility of the material	Constituent may exist dissolved in a material or as separate exclusions or phases, e.g. lead in copper alloys. Undissolved substances potentially migrate more easily than completely dissolved substances. For example, for organic materials, a relevant substance that has low solubility in the material and high solubility in water could represent a worst case.	Low	Low	Low	Low
Material related	Molecular weight	Molecular weight of the substances forming the matrix may indicate how much free space there is for diffusion.	Low	Low	N.R.	N.R.
	Crystallinity	Measure of internal degree of ordering. For organic materials, this describes the organisation of polymer chains and is given as a percentage. Other materials are considered mainly crystalline or amorphous with the type of ordering varying. Crystallinity relates to other properties such as density.	Low Low crystallinity relates to lower density and more space for migration.	Low	Varies	Varies
	Viscosity	An indicator of resistance to shear forces and potentially also indicates resistance to diffusion.	Low	Low	N.R.	N.R.
	Phases present	If a migrating substance needs to pass through different phases of varying diffusivity it may affect migration.	Varies	Varies	Varies	Varies

Category	Parameter	Additional description and remarks	Material type (EUPL)			
			Organic	Cementitious	Metallic	Enamels, etc.
	Polarity	Polarity differences compared with the matrix affect migration. This may also relate to solubility of the migrating substance to the material matrix.	Varies	Varies	N.R.	N.R.
	Density	Describes how tightly atoms and molecules in a material are packed and how much free space there is for diffusion. Relates to crystallinity, with higher ordering usually indicative of higher density. Low density materials may have more space for diffusion.	Low	Low	Varies	Varies
	Presence of copolymers or additives	Affects structure and hence may affect diffusivity.	Varies	Varies	N.R.	N.R.
	Tacticity	Affects structure and hence may affect diffusivity.	Varies	Varies	N.R.	N.R.
	State of degradation		High	High		
	Molecular structure		Varies	Varies	N.R.	N.R.
	Degree of cross-linking	Expressed as a percentage. Describes bonding between polymers. Higher value indicates more resistance to diffusion.	Low	Low	N.R.	N.R.
Environment related	Temperature	The water temperature affects migration. Higher temperature typically increases migration for organic substances. For metals, it is more complicated.	High	High	Varies	High

Category	Parameter	Additional description and remarks	Material type (EUPL)			
			Organic	Cementitious	Metallic	Enamels, etc.
	S/V ratio	The relative surface available for migration. If migration is considered to occur at a constant rate, then a higher S/V ratio results in higher migration.	High if no saturation of the liquid is expected. Low if saturation of the liquid is expected. The S/V range given in EN 12873-1 should be respected. Water solubility and partition coefficients may be used to estimate saturation and migration potential.	High	High	High
	Water type	Water type may affect migration and formation of NIAS via hydrolysis depending on its properties, e.g. chlorinated water may increase migration compared with non-chlorinated water.	Varies	Varies	Varies	Varies

NB: N.R., not relevant.

5.1.3. Organic materials

According to Section 1.4 of Annex IV to Commission Implementing Decision (EU) 2024/365, physical migration testing for starting substances should encompass the following.

- The starting substance.
- Any substance constituent, and each NIAS identified in accordance with points 1.3–1.5 in Table 1 of Annex I and point 3 in Table 2 of Annex II to Commission Implementing Decision (EU) 2024/365. NB: Each individual constituent should be identified. If for a technical/scientific reason this cannot be done, a justification should be provided. When the application is submitted the details of the manufacturing process must be available; therefore, applicants should know what type of impurities and degradation products might be migrating and, hence, tested for.
- Any starting substance that functions as a monomer or other reactant of a main polymer in the material.

Migration testing of starting substances for organic materials should be conducted in accordance with **standard EN 12873-1 or EN 12873-2** (Static immersion). If this method is inapplicable, other scientifically valid methods may be used on condition that a full scientific justification for their selection is provided together with the study results. The scientific justification for demonstrating the equivalence of a test method must include:

- description of the test protocol;
- materials used in the test (test waters, containment vessels, pipes, hoses, etc.);
- S/V ratio corresponding to standard EN 12873;
- contact time;
- results with analytical methods and measured concentrations;
- calculation of the migration rate and evaluation of its correspondence to the migration rate calculated using standard EN 12873;
- an uncertainty analysis of the study and the results.

A full migration test report complying with the requirements of standard EN 12873-1 (or -2) must be provided by the applicant.

5.1.3.1. Test pieces

The full test report must include a sufficiently detailed description of the test pieces (and a reference if applicable, e.g. an inert sample or blank sample) used in the migration test, including:

- Name of the manufacturer of the test pieces;
- manufacturing and processing details of the test piece(s) as in Section 3.3.2 for bulk pieces and Section 3.3.2.1 for coatings (e.g. polymerisation process, fabrication method, relevant additional processing steps);
- dosage of the applied-for entry to produce the test piece(s);

- measured content (% w/w) of the applied-for entry in the matrix material, that is, percentage of starting substance in the test piece;
- chemical and physical composition properties/structure (e.g. formulation, analytical data of chemical composition, physical shape, single or multilayer, dimensions, thickness, crystallinity, viscosity, density, polarity, tacticity, presence of copolymers, molecular structure, degree of cross-linking, resin type, repeat unit composition of main polymer, matrix molecular weight);
- if the applied-for entry is used in multilayer materials, it is advisable to test only a single layer material containing the substance as a first choice; only if the layers cannot be separated because of, for example, the manufacturing process, should the multilayer material be tested;
- treatment prior to testing, including cleaning of the surface from residual products and substances from processing;
- storage period: multilayer material test pieces must undergo a minimum 30-day period of storage at room temperature prior to testing;
- any other relevant information.

The test pieces must be representative of the applied-for uses under the worst foreseeable conditions (see Section 5.1.2), that is, the maximum concentration of the applied-for entry (starting substance) should be used in their preparation.

5.1.3.2. Migration testing

By default, only certain additives will migrate from organic materials in significant quantities. In addition, residual monomers or polymerised parts may migrate.

According to standard EN 12873-1 (or -2), a static immersion test consists of immersing (contacting) a test material/product made with the substance being assessed and a specified contact surface area into a specified volume of test water under defined repeated use conditions. The test material/product should represent the final material described for the intended use or, where this is impossible, be as close to the final material as possible. The choice of the actual geometry of the test material/product can vary (e.g. a pipe, pipe segment or plate). Migration can be determined from the material or product or, if this is impracticable, from a test piece representative of the material or product. In addition to the test piece information (see Section 5.1.3.1), the full migration test report provided by the applicant, to comply with the requirements of standard EN 12873-1 (or -2), should include the following information:

- test information and relevant testing conditions – pretreatment, duration, water type(s), S/V ratio used (it is advisable to adhere to S/V ratios as defined in standard EN 12873-1 or -2), temperature of water test (25, 65 or 85 °C);
- test results – sampling, analysed concentrations (see Section 5.1.7), measurement uncertainties;
- description of any deviations from the specified procedure;
- list of disturbances – any factors that may have affected the results, such as any incidents or any operating details not specified in the standard used;
- any other information considered relevant.

5.1.4. Metallic materials

According to Section 1.4 of Annex IV to Commission Implementing Decision (EU) 2024/365, physical migration testing for metallic compositions should encompass each metallic composition constituent and impurity identified in accordance with points 1.3.2 and 1.3.3 in Table 1 of Annex I to Commission Implementing Decision (EU) 2024/365, unless:

- the metallic composition constituent is one of the following elements: phosphorus, silicon, sulphur or tin;
- the metallic composition impurity is one of the following elements: aluminium, iron, manganese, phosphorus, silicon, tin or zinc ⁽³³⁾.

Metals and metallic alloys in contact with water are subject to corrosion via electrochemical reactions, resulting in the oxidation and subsequent release of metal ions into the water. Because their interaction with water differs from the other types of materials relevant to the DWD, testing for the hygienic fitness of metallic compositions also differs.

The corrosion resistance of a metal or a metallic alloy in various environments depends on its chemical composition and also on its microstructure (crystallography, grain size and shape, grain heterogeneity, second phases, impurity inclusions), residual stress and surface condition. The corrosion of metals in contact with water, when only the water-metal interaction is considered, is mainly uniform or localised corrosion (see glossary of key terms). In uniform corrosion, the corrosion occurs uniformly over the contact surface, resulting in significant concentrations of metals in the water. In contrast, some alloys are 'passive', that is, less susceptible to general corrosion because of a protective passive layer on the surface. Local corrosion can occur on passive alloys. In local corrosion, the corrosion is concentrated in small areas of the contact surface. From a hygienic perspective, uniform corrosion is more important, as local corrosion has a minor impact on the release of metals into the water but can have significant impact on the technical function of the alloy. Corrosion type (i.e. uniform or local) affects the suitability of the tests used to assess the hygienic fitness of water contact materials and subsequent risk acceptance.

5.1.4.1. Test pieces

For the testing of uniform corrosion of a metallic composition, according to standard EN 15664-1 or similar, the test pieces must be, if possible:

- representative of the metallic composition that is applied for;
- representative of the worst-case composition in terms of corrosion behaviour; it should be noted that the applicant's justification for the selection of worst-case corrosion behaviour (see information and guidelines below) should be attested by a written report based on, for example, the results of scientific research, experience of previous use or expert judgement.

The full migration test report must include a detailed description of the test pieces equivalent to that in standard EN 15664-1 (and a reference if applicable, e.g. an inert sample or blank sample) used in the migration test, including:

⁽³³⁾ Applicants are reminded that certain elements (calcium, magnesium, potassium, sodium) appear in Annex V to Commission Implementing Decision (EU) 2024/367 with the indication 'Not applicable' in the place of an MTC_{tap} value. 'Not applicable' means that the metal is not hazardous and no MTC_{tap} value applies, and so it can be excluded from the list of relevant chemical species.

- sampling and name of the manufacturer of the test pieces;
- compliance with standard EN 15664-1;
- manufacturing and processing details, as set out in Section 3.3.3 for bulk pieces and Section 3.3.3.1 for platings;
- chemical composition of the test piece(s);
- technical function of each constituting element: this involves selecting the worst foreseeable concentration for the test piece (see Section 5.1.2 and below);
- physical (single or multilayer, dimensions) composition;
- surface roughness of the inner surface (point 5.1.5 of Annex IV to Commission Implementing Decision 2024/365); surface finishing of the test piece should be such that the parameter $R_z = 4\text{--}10\text{ }\mu\text{m}$ for the test piece (EN ISO 21920-2);
- treatment prior to testing, including cleaning the surface to remove residual products and substances from processing, if relevant;
- storage period;
- any other relevant information, such as results of a study of the test piece surface after testing (i.e. post-mortem study).

It should be noted that it is difficult to give exhaustive guidelines of worst-case corrosion for all alloys. Different elements have different effects on corrosion and the release of metals into the water. Some elements, such as aluminium, can have both beneficial and deleterious effects, either contributing to corrosion resistance or lowering it, depending on the alloy. Some elements have better corrosion inhibition properties in the lower concentration range, while others exhibit better corrosion inhibition properties in the higher concentration range. The combination of different alloying elements is also important. Nonetheless, some general guidelines for determining the worst-case composition of a test piece of a metallic composition, particularly copper alloys, are as follows:

- all constituent elements and impurities must be within the range declared for the composition unless otherwise specified below;
- zinc, copper and tin must be within the declared range;
- the concentration of arsenic must be at least 66 % of the declared range;
- the concentration of minor (< 10 %) constituents in the test piece that inhibit corrosion more effectively at the higher end of the declared range (e.g. aluminium, phosphorus, silicon), based on the possible interactions between the main and minor alloying elements, should be less than 50 % of the declared range;
- the concentration of minor (< 10 %) constituents in the test piece that inhibit corrosion more effectively at the lower end of the declared range, based on the possible interactions between the main and minor alloying elements, should be more than 50 % of the declared range;
- the concentrations of all other minor (< 10 %) constituents must be more than 80 % of the declared range;

- the content of impurities that must be analysed in the migration water should be more than 60 % of the declared maximum content.

The above guidelines are proposed because the change in the relative concentration, and thus the effect, is greater for minor constituents than for major constituents. However, because of the complexity of corrosion phenomena, each composition should be analysed individually to determine the content of individual constituents.

A description of the analytical methods used in the identification and quantification of the constituents of the metallic composition test piece, as well as impurities and crystalline structure, that is, detected phases, must be provided. The description should consist of the experimental protocols followed and the relevant interpretation of the results. This information should be sufficient to allow the methods to be reproduced. Standardised analytical methods must be used when possible. Suitable analytical techniques may include ICP-MS, X-ray fluorescence (e.g. standards EN 15063-1 and -2), X-Ray diffraction, AAS and optical emission spectroscopy (OES) (e.g. spark OES (S-OES) standard EN 15079, inductively coupled plasma OES (ICP-OES) standard EN 15605).

The analytical methods used should be validated and documented in accordance with standard EN ISO/IEC 17025 or other equivalent standards accepted at the international level, where possible, according to the criteria for analysing relevant chemical species in water set out in Section 5.1.7. The analytical criteria for all relevant analyses in the context of the DWD are similar.

Techniques such as optical microscopy or electron microscopy may be used to provide additional structural information from the test piece and its surface both before and after testing, including light micrographs and electron micrographs. In addition, phase diagrams or potential/pH (E-pH) diagrams can be provided. Phase diagrams and E-pH diagrams may be experimental or computational (e.g. FactSage, ThermoCalc, MTDATA, HSC Chemistry) but relevant source information such as model and databases used must be given (Andersson et al., 2002; Davies et al., 2002; Bale et al., 2016; Roine, 2021).

Specific criteria concerning the composition of the reference material are described below.

Lastly, it should be noted that these guidelines for test piece composition do not apply when testing 'passive' alloys.

5.1.4.2. Migration testing

The following tests, according to whether a metallic composition is passive or not, should be used to determine hygienic fitness:

- long-term migration tests according to standard **EN 15664-1** using test waters specified in standard **EN 15664-2**;
- for compositions exhibiting a tendency to passivate, an electrochemical test to determine passivity and passive film stability (e.g. standard **EN 16056** for steels/passive alloys) is suitable for passive alloys.

According to standard EN 15664-1, the test is performed using a test rig consisting of test lines (containing a test pipe or test pieces, an inert material (i.e. a control line), the test water inlet and discharge arrangements (see further detail below)). Water is set to flow in a pattern of alternating stagnant and flow periods at a controlled temperature through the test material for a minimum of 26 weeks. Test parameters and water compositions are analysed during the duration of the test. As the migration test, according to standard EN 15664-1, must be conducted

using either test pipes or test pieces of specified size (inner diameter 13 mm) of which the whole surface area (100 %) is exposed to water, the test sample represents the worst-case conditions in terms of the highest S/V ratio (ca 30 dm⁻¹), thus maximising the migration (metal release) rate.

The test according to standard EN 16056 is an electrochemical test in which the relationship between the current (corrosion rate) and the potential between the sample material electrode and a reference electrode is investigated within specified ranges. From the changes in current (and current density) in relation to potential, the corrosion rate (and passivity) of the test material can be determined.

5.1.4.3. Creation of new metallic composition category by testing a reference material

Every metallic composition applied for should belong to a 'metallic composition category'. For compositions tested according to standard EN 15664-1 or similar, these categories determine the constituent elements and impurities and their ranges that the compositions belonging to them must fulfil: to belong to a category, a composition must have the same alloying elements and be within the ranges determined for the category. In contrast, the category 'passive materials' encompasses a wide variety of different alloys. These alloys belong to a single category because their safety of use depends on the stability of the passive layer, which depends on different factors for each alloy but results in a similar resistance, as demonstrated by an electrochemical test conducted according to standard EN 16056 with stainless steel as the reference material.

For metallic compositions that do not fall within one of the existing metallic composition categories, a new category must be created by:

- setting out the constituents and impurities and their accepted limits;
- setting out and testing a reference material of a restricted composition with three test waters as specified in standard EN 15664-2.

The reference material has the same general requirements as regular test pieces (see Section 5.1.4.1). In addition, it should be ensured that the reference material is the most corrosive composition. For instance, for the copper–zinc alloy category, the reference material is the one with the highest zinc content, that is, the least noble composition.

While it may be difficult to give exhaustive guidelines concerning metallic composition categories and each new category requires expert review, some general guidelines for creating new categories are as follows.

- A category is defined by the alloying elements. Alloying elements have a technological function in the alloy, and the minimum and maximum value of the content have to be defined. Other elements present in the alloy are impurities, for which only the maximum value has to be stated.
- The categories and their respective compositional boundaries must not overlap. It is important that for each alloy only one category applies. Thus, it is important for an applicant to check carefully whether their alloy is covered by an existing category or whether the creation of a new category is required. A composition cannot belong to more than one category.

- Lead-free ($\leq 0.2\%$) and lead-containing ($> 0.2\%$) alloys and their categories should be distinguished. A category in which the lead concentration range extends from lead free to lead containing (e.g. from 0.1% to 0.3%) is not possible.

5.1.4.4. Adding a composition to an existing metallic composition category

Where the constituents of a metallic composition candidate for approval are found to fall within an existing metallic composition category, an alloy can be accepted into that existing category by absolute testing with the worst-case test water(s). The worst-case test water is the one exhibiting the highest metal releases when the reference material for the category is tested. The worst-case test water for each category is shown in Annex IV to this guidance. If the worst-case test water is not obvious (e.g. for lead release, the worst-case test water is test water 1 and for the nickel release it is test water 2) several (up to the three test waters according to standard EN 15664-2) can be defined as worst case. In this case, according to standard EN 15664-1, the test must be completed with all worst-case test waters.

5.1.4.5. Reporting

The test report for the **long-term leaching test** and **electrochemical test** must provide the following information on the test set-up:

- test reports containing full details as specified in the relevant standard used (EN 15664-1 or EN 16056);
- test piece information as laid out in Section 5.1.4.1;
- test information and relevant testing conditions – pretreatment, duration, water type(s), etc.;
- list of disturbances – any factors that may have affected the results, such as any incidents or any operating details not specified in the standard used;
- any other information considered relevant.

5.1.5. Cementitious materials

According to Section 1.4 of Annex IV to Commission Implementing Decision (EU) 2024/365, physical migration testing for organic cementitious constituents should encompass:

- the organic cementitious constituent;
- any substance constituent, and each NIAS identified in accordance with points 1.3–1.5 in Table 1 of Annex I and point 3 in Table 2 of Annex II to Commission Implementing Decision (EU) 2024/365;
- any organic cementitious constituent that functions as a monomer or other reactant of a main polymer in the material.

Migration testing of cementitious materials should adhere to the same principles as the tests for organic and inorganic materials, with the migration testing conducted as a static immersion test. Different parts of standard **EN 14944** (particularly parts 3 or 4) describe a method for testing the migration rate of substances, that is, organic cementitious constituents from cementitious materials. **A full study report of the migration test complying with EN 14944 must be provided.**

All the requirements listed for organic materials (see Section 5.1.3) apply, with the following additional recommendations.

- The S/V ratios set out in standard EN 14944 should be used.
- Samples should be preconditioned with demineralised water as specified in standard EN 14944 (solution of calcium chloride and sodium bicarbonate) containing anhydrous calcium in five sequential periods: one 72-hour preceded by four 24-hour periods). The pH of the water used for preconditioning in contact with the sample must be lower than 9.5 after the preconditioning has taken place.
- For linings, porosity should be assessed, for example as described in standard EN 14944-3, to assess cementitious behaviour.

5.1.5.1. Test pieces

The test piece should be representative of the worst foreseeable conditions of use. The description of a cementitious materials test piece should include, in addition to the requirements given for organic materials (see Section 5.1.3.1):

- when the test piece was produced (this should be within 28–90 days of beginning the test);
- an indication of whether the test piece is bulk or lining;
- the content of cement of the applied-for entry in the test piece;
- the constituents of the test piece (cement type, specification of alternative binders and inorganic additions, e.g. slag, fly ash, silica fume, volcanic ashes, lime, mud, water, aggregates, admixtures, fibres, polymer modifiers, formwork release agents, curing compounds);
- the contents of the constituents of the test piece;
- the drying conditions of the test piece;
- the water to cement ratio of the test piece;
- the porosity of the test piece or the porosity of the lining layer in the case of a lining;
- any treatment prior to testing such as cleaning the test piece surface to remove residual products and substances from processing, if relevant;
- any other relevant information.

5.1.6. Enamels, ceramics and other inorganic materials

According to Section 1.4 of Annex IV to Commission Implementing Decision (EU) 2024/365, physical migration testing of enamels, ceramics and other inorganic compositions should encompass each inorganic composition constituent and impurity identified in accordance with points 1.3.2 and 1.3.3 in Table 1 of Annex I to Commission Implementing Decision (EU) 2024/365, unless the inorganic composition constituent is one of the following elements: calcium, carbon, fluorine, iron, magnesium, nitrogen, phosphorus, potassium, silicon, sodium, tin or zinc.

The general provisions considered for organic materials (see above) apply to enamels, ceramics and other inorganic materials: migration test by static immersion (e.g. standard EN 12873-1) and fully demineralised water, at the temperature of intended use when possible. In contrast to starting substances, these inorganic materials are accepted on to the EUPL according to their composition. Information on the test piece, the migration test set-up and the results and any respective uncertainties should be provided in a **full study report complying with standard EN 12873-1** as described for organic materials (see Section 5.1.3). Specific provisions for these inorganic materials are:

- in most cases the test equipment must not contain any glassware;
- however, migration testing of carbon-containing ceramics and subsequent analysis of polycyclic aromatic hydrocarbons should be done exclusively in glass containers or receptacles.

5.1.6.1. Test pieces

The test piece should be representative of the worst foreseeable conditions of intended use. Information on the test piece should be provided as described for organic materials (see Section 5.1.3) and include:

- the manufacturer of the test piece;
- processing and manufacturing details, as given in Section 3.3.5 for bulk pieces and Section 3.3.2.1 for enamels and other ceramic or inorganic coatings; potential compliance with SFS-EN ISO 28764 should be mentioned;
- composition of the test pieces along with relevant analytical details of the compositional analysis;
- treatment prior to testing, such as cleaning the test piece surface to remove residual products and substances from processing, if relevant;
- any other relevant information.

5.1.7. Analysis of water originating from migration

This section offers guidance on how to conduct the analysis of the migration water or contact water to fulfil the requirements set out for the acceptance of starting substances, compositions and constituents on to the EUPL.

There are two types of analyses that applicants must conduct:

- targeted analyses aiming to accurately quantify substances in the water;
- screening analyses aiming to identify unintentional substances in the water for accurate quantification via dedicated analysis later.

The targeted analytical method used should:

- be validated and documented in accordance with standard EN ISO/IEC 17025 or other equivalent standards accepted at international level where possible;
- comply with minimum performance criteria for all the validation characteristics (selectivity, linearity, trueness (bias), precision (repeatability, intermediate precision and reproducibility),

range of applicability, quantification limit, detection limit (see below); more information on these guidelines is available in standards ISO 8466-1 and ISO 8466-2 and in Bratinova et al. (2009);

- as a minimum, be capable of measuring concentrations equal to the existing MTC_{tap} ⁽³⁴⁾ set for the substance with LOQ, as defined in Article 2(2) of Commission Directive 2009/90/EC ⁽³⁵⁾, of 30 % or less of the relevant MTC_{tap} and an uncertainty of measurement no greater than 50 % ($k = 2$).

An appropriate screening analysis (e.g. standard EN 15768, see Section 3.5.1) must always be conducted for organic starting substances to demonstrate the presence or lack thereof of unintentional substances. If the screening method discovers previously non-quantified substances, they must be quantified with a targeted method, if possible. If suitable targeted methods and/or reference substances do not exist, semi-quantitative methods may be used.

There may be practical limitations on the parameters of the screening method and technique used in the screening analysis, such as accuracy or LOQ, so that, for example, the concentration of the target substance in water should be maximised, and this should be considered in the test design, for example by specifying a maximum S/V ratio.

Where there are no methods that comply with the minimum performance criteria for the validation parameters, the analysis should be based on best available techniques and the applicant should provide full scientific justification for the technique used.

The applicant must use analytical methods that are the most suitable, that account for the specific limits set by Commission Implementing Decision (EU) 2024/367, standards and this guidance, and that are appropriate for the matrix and the nature of the analyte to be measured. The choice of appropriate method is the responsibility of the applicant.

To enable accurate quantification of the substance in water, for dedicated analyses the following information on the validation parameters should be provided.

- **Migration method and sampling** (see Commission Implementing Decision (EU) 2024/368). Full details of the sampling method must be provided to ensure sample representativeness and clarify uncertainties related to solvation, volatility, precipitation, etc. The sampling details should include the migration period from which the analysis was taken and, for metals, the stagnation period (e.g. 4 hours) prior to sampling. Other parameters, for example temperature, should be provided. Full descriptions of procedures for solvent extraction, etc., where relevant, should be given. When trends in migration rate are needed, the results from the relevant composition(s) in water after the n th migration period must be provided. For metallic compositions, sampling of the test water and of the contact water should be carried out in accordance with standard EN 15664-1. Full details of sampling of the test water must be provided to ensure sample representativeness.
- **Analytical method.** The principle of the analytical method, including the analytical technique, equipment and detector type and standard operating procedure that would allow reproduction of the analysis, should be provided (including instrumental parameters used during analysis such as temperature, flow rate and column type). It is recommended that

⁽³⁴⁾ This should be an MTC_{tap} value that is shown in Annex V to Commission Implementing Decision (EU) 2024/367 or an MTC_{tap} value for the relevant chemical species in the applicable material type that is set in the corresponding annex to that decision following a decision by the Commission on an application for a starting substance, composition or constituent that has been submitted to ECHA and the applicant submits at least any new or updated information available as from the date of the Commission's decision.

⁽³⁵⁾ Commission Directive 2009/90/EC on the technical specifications for chemical analysis and monitoring of water status.

established techniques appropriate for the substance are used, such as AAS, ICP-MS for metals or GC-MS. The calibration details of the technique, original measurement data in graphical (chromatograms and/or spectrograms) and tabulated form, blank analyses, reference samples, correlation coefficients and all other relevant data needed for a proper evaluation of the method and the migration data must be provided. The method may be used by accredited laboratories (to standard ISO/IEC 17025) and should, therefore, use generally available equipment.

- **Method validation parameters.** At least the following method validation parameters need to be reported:
 - **Detection/quantification limit.** The detection and/or quantification limits of the instrumental (GC, AAS, HPLC, etc.) and global methods (solid-phase microextraction GC, solid-phase extraction HPLC) (in µg/l) must be defined. The way in which the detection limit was established must be set out only if the concentration is near the LOQ of the global method.
 - **Linearity.** The operating range of the equipment should be provided (in µg/l) at least in the range 0.2–2 times the MTC_{tap} for metallics, enamels, ceramics and other inorganics and scaled to the concentrations directly analysed from migration water before being converted for organic and cementitious materials. In the scaling, the CF for calculating the C_{tap} for the worst-case use applied for and the order of magnitude of the concentration of the substance directly analysed from the migration water should be considered. By way of example, if the MTC_{tap} for a starting substance is 2.5 µg/l and analysed concentrations are within 100–300 µg/l, the linearity should be reported for a range 100 times 2.5 µg/l (i.e. 0.2 and 2 times 250 µg/l), resulting in a linearity of between 50 and 500 µg/l.
 - **Precision and trueness (bias) of test method.** The repeatability (r) and the intermediate precision or reproducibility (R) of the method at a concentration representative of typical migration rates for the substance under the test conditions must be provided. For example, the repeatability of the method can be obtained from the standard deviation of triplicate migration tests or recovery tests (see below).
 - **Recovery.** The percentage of substance recovery, as determined in recovery experiments under the time–temperature conditions of the migration tests, should be included, when applicable. To obtain data on the suitability of the analytical method, and on the stability of the substance, recovery experiments (in triplicate) should be performed with the test water spiked with the substance. This may not be feasible for some substances. The spiked test water should be kept for the same duration and at the same temperature in containers the same as or equivalent to those used in the migration experiments. If the recovery is below 90 %, the reasons for this must be explained.
 - **Uncertainty.** The expanded measurement uncertainty needs to be reported at the relevant concentration levels, including how it was determined.

It should be noted that the terms used here differ slightly from those used in the IUCLID format for reporting analytical information. The DWD IUCLID user manual and IUCLID helptexts can be consulted to verify where each piece of information should be reported.

In the case of impurities and anticipated reaction and degradation products arising from the intended use, including those that are in the matrix and those that are formed after migration (e.g. hydrolysis products), these need to be considered when analysing the migration water.

The applicant may have the opportunity to use internationally recognised standard methods (e.g. EN ISO 17294-2, EN ISO 11885, EN ISO 15586, ISO 18862), when appropriate, or a guide to choosing suitable analytical methods for their starting substance, composition or constituent.

5.1.7.1. Metallic compositions

Analysis of the migration water for metal concentrations refers to the compositions tested with the test rig in accordance with standard EN 15664-1.

For metallic compositions tested in accordance with standard EN 15664-1, the elements analysed according to this methodology are as follows.

- If a new reference material is tested, according to standard EN 15664-2, the three contact waters must be analysed for all elements exceeding 0.02 % (by mass) in the composition with the exception of (see Section 5.1.4):
 - sulphur, tin, silicon and phosphorus, if present as constituents;
 - iron, tin, manganese, aluminium, silicon, phosphorus and zinc, if present as impurities in the alloy.
- For testing of a metallic composition falling under an existing metallic composition category, it may be necessary to analyse only one type of water that was considered the most corrosive and determined when setting up the category, and the analysis of contact water(s) may be limited to the elements specified for the relevant category. Annex IV to this guidance shows the current categories, elements of relevance and the most critical testing waters.

5.2. Calculation of the migration rate and C_{tap} (point 5.2 in Annex IV to Commission Implementing Decision (EU) 2024/365)

This section describes how to calculate the migration rate (organics, cementitious materials, enamels, ceramics and other inorganics) and how to use that to calculate the C_{tap} value. Determination of the C_{tap} value for metallic materials is also described.

The migration rate is calculated from the analytically measured substance concentration in the migration test water using the test time and exposed surface area. This test migration rate can be transformed to a migration rate for the intended use by applying an appropriate conversion procedure, which takes into consideration the S/V ratio of the actual product and the real contact time expressed in days.

5.2.1. Organic, cementitious and inorganic materials

For these material types the expected average C_{tap} value must be calculated following the procedures in standards **EN 12873-1 (or -2)** or **EN 14944-3 (or -4)**, as applicable. While, for risk acceptance purposes, the migration rate should be calculated using the measured concentration from the third (or later) migration period, it can also be calculated individually for each period.

The concentration (C_n^T) measured in the migration water must be converted into migration rates (M_n^T) (EN 12973-1, EN 14944-3):

$$M_n^T \left[\frac{\mu\text{g}}{\text{dm}^2 \times \text{day}} \right] = \frac{C_n^T \left[\frac{\mu\text{g}}{\text{L}} \right]}{t[\text{day}] \times (S/V)[\text{dm}^{-1}]}$$

where:

- C_n^T is the concentration of the measured chemical species in the migration water ($\mu\text{g/l}$), for the following conditions:
 - T is the test temperature ($23\text{ }^\circ\text{C} \pm 2\text{ }^\circ\text{C}$ or another temperature set in the European standards);
 - n is the sequence number of the migration period (3 for the third stagnation period);
- t is the duration of the migration period in days (3 days in cold water, 1 day in hot or warm water);
- S/V is the ratio of wetted surface of the material to the volume of the test water (dm^{-1}) used for the test (may range from 5 to 40 dm^{-1} according to standards EN 12873-1 and -2).

M_n^T must be subsequently converted to the 'real' C_{tap} value by multiplying the migration rate by a conversion factor ($\text{CF} = 20$ by default or as specified for the product(s) relevant to the intended use of the applied-for entry; see the corresponding Annex to Commission Implementing Decision (EU) 2024/368):

$$C_{\text{tap},n} \left[\frac{\mu\text{g}}{\text{l}} \right] = M_n^T \left[\frac{\mu\text{g}}{\text{dm}^2 \cdot \text{day}} \right] \times \text{CF} \left[\frac{\text{day}}{\text{dm}} \right]$$

If the applicant deviates from these guidelines, they must provide scientific justification for doing so.

5.2.2. Metallic materials

For metals and metallic alloys, C_{tap} is determined by multiplying the average concentrations of metals in the contact water (for a specified test duration in weeks) by the appropriate correction factors for the relevant product group (see Annex II). This C_{tap} value can then be compared with the corresponding MTC_{tap} value of the element in question.

Migration testing in accordance with **standard EN 15664-1** (or equivalent) must be conducted using either pipes or pipe segments of a specified size and of which the whole surface area (100 %) is exposed to water. The test sample thus represents the highest S/V ratio (ca 30 dm^{-1}), the highest migration rate and the highest concentration likely to be analysed at the tap.

In the case of metallic compositions, an estimation of migration rate is necessary for compositions that are tested in accordance with standard EN 15664-1 and are intended for use in product groups A, B or C. Metallic compositions that are tested and accepted in accordance with standard EN 16056 and the criteria for passive alloys or which are intended for use in product group D have negligible releases and thus a low migration rate is assumed.

5.2.2.1. Determination of C_{tap} from physical testing in accordance with standard EN 15664-1

The C_{tap} values for metallic compositions are determined by:

- The average of the mean of the equivalent pipe concentrations $\text{MEP}_a(T)$. Mean of the equivalent pipe concentration, $\text{MEP}_n(T)$ for each of the three test lines is calculated as defined in EN 15664-1. T is the operation period in weeks;
- a multiplication factor 'a' for the relevant product group (see Annex II to this guidance).

For tests conducted using pipes:

- determine which product group (A, B, C or D) the intended use belongs to (see Annex II to this guidance);
- multiply the $(MEP_a(T))$ with the percentage associated with the product group to establish the true migration for the intended use.

For tests conducted using pipe-shaped test pieces:

- calculate the pipe equivalent concentration as instructed in standard EN 15664-1 to compensate for the reduction in surface area;
- follow the steps given for pipe tests above.

From this, $C_{tap} = MEP_a(T) \times a$

It should be noted that, for the purposes of risk acceptance, C_{tap} values determined for weeks 16, 21 and 26 are the most important.

5.3. Alternative methods (adaptations of migration testing)

Alternative methods (adaptations) are available for estimating the migration rate from a DWCM. They can be used when physical testing cannot be performed or when the outcome of such testing might not be reliable, for example when the concentration of a substance in a material is low, the substance degrades or hydrolyses or the analytical LOQ does not allow the estimation of the concentration of the substance in water. Conditions for using adaptations of migration testing are stipulated in Section 2 of Annex IV to Commission Implementing Decision (EU) 2024/365, 'General rules for adaptation of information and testing with regard to the migration'.

Alternative methods need to provide a reasonably conservative estimation of the rate of migration of a substance into drinking water. In this guidance, alternative methods are divided into:

- modelling approaches;
- other methods such as full transfer calculation and scientific dossier.

Physical testing is the method the applicant must use to estimate migration of starting substances, unless they can demonstrate that an alternative method such as modelling is more suitable for the specific case (e.g. because analytical methods are too uncertain to accurately determine the concentration of the substance in physical testing).

5.3.1. Modelling

Modelling is applicable only to starting substances for organic materials, since currently suitable models are available only for (some) organic materials. Modelling can be used both for the applied-for entry and for NIAS and modelling of an applied-for entry does not exclude the use of physical testing for NIAS or vice versa. Any use of modelling should be properly justified.

Migration modelling is specifically envisaged in Section 4.1 (Specific migration) of the EFSA for guidance note on the preparation of an application for the safety assessment of a substance to

be used in plastic FCMs ⁽³⁶⁾. In the context of FCMs, modelling instead of physical testing is commonly used in applying for listing on the Union list of plastic FCMs in Regulation (EU) No 10/2011.

The accepted modelling is based on diffusion and partitioning mechanisms (diffusion-based modelling) and can be used only for neutral organic substances in organic materials, in particular for plastics such as polyethylene, polypropylene, polystyrene and polyvinyl chloride, for which the information needed to perform the assessment is available and, according to CEN/TR 16364, diffusion models are proven to provide a reliable estimation of migration. Substances that concentrate at the surface of manufactured products, such as antistatic agents and lubricants, are excluded from the diffusion modelling approach, as are electrolytes, salts, oxides and metals.

Migration modelling must be performed in accordance with **CEN/TR 16364** ⁽³⁷⁾. This 2012 technical report defines the criteria for performing the migration modelling and boundaries of applicability to estimate the most relevant input parameters, how to report the outcome of modelling, etc. It also sets the requirements for the validation of migration models. Any deviations from this technical report must be justified. However, we need to highlight that more recent publications on how to estimate input parameters such as diffusion and partition coefficients are available, such as the background document of the PLASI project ⁽³⁹⁾ mentioned below.

Although physical testing is the first option when applying for the inclusion of a starting substance in the EUPL for DWCMs, modelling can be a valuable alternative, for example in cases where there is a need to evaluate the migration rates of substances present in materials at low levels (e.g. impurities) or when physical testing might result in a non-detectable concentration due to the LOQ or (rapid) transformation (e.g. hydrolysis) of the substance in water.

Any modelling should provide a reasonably conservative estimation of the migration rate of the starting substance from the material into drinking water. This means that the input parameters to be used in the software tool need to be selected to overestimate, rather than underestimate, the migration rate in the last migration period for evaluation.

The main input parameters needed to estimate migration rates via modelling are listed in Chapter 7 of CEN/TR 16364 as follows.

- **Diffusion coefficient** (describing the diffusion of the substance within the organic material) and **partition coefficient** (describing the partitioning of the substance between the solid surface layer and the water at equilibrium). If available, reliable real values, for example from experimental data, it can be used. If reliable real values are not available, both diffusion and partition coefficients can be estimated as upper limit values with different degrees of conservatism. CEN/TR 16364 presents worst-case values for both that can be used if there are no other data. Apart from the CEN/TR 16364 technical report, other references can be consulted for the estimation of the abovementioned coefficients, such as JRC guidance on modelling for plastic materials ⁽³⁸⁾ and the background document of the PLASI project ⁽³⁹⁾.
- **Characteristics of the final material** (products) in contact with drinking water. This covers the concentration of the substance in the material, geometry of the material/product, thickness, etc. The amount added during formulation can be used or the residual

⁽³⁶⁾ *Note for guidance for the preparation of an application for the safety assessment of a substance to be used in plastic food contact materials* (<https://efsa.onlinelibrary.wiley.com/doi/epdf/10.2903/j.efsa.2008.21r>).

⁽³⁷⁾ CEN/TR 16364: Influence of materials on water intended for human consumption – Influence due to migration – Prediction of migration from organic materials using mathematical modelling (https://standards.iteh.ai/catalog/standards/cen/7f12962b-a600-47f8-8511-183efe3738cd/cen-tr-16364-2012?srsltid=AfmBOooQpOHe3n4JRAY1mB726oETm5CWPufyu5LR_xRwSnBTkCeAsLE6).

⁽³⁸⁾ <https://publications.jrc.ec.europa.eu/repository/handle/JRC98028>.

⁽³⁹⁾ <https://echa.europa.eu/plastic-additives-initiative>.

concentration measured in the product, the latter being the only suitable option for monomers (residual concentration measurement).

- **Temperature of the system.** This applies to the water and test material.
- **Simulation scenario.** This covers the duration of the contact with water, that is, the length of the migration period (72 hours for cold water, 24 hours for warm or hot water) and the number of migration periods simulated in accordance with standard EN 12783-1 or -2; each period has to be simulated including the pre-period 24-hour stagnation (washing processes before and after this 24-hour stagnation period can be neglected). For the applied-for entry, this has to reflect the provisions set out for migration testing, that is, the 3rd period to be used for migration rate estimation in the case of cold water testing, the 7th period in the case of warm or hot water testing, or, in the case of a prolonged testing scenario of 31 days, the 9th or 22nd contact period.

The outcome of the modelling should be expressed in terms of migration rate ($\mu\text{g}/\text{dm}^2 \times \text{days}$), concentration of the substance in test water ($\mu\text{g}/\text{l}$), residual content of the substance in the material (mg/kg), in the material (see Chapter 9 of CEN/TR 16364).

The elements that the applicant is requested to provide on the migration modelling are listed in Chapter 10 of CEN/TR 16364 and are reproduced below:

- the organic material under test;
- the modelling software used;
- values of all the inputs and their source or derivation;
- the concentration of the starting substance and the relative density (specific gravity) and thickness of the material;
the chemical identity and molecular weight of the starting substance;
- the diffusion coefficient of the starting substance in the organic material (and, if applicable, the estimation procedure);
- the partition coefficient between the organic material and water of the starting substance (and, if applicable, the estimation procedure);
- full details of the simulated migration test, including the temperature, duration and number of migration periods;
- the S/V ratio.

The principles of modelling are described in Annex 1 to CEN/TR 16364. Apart from the equations reported there, one element of relevance explained in Annex 1 to CEN/TR 16364 is the derivation of key parameters (diffusion and partition coefficients) needed to model the migration from the plastic matrix into drinking water. Several equations are proposed, together with different degrees of expected conservatism. For their estimation, applicants are also advised to consult the most recent publications and guidance, namely JRC's guidance on modelling for plastic materials and the background document of the PLASI project, both of which are referenced above.

Annexes B and C to CEN/TR 16364 report examples and the validation mechanism, respectively.

Lastly, commercial and open-source software for conducting migration modelling in accordance with CEN/TR 16364 is available.

5.3.2. Other methods

5.3.2.1. Organic and cementitious materials (full transfer calculation)

For organic and cementitious materials, the amount of substance liable to migrate is calculated on the assumption that all of the substance in the material, that is, 100 %, can migrate:

$$C_{\text{tap}} = Q \times S/V \times L_p \times D$$

where:

- **C_{tap}** (mg/l) is the maximum possible migration of the substance;
- **Q** (mg/kg of material) is the quantity of substance (applied-for entry or NIAS) in the finished material, or alternatively the quantity of substance used to manufacture 1 kg of material;
- **S/V** (dm⁻¹) is the ratio of the surface area of the product or component made from the material to the volume of water;
- **L_p** (dm) is the thickness of the material;
- **D** (g/cm³ or kg/dm³) is the density of the material.

5.3.2.2. Metallic materials: scientific dossier ⁽⁴⁰⁾

When considering only contact between one metallic alloy and water, the corrosion of the alloy and subsequent migration of metallic elements into the water depend on the electrochemical potential difference between the water and the alloy. The electrochemical potential of the alloy depends on the constituent elements, impurities, their respective contents and the microstructure of the alloy. Based on these considerations, in some cases physical testing of a metallic composition may be waived (Section 2 of Annex IV to Commission Implementing Decision (EU) 2024/365) via the provision of a 'scientific dossier'.

The scientific dossier must comprehensively address the chemical corrosion behaviour of the materials. The dossier will be used in the approval process. It is recommended that the dossier is carefully prepared and contains all available relevant scientific knowledge explaining the corrosion behaviour of the composition. Corrosion experts should be consulted in the generation of the dossier.

A scientific dossier may be used in the following cases.

- **For ferrous materials** used under permanent water flow. The use of ferrous materials has to be limited to certain water compositions (e.g. high oxygen concentration and redox potential) and flow conditions (no stagnating water, turbulent flow). These have to be addressed in the scientific dossier.

⁽⁴⁰⁾ 4MS Initiative, Common approach on metallic materials – Part C
(<https://www.umweltbundesamt.de/en/topics/water/drinking-water/distributing-drinking-water/approval-harmonization-4ms-initiative>).

- **For some copper alloys**, a certain test piece may be representative of several different materials. In this case it is sufficient to test a representative test piece in accordance with EN 15664-1 in order to accept different materials. This is only possible if:
 - due to their composition and metallic structure the materials exhibit comparable corrosion behaviour, with respect to the release of metals into drinking water, to the composition of the test specimens;
 - the materials belong to the same category;
 - the materials do not differ in alloying elements (constituents);
 - the composition of the test specimens used meets the requirements of the entire range of materials;
 - the constituents of the alloy composition have an MTC_{tap} value greater than or equal to 100 µg/l (i.e. copper, aluminium, iron, zinc, tin).

5.4. Identification of relevant chemical species

5.4.1. Screening for non-intentionally added species

Analytical screening methods are qualitative and semi-quantitative methods that can be used to give rough indications of unknown substances that could migrate at levels giving rise to potential toxicological concern. These substances may arise from interactions between the applied-for substance and the matrix during processing and testing, etc. Their identification may be based on, for example, using data libraries and comparing the analytical results.

For the purposes of applications under the DWD, non-target approaches based on chromatography and MS (i.e. GC-MS and liquid chromatography with high-resolution MS) can be used to screen and identify NIAS that are not identified elsewhere (see Commission Implementing Decision (EU) 2024/368). For organic chemicals containing polar, ionisable or ionic functional groups, modern chromatographic techniques such as super-critical fluid chromatography and reversed-phase liquid chromatography coupled with hydrophilic interaction liquid chromatography may be more suitable.

5.4.2. Selection of relevant chemical species for toxicological assessment

As explained in Section 3 of Annex IV to Commission Implementing Decision (EU) 2024/365, 'relevant chemical species are those that are covered by the requirements set in Annex V [of that decision] in order to demonstrate that the starting substance, composition or organic cementitious constituent meets the acceptance criteria set in Annex VI [of that decision].' An overview of which chemical species may be relevant is provided in Table 3.

Table 3. Relevant chemical species for toxicological assessment of different material types

Material type	Relevant chemical species
Organic	• Monomers or other reactants of a main polymer in the material • Chemical species that show one of the human health hazards listed under Section 1.1 of Annex VI to Commission Implementing Decision (EU) 2024/365 irrespective of their levels of migration into drinking water • Chemical species that have been tested for migration and have been found to migrate into water intended for human consumption with a C_{tap} value exceeding 0.1 µg/l
Metallic	• Metallic composition constituents that have been tested in accordance with Section 5.1.4 • Metallic composition impurities that have been tested in accordance with Section 5.1.4
Cementitious	• Monomers or other reactants of a main polymer in the material • Chemical species that show one of the human health hazards listed under Section 1.1 of Annex VI to Commission Implementing Decision (EU) 2024/365 irrespective of their levels of migration into drinking water • Chemical species that have been tested for migration and have been found to migrate into water intended for human consumption with a C_{tap} value exceeding 0.1 µg/l
Enamel, ceramic and other inorganic	• Inorganic composition constituents that have been tested in accordance with Section 5.1.6 • Inorganic composition impurities that have been tested in accordance with Section 5.1.6

6. Toxicological data for the relevant chemical species

6.1. Considerations before initiating testing

The toxicity of all relevant chemical species (identified in accordance with Section 3 of Annex IV to Commission Implementing Decision (EU) 2024/365) should be evaluated in toxicity studies to assess whether their migration into drinking water poses a risk to consumers. The need to provide toxicity data in the context of the DWD is dependent on the level of migration of the relevant chemical species following the principle of the higher the migration, the greater the amount of toxicological data are required (see Table 4). According to Section 2 of Part 1.5 of Annex V to Commission Implementing Decision (EU) 2024/365, in addition to the standard information specified in Tables 1–3 of Annex V to Commission Implementing Decision (EU) 2024/365, 'Any other relevant toxicological information that is available shall be identified and considered.'

6.1.1. Situations where no standard toxicological information or testing is required

There are certain situations where no standard toxicological information for or testing of a relevant chemical species is required, as explained in Section 1 of Annex V to Commission Implementing Decision (EU) 2024/365.

- A parametric value for the relevant chemical species is set under Annex I to the DWD. It should be noted that parametric values are the basis of certain MTC_{tap} values for elements and ions presented in Annex V to Commission Implementing Decision (EU) 2024/367.
- An MTC_{tap} value for the relevant chemical species and for the specific material type is available in the corresponding Annex to Commission Implementing Decision (EU) 2024/367 **following a decision by the Commission on an application** for a starting substance, composition or organic cementitious constituent that has been submitted to ECHA. In such cases, the applicant should at least submit any new or updated toxicological information that may have become available from the date of the Commission's decision.
- The relevant chemical species is classified as (1) carcinogenic, mutagenic or reproductive toxicity, categories 1A or 1B, (2) ED for human health category 1, (3) persistent bioaccumulative and toxic, (4) very persistent and very bioaccumulative, (5) persistent mobile and toxic, or (6) very persistent and very mobile, as established under the CLP regulation ⁽⁴¹⁾.
- The relevant chemical species is already identified as a substance of very high concern under the candidate list established under Article 59 of the REACH regulation, except those identified on the basis of Article 57(f) of the REACH regulation only for the environment ⁽⁴²⁾.
- The relevant chemical species is authorised as an active substance under the BPR on the basis of an agency opinion setting a defined safety threshold for the oral route and used in that way in material(s) in contact with water under product type 6.
- In addition, no standard information or testing is required for a relevant chemical species, to the extent that an SML is set under the plastic FCM regulation for less than 15 years from the date of submission of the application under Article 3 of Commission Delegated Regulation (EU) 2024/369. It should be noted that SML values are the basis of certain MTC_{tap} values for

⁽⁴¹⁾ <https://eur-lex.europa.eu/legal-content/en/TXT/?uri=CELEX%3A32008R1272>.

⁽⁴²⁾ <https://echa.europa.eu/candidate-list-table>.

elements and ions presented in Annex V to Commission Implementing Decision (EU) 2024/367.

In all cases covered above, applicants should not carry out vertebrate animal testing for the purpose of meeting the information requirements of a DWD application to be submitted to the agency.

6.1.2. General considerations where standard toxicological information or testing is required

Before any testing is initiated, the applicant should follow the general principles, explained in Figures 1 and 2 in Section 1.2.2 of this guidance. In particular, for each information requirement, steps 4 (collect) and 5 (evaluate all the available information on toxicological properties of relevant chemical species) need to be considered first to conclude on the need to conduct further testing. The test material used in toxicity studies carried out to fulfil the information requirements should be representative of the relevant chemical species. All available information should be scrutinised for evidence that may indicate severe effects, serious specific systemic or target organ toxicity (e.g. neurotoxicity or immunotoxicity) or delayed effects or cumulative toxicity. If there is concern that an effect is not adequately covered by existing EU methods or OECD test guidelines, specialised *in vitro* or *in vivo* study protocols may be used. It should be kept in mind, however, that the OECD test guideline methods are preferred, and their replacement is exceptional. Whenever deviating from EU methods or OECD test guidelines, a scientifically valid justification should be provided. Any specialised study protocols should be designed on a case-by-case basis with the aim of enabling an adequate hazard characterisation, including the dose-response, threshold for the toxic effect and an understanding of the nature of the toxic effects.

6.2. General considerations for reporting animal data

The doses tested, including the highest dose tested, should be selected on the basis of the results of short-term testing.

Doses causing excessive toxicity (e.g. high mortality rate, severe suffering) should not be considered relevant for evaluation. Further guidance on the dose selection for repeated dose toxicity testing is provided in detail in the EU methods and OECD test guidelines. Concurrent control group(s) should be included in the study design, as required by the respective test guidelines, and should always be considered first in the analysis of the results of the study. Historical control data fulfilling the requirements listed below should be provided, especially in cases in which the comparison of the test results with the concurrent control data cannot result in an unambiguous conclusion on the effects observed.

Where submitted, historical control data should be from the same animal species and strain, maintained under similar conditions in the same laboratory, and should be from contemporaneous studies (e.g. within a period of up to around 5 years of the study). Additional historical control data from other laboratories may be reported separately as supplementary information.

Recommendations for the use of historical control data, provided in the relevant EU methods and OECD test guidelines should be followed. The information on historical control data provided should include:

- identification of animal species and strain and name of the supplier and specific colony identification if the supplier has more than one geographical location;
- name of the laboratory where and the dates when the study was performed;

- description of the general conditions under which animals were maintained, including the type or brand of diet and, where possible, the amount consumed;
- approximate age, in days, and weight of the control animals at the beginning of the study and at the time of sacrifice or death;
- description of the control group mortality pattern observed during or at the end of the study, and other pertinent observations (such as diseases, infections);
- name of the laboratory and the examining scientists responsible for gathering and interpreting the pathological data from the study;
- for carcinogenicity studies, a statement of the nature of the tumours that may have been combined to produce any of the incidence data.

The historical control data should be presented on a study-by-study basis, giving absolute values plus percentage and relative or transformed values where these are likely to be helpful for the evaluation. If combined or summary data are submitted, these should contain information on the range of values, the mean, median and, if applicable, standard deviation.

6.3. Approach to toxicity testing – principles

As stated under Section 2 of Part 1.4 of Annex V to Commission Implementing Decision (EU) 2024/365 'a stepwise approach for toxicological testing is applied based on the C_{tap} of a relevant chemical species in water intended for human consumption.' The approach is summarised in Table 4.

Table 4. Stepwise approach to toxicity testing

C_{tap}	Toxicity data required	Additional considerations
Low migration tier $C_{\text{tap}} < 2.5 \mu\text{g/l}$	Genotoxicity/mutagenicity studies (see Section 6.4)	If there is indication that one or more mechanisms/modes of action of the substance is associated with (developmental) neurotoxicity and/or endocrine disruption and/or (developmental) immunotoxicity, corresponding additional data should be generated, as described under point 8.5 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365
Medium migration tier $2.5 \leq C_{\text{tap}} < 250 \mu\text{g/l}$	As above, plus: • Data to demonstrate the absence of potential for accumulation in man (see Section 6.5.1) • 90-day oral toxicity study in rodents (see Section 6.6.1) • Reproduction/developmental toxicity screening study (see Section 6.7.3.1)	
High migration tier $C_{\text{tap}} \geq 250 \mu\text{g/l}$	As above ^(a) , plus: • Study on ADME (see Section 6.5.2) • Extended one-generation reproductive toxicity study (Section 6.7.3.2) • Study on prenatal developmental toxicity (Section 6.7.4) • Studies on long-term toxicity/carcinogenicity (Section 6.8.2)	

^(a) The 90-day repeated dose toxicity study required at medium migration level might not be needed, as it is covered by the long-term toxicity/carcinogenicity study.

NB: ADME, absorption, distribution, metabolism and excretion.

Column 1 of Tables 1–3 of Annex V to Commission Implementing Decision (EU) 2024/365 'establishes the standard information for relevant chemical species'.

Column 2 of Tables 1–3 of Annex V to Commission Implementing Decision (EU) 2024/365 'lists specific rules according to which the standard information and testing may be omitted'.

In addition to the specific rules for adaptation given in column 2 of the respective standard information requirement, standard information and testing may be adapted according to the general rules of adaptation, as specified under Section 2 of Part 2 of Annex V to Commission Implementing Decision (EU) 2024/365.

6.4. Genotoxicity/mutagenicity (point 6.1 of Table 1 of Annex V to Commission Implementing Decision (EU) 2024/365)

The testing methodology on genotoxicity/mutagenicity is described in column 1 of Table 1 of Annex V to Commission Implementing Decision (EU) 2024/365. This information **is required for all migration levels**.

According to OECD (2017), two types of genetic toxicology studies are considered (in order of importance): (1) those measuring direct, irreversible damage to the DNA that is transmissible to the next cell generation (i.e. mutagenicity) and (2) those measuring early, potentially reversible effects to DNA or the effect on mechanisms involved in the preservation of the integrity of the genome (genotoxicity).

The aim of genotoxicity testing is to predict the genotoxic potential of substances and to elucidate the potential mechanism of action.

Appropriate concentration/dose levels, depending on the test requirements, should be used in either *in vitro* or *in vivo* assays. A concentration-/dose-based approach should be adopted, with selection of higher concentration-/dose-level tests being dependent on test-specific guideline requirements and interpretation of results at each stage.

At least one *in vitro* test for gene mutations in bacteria (see Section 6.4.2.1.1), one *in vitro* mammalian chromosomal aberration study or *in vitro* mammalian micronucleus study (see Section 6.4.2.1.2) and, if the first two tests report negative results, one *in vitro* test for gene mutation in mammalian cells (see Section 6.4.2.1.3) is required.

In general, substances that are negative in the full set of *in vitro* tests specified in Section 6.1.1 of Table 1 of Annex V to Commission Implementing Decision (EU) 2024/365 are considered to be non-genotoxic. Occasionally, knowledge about the metabolic profile of a substance may indicate that the standard *in vitro* tests are not able to detect a potential genotoxic effect, and a further *in vitro* test may be needed to ensure that mutagenicity potential is adequately explored (e.g. use of an alternative to rat liver S9 mix, a reducing system, a metabolically active cell line or genetically engineered cell lines might be judged appropriate).

6.4.1. Collection and evaluation of available information

For the assessment of existing information further guidance is available in DWD Guidance Volume II and in the ECHA guidance on the application of the CLP criteria ⁽⁴³⁾.

As specified in column 2 of point 6.1.1.1 of Table 1 of Annex V to Commission Implementing Decision (EU) 2024/365:

⁽⁴³⁾ <https://echa.europa.eu/web/guest/guidance-documents/guidance-on-clp>.

The *in vitro* gene mutation study in bacteria does not need to be conducted if this test is not applicable for the relevant chemical species. In this case, the applicant shall provide a justification and perform an *in vitro* study referred to in point 6.1.1.3. (*in vitro* gene mutation test in mammalian cells). The study does not need to be conducted for nanoforms where it is not appropriate. In this case, other studies involving one or more *in vitro* mutagenicity studies in mammalian cells shall be provided.

As specified in column 2 of point 6.1.1.2 of Table 1 of Annex V to the above decision, *in vitro* mammalian chromosomal aberration study or *in vitro* mammalian micronucleus study 'does not need to be carried out if an adequate data from an *in vivo* cytogenicity test is available'.

If gene mutation or clastogenicity/aneuploidy is detected in a battery of tests consisting of the Ames test and the *in vitro* micronucleus test or *in vitro* mammalian chromosomal aberration study, no further *in vitro* testing needs to be conducted.

As specified in column 2 of point 6.1.1.3 of Table 1 of Annex V to the above decision, the *in vitro* gene mutation study in mammalian cells 'does not need to be provided if there are adequate data from a reliable *in vivo* mammalian gene mutation test'.

6.4.2. Generation of new test data

If, after assessment of the data, further testing is needed to assess the potential for genotoxicity *in vitro*, the test methods discussed in the following chapter must be used. In addition to those test methods, new OECD validated tests for genotoxicity should be taken into account, once available. Therefore, the OECD test guidelines programme, as well as non-animal test methods that have been validated by the OECD or the European Union Reference Laboratory for Alternatives to Animal Testing, should be regularly consulted for possible updates.

6.4.2.1. Genetic toxicity *in vitro* (point 6.1.1 of Table 1 of Annex V to Commission Implementing Decision (EU) 2024/365)

The test guidelines to follow for the investigation of *in vitro* genotoxicity are listed below. These should be used taking account of the considerations described here but also taking account of the existing information for this end point and its assessment.

If there are indications of micronucleus formation in an *in vitro* micronucleus assay, further *in vitro* testing with appropriate staining procedures should be conducted to clarify whether there is an aneugenic or clastogenic response. Further investigation of the aneugenic response may be considered to determine whether there is sufficient evidence for a threshold mechanism and threshold concentration for the aneugenic response (particularly for non-disjunction).

It is known that there are classes of substances to which the bacterial reverse mutation test is not suited. This relates to, for example, biocidal substances and antibiotics that are highly toxic to bacteria, as well as to substances with low bioavailability that do not pass easily through the bacterial wall such as nanomaterials and larger particles (*Salmonella* and *Escherichia* bacteria lack the mechanisms (e.g. endocytosis) to incorporate particles). In such cases, the results from a bacterial reverse mutation test are not reliable, and at least one *in vitro* mammalian cell gene mutation test (using either the thymidine kinase gene or the *Hprt* or *Xprt* gene) should be provided. In these cases, the test strategy should be modified, and a scientific justification should be given.

For substances bearing structural alerts that have given inconclusive/ambiguous results *in vitro*, additional *in vivo* data will be needed to exclude genotoxicity (see Section 6.4.2.2).

For substances bearing structural alerts that have given negative results in the standard test battery, additional testing may be required if the standard tests have not been optimised for these alerts. Accepted modifications to standard test guidelines/methods have been developed to enhance test sensitivity for specific substance classes, as described in the corresponding guidelines. Expert judgement should determine their appropriateness for a given substance. For example, protocol modifications for the Ames test might be appropriate for substances such as gases, volatile liquids, azo dyes, diazo compounds, glycosides and petroleum oil-derived products, which should be regarded as special cases.

Furthermore, some new *in vitro* test methods have been included in the OECD work programme with the aim of developing detailed review papers on test protocols and potentially OECD test guidelines as follows.

- ToxTracker uses stem cells and measures the expression of reporter genes involved in several genotoxic and non-genotoxic pathways linked to carcinogenicity. Extensive technical validation (> 400 compounds) showed that ToxTracker is a robust *in vitro* assay for the accurate prediction of *in vivo* genotoxicity with the ability to provide insight into the mode of action of genotoxic agents (Hendriks et al., 2012, 2016; EFSA Scientific Committee et al., 2017). According to the recently published OECD validation report, the interlaboratory validation confirmed the accuracy of ToxTracker in correctly identifying genotoxic compounds with a sensitivity of 87 % and a specificity of 90 % (Hendriks et al., 2023);
- The *in vitro* γ H2AX / phospho-histone H3 assay is based on the phosphorylation states of specific histones (H2A and H3) used as early biomarkers of cellular response to DNA damage. It is claimed to have the ability to provide information on different mutagenicity modes of action, including discrimination between clastogenic and aneugenic effects.

The choice of an additional *in vitro* study or study plan modifications depends on the chemical nature, the known reactivity and the metabolism data on the structurally alerting relevant chemical species.

However, for substances showing inconclusive/ambiguous results *in vitro*, additional *in vivo* data will be needed (see Section 6.4.2.2).

6.4.2.1.1. *In vitro* gene mutation study in bacteria (point 6.1.1.1 of Table 1 of Annex V to Commission Implementing Decision (EU) 2024/365)

- EU method B.13/14: Mutagenicity – Reverse mutation test using bacteria.
- OECD TG 471: Bacterial reverse mutation test.

6.4.2.1.2. *In vitro* cytogenicity study in mammalian cells (point 6.1.1.2 of Table 1 of Annex V to Commission Implementing Decision (EU) 2024/365)

- EU method B.49: *In vitro* mammalian cell micronucleus test.
- OECD TG 487: *In vitro* mammalian cell micronucleus test.
- EU method B.10: Mutagenicity *in vitro* mammalian chromosome aberration test.
- OECD TG 473: *In vitro* mammalian chromosome aberration test.

Compared with the *in vitro* mammalian chromosome aberration test, the *in vitro* cell micronucleus test can, with the current state of knowledge, be considered the preferred method

for examining *in vitro* cytogenicity in mammalian cells because of its increased sensitivity and ability to identify aneugens.

6.4.2.1.3. *In vitro* gene mutation study in mammalian cells (point 6.1.1.3 of Table 1 of Annex V to Commission Implementing Decision (EU) 2024/365)

- EU method B.17: Mutagenicity – *In vitro* mammalian cell gene mutation test using the *Hprt* and *Xprt* genes.
- OECD TG 476: *In vitro* mammalian cell gene mutation test using the *Hprt* and *Xprt* genes.
- EU method B.67: *In vitro* mammalian cell gene mutation tests using the thymidine kinase gene.
- OECD TG 490: *In vitro* mammalian cell gene mutation tests using the thymidine kinase gene.

6.4.2.2. Genetic toxicity *in vivo* (point 6.1.2 of Table 1 of Annex V to Commission Implementing Decision (EU) 2024/365)

6.4.2.2.1. Collection and evaluation of available information

For the assessment of existing information, further guidance is available in DWD Guidance Volume II and in the ECHA guidance on the application of the CLP criteria.

The *in vivo* genotoxicity study (or studies) does not generally need to be conducted if the results are negative for the three *in vitro* tests and if no metabolites of concern are formed in mammals.

6.4.2.2.2. Generation of new test data

Follow the principles explained in Section 6.4.2.

6.4.2.2.3. *In vivo* studies in somatic cells (point 6.1.2.1 of Table 1 of Annex V to Commission Implementing Decision (EU) 2024/365)

As set out in the above point in the decision:

An appropriate *in vivo* mammalian somatic cell genotoxicity study ... shall be carried out if there is a positive result in any of the *in vitro* genotoxicity study referred to in point 6.1.1. which gives rise to a concern. The study shall address the chromosomal aberration concern or the gene mutation concern or both, as appropriate.

A particular *in vivo* test should be conducted only when it can be reasonably expected from all the properties of the test substance and the proposed test protocol that the specific target tissue will be adequately exposed to the test substance and/or its metabolites. Provided that evidence for adequate availability of the substance or its metabolites to the target tissue/cells specified in the corresponding test guidelines is shown, there are several options for the *in vivo* testing, as explained below.

If *in vivo* testing with the comet assay and the micronucleus test is required, the two tests should be combined into a single short-term study with appropriate modification of treatment and sampling times (see, for example, Bowen et al., 2011). The combination should not impair the validity of and the results from each individual study. Careful consideration should be given to the dosing and tissue sampling for comet analysis alongside the requirements of tissue sampling

for the mammalian erythrocyte micronucleus test. The following assays can be integrated into repeated dose toxicity studies described in Section 6.6 of this guidance:

- the comet assay;
- the *in vivo* micronucleus test;
- the transgenic rodent somatic and germ cell gene mutation assay;
- the mammalian erythrocyte *Pig-a* gene mutation assay.

In the interest of ensuring that the number of animals used in genotoxicity tests is kept to a minimum, both males and females should not automatically be used. In accordance with standard guidelines, testing in only one sex is possible when the substance has been investigated for general toxicity and no sex-specific differences in toxicity have been observed.

If the *in vitro* mammalian chromosome aberration test or the *in vitro* micronucleus test is positive for clastogenicity, an *in vivo* test for clastogenicity using somatic cells, such as the chromosomal aberration test in rodent bone marrow or the micronucleus test in rodent erythrocytes, should be conducted. The *in vivo* micronucleus test is the preferred test, as:

- it allows the investigation of both clastogenicity and aneugenicity;
- its end point is simple and easy to identify;
- it detects the genetic alteration remaining after mitosis in 'daughter cells'.

Evidence of exposure of the bone marrow to a test substance may include a depression of the immature to mature erythrocyte ratio or measurement of the plasma or blood levels of the test substance. The evidence of exposure of the bone marrow to the test substance should be provided with the robust study summary in the application.

In the event of a positive result in the *in vivo* micronucleus assay, an appropriate procedure to identify an aneugenic and/or clastogenic response, for instance fluorescence *in situ* hybridisation should be used in line with OECD TG 474.

If any of the *in vitro* gene mutation tests is positive, an *in vivo* test to investigate the induction of gene mutation should be conducted, such as the transgenic rodent somatic and germ cell gene mutation assay or the mammalian erythrocyte *Pig-a* gene mutation assay.

When conducting new *in vivo* genotoxicity studies, only relevant exposure routes and methods (*such as* admixture in the diet or drinking water or gavage) should be used. However, if positive results from reliable *in vivo* genotoxicity studies, using less appropriate routes, for example intraperitoneal application, are available, these can be considered for hazard identification, as long as their relevance for the purpose of Commission Implementing Decision (EU) 2024/365 can be scientifically justified (e.g. by providing evidence of systemic exposure).

The test guidelines and methods available for assessing the *in vivo* genotoxic potential in somatic cells of a substance are listed below and reflect the current state of knowledge. The choice of the most appropriate test should reflect the considerations described in this section and future recommendations or changes in the OECD test guidelines programme for this end point.

Test methods for in vivo genotoxicity in somatic cells:

- EU method B.12: Mutagenicity – *In vivo* mammalian erythrocyte micronucleus test;
- OECD TG 474: Mammalian erythrocyte micronucleus test;
- EU method B.11: Mutagenicity – *In vivo* mammalian bone-marrow chromosome aberration test;
- OECD TG 475: Mammalian bone marrow chromosome aberration test;
- OECD TG 470: Mammalian erythrocyte *Pig-a* gene mutation assay;
- EU method B.58: Transgenic rodent somatic and germ cell gene mutation assays (does not include the latest update to OECD TG 488);
- OECD TG 488: Transgenic rodent somatic and germ cell gene mutation assays;
- EU method B.62: *In vivo* mammalian alkaline comet assay;
- OECD TG 489: *In vivo* mammalian alkaline comet assay.

6.4.2.2.4. *In vivo* studies in germ cells (point 6.1.2.2 of Table 1 of Annex V to Commission Implementing Decision (EU) 2024/365)

As set out in the above point in the decision:

An appropriate *in vivo* mammalian germ cell genotoxicity study shall be carried out if there is a positive result in an available *in vivo* mammalian somatic cell genotoxicity study, which gives rise to concern. The study shall address the chromosomal aberration concern or the gene mutation concern or both, as appropriate.

The study does not need to be conducted if there is clear evidence that neither the relevant chemical species nor its metabolites reach the germ cells.

The potential to affect germ cells should always be considered for relevant chemical species classified as germ cell mutagen category 2 for which data on germ cells are not available or conclusive, unless the relevant chemical species also meets the criteria for classification as a carcinogen, category 1A or 1B. The first step is to carry out an appraisal of all the available toxicokinetic and toxicodynamic properties of the test substance. Expert judgement is needed at this stage to consider whether there is sufficient information to conclude that the relevant chemical species poses a mutagenic hazard to germ cells. If this is the case, it can be concluded that the relevant chemical species may cause heritable genetic damage, and no further testing is justified. If the appraisal of mutagenic potential in germ cells is inconclusive, additional investigation will be necessary. In the event that additional information about the toxicokinetics of the relevant chemical species would resolve the question, toxicokinetic investigation (i.e. not a full toxicokinetic study) tailored to address this is required. The type of mutation produced in earlier studies, namely gene mutations, numerical chromosomal or structural chromosomal changes, should be considered when selecting the appropriate assay.

A study for the presence of DNA adducts in gonad cells may also be considered. If germ cell testing is to be undertaken, expert judgement should be used to select the most appropriate test strategy.

Alternatively, other methods can be used if deemed appropriate by expert judgement. These may include the mammalian spermatogonial chromosome aberration test (OECD TG 483) or gene mutation tests with transgenic animals (OECD TG 488).

The comet assay, as described in OECD TG 489 is, at present, not considered appropriate for measuring DNA strand breaks in mature germ cells. However, the feasibility of the analysis of cells from the gonads using the comet assay has been demonstrated in the literature (Zheng and Olive, 1997; Cordelli et al., 2003; Speit et al., 2009; Dirven et al., 2023) and in an OECD project 'Update of TG 489 comet assay for gonadal cells to study germ cell specific genotoxic effects'. When performing the OECD TG 489 study, the applicant may consider collecting the male gonadal cells from the seminiferous tubules, in addition to the other somatic tissues, and preparing slides of the male gonadal cells and storing them for up to 2 months at room temperature, in dry conditions and protected from light. Following the generation and analysis of data on somatic cells in the comet assay (Section 6.4.2.2.3), the applicant should consider analysing the slides prepared with gonadal cells. As specified in paragraph 10 of OECD TG 489, the inclusion of such an examination may reveal information relevant to the overall assessment of germ cell mutagenicity, for instance with respect to gonad exposure to the relevant chemical species.

To minimise animal use, the possibility of combining germ cell genotoxicity tests and reproductive toxicity tests should be considered. To avoid testing on vertebrate animals, it is recommended to – wherever possible and providing appropriate sampling times are used – include the collection of germ cell samples when conducting tests that investigate somatic tissues. Germ cell samples should then be appropriately stored for later analysis, if needed.

The test guidelines available for assessing the *in vivo* germ cell mutagenicity of a substance are listed below and reflect the current state of knowledge. The choice of the most appropriate test should reflect the considerations described in this section and future recommendations or changes within the OECD test guidelines programme for this end point

Test methods for *in vivo* genotoxicity in germ cells

- *In vivo* test methods for genotoxicity in germ cells, EU method B.23: Mammalian spermatogonial chromosome aberration test;
- OECD TG 483: Mammalian spermatogonial chromosome aberration test;
- EU method B.58: Transgenic rodent somatic and germ cell gene mutation assays (does not include the latest update to OECD TG 488);
- OECD TG 488: Transgenic rodent somatic and germ cell gene mutation assays.

6.5. Toxicokinetic and metabolism studies in mammals (points 6.2, 7.1 and 8.1 of Tables 1, 2 and 3 of Annex V to Commission Implementing Decision (EU) 2024/365)

Column 1 of Tables 1, 2 and 3 of Annex V to Commission Implementing Decision (EU) 2024/365 specifies the information requirements on toxicokinetics as follows:

- low migration tier ($C_{tap} < 2.5 \mu\text{g/l}$) – no data requirement unless there is concern about at least one of the following hazard classes defined in Annex I to Regulation (EC) No 1272/2008: specific target organ toxicity – repeated exposure (STOT RE), carcinogenicity, mutagenicity or reproductive toxicity, or endocrine disruption for human health (point 6.2);

- medium migration tier ($2.5 \leq C_{\text{tap}} < 250 \mu\text{g/l}$) – data to demonstrate the absence of potential for accumulation in humans (point 7.1.1);
- high migration tier ($C_{\text{tap}} \geq 250 \mu\text{g/l}$) – study on absorption, distribution, metabolism and excretion (ADME) (point 8.1.1).

In principle, **accumulation is undesirable** but not automatically associated with any toxic effect. In cases in which accumulation potential has been demonstrated or lack of accumulation potential has not been fully demonstrated, it remains the responsibility of the applicant to provide **evidence that any (potential) accumulation will not be associated with toxic effects even after long-term exposure**. If, for example, this cannot be demonstrated at the medium migration tier, according to Annex VI, point 1.4(b), an MTC_{tap} of $2.5 \mu\text{g/l}$ may be applied or higher migration tier studies (e.g. a toxicokinetic study and/or a long-term repeated dose toxicity study, may be needed to demonstrate a lack of accumulation or toxic effect, according to the provisions of point 7.4 of Table 2 of Annex V.

6.5.1. Data to demonstrate the absence of potential for accumulation in humans

Accumulation of a substance in the body occurs when the absorption of that substance is faster than the elimination (e.g. by metabolism and excretion). Therefore, the longer the biological half-life of a substance, the greater the body burden of that substance. Accumulation in various organs and tissues may be related to the chemical structure and physico-chemical properties of the substance.

For organic substances, a first step in demonstrating the absence of potential accumulation in humans is a screening assessment based on the hydrophilicity and volatility of the substance.

Hydrophilicity refers to substances that are water soluble enough to be readily eliminated by urinary excretion or have a low gastrointestinal uptake. Hydrophilicity is usually measured (or estimated) in the screening phase as lipophilicity, that is, the ratio between *n*-octanol and water at equilibrium (*n*-octanol–water partition coefficient, K_{ow}).

Volatility refers to substances that are volatile enough to be readily eliminated by exhalation. Volatility can be measured (or estimated) as the ratio between *n*-octanol and the air phase at equilibrium (*n*-octanol–air partition coefficient, K_{oa}).

Thresholds for screening substances that do not have the potential to accumulate in mammals, have been set to a $\log K_{\text{ow}} < 2$ and a $\log K_{\text{oa}} < 5$ (Armitage and Gobas, 2006; Kelly et al., 2007; Gobas et al., 2009; McLachlan et al., 2011; Goss et al., 2013). Data on the hydrophilicity/lipophilicity and volatility of organic substances should ideally be measured but can also be estimated using *in silico* models. Baskaran et al. (2021) have compiled all K_{oa} values reported in the published literature. Their dataset includes more than 2 500 experimentally derived values and more than 10 000 estimated values for K_{oa} , in total covering over 1 500 distinct molecules. More details on which information can be used to determine $\log K_{\text{ow}}$ and $\log K_{\text{oa}}$ is provided in ECHA *Guidance on Information Requirements and Chemical Safety Assessments – Chapter R.11: PBT/vPvB assessment* ⁽⁴⁴⁾ (Section 4.1.2.10, Appendix R11-5 and Section 4.1.2.8).

It has to be noted that the approach described above does not consider physiological processes, including excretion and/or biotransformation, that may finally result in relatively low concentrations of lipophilic compounds. Alternatively, biotransformation may lead to formation

⁽⁴⁴⁾ https://echa.europa.eu/documents/10162/13632/information_requirements_r11_en.pdf/a8cce23f-a65a-46d2-ac68-92fee1f9e54f.

of metabolites with a similar or higher accumulation potential, or a higher affinity for proteins than the parent substance, potentially reaching higher concentrations in the organism under repeated exposure.

If the data do not demonstrate the absence of potential for accumulation in humans (or if the screening assessment is not applicable to a substance), further information on the accumulation potential is needed.

Additional information may be obtained through several different studies as described in OECD TG 417 on toxicokinetics.

Because substances with similar partitioning properties may have completely different bioaccumulation potential, due to metabolism and excretion being the main drivers of their fate in the organism, an *in vivo* toxicokinetic study in rats is the preferred option. In this case, a repeated oral exposure over 14 days, with an expected lowest observed adverse effect level / no observed adverse effect level (NOAEL) from a repeated dose toxicity study, represents the optimal design for a toxicokinetic study. To avoid (unnecessary) animal testing, data on accumulation can also be collected as part of the evaluation in other toxicological *in vivo* studies. For example, it may also be possible to evaluate the potential for accumulation from the data generated from an OECD TG 408 study with a recovery phase and where no specific toxicokinetic data, such as blood and urine concentrations of the test substance, are available. In this case, it must be shown that all significant effects caused by exposure to the test substance during the 90-day administration phase of the study will be reversible during a recovery phase of 2–4 weeks. The reversibility during the recovery phase indicates that the test substance may have no accumulation potential. However, any deviation from the optimal study design should be addressed by an additional assessment factor (AF) when deriving a DNEL. For example, an AF of 3 is applied for minor deviations if a toxicokinetic study exists but was based on a single exposure, or an AF of 5 is applied when there is only evidence from *in vitro* to *in vivo* extrapolation modelling data. Using an OECD TG 408 study for evaluating the potential for accumulation, an AF of 3–5 should be applied to derive a DNEL. These AFs need to be considered in the derivation of the DNEL value under Section 4.2.2.3 of DWD Guidance Volume II.

When validated OECD test guidelines and assessment frameworks for mammalian *in vitro* metabolism are available, it will also be possible to also consider *in vitro* data in the assessment. While an OECD test guideline is available for fish hepatocytes and liver S9 subcellular fraction (OECD TG 319A and 319B), this is not yet the case for mammalian cells. Using appropriate *in vitro* experimental models (e.g. primary hepatocytes, well characterised liver cell lines, subcellular fractions) for the estimation of biotransformation rate and robust *in vitro* to *in vivo* extrapolation models, accumulation in humans can potentially be assessed.

Further guidance can be found in ECHA *Guidance on Information Requirements and Chemical Safety Assessments – Chapter R.11: PBT/vPvB assessment* ⁽⁴⁵⁾. The ECHA guidance should be interpreted with a special focus on the accumulation potential in humans.

6.5.2. Study on absorption, distribution, metabolism and excretion (ADME)

The main objective of the toxicokinetic study is to obtain adequate information on the fate of a substance in the organism, that is, its ADME as a function of dose and time. Data on these toxicokinetic parameters can be derived from *in vivo* studies and, to a limited extent, from standardised *in vitro* / *ex vivo* studies e.g. the hepatic clearance assay, the protein binding assay and the Caco-2 cell permeability assay (Rotroff et al., 2010; Wetmore et al., 2012; Paini et al., 2020; Hagiwara et al., 2022; Panse et al., 2022). Furthermore, the evaluation of non-hepatic metabolism relevant to the route of exposure, for example very rapid hydrolysis at

⁽⁴⁵⁾ https://echa.europa.eu/documents/10162/13632/information_requirements_r11_en.pdf/a8cce23f-a65a-46d2-ac68-92fee1f9e54f.

gastrointestinal level (e.g. using digestive fluid simulants), should be included in an overall assessment of bioaccumulation potential. These assays are, however, non-stand-alone methods but can be used in a weight of evidence approach. Possible constraints related to the experimental models used should be considered and clearly described (e.g. metabolic competence of the cellular model used for measuring hepatic clearance; intrinsic variability or limitations in evaluating substances actively transported by Caco-2 cells).

Bearing in mind that the currently available *in vitro* assays were not developed as a one-to-one replacement for *in vivo* ADME assays, conclusions on the ADME properties of the substance can be drawn only in the context of what the respective *in vitro* assays were developed to evaluate (applicability domain). However, given the current progress in the field, advances in *in silico* and *in vitro* test batteries may allow a conclusive *in vitro* ADME assessment in the future.

OECD TG 417 describes several choices for setting up a toxicokinetic study. The design of the study should, in principle, include the following:

- an evaluation of the rate and extent of oral absorption – maximum serum concentration ($C_{\max}$), area under the curve (AUC), time to maximum plasma concentration ($T_{\max}$) – and other appropriate parameters, such as bioavailability;
- the potential for bioaccumulation;
- plasma half-lives ($t_{1/2}$) ⁽⁴⁶⁾;
- the distribution in major organs and tissues;
- information on the distribution in blood cells;
- the chemical structure and the quantification of metabolites in biological fluids and tissues;
- the different metabolic pathways;
- the route and time course of excretion of the relevant chemical species, including metabolites;
- investigations on whether and to what extent enterohepatic circulation takes place.

In the context of the DWD, the toxicokinetic study should investigate as a minimum the following aspects:

- $C_{\max}$
- $T_{\max}$
- bioavailability
- AUC
- $t_{1/2}$

⁽⁴⁶⁾ Minimising the volume of blood collection during *in vivo* studies, so called micro-sampling, can reduce pain and distress in animals. It can also eliminate or reduce control groups in rodent studies (<https://www.fda.gov/files/drugs/published/S3A-Guidance--Note-for-Guidance-on-Toxicokinetics--The-Assessment-of-Systemic-Exposure-in-Toxicity-Studies--Focus-on-Microsampling.pdf>).

- distribution in major organs and tissues (including bones).

The following are the test methods for toxicokinetics:

- EU method B.36: Toxicokinetics;
- OECD TG 417: Toxicokinetics (adopted 22 July 2010).

6.6. Repeated dose toxicity (points 7.2 and 8.2 in Tables 2 and 3 of Annex V to Commission Implementing Decision (EU) 2024/365)

Column 1 of Tables 2 and 3 of Annex V to Commission Implementing Decision (EU) 2024/365 specifies the testing methodology for repeated dose toxicity. The information is required for relevant chemical species with:

- medium migration tier ($2.5 \leq C_{\text{tap}} < 250 \mu\text{g/l}$) – a subchronic repeated dose toxicity study (90-day) in one animal species (rodents), male and female, via oral route of administration, is the minimum information requirement;
- high migration tier ($C_{\text{tap}} \geq 250 \mu\text{g/l}$) – long-term repeated dose toxicity (≥ 12 months) **or** combined chronic toxicity/carcinogenicity study.

Repeated dose toxicity testing provides information on the adverse effects as a result of repeated or prolonged exposure. In general, only one route of administration is necessary.

Since the most relevant route of exposure of the general population to drinking water is oral, it is necessary to conduct toxicity studies via the oral route.

Scientifically valid justification for replacing the oral route with another significant route needs to be provided.

To reduce the amount of testing carried out on vertebrates and in particular the need for free-standing, single-endpoint studies, the design of the EU methods or OECD test guidelines for repeated dose toxicity studies should take account of the potential to investigate several parameters within the framework of one study (e.g. kinetic data generation, micronucleus formation, neurotoxicity, immunotoxicity) to the extent permitted by the relevant EU method or OECD test guideline and without jeopardising the relevance and reliability of the information required.

The following are the test methods for toxicokinetics:

- EU method B.36: Toxicokinetics;
- OECD TG 417: Toxicokinetics.

6.6.1. Subchronic repeated dose toxicity study (90-day) in one animal species (rodents), male and female, via oral route of administration (point 7.2.1 of Table 2 of Annex V to Commission Implementing Decision (EU) 2024/365)

The subchronic (90-day) repeated dose toxicity study is standard information required for relevant chemical species migrating into drinking water in concentrations from $2.5 \mu\text{g/l}$ to $250 \mu\text{g/l}$.

6.6.1.1. Collection and evaluation of available information

For the assessment of existing information, guidance is available in DWD Guidance Volume II and in the ECHA guidance on the application of the CLP criteria ⁽⁴⁷⁾.

As specified in column 2 of Table 2 of Annex V to Commission Implementing Decision (EU) 2024/365, the study does not need to be conducted, if the following specific rules for adaptation can be applied:

- a reliable short-term toxicity study (28 days) or a repeated dose toxicity study with the reproduction/developmental toxicity screening test is available showing severe toxicity effects according to the criteria for classifying the relevant chemical species as STOT RE (Regulation (EC) No 1272/2008), for which the observed No Observed Adverse Effect Level (NOAEL)-28 days, with the application of an appropriate uncertainty factor allows the extrapolation towards the NOAEL-90 days for the same route of exposure; or
- a reliable chronic toxicity study is available, in which an appropriate animal species and route of administration were used; or
- the relevant chemical species is unreactive, insoluble, not bioaccumulative and there is no evidence of absorption and no evidence of toxicity in a 28-day 'limit test'.

The term 'unreactive' in the above quotation from the legal text may relate to the inherent chemical reactivity of the relevant chemical species (e.g. structural alerts) and must be justified. The term 'insoluble' should be characterised in terms of an amount (or LOD) per unit volume and must be justified. The term 'no evidence of absorption' requires experimental evidence of the lack of absorption (i.e. toxicokinetic studies) in order to omit the study. Furthermore, 'no evidence of toxicity in a 28-day limit test' can be interpreted as meaning that there has to be at least a 28-day limit test available in order to waive the 90-day study and that this 28-day study should not show any sign of toxicity at a dose of 1 000 mg/kg body weight per day.

6.6.1.2. Generation of new test data

If, after the analysis of the available data, further testing is needed to assess systemic (target organ) toxicity after subchronic (90-day) administration, the test methods described below should be used. The study should be performed in a single rodent species, preferably the rat, via the oral route (for application of corrosive substances, please see Section 6.7.2.1.3). The use of non-rodent species must be justified. The dose levels should be spaced sufficiently to produce a gradation of toxic effects. For the high dose level, it must be demonstrated that the highest possible dose level without severe suffering or death was either (1) achieved based on the actual results of the main study or (2) justified based on the results of the range-finding study. Alternatively, the limit dose concept has to be used, or it has to be demonstrated that the physico-chemical properties of the test substance limit the achievable dose.

Further information on dose selection can be found in ECHA document *Advice on dose-level selection for the conduct of sub-acute and sub-chronic assays under REACH* ⁽⁴⁸⁾.

The aim of the subchronic toxicity study is to collect information for hazard identification (i.e. for characterising the major toxic effects and target organs of the relevant chemical species and, hence, for detecting triggers for further studies) and for risk assessment (i.e. for assessing a NOAEL and the dose-response relationship).

⁽⁴⁷⁾ <https://echa.europa.eu/web/guest/guidance-documents/guidance-on-clp>.

⁽⁴⁸⁾ [63e6d895-a80c-f6f4-99c8-c5a3248ce02a](https://echa.europa.eu/web/guest/guidance-documents/guidance-on-clp).

Therefore, the information to be provided and evaluated should be sufficient to permit the identification of effects following repeated exposure to the relevant chemical species and, in particular, to further establish or indicate:

- the relationship between dose and adverse effects;
- the toxicity of the relevant chemical species including where possible the NOAEL;
- the target organs, where relevant (including immune, nervous and endocrine systems);
- the time course and characteristics of adverse effects with full details of behavioural changes and possible pathological findings at necropsy;
- specific adverse effects and pathological changes produced;
- where relevant, the persistence and reversibility of certain adverse effects observed, following discontinuation of dosing;
- where possible, the mode of toxic action;
- relevant critical end points at appropriate time points for setting reference values, where necessary.

In the 90-day study, potential neurotoxic and immunotoxic effects (see also Sections 6.10.1 and 6.10.3 of this guidance), genotoxic effects (see also Section 6.4 of this guidance) and effects potentially related to changes in the endocrine system (see also Section 6.10.2 of this guidance) must be carefully considered during the design and conduct of the test and be reported, taking into account potential limitations when modifying test protocols to investigate specific effects.

Information on the mode of action of structurally similar substances should also be considered in the design of repeated dose toxicity tests.

6.6.1.3. Test methods for subchronic repeated dose toxicity via oral route

- EU method B.26: Subchronic oral toxicity test. Repeated dose 90-day oral toxicity study in rodents.
- OECD TG 408: Repeated dose 90-day oral toxicity study in rodents.

6.6.2. Long-term repeated dose toxicity (≥ 12 months) (point 8.2.1 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365)

The long-term toxicity study is standard information required for substances migrating into drinking water at concentrations equal to or above 250 µg/l.

6.6.2.1. Collection and evaluation of available information

For the assessment of existing information, guidance is available in DWD Guidance Volume II and in the ECHA guidance on the application of the CLP criteria ⁽⁴⁹⁾.

⁽⁴⁹⁾ <https://echa.europa.eu/web/guest/guidance-documents/guidance-on-clp>.

As specified in column 2 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365, 'this study does not need to be conducted if the combined chronic toxicity/carcinogenicity study referred to in point 8.4.1 is provided.'

6.6.2.2. Generation of new test data

If, after the analysis of the available data, further testing is needed to assess systemic (target organ) toxicity after chronic (≥ 12 months) administration, the test methods described below should be used. Where new studies need to be conducted to fulfil this information requirement and the information requirement for carcinogenicity under point 8.4 in Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365, the combined chronic toxicity/carcinogenicity study should be performed.

The test is required to be conducted in one rodent, the rat being the preferred animal species.

As indicated in OECD TG 452, the long-term chronic toxicity study provides information on the possible health hazards likely to arise from repeated exposure over a considerable part of the lifespan of the animal species used. The study will provide information on the toxic effect of the substance and indicate the target organs and can also provide an estimate of the NOAEL that can be used in hazard characterisation.

The results of the study should be sufficient to permit the identification of the systemic (target organ) effects, following repeated exposure to the relevant chemical species, in particular:

- identification of the chronic toxicity of the relevant chemical species;
- identification of target organs;
- characterisation of the dose–response relationship;
- identification of the NOAEL or point of departure for estimation of a benchmark dose;
- provision of data to test the hypothesis regarding mode of action.

When a new long-term toxicity study is to be performed, consideration should be given to carrying out a combined chronic toxicity and carcinogenicity study (OECD TG 453), rather than separate execution of a chronic toxicity study (OECD TG 452) and carcinogenicity study (OECD TG 451). The combined test is more efficient in terms of time and cost than conducting two separate studies, without compromising the quality of the data in either the chronic phase or the carcinogenicity phase. Careful consideration should, however, be given to the principles of dose selection when undertaking a combined chronic toxicity and carcinogenicity study (OECD TG 453), and it is also recognised that separate studies may be required under certain regulatory frameworks.

Detailed guidance on and discussion of the principles of dose selection for chronic toxicity and carcinogenicity studies can be found in the OECD Guidance Document 116 (2012)⁽⁵⁰⁾. By selecting appropriate dose levels, a balance should be achieved between hazard screening on the one hand and characterisation of low-dose responses and their relevance on the other. This is particularly relevant in the situation where a combined chronic toxicity and carcinogenicity study (OECD TG 453) is to be carried out. In selecting the dose, the investigator should also consider and ensure that the data generated are adequate to fulfil the regulatory requirements

⁽⁵⁰⁾ https://read.oecd-ilibrary.org/environment/guidance-document-116-on-the-conduct-and-design-of-chronic-toxicity-and-carcinogenicity-studies-supporting-test-guidelines-451-452-and-453_9789264221475-en#page1.

across OECD countries as appropriate (e.g. hazard and risk assessment, classification and labelling, endocrine disruptor (ED) assessment).

When a combined chronic toxicity and carcinogenicity study is performed, the results of the systemic part of the study should provide sufficient information on the systemic toxicity of the relevant chemical species, as listed above.

Correspondingly, the results of the carcinogenicity part of the study should provide sufficient information to:

- identify carcinogenic effects resulting from long-term exposure to the relevant chemical species;
- establish the animal species, sex and organ specificity of tumours induced;
- establish the dose–response relationship and mode of action;
- identify (where applicable) the maximum dose eliciting no carcinogenic effect;
- determine (where possible) the mode of action and human relevance of any carcinogenic response identified.

6.6.2.3. Test methods for long-term repeated dose toxicity

- EU method B.30: Chronic toxicity studies.
- EU method B.33: Combined chronic toxicity / carcinogenicity studies.
- OECD TG 452: Chronic toxicity studies.
- OECD TG 453: Combined chronic toxicity / carcinogenicity studies.

6.7. Reproductive toxicity (points 7.3 and 8.3 of Tables 2 and 3 of Annex V to Commission Implementing Decision (EU) 2024/365)

Column 1 of Tables 2 and 3 of Annex V to Commission Implementing Decision (EU) 2024/365 specifies the testing methodology for reproductive toxicity. The information required for the different migration tiers is as follows:

- medium migration tier ($2.5 \mu\text{g/l} \leq C_{\text{tap}} < 250 \mu\text{g/l}$):
 - reproduction/developmental toxicity screening study in one animal species (OECD TG 421 or 422), is the minimum information requirement (see Section 6.7.3.1);
- high migration tier ($C_{\text{tap}} \geq 250 \mu\text{g/l}$):
 - extended one-generation reproductive toxicity study via oral route of administration (OECD TG 443) (see Section 6.7.3.2);
 - prenatal developmental toxicity study, in rat unless another animal species can be justified as more appropriate, via oral route of administration (OECD TG 414) (see Section 6.7.4);

- depending on the outcome of the above study, further prenatal developmental toxicity study, in a second animal species, via oral route of administration or a mechanistic study (OECD TG 414) (see Section 6.7.5).

Possible effects on sexual function and fertility in adult males and females, the development of the offspring and adverse effects on or mediated via lactation should be investigated and reported, as recommended in Sections 3.7.1.3–3.7.1.5 of Annex I to the CLP regulation,.

For further details, refer to ECHA *Guidance on Information Requirements and Chemical Safety Assessment – Chapter R.7a: Endpoint specific guidance*, Section R.7.6.

Effects that become accentuated over generations should be reported.

6.7.1. Collection and evaluation of available information

For the assessment of existing information, further guidance is available in DWD Guidance Volume II and in the ECHA guidance on the application of the CLP criteria ⁽⁵¹⁾.

As specified in column 2 of Tables 2 and 3 of Annex V to Commission Implementing Decision (EU) 2024/365, the reproductive toxicity studies:

do not need to be conducted if the relevant chemical species is of low toxicological activity (no evidence of toxicity seen in any of the tests available, in which a sufficiently comprehensive and informative dataset has been used), it can be proven from toxicokinetic data that no systemic absorption occurs via the oral route of exposure, e.g. plasma/blood concentrations below detection limit using a sensitive method and the relevant chemical species and its metabolites are absent in urine or bile.

6.7.2. Generation of new test data

If after the analysis of the data further testing is needed to assess reproductive toxicity, the test methods described below should be used. If there are specific concerns that are not addressed by the standard information requirements, additional testing might be needed to produce the information necessary for hazard identification, for example to draw conclusions on the immunotoxicity, ED properties or neurotoxicity (see Sections 6.10.1–6.10.3 of this guidance).

Where new testing is needed, please also see the general information under Section 6.1.

6.7.2.1. Preliminary considerations

When planning any reproductive toxicity studies, dose levels, vehicle, study design, and animal species and strain need to be considered, as presented below.

6.7.2.1.1. Dose range-finding studies

In selecting dose levels, information from existing studies should be considered, as well as information from any dose range-finding studies that may need to be conducted. The results of the dose range-finding studies should be reported to provide sufficient information about and justification for the doses selected for testing. The findings from a range-finding study may also support the interpretation of the results from the main study.

⁽⁵¹⁾ <https://echa.europa.eu/web/guest/guidance-documents/guidance-on-clp>.

6.7.2.1.2. Selection of vehicle

Most test methods provide guidance on vehicle selection. If any vehicle other than water is used, a scientific justification is needed. Any vehicle used should not reduce the solubility, absorption or bioavailability of the test substance. The vehicle itself should not cause any adverse effects, must not react with the substance, and should not interfere with the toxicokinetics of the substance or significantly affect the nutritional status of the animals. All these may interfere with the interpretation of the study results and may invalidate the study. The control group should receive the same vehicle at the same dosing volume as the treated groups.

6.7.2.1.3. Route of administration

According to the test methods for reproductive toxicity, the oral route (gavage, in the diet or in drinking water) is the default route, except for gases. For the extended one-generation reproductive toxicity study (EU method B.56, OECD TG 443) dietary administration may be an appropriate route for modelling human exposure. If a route of administration other than oral is used, a scientifically valid justification is required.

In vivo testing at concentration/dose levels causing corrosivity should be avoided. For irritating substances, the vehicle should be chosen to minimise gastrointestinal irritation. For some substances, dietary administration may allow adequate dosing without irritation compared with administration via gavage. In certain cases, irritation/corrosivity may be avoided by testing the neutral salts of alkaline or acidic substances in order to allow investigation of their intrinsic properties at adequate dose levels.

Gavage dosing provides accurate information on dose levels, and the resulting toxicokinetics generally follow daily bolus dosing with high maximum concentration in serum ($C_{\max}$). Depending on the elimination rate, daily periods with essentially no exposure are possible. Toxicity requiring high $C_{\max}$ values can be observed.

Incorporating the test substance into the diet or dissolving in drinking water provides less accurate information on dose levels because of potential loss of material due to spillage. However, the blood levels are steadier for many hours because feed and water consumption is distributed throughout the day. Toxicity levels requiring longer effect levels per day are more easily observed. Dietary or drinking water administration is not recommended if palatability issues are observed, as reduced food or water consumption and subsequent effects may be the result of the taste of the relevant chemical species and not its toxic properties.

6.7.2.1.4. Selection of animal species

The most common animal species for reproductive toxicity testing is the rat. There is often good historical background information on various rat strains that may be used to support the interpretation of results. The strain selected should have an adequate fecundity and not too high an incidence of spontaneous malformations (e.g. compared with existing historical control data) or any other specific feature that may reduce the adequacy of the strain for studying reproductive toxicity. To facilitate integrated data interpretation together with data from other studies, it is recommended that the same (rat) strain is used in reproductive toxicity testing and in repeated dose toxicity studies.

If there is information available regarding the sensitivity of animal species and strains, the most sensitive animal species and strain should be used. There is no need to demonstrate human relevance, which is assumed unless demonstrated otherwise. In choosing the appropriate animal species and strain, consideration must be given to its suitability for the test protocol and to the availability of background information on the animal species and strain. If the default animal

species referred to in a test method is not used, the selection of the animal species/strain should be robustly justified.

More information on animal species selection for the prenatal developmental toxicity studies is given in Section 6.7.4 of this guidance.

6.7.2.1.5. Dose level selection

ECHA has published *Advice on dose-level selection for the conduct of reproductive toxicity studies (OECD TGs 414, 421/422 and 443) under REACH* ⁽⁵²⁾. Based on this document, the dose levels should be sufficiently spaced to produce a gradation of toxic effects. If there is no evidence of toxicity at a dose of at least 1 000 mg/kg body weight per day in repeated dose studies, the necessary study/studies could be conducted using a control group and a single dose of at least 1 000 mg/kg body weight per day. However, should evidence for reproductive or developmental toxicity be found at this limit dose, further studies at lower dose levels will be required to identify a NOAEL. The conditions for applicability of a limit test are provided in the individual test methods for reproductive toxicity.

In selecting dose levels, information should be considered from existing studies, as well as from any dose range-finding studies that may need to be conducted. Toxicokinetic information may provide reasons for adjusting, for example, the dosing route and regimen. Furthermore, toxicity and toxicokinetics in pregnant animals may differ from those in non-pregnant animals. This may create challenges in selecting the highest dose level for the study, because the sensitivity of the animals may differ in the various phases of the study.

It is important to obtain information about the reproductive toxicity profile of a relevant chemical species, including the spectrum of reproductive toxicity effects related to different dose levels as well as information to allow the evaluation of the severity of reproductive toxicity of a relevant chemical species.

The highest dose level should produce sufficient levels of toxicity to provide adequate information on reproductive toxicity for the purposes of classification and labelling (including categorisation), risk assessment and identification of endocrine activity. For classification and labelling, it is important that the doses tested are high enough to enable a conclusion to be drawn on a lack of reproductive toxic properties warranting a classification in category 1B or category 2 if clear evidence warranting a category 1B classification on reproductive toxicity is not observed (see the CLP regulation criteria) ⁽⁵³⁾. Therefore, the top dose selection should have the aim of obtaining clear evidence of reproductive toxicity (adverse effects on reproduction) without

⁽⁵²⁾ https://echa.europa.eu/documents/10162/17220/211221_echa_advice_dose_repro_en.pdf/27159fb1-c31c-78a2-bdef-8f423f2b6568?t=1640082455275.

⁽⁵³⁾ Sections 3.7.2.5.7–3.7.2.5.9 of Annex I to the CLP regulation state the following on the limit dose and very high dose levels: 'There is general agreement about the concept of a limit dose, above which the production of an adverse effect is considered to be outside the criteria which lead to classification, but not regarding the inclusion within the criteria of a specific dose as a limit dose. However, some guidelines for test methods, specify a limit dose, others qualify the limit dose with a statement that higher doses may be necessary if anticipated human exposure is sufficiently high that an adequate margin of exposure is not achieved. Also due to animal species differences in toxicokinetics, establishing a specific limit dose may not be adequate for situations where humans are more sensitive than the animal model.' Section 3.7.2.5.8 states: 'In principle, adverse effects on reproduction seen only at very high dose levels in animal studies (for example doses that induce prostration, severe inappetence, extensive mortality) would not normally lead to classification, unless other information is available, e.g., toxicokinetics information indicating that humans may be more susceptible than animals, to suggest that classification is appropriate. Please also refer to the section on maternal toxicity (3.7.2.4) for further criteria in this area.' And Section 3.7.2.5.9 continues: 'However, specification of an actual "limit dose" will depend upon test method that has been employed to provide the test results, e.g. in the OECD Test Guideline for repeated dose toxicity studies by oral route, an upper dose of 1 000 mg/kg has been recommended as a limit dose, unless expected human response indicates the need for a higher dose level.'

excessive toxicity and severe suffering in parental animals (e.g. prostration, severe inappetence, excessive mortality), which would compromise the interpretation of reproductive effects.

OECD TGs 421/422, 414 and 443 outline aspects to be considered in dose level setting. Common to all these test guidelines is that the lowest dose should not produce any evidence of either maternal or developmental toxicity (and allow setting of the NOAEL). Dose level selection should ensure that any dose-related effect is demonstrated and enable the establishment of NOAELs for the most sensitive end point. To demonstrate a dose response, the mid-dose level is expected to produce observable toxic effects. However, there are some differences in the specifications for the top dose level (see below). Irrespective of the specifications in OECD text guidelines regarding selection of the top dose, for classification and labelling, as explained above, it is critical that the doses tested are high enough to enable a conclusion on a lack of reproductive toxic properties warranting a classification in category 1B or category 2 if clear evidence on reproductive toxicity is not observed.

The OECD TG 421/422 main specification for the top dose is:

- '[t]he highest dose level should be chosen with the aim of inducing toxic effects but not death or severe suffering.'

The OECD TG 443 main specification for the top dose is:

- '[i]f dose levels are based on toxicity, the highest dose should be chosen with the aim to induce some systemic toxicity, but not death or severe suffering of the animals.'

The OECD TG 414 main specification for top dose is:

- 'the highest dose should be chosen with the aim to induce some developmental and/or maternal toxicity (clinical signs or a decrease in body weight) but not death or severe suffering.'

There is a need to study various aspects in parents and their offspring, as set out in OECD TG 443. The study should be designed to ensure adequate assessment of the effects on sexual function and fertility, that is, the dose levels should not be reduced in order to get a sufficient number of offspring for the assessment of developmental toxicity. Even if the number of offspring is reduced because of effects on sexual function and fertility, any offspring available at that those level should be investigated for adverse effects on development. Results obtained at lower dose levels can also still be used to assess potential adverse effects on development.

It is also important that toxicity in both female and male animals is observed in order to ensure that reproductive toxicity in either sex is not overlooked. If existing information, including results from a dose range-finding study, show that sensitivity between male and female animals differs significantly, the dose setting should take these differences into account. The less sensitive sex should be tested at higher doses than the more sensitive sex.

All the OECD test guidelines require that appropriate dose level setting is demonstrated.

Dose level selection must be justified and documented to allow independent evaluation of the choice.

6.7.3. Reproductive toxicity (fertility)

6.7.3.1. Reproductive/developmental toxicity screening study (point 7.3.1 of Table 2 of Annex V to Commission Implementing Decision (EU) 2024/365)

According to column 1 of Table 2 of Annex V to Commission Implementing Decision (EU) 2024/365, reproductive/developmental toxicity screening tests (OECD TGs 421 and 422) are standard information required for substances migrating into drinking water with a C_{tap} value equal to or above 2.5 µg/l and below 250 µg/l.

According to the first indented point of column 2, 'the study does not need to be conducted if a reliable extended one-generation reproductive toxicity study is available, in which an appropriate animal species and route of administration were used.'

Additional conditions under which the study does not need to be carried out and reported are given in Section 6.7.1.

The purpose of the screening study is to provide initial information on the effects on male and female reproductive performance and on developmental toxicity during and shortly after birth, and also on certain additional parameters for measuring endocrine disruption, including anogenital distance, nipple/areola retention and thyroid hormone levels, as given in the revised OECD test guidelines 421 and 422(2016). These screening tests are not meant to provide complete information on all aspects of reproduction and development.

However, an evaluation of the screening tests (OECD TGs 421 or 422) has confirmed that these tests are useful for initial hazard assessment and can contribute to decision-making on further test requirements (Reuter et al., 2003; Gelbke et al., 2004; Beekhuijzen et al., 2014). In certain cases, further testing is required (see Section 6.7.3.3).

The following test methods should be used:

- EU method B.63: Reproduction/developmental toxicity screening test;
- OECD TG 421: Reproduction/developmental toxicity screening test;
- EU method B.64: Combined repeated dose toxicity study with the reproduction/developmental toxicity screening test;
- OECD TG 422: Combined repeated dose toxicity study with the reproduction/developmental toxicity screening test.

6.7.3.2. Extended one-generation reproductive toxicity study, oral route of administration (point 8.3.1 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365)

According to Column 1 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365 the extended one-generation reproductive toxicity study (EOGRTS) is standard information required for substances migrating into drinking water at concentrations equal to or above 250 µg/l. If another reproductive toxicity test is used, scientific justification should be provided.

As specified in column 2 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365, 'two-generation reproductive toxicity studies [EU method B.35, OECD TG 416] that

were initiated before 13 May 2015 shall be considered appropriate to address this standard information requirement.'

Additional conditions under which the study does not need to be provided are given in Section 6.7.1.

6.7.3.3. Study design for the extended one-generation reproductive toxicity study

The basic study design focuses on evaluating the sexual function and fertility of parental animals (F0 animals) and of defined parameters measuring the postnatal development of F1 animals until adulthood (see EU method B.56 and OECD TG 443). The basic study design does not include mating of F1 animals (extension of cohort 1B) or cohorts for developmental neurotoxicity (cohorts 2A and 2B) or developmental immunotoxicity (cohort 3).

According to column 2 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365:

An Extended One-Generation Reproductive Toxicity Study with the extension of cohort 1B to include the F2 generation [shall be submitted] where any of the following conditions are met:

- (a) the relevant chemical species displays genotoxic effects in somatic cell mutagenicity tests *in vivo* which could lead to its classification as mutagen category 2;
- (b) there are indications that the internal dose for the relevant chemical species and/or any of its metabolites will reach a steady state in the test animals only after an extended exposure;
- (c) there are indications of one or more relevant modes of action related to endocrine disruption from available *in vivo* studies or non-animal approaches.

An Extended One-Generation Reproductive Toxicity Study including cohorts 2A/2B (developmental neurotoxicity) and/or cohort 3 (developmental immunotoxicity) shall be included in case of particular concerns on (developmental) neurotoxicity or (developmental) immunotoxicity justified by any of the following:

- (a) existing information on the relevant chemical species itself derived from relevant available *in vivo* or non-animal approaches (e.g., abnormalities of the central nervous system (CNS), evidence of adverse effects on the nervous or immune system in studies on adult animals or animals exposed prenatally);
- (b) specific mechanisms/modes of action of the relevant chemical species with an association to (developmental) neurotoxicity and/or (developmental) immunotoxicity (e.g., cholinesterase inhibition or relevant changes in thyroidal hormone levels associated to adverse effects);
- (c) existing information on effects caused by substances analogous to the relevant chemical species being studied, suggesting such effects or mechanisms/modes of action.

With regard to assessment of developmental neurotoxicity by triggering cohorts 2A and 2B (developmental neurotoxicity) in the EOGRTS design, the ECHA Committee for Risk Assessment has recently published a guidance note⁽⁵⁴⁾, stating that the adverse effects on the nervous system (e.g. brain), investigated or detected at any point in the lifespan of the organism exposed during the developmental period, covering both prenatal and postnatal development until sexual maturation (determined by preputial separation and vaginal opening), should be addressed

⁽⁵⁴⁾ https://echa.europa.eu/documents/10162/17090/rac_clh_guidance_note_neurotoxicity_en.pdf/96717ed9-55d3-10e0-785b-093d07e267f3?t=1665023711575.

under developmental toxicity (developmental neurotoxicity), even if the exposure had also continued after sexual maturation.

6.7.4. Prenatal developmental toxicity study in one animal species (point 8.3.2 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365)

A prenatal developmental toxicity study (EU method B.31, OECD TG 414), conducted in one animal species (in rat, unless another animal species can be justified as being more appropriate), is standard information required for substances migrating into drinking water at concentrations equal to or above 250 µg/l.

As specified in column 1 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365 the study shall be performed 'in rat, unless another animal species is justified to be more appropriate, oral route of administration'. The prenatal developmental toxicity studies reported, taken together with other relevant data and information on the relevant chemical species (e.g. the developmental parameters of the EOGRTS and OECD TG 426, when available), must be sufficient to permit the assessment of potential hazardous properties and risks to the offspring following exposure to the relevant chemical species during development.

Conditions under which the study does not need to be provided are given in Section 6.7.1.

The prenatal developmental toxicity study (EU method B.31, OECD TG 414) provides a focused evaluation of potential effects on prenatal development, although only effects that are manifest before birth can be detected. Detailed information on external, skeletal and visceral malformations and variations and also on other prenatal developmental effects are provided. Caesarean section allows precise evaluation of the number of fetuses affected.

The prenatal developmental toxicity study should be sufficient to:

- identify direct and indirect effects on embryonic and fetal development resulting from exposure to the relevant chemical species;
- identify any maternal toxicity;
- establish the relationship between observed responses and dose in both dam and offspring;
- establish NOAELs for maternal toxicity and pup development;
- provide additional information on adverse effects in pregnant compared with non-pregnant females;
- provide additional information on any enhancement of general toxic effects on pregnant animals.

The decision on animal species to be tested primarily depends on consideration of all available information, including the type of substance to be tested. According to the test methods (EU method B.31, OECD TG 414), the rat is the preferred rodent species and the rabbit the preferred non-rodent species. Since most of the toxicity studies (e.g. acute, repeated dose and toxicokinetic studies) are conducted in the rat, it may be considered that the prenatal developmental toxicity study should also be conducted in that animal species. Findings from previous studies may be useful in dose selection or the identification of additional end points for evaluation. In addition, the outcome of the prenatal developmental toxicity study may be helpful

in the interpretation of other reproductive toxicity studies, for which the rat is generally the preferred animal species.

In certain cases, the rabbit might be selected as the animal species for the prenatal developmental toxicity study. This may be done, for example, if the rabbit is considered to be a more sensitive species than the rat for that specific substance. The selection of the animal species for the prenatal developmental toxicity study should be made taking into account substance-specific aspects. If an animal species other than the rat or the rabbit is selected, the selection should be justified. The results from prenatal developmental toxicity studies are considered relevant to humans unless there is substance-specific toxicokinetic or toxicodynamic evidence showing otherwise.

Malformations and variations and external skeletal and visceral anomalies should be reported separately and combined in such a way that all relevant changes that are observed to occur in characteristic patterns in individual fetuses, or those that could be considered to represent different grades of severity of the same type of change, are reported in a concise manner.

Diagnostic criteria for malformations and variations should be given in the report. The terminology should follow that presented in Appendix I to OECD Guidance Document 43 and via the DevTox project (<http://www.devtox.org>). Further guidance on the conditions for historical control data are provided in OECD Guidance Document 43 on mammalian reproductive toxicity testing and assessment ⁽⁵⁵⁾.

Information on prenatal developmental toxicity coming from one- or multigeneration studies (such as OECD TGs 443, 416, 426, 421, 422) is not equivalent to that from the prenatal developmental toxicity study. The results from e.g. OECD TG 443 and 416 studies do not provide confidence to conclude that there is no prenatal developmental toxicity. Structural malformations and variations are not specifically investigated in one- and multigeneration studies. Therefore, information from one- or multigeneration studies does not cover the information on prenatal developmental toxicity in rodent species. However, in addition to information on prenatal developmental toxicity, information on effects due to exposure during peri- and postnatal developmental periods that is obtained from one- or multigeneration studies (e.g. OECD TGs 426, 443 and 426) is also relevant for developmental hazard identification and should be assessed to conclude on classification and labelling for developmental toxicity (see section 3.7.1.4. of the CLP regulation).

For the generation of new test data, the latest update of the test methods for prenatal developmental toxicity, given below, should be used.

The prenatal developmental toxicity study also provides information that may be relevant for clarifying endocrine activity. OECD TG 414 has been updated to include thyroid hormone and thyroid-stimulating hormone analysis in dams (T4, T3 and TSH) and measurement of anogenital distance (by sex and related to weight) in fetuses in rats. Some findings, such as alterations in fetal weight or placental weight, taken together with litter size, can also be considered beyond the assessment of reproductive toxicity, for example in the assessment of endocrine disruption.

OECD TG 414 may detect and provide diagnostic information on the effects of substances with oestrogen, androgen, thyroid and steroidogenic (EATS)-related modes of action. Further examples of parameters, diagnostic for 'EATS-mediated' and parameters 'sensitive to, but not diagnostic of, EATS' are provided in Tables 13–14 of the ECHA/EFSA *Guidance for the identification of endocrine disruptors in the context of Regulations (EU) No 528/2012 and (EC) No 1107/2009* ⁽⁵⁶⁾.

⁽⁵⁵⁾ [https://one.oecd.org/document/ENV/JM/MONO\(2008\)16/en/pdf](https://one.oecd.org/document/ENV/JM/MONO(2008)16/en/pdf).

⁽⁵⁶⁾ <https://echa.europa.eu/fi/-/guidance-on-identifying-endocrine-disruptors-published>.

The following test methods for prenatal developmental toxicity should be used:

- EU method B.31: Prenatal developmental toxicity study;
- OECD TG 414: Prenatal developmental toxicity study (2018 revision).

6.7.5. Further prenatal developmental toxicity study in a second animal species, oral route of administration or mechanistic study (point 8.3.3 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365)

As indicated in column 2 of point 8.3.3 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365, 'a decision on the need to perform additional studies on a second animal species or mechanistic studies should be based on the outcome of the first test (point 8.3.2) and all other relevant available data (in particular rodent reproductive toxicity studies).'

A second prenatal developmental toxicity study in another animal species (rabbit) does not need to be performed if no prenatal developmental effects are observed in the study conducted in the first animal species and if no indication of pre- and/or postnatal developmental toxicity is observed in a one- or multigeneration reproductive toxicity study (performed in the rat) at the highest dose tested.

The study should be carried out in accordance with OECD TG 414 (revised in 2018), and the preferred animal species is the rabbit. The study should be carried out by the oral route of administration.

Apart from the results of the prenatal developmental toxicity study, all other relevant and available data, including indications from repeat dose toxicity studies (90-day studies), ADME studies, and multigeneration developmental neurotoxicity or extended one-generation studies, further neurotoxicity studies and, if possible, the mode of action of the test-relevant chemical species should be considered when deciding on an additional prenatal developmental toxicity study on a second animal species. Knowledge of structural analogues for the relevant chemical species should also be included in the assessment.

It should be noted that, currently, there are no mechanistic studies available that can replace the *in vivo* prenatal developmental toxicity study.

6.8. Carcinogenicity (point 8.4 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365)

The carcinogenicity study identifies the carcinogenicity potential of the relevant chemical species in laboratory animals to facilitate the extrapolation of potential risks to humans. The studies should be sufficient to establish the organ specificity of tumours induced, to establish the dose-response relationship and, for non-genotoxic carcinogens, to identify doses eliciting no adverse effects (threshold dose).

6.8.1. Collection and evaluation of available information

For the assessment of existing information, further guidance is available in DWD Guidance Volume II and in the ECHA guidance on the application of the CLP criteria ⁽⁵⁷⁾.

According to point 8.4 in column 1 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365, a carcinogenicity study is standard information required for relevant chemical

⁽⁵⁷⁾ <https://echa.europa.eu/web/guest/guidance-documents/guidance-on-clp>.

species migrating into drinking water at concentrations equal to or above 250 µg/l. However, as specified in column 2 of Table 3:

the study does not need to be conducted where all of the following conditions are fulfilled:

- (a) no genotoxic potential is identified in genotoxicity tests; and
- (b) the sub-chronic and long-term (≥ 12 months) toxicity studies show no evidence of toxicity at the limit dose level.

6.8.2. Combined chronic toxicity/carcinogenicity study (point 8.4.1 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365)

As indicated in point 8.4 of column 1 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365, if a new carcinogenicity study needs to be performed, it has to be a combined chronic toxicity / carcinogenicity study. For further considerations on the data generation, see Section 6.6.2.2 of this guidance.

The following test methods should be used:

- EU method B.33: Combined chronic toxicity / carcinogenicity test;
- OECD TG 453: Combined chronic toxicity / carcinogenicity studies.

6.9. Appropriate additional studies, referred to under point 6.2 of Table 1 and point 7.4 of Table 2 of Annex V to Commission Implementing Decision (EU) 2024/365

As stated in column 1 of points 6.2 and 7.4, additional testing to that listed in Tables 1 and 2 of Annex V to Commission Implementing Decision (EU) 2024/365 must be conducted, if there is concern about one of the hazards listed in column 2. A concern means that there is a reasonable ground to expect that the relevant chemical species might have such effects.

A list (not exhaustive) of the concerns that may trigger an extended one-generation reproductive toxicity study is given below.

From a screening study or equivalent:

- changes in reproductive or other endocrine organ weight in intact animals;
- effects on spermatogenesis or folliculogenesis *in vivo* and/or histopathological findings in reproductive organs and/or accessory sex organs;
- effects on histopathology of the thyroid;
- changes in thyroid hormones;
- effects on sperm parameter analysis or the oestrous cycle;
- biologically relevant changes in hormone levels *in vivo* (related to reproductive toxicity);
- reduced mating, fertility or litter size;
- increased incidence of abortions compared with controls;

- changes in gestation length;
- reduced survival of offspring;
- reduced body weight of offspring independent of litter size;
- reduced levels of maternal care;
- changes in anogenital distance unrelated to body weight/size;
- changes in nipple/areola retention;
- indication of other endocrine-disrupting modes of action related to reproductive toxicity.

From a repeated dose toxicity study:

- changes in reproductive or other endocrine organ weight in intact animals;
- effects in spermatogenesis or folliculogenesis *in vivo* and/or histopathological findings in reproductive organs and/or accessory sex organs;
- effects on sperm parameters analysis or oestrous cycle
- biologically relevant changes in hormone levels (related to reproductive toxicity);
- indication of other endocrine disrupting modes of action related to reproductive toxicity.

Additional guidance on the evaluation of triggers can be found in point (iii) of step 4.4 of Section R.7.6.2.3.2 of *Guidance on information requirements and chemical safety assessment – Chapter R.7a: Endpoint specific guidance* (version 6.0, July 2017) ⁽⁵⁸⁾.

6.10. Additional toxicity properties (point 8.5 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365)

As stated in column 2 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365, 'if there is indication for one or more mechanisms/modes of action of the relevant chemical species with an association to (developmental) neurotoxicity and/or endocrine disruption and/or (developmental) immunotoxicity, corresponding additional data shall be generated in accordance with this point, unless already fully covered by the information under point 8.3.1.'

6.10.1. Appropriate neurotoxicity information or study, including developmental neurotoxicity (point 8.5.1 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365)

Additional information on specific studies may be needed if there is any evidence, for example knowledge of the mechanism of action or from repeated dose toxicity studies, that the relevant chemical species may have neurotoxic or developmental neurotoxic properties.

⁽⁵⁸⁾ https://echa.europa.eu/documents/10162/17224/information_requirements_r7a_en.pdf/e4a2a18f-a2bd-4a04-ac6d-0ea425b2567f.

According to point 8.5.1 of Table 3, 'If anticholinesterase activity is detected a test for response to reactivating agents shall be generated.'

If additional information on neurotoxicity needs to be generated, applicants are advised to follow the principles, described in *Guidance on the biocidal products regulation – Volume III: Human health – Part A: Information requirements*, Section 1.13.2⁽⁵⁹⁾, and with regard to developmental neurotoxicity the ECHA Committee for Risk Assessment guidance note⁽⁶⁰⁾.

6.10.2. Appropriate information or study on endocrine disruption (point 8.5.2 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365)

According to the above point, 'If there is any evidence from in vitro studies or repeated dose or reproduction toxicity studies, that the relevant chemical species may have endocrine disrupting properties for human health', additional information or specific studies should be generated 'in order to elucidate the mode/mechanism of action and provide sufficient evidence for relevant adverse effects'.

If the assessment of the available information suggests that the relevant chemical species has an endocrine-disrupting property and additional information needs to be generated, applicants are advised to follow the steps in the assessment strategy, described in *Guidance on the biocidal products regulation – Volume III: Human health – Part A: Information requirements*, Sections 1.13.3 and 1.13.3.1⁽⁶¹⁾.

In addition, OECD Guidance Document 150 (2012, updated 2018)⁽⁶²⁾ and the ECHA/EFSA *Guidance for the identification of endocrine disruptors in the context of Regulations (EU) No 528/2012 and (EC) No 1107/2009*⁽⁶³⁾, where the testing strategy is further elaborated, should be considered.

The applicant should include a collation of the relevant information and report the lines of evidence for each modality, as recommended in the ECHA/EFSA guidance.

6.10.3. Appropriate immunotoxicity information or study, including developmental immunotoxicity (point 8.5.3 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365)

According to the above point, 'If there is any evidence, from skin sensitisation and repeated dose or reproduction toxicity studies, that the relevant chemical species may have immunotoxic properties', then additional information or specific studies should be generated 'in order to elucidate the mode/mechanism of action and provide sufficient evidence for relevant adverse effects'.

If the assessment of the available information suggests that the relevant chemical species have immunotoxicity properties and additional information needs to be generated, the applicants are advised to follow the principles, described in *Guidance on the biocidal products regulation –*

⁽⁵⁹⁾ https://echa.europa.eu/documents/10162/2324906/bpr_guidance_vol_iii_part_a_en.pdf/05e4944d-106e-9305-21ba-f9a3a9845f93?t=1648525287369.

⁽⁶⁰⁾ https://echa.europa.eu/documents/10162/17090/rac_clh_guidance_note_neurotoxicity_en.pdf/96717ed9-55d3-10e0-785b-093d07e267f3?t=1665023711575.

⁽⁶¹⁾ https://echa.europa.eu/documents/10162/2324906/bpr_guidance_vol_iii_part_a_en.pdf/05e4944d-106e-9305-21ba-f9a3a9845f93?t=1648525287369.

⁽⁶²⁾ <https://www.oecd-ilibrary.org/docserver/9789264304741-en.pdf?expires=1649583126&id=id&accname=quest&checksum=9DFC6220D35A070704403369E05EA5FE>

⁽⁶³⁾ <https://echa.europa.eu/fi/-/guidance-on-identifying-endocrine-disruptors-published>.

Volume III – Human health – Part A: Information requirements, Section 1.13.4 ⁽⁶⁴⁾. In addition, *in vitro* assays that evaluate potential immunotoxic effects (i.e. OECD TG 444A (IL-2 Luc assay)) may also provide useful information. The IL-2 Luc assay is an important method for evaluating the immunotoxic potential of chemicals as a part of a battery (JaCVAM, 2020) ⁽⁶⁵⁾. It provides a rapid screening tool that can be used as part of a systematic assessment of immunotoxicity when combined with other immunotoxicity tests. However, the following limitations should be noted: (1) the method cannot detect immunotoxicity associated with the inhibition of DNA synthesis and cell division and (2) the assay cannot detect test chemicals that require metabolic activation to form an immunotoxic metabolite.

6.11. Appropriate mechanistic data or studies (point 8.5.4 of Table 3 of Annex V to Commission Implementing Decision (EU) 2024/365)

According to the above point, 'This standard information or testing shall be generated if necessary to clarify effects reported in toxicity studies.'

These data may be relevant in the weight of evidence assessment along with other information to assess the toxicological properties of a relevant chemical species, as they can provide information on the mode of action of the substance.

Studies of the mechanisms of toxicity / mode of action may provide useful supplementary information when there are indications that the relevant chemical species may have effects on, for example, carcinogenicity (genotoxic and non-genotoxic modes of action are relevant for the classification of carcinogens in accordance with the CLP regulation), reproduction, neurotoxicity or immunotoxicity. Such studies may in some cases be used in concluding that the effects observed in experimental animals are not relevant for humans. For ED identification, mechanistic studies may be needed. In addition, information on the modes of action / mechanisms may clarify observed differences in effects between sexes (potential information on endocrine activity), differences in toxicity between different life stages (e.g. sensitivity during developing or elderly animals) or an underlying cause (e.g. immunotoxicity) of other effects.

As a general principle, the effects observed in animal studies are considered relevant for humans unless there is sufficient information to prove the contrary. In order to conclude that the adverse effects are not relevant for humans, it is necessary to establish that the adverse effects are caused by a mode of action that is not relevant for humans, and it must also be possible to exclude other modes of action from the adverse effects observed.

⁽⁶⁴⁾ https://echa.europa.eu/documents/10162/2324906/bpr_guidance_vol_iii_part_a_en.pdf/05e4944d-106e-9305-21ba-f9a3a9845f93?t=1648525287369.

⁽⁶⁵⁾ https://jacvam.jp/files/list/12/12_01_peer%20review%20report.pdf.

7. Testing for nanoforms

7.1. Overview

This section focuses on nanoforms of substances potentially coming into contact with drinking water as a result of the use of DWCMs. Nanoforms of relevance to DWD applications include any nanoparticle (the applied-for entry or NIAS) that migrates into the drinking water.

In the scope of the DWD, the term nanoform⁽⁶⁶⁾, as defined in the glossary, is used to describe nano-sized applied-for entries aligning with the REACH regulation. Consequently, for the DWD, nanoforms concern mostly substances and particulate matter incorporated into conventional DWCMs. Nevertheless, when applying for nanoforms, apart from the nanoform itself, the material matrix into which the nanoform is incorporated should also be described.

If a substance is added to a material in different forms (i.e. in different nanoforms differing in, for example, shape and size and/or in non-nanoforms), each form requires a separate application to be submitted to ECHA.

Figure 18 and Figure 19 illustrate the main steps to be followed in deciding whether nanoform-specific advice should be followed or, because of the conclusions reached on the nanoform's properties, the guidance provided in Chapters 1–6 of this guidance should be followed instead.

For **applied-for entries that fulfil the definition of a nanoform**, the following approach should be taken (step 1 in Figure 18).

1. Provide the appropriate nanoform characterisation information, in accordance with point 1.4. of Annex I to Commission Implementing Decision (EU) 2024/365.
2. Considerations of water solubility:
 - If the water solubility of the nanoform is high (> 33.3 g/l), an assessment of the migration of nanoparticles into drinking water is not required. Follow the information provided in Chapters 1–6 of this guidance.
 - If the water solubility is < 33.3 g/l, go to point 3.
3. Considerations of half-life ($t_{1/2}$) in water (dissolution test):
 - If the $t_{1/2}$ in water is ≤ 10 minutes, – an assessment of the migration of nanoparticles into drinking water is not required. Follow the information provided in Chapters 1–6 of this guidance.
 - If the $t_{1/2}$ in water is ≥ 10 minutes, an assessment of the migration of nanoparticles into drinking water is required. Go to step 2 in Figure 19.

For applied-for entries that do not meet the definition of nanoform, but **contain nanoparticle constituents or impurities at ≥ 0.1 % (w/w) in the substance and ≥ 0.02 % (w/w) in the formulation**, the following approach should be taken (step 1 in Figure 18).

⁽⁶⁶⁾ Outside (and also within) regulatory texts, for example in the scientific literature, such terms as 'nanomaterial' and 'nanoparticle' are used instead of nanoforms. These terms may overlap and cause confusion. Thus, for DWD applications the term 'nanoform' and this guidance should be adhered to.

1. Considerations of water solubility of nanoparticle constituents or impurities:
 - If the water solubility is high (> 33.3 g/l), an assessment of the migration of nanoparticles into drinking water is not required. Follow the information provided in Chapters 1–6 of this guidance.
 - If the water solubility is < 33.3 g/l, go to point 2.
2. Considerations of half-life ($t_{1/2}$) in water (dissolution test) of nanoparticle constituents or impurities:
 - If the $t_{1/2}$ in water is ≤ 10 minutes, an assessment of the migration of nanoparticles into drinking water is not required. Follow the information provided in Chapters 1–6 of this guidance.
 - If the $t_{1/2}$ in water is ≥ 10 minutes, an assessment of the migration of nanoparticles into drinking water is required. Go to step 2 in Figure 19.

When assessment of the migration of nanoparticles into drinking water is required, the following approach should be taken (step 2 in Figure 19).

1. Identify if there are necessary analytical methods for nanoparticle quantification in the migration water.
 - If **yes**, perform physical migration testing, using the applied-for entry as the test material.
 - If **no**, perform physical migration testing using the analytical method for migration of the chemical substance or, if not possible (justification is needed), perform migration modelling as in Section 5.3.1.
2. Identify the relevant chemical species (Section 3 of Annex IV to Commission Implementing Decision (EU) 2024/365).
3. Identify, for each relevant chemical species if it migrates as nanoparticles (1–100 nm):
 - If **yes**, follow the guidance in this chapter and use the nanoparticle-relevant chemical species as the test material to generate toxicological information relevant to the migration tier identified.
 - If **no**, follow the information provided in Chapters 1–6 of this guidance.
 - If **uncertain**, when using an analytical method that is unable to distinguish between nano- and non-nanoparticles, toxicological information, relevant to the migration tier identified, should be provided for both nanoforms/nanoparticles and non-nanoforms/non-nanoparticles as the worst-case approach.

When assessing migration of nanoparticles, certain theoretical considerations may also be taken into account, such as the size distribution and nature of inclusion in the material (see also Section 7.4.2). Apart from physical migration testing, diffusion modelling may also be considered (see also Sections 5.3.1 and 7.4.2).

Figure 18. Decision tree for deciding on the testing strategy for nanoforms (step 1)

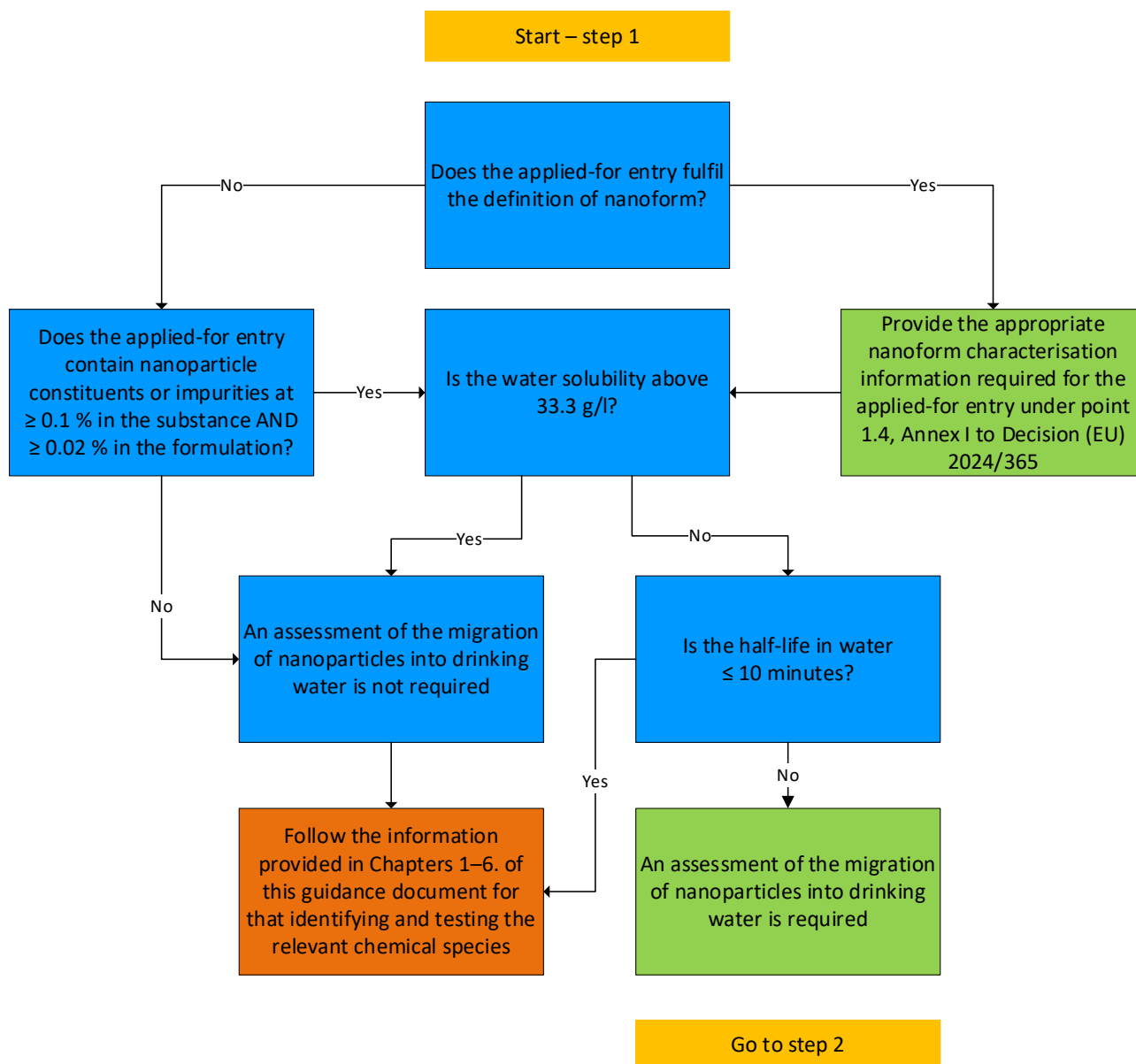
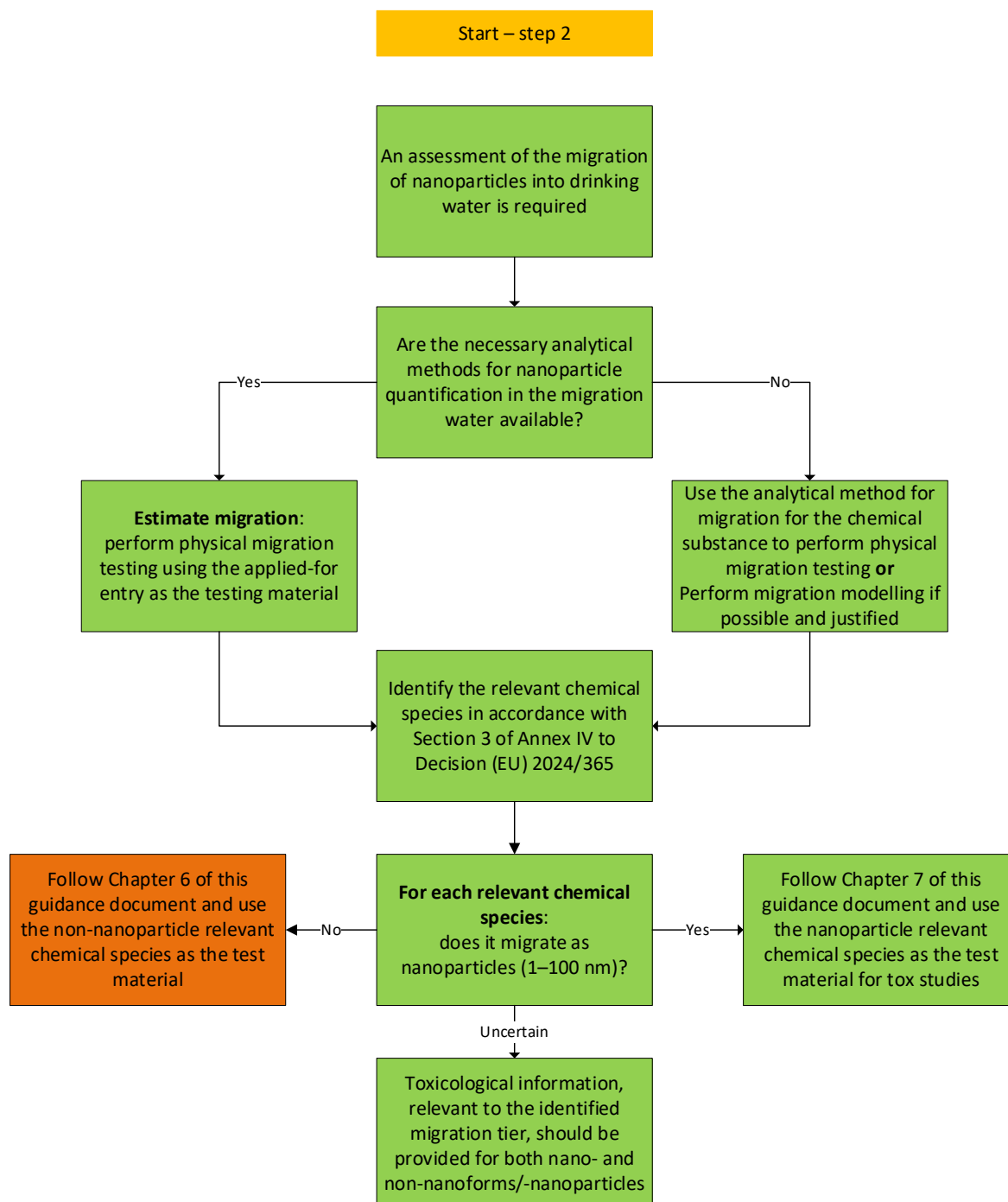


Figure 19. Decision tree for deciding on the testing strategy for nanoforms (step 2)

7.2. Substance identification and characterisation of nanoforms (point 1.4 in Annex I to Commission Implementing Decision (EU) 2024/365)

According to Article 3(6) of Commission Delegated Regulation (EU) 2024/369, a separate DWD application must be submitted for each nanoform; consequently, a separate notification or intention must be submitted in advance of the DWD application, in accordance with Article 2 of

the same regulation. These nanoforms should be characterised pursuant to point 1.4 of Annex I to Commission Implementing Decision (EU) 2024/365.

Chapter 4 of the ECHA *Guidance for identification and naming of substances under REACH and CLP* should be applied for the purpose of identification of the substance. Commission Regulation (EU) 2018/1881 of 3 December 2018 amended the REACH regulation with regard to Annexes I, III, VI, VII, VIII, IX, X, XI and XII to address specifically nanoforms of substances.

For each substance, the information given to identify the substance should be sufficient to enable each substance to be identified and the different nanoforms to be characterised. If it is not technically possible or if it does not appear scientifically necessary to give information on one or more of the required items, the reasons should be clearly stated.

7.3. Physico-chemical properties beyond granulometry

Information specific on nanoforms, such as dissolution rate, particle aggregation, agglomeration and particle surface chemistry, may already be available under the REACH regulation (as amended by Regulation (EU) 2018/1881); these properties can be reported in the DWD application to support the assessment ⁽⁶⁷⁾.

7.3.1. Consideration of solubility and dissolution rate

The Scientific Committee on Emerging and Newly Identified Health Risks has stated that many nanomaterials have considerable solubility and that for 'these materials the interaction with living systems remains close enough to the bulk chemical agent to justify the use of well-established toxicological testing procedures and approaches' (SCENIHR, 2006). Water solubility may give an initial indication of the (none) biopersistence of a nanomaterial (Arts et al., 2015). EFSA guidance on technical requirements (EFSA Scientific Committee et al., 2021) considers a solubility of 33.3 g/l or above as high solubility in water, which is also in line with the Scientific Committee on Consumer Safety guidance (SCCS, 2019) and with the categories for degrees of solubility proposed by JECFA (2006).

Based on achieving a pragmatic and consistent approach among different regulatory bodies, a similar value for water solubility is also recommended here to decide whether this guidance is applicable (see Figure 18).

The latest approaches for the risk assessment of nanomaterials recommend a strategy in which the dissolution rate and equilibrium in water is a first key element (Dekkers et al., 2016; EFSA Scientific Committee et al., 2021). Dissolution rate is a kinetic process that reveals not only whether a substance dissolves but also how long it takes to dissolve. Dissolution rate depends on the size and surface properties of the particles and fibres, the nature of media where the release of ions occurs, temperature, pH and the presence of substances interacting with the nanoform's surface. The pH and temperature should be specified and relate to biologically relevant conditions. Another important factor in the dissolution of nanoparticles is aggregation (Utembe et al., 2015). Aggregation decreases the available external surface area, thereby reducing the extent of dissolution (Borm et al., 2006). It has been observed that the dissolution rate decreases as the aggregation state of nanoparticles increases.

To determine the dissolution rate in water, the protocol proposed by EFSA is recommended. In the EFSA guidance on technical requirements (EFSA Scientific Committee et al., 2021) a high dissolution rate in water is defined as a half-life of 10 minutes or less, corresponding to a dissolved fraction of than 88 % or more in 30 minutes. In this context, the EFSA guidance on technical requirements refers to a 'concentration corresponding to exposure at maximum use

⁽⁶⁷⁾ Additional information is available at <https://euon.echa.europa.eu/reach-test-methods-for-nanomaterials>.

level'. A similar threshold for dissolution rate in water is proposed in this guidance (see Figure 18).

7.3.2. Methods available for solubility and dissolution testing

Different analytical techniques/methods and their suitability for measuring solubility of nanomaterials have been reviewed (Tantra et al., 2016). The solubility of nanomaterials in water should be determined in accordance with OECD TG 105 with specific consideration of nanoforms (OECD, 2020). The pH and temperature should be specified for solubility testing and relate to biologically relevant conditions (Avramescu et al., 2017).

In the solubility/dissolution kinetics experiments, the solid phase needs to be separated from the liquid phase to avoid artefactually high solubility values by also analysing solid materials. The presence of particles can be checked qualitatively by, for example, Tyndall analysis, and solid nanoparticles can be removed by ultracentrifugation or ultrafiltration (Tantra et al., 2016).

So far, no specific OECD testing guideline is available for determination of solubility and dissolution rate of nanomaterials in water, but some guidelines may be adapted, for example OECD TG 105 (Misra et al., 2012; Tantra et al., 2016) and ISO/TR 19057:2017. There is also ongoing work on the OECD WNT project 1.5 on the determination of the solubility and dissolution rate of nanomaterials in water and relevant synthetic biological media, aiming to provide harmonised approaches for testing the solubility and dissolution rate of nanomaterials.

Further guidance on solubility and dissociation rate can be found in Sections 2.1.1.1 and 2.1.1.2 of ECHA *Guidance on Information Requirements and Chemical Safety Assessment – Appendix R7-1 for nanomaterials applicable to Chapter R7a: Endpoint specific guidance* ⁽⁶⁸⁾ and in EFSA *Guidance on technical requirements for regulated food and feed product applications to establish the presence of small particles including nanoparticles* ⁽⁶⁹⁾.

As long as no standard methods are available, detailed description and justification of the method including adequate controls (and reference materials, if possible) is required.

7.4. Migration

In the scope of the DWD, to test the migration of nanoform-containing materials, the standard test methods referred to in Chapter 5 of this guidance may be used. Apart from the standard testing procedures, any nano-specific measures (e.g. adaptation of migration medium with stabilisers) taken in testing should be justified and reported.

To enable proper assessment of nanoform migration, at least the following information should be provided.

7.4.1. Description of nanoform structure, type of inclusion in matrix and concentration

The exact nature of the nanoform or nanoparticle and the nature of its inclusion in the material should be provided, either as part of the substance identity information or in a specific migration testing description. The concentration of the nanoform in the matrix must be reported.

Nanoforms or nanoparticles should be characterised in accordance with the requirements set out in point 1.3 of Annex I to Commission Implementing Decision (EU) 2024/365. Furthermore, the

⁽⁶⁸⁾ https://echa.europa.eu/documents/10162/17224/appendix_r7a_nanomaterials_en.pdf/1bef8a8a-6ffa-406a-88cd-fd800ab163ae?t=1633348005491.

⁽⁶⁹⁾ <https://efsa.onlinelibrary.wiley.com/doi/epdf/10.2903/j.efsa.2021.6769>.

nature of inclusion in the host matrix should be demonstrated via techniques such as transmission electron microscopy or scanning electron microscopy. Initial concentration in a matrix should be demonstrated using techniques such as X-ray diffraction or energy-dispersive X-ray spectroscopy combined with transmission electron microscopy.

7.4.2. Testing

The standard migration test methods referred to in Chapter 5 of this guidance for non-nanoform materials can be used, unless there is a specifically requirement to use other methods due to the properties of the nanoform.

Before actual testing for nanoparticle migration, it may be useful to consider the likelihood of such migration in theory. According to the literature, the likelihood of nanoparticle migration changes according to particle size and the nature of their inclusion in the host material. Previous work suggests that only nanoparticles < 5 nm are able to diffuse at potentially meaningful rates within a material, thus allowing migration into water. In addition, when a nanoparticle is completely embedded in a host matrix, the likelihood of migration is much smaller than for nanoparticles located at the surface of the host material and thus only partially embedded. To demonstrate size distribution and the nature of inclusion within a material, microscopic techniques (scanning electron microscopy, transmission electron microscopy) may be employed. For nanoparticles completely embedded in a polymeric matrix, migration modelling or theoretical justification of non-migration may be applicable (Franz et al., 2017, 2020; Störmer et al., 2017).

Nanoforms that migrate into water may remain as nanoforms or degrade or dissolve due to their potential instability. This presents experimental uncertainties by introducing experimental artefacts and distorting analysis. The degraded and dissolved nanoforms are treated as non-nanoform substances contributing to the overall measurable concentration of the substance in water. It is important for applicants to establish in which form the nanoforms migrate. For this purpose, the methods described in EFSA guidance on the technical requirements for establishing the presence of small particles ⁽⁷⁰⁾ may prove valuable.

In certain cases, diffusion modelling of nanoform migration may also be an option. Similarly to bulk form modelling, nano-modelling must fulfil the conditions regarding the modelling itself, including reliable input parameters, and the conditions laid down in Section 2.1 of Annex IV to Commission Implementing Decision (EU) 2024/365.

7.4.3. Analysis

To analyse migrated nanoforms, suitable analytical methods capable of distinguishing nanoforms from other solutes (ionic or molecular) should be used, for example single-particle ICP-MS (spICP-MS). When possible, the analytical results for a blank sample migration water (i.e. a sample equivalent to the nanoform-containing sample but without the nanoform) should be provided for comparison. This enables estimation of whether or not there is migration of nanoforms.

The challenge posed by nanoforms is their distinctiveness and degree of separation from their non-nanoform counterparts. Suitable analytical methods capable of distinguishing between nanoforms and non-nanoforms, such as spICP-MS should be used. Otherwise, techniques such as electron microscopy, analytical centrifugation or fractionation may be used for separation and combined with analytical methods ⁽⁶⁹⁾. Nevertheless, it may prove difficult to find targeted, reliable, validated methods for nanoforms or nanoparticles. In these cases, combinations of analytical techniques, modelling and theoretical approaches may be used to demonstrate

⁽⁷⁰⁾ <https://efsa.onlinelibrary.wiley.com/doi/epdf/10.2903/j.efsa.2021.6769>.

migration levels. Then the level of concern about nanoforms or nanoparticles may be assessed taking into account toxicological knowledge.

7.5. Toxicological data

As explained in Figure 18, for nanoforms that migrate the drinking water and for which solubility is less than 33.3 g/l and the half-life measured by the water dissolution test is 10 minutes or less, toxicological data need to be provided.

The testing methodology for nanoforms is based on the migration concentration, in the same way as for non-nanoforms, and should follow the principles of the toxicological assessment described in Section 6.4 of this guidance with some modifications, as described in Section 7.5.2.

7.5.1. Changes in and adaptations to the testing methodology for nanoforms

7.5.1.1. Choice of experimental conditions for toxicological tests

Overall, the choice of experimental conditions determines the relevance of the results obtained in a study and the weight that can be attributed to the results. The three major parameters related to experimental conditions are as follows: (1) the treatment conditions (in terms of maximum dose studied and range of doses, duration, frequency, means of exposure, etc.); (2) choice of test system (cell type or animal species); and (3) possible interference with the biological parameter measured. Further information can be found in an opinion from the French Agency for Food, Environmental and Occupational Health & Safety on a specific health risk assessment guide for nanomaterials in food products⁽⁷¹⁾ and in EFSA guidance on risk assessment of nanomaterials⁽⁷²⁾.

7.5.1.2. Treatment conditions

Even for the same nanoform, the aggregation/agglomeration characteristics of a dispersion can modify the level of exposure of cell models. This can then influence the intracellular concentration of the nanoform and therefore possibly the test result. The treatment conditions (cells/ml or cells/cm², treatment volume, well surface area, etc.) may also affect the intracellular concentration and consequently the results of a study. When assessing the hazard associated with nanoforms, it is therefore important to establish standardised test procedures covering these points.

The question of the most relevant metric to describe the concentration of a nanoform is still a subject of debate. In *in vitro* tests, the nanomaterial's mass and surface area per unit area of the culture medium are most commonly used (while it may seem optimal, expression of the surface concentration, i.e. particle surface area/unit area of the culture medium, is rarely used). Using particle number concentration has more challenges because the analytical techniques applied to monitor dosimetry give very variable results (Peterson et al., 2019). In *in vivo* tests, because of the need to use a mass reference, the dose is always expressed in units of mass of the nanoform (e.g. mg/kg of body weight of the animal species studied). Dispersion preparations and exposure dosimetry can refer to the recommendations of Deloid et al. (2017). Additional information on sample preparation and dosimetry can be obtained from OECD Guidance Document 36⁽⁷³⁾.

⁽⁷¹⁾ <https://www.anses.fr/en/system/files/ERCA2016SA0226EN.pdf>.

⁽⁷²⁾ <https://efsa.onlinelibrary.wiley.com/doi/epdf/10.2903/j.efsa.2021.6768>.

⁽⁷³⁾ <https://www.oecd-ilibrary.org/docserver/ed430e1d-en.pdf?expires=1721987296&id=id&accname=guest&checksum=72827E04A06E8C020AD495CC63C806FA>.

7.5.1.2.1. Relevance of dose levels

The choice of relevant dose/concentration levels for use in hazard characterisation differs for *in vivo* and *in vitro* studies. This can make it difficult to compare adverse/toxic responses obtained from *in vivo* and *in vitro* experiments. For example, with some nanoforms, the doses used in *in vitro* experiments may be sufficient to induce pro-inflammatory, toxic and/or genotoxic effects in cell models, whereas the same dose used *in vivo* may only mobilise antioxidant defences without any subsequent adverse effect. In addition, it should be noted that the solubility of nanoparticles in biological media (e.g. gastric juice) could be higher than in *in vitro* conditions. Overall, it can be difficult to extrapolate between *in vitro* and *in vivo* tests, since *in vitro* studies are often carried out on exposures with treatment times of 24 hours or less at relatively high concentrations, whereas any *in vivo* effects are expected over the medium/long term, because of the time needed for absorption and/or potential accumulation of nanomaterials in certain organs. Thus, for the same biological parameter studied, the effects observed in *in vitro* and *in vivo* tests may be different or even contradictory, which raises the question of the predictivity of *in vitro* experiments. Therefore, data on nanoparticles' toxicity should be derived from *in vivo* studies as a priority and, to some limited extent, from standardised *in vitro* / *ex vivo* studies.

This is particularly important, since there is a growing trend towards adopting alternatives to animal testing, and in particular towards cell culture models, which need to be as predictive as possible. It is therefore necessary to work on the experimental design of the *in vitro* experiment, taking care to implement chronic/repeated exposures at lower concentrations, with nanoforms prepared in a dispersant that is representative of that encountered in *in vivo* conditions. For example, to improve the predictive power of *in vitro* tests with regard to the intestinal toxicity of nanoforms, Marucco et al. (2020) emphasised the importance of using a simulant of the human digestive system as the dispersion medium. Furthermore, Drasler et al. (2017) have summarised the criteria important for producing reliable and robust data from *in vitro* nanotoxicological assays, including: (1) detailed characterisation of the nanomaterial/nanoparticles, including physico-chemical properties before, during and after the testing; (2) use of comparable and realistic dose metrics and test conditions; and (3) implementation of chemical positive and negative controls and reference nanomaterial/nanoparticles allowing comparison between studies, both intra- and interlaboratory.

If *in vitro* results have to be used to predict the *in vivo* effects of nanomaterials, their relevance will be more certain if the *in vitro* dose is extrapolated from a relevant *in vivo* concentration (availability of toxicokinetic data). A 'starting point' dose for *in vitro* studies can be set by extrapolating *in vivo* organic burden data to determine the dose to be used *in vitro*. This approach reduces the difficulties in setting exposure doses for *in vitro* models. For *in vivo* tests in animals, it is also important to avoid the use of excessive doses, considering the method of administration, which could lead to irrelevant results. In all cases, it is therefore necessary to justify the dose levels used in the assessment tests. Further guidance on the stability of the dispersion and selection of appropriate doses for the testing of materials containing nanoparticles or nanomaterials are provided in EFSA guidance on the technical requirements for establishing the presence of small particles ⁽⁷⁴⁾ and EFSA guidance on risk assessment of nanomaterials ⁽⁷⁵⁾.

7.5.1.3. Choice of test system

7.5.1.3.1. Cell model

As explained above, due to the limitations on comparing/extrapolating *in vitro* data to adverse effects that could be seen *in vivo* after exposure to nanoparticles, *in vivo* studies should be prioritised for deriving nanotoxicological data. However, when standardised *in vitro* studies are

⁽⁷⁴⁾ <https://efsa.onlinelibrary.wiley.com/doi/epdf/10.2903/j.efsa.2021.6769>.

⁽⁷⁵⁾ <https://efsa.onlinelibrary.wiley.com/doi/epdf/10.2903/j.efsa.2021.6768>.

used, the choice of a cell model is a crucial parameter, as the sensitivity of cell lines (human, animal, etc.) to exposure to nanoparticles varies. Depending on their origin, the cell models may use different metabolic pathways, different receptors on their membrane surface or DNA repair capabilities and may have different antioxidant statuses. For example, since the genotoxicity of nanomaterials depends largely on oxidative mechanisms, if cells with poor or no antioxidant systems are used, the effects may be exacerbated.

An analysis of the applicability of *in vitro* methods to nanoparticles and recommendations for the adaptations needed is available in the OECD testing programme for the safety of manufactured nanomaterials (2018) ⁽⁷⁶⁾.

7.5.1.3.2. Choice of animal model

In general, compared with cell-based models, there are fewer restrictions on the choice of an animal model, and the animal species recommended in the guidelines can be used. For oral exposure, the general recommendations in the respective guidelines can also be applied for nanoforms.

7.5.2. Specific advice for individual end points

7.5.2.1. Mutagenicity testing of nanoforms

The testing methodology for mutagenicity, described in Section 6.4 of this guidance, should be followed together with the recommendations provided in this section.

It is recommended that genotoxicity tests are conducted for nanoforms in accordance with the available OECD test guidelines, as for any other forms of a substance, with the exception of specific adaptations required for nanomaterials, as mentioned here and in the OECD guidance on sample preparation and dosimetry ⁽⁷⁷⁾.

Different groups have published reviews on the genotoxic assessment of nanomaterials (Pfuhler et al., 2013; Magdolenova et al., 2014;), and the most recent comprehensive review (Elespuru et al., 2018) extended the previous works with critical analysis of published data. All these reviews agreed on a number of recommendations that are described in this section (and also mentioned in recent EFSA guidance on risk assessment of nanomaterials ⁽⁷⁸⁾ and SCCS guidance on the safety assessment of nanomaterials in cosmetics ⁽⁷⁹⁾).

7.5.2.2. Bacterial (Ames) mutagenicity assays are not recommended

The bacterial reverse mutation (Ames) test (OECD TG 471/EU method B.12/13) detects point mutations in *Salmonella* serotype Typhimurium and *Escherichia coli*. However, the strains of bacteria used in the standard assays do not appear to have the ability to take up/internalise nanoparticles, as they lack mammalian mechanisms of endocytosis, pinocytosis, and phagocytosis (Warheit and Donner, 2010; Doak et al., 2012; Elespuru et al., 2018). Therefore, the use of the Ames test is not recommended for the investigation of the genotoxicity of nanoforms unless there is evidence that the nanoform penetrates the cell wall of the bacteria or for indirect genotoxic effects due to generation of reactive oxygen species outside the bacteria.

Column 2 of Table 1 of Annex V to Commission Implementing Decision (EU) 2024/365 states that 'the study does not need to be conducted for nanoforms where it is not appropriate. In this

⁽⁷⁶⁾ [https://one.oecd.org/document/ENV/JM/MONO\(2018\)4/en/pdf](https://one.oecd.org/document/ENV/JM/MONO(2018)4/en/pdf).

⁽⁷⁷⁾ <file:///C:/e/ECHA/data/users/u17301/Roaming%20Profile/Downloads/ed430e1d-en.pdf>.

⁽⁷⁸⁾ <https://efsa.onlinelibrary.wiley.com/doi/epdf/10.2903/j.efsa.2021.6768>.

⁽⁷⁹⁾ https://health.ec.europa.eu/system/files/2020-10/sccs_o_233_0.pdf.

case other studies involving one or more *in vitro* mutagenicity studies in mammalian cells shall be provided.'

If the Ames test is not appropriate for the reasons explained above, the applicant must provide one test detecting gene mutation and one test detecting chromosomal aberrations.

- **A test detecting gene mutation.** The *in vitro* mammalian cell gene mutation test using the thymidine kinase gene (in accordance with OECD TG 490, latest update 2016) or the *in vitro* mammalian cell gene mutation test using the *Hprt* and *Xprt* genes (in accordance with OECD TG 476, latest update 2016). The considerations explained under 'Interference with measured parameters' should also be taken into account when designing the experiment.
- **A test detecting chromosomal aberrations.** The *in vitro* mammalian cell micronucleus test (in accordance with OECD TG 487, latest update 2016) or the *in vitro* mammalian chromosomal aberration test (in accordance with OECD TG 473, latest update 2016).

7.5.2.3. Verification of uptake into target cells

The verification of cellular uptake by appropriate methods is highly advisable when using *in vitro* genotoxicity tests. The importance of cellular uptake was pointed out in the 2013 Nanogenotox report⁽⁸⁰⁾, the 2014 OECD report on the genotoxicity of manufactured nanomaterials⁽⁸¹⁾, and recent EFSA guidance on risk assessment of nanomaterials⁽⁸²⁾ and the SCCS guidance on the safety assessment of nanomaterials in cosmetics⁽⁸³⁾. Several parameters (e.g. agglomeration and protein coating) can influence cellular uptake.

In vitro tests should be performed on cell lines that have demonstrated the ability to take up nanoparticles. A recent JRC report (Bogni et al., 2020) showed that several cell lines (Caco-2, A549, CHO, V79 and TK6) demonstrated the ability to internalise the tested nanomaterials (gold, silica and silver nanoparticles). If a test is performed on a cell line for which the uptake capacity of nanoparticles is unknown, the test data should demonstrate the uptake ability of the cell line studied. Such data can be generated using optical- and electron-based microscopy techniques (e.g. fluorescence microscopy, transmission electron microscopy) (Peckys and De Jonge, 2011; Van Der Zwaag et al., 2016; Donahue et al., 2019). It is noted that the fluorophore used for labelling of nanomaterial may influence the uptake of and/or detach from the nanomaterial. These would complicate the interpretation of the analysis. The measurement of intracellular nanomaterial can be challenging depending on the nature of the nanomaterial studied.

7.5.2.4. Recommendations to avoid interference with uptake or end point analysis

Nanoforms can interfere with the components of the media used in (cyto)toxicity tests, particularly with the reagents used (dyes, fluorescent agents, etc.). Such interference can lead to bias in the response.

Incompatibilities with certain experimental conditions may also be observed, particularly for the *in vitro* genotoxicity tests.

An *in vitro* mutagenicity test usually includes one experiment in the absence and one in the presence of a metabolic activation system (S9mix). The use of S9mix in *in vitro* studies can affect the outcome of the tests, since S9 can induce the formation of mutagenic metabolites (if the nanomaterial can be metabolised, e.g. if it is composed of organic materials) like any other

⁽⁸⁰⁾ https://www.anses.fr/en/system/files/nanogenotox_final_report.pdf.

⁽⁸¹⁾ [https://www.oecd.org/officialdocuments/publicdisplaydocumentpdf/?cote=env/jm/mono\(2014\)34&doclanguage=en](https://www.oecd.org/officialdocuments/publicdisplaydocumentpdf/?cote=env/jm/mono(2014)34&doclanguage=en)

⁽⁸²⁾ <https://efsa.onlinelibrary.wiley.com/doi/epdf/10.2903/j.efsa.2021.6768>.

⁽⁸³⁾ https://health.ec.europa.eu/system/files/2020-10/sccs_o_233_0.pdf.

chemical tested. The addition of proteins (contained in S9) in the culture medium can also modify the cellular uptake of nanomaterials (Doak et al., 2012; Pfuhler et al., 2013; Magdolenova et al., 2014). It is recommended that *in vitro* tests are performed in the **presence** of S9 for only nanoforms composed of organic materials and not for those composed of inorganic materials (biotransformation is not expected). The *in vitro* test should be performed in the **absence** of S9 for **all** nanoforms, that is, for inorganic and organic nanomaterials.

Cytotoxicity measurements described in relevant test guidelines (e.g. relative population doubling, cloning efficiency or relative total growth) are appropriate for determining the top concentration that should be used for *in vitro* tests of nanomaterials. Moreover, cytotoxicity should be assessed concurrently with genotoxicity and not in separate experiments (Elespuru et al., 2018). Since a variety of proliferation or cytotoxicity assays employ colourimetry or fluorometry, dye interference should be checked beforehand. To establish the cytotoxicity of nanomaterials without any ambiguity, therefore, the recommendations below should be followed:

- test for possible interference with the reagents and test conditions, starting with cell-free controls to ensure that the nanoparticles do not interfere with the reagents or with the type of measurement used in the test;
- use at least two types of tests in parallel, based on different principles.

According to the 2014 OECD report on genotoxicity of manufactured nanomaterials ⁽⁸⁴⁾, 'The intent when applying nanomaterials to a cell culture medium is to create conditions that are comparable, to the extent possible, with the biological and physiological conditions within the *in vivo* system.'

According to column 2 of Table 1 of Annex V to Commission Decision 2024/365, a micronucleus test (OECD TG 487) or a chromosomal aberration test (OECD TG 473) is required. The Nanogenotox report concluded that the guideline for the testing of chemicals with the *in vitro* mammalian cell micronucleus test (OECD TG 487) is 'applicable for nanomaterials but may need some adaptation in order to provide predictive results *in vivo*' ⁽⁸⁵⁾.

In the standard protocol for the *in vitro* micronucleus test (OECD TG 487), cytochalasin B is often used to block cells at a certain phase of mitosis. However, this substance inhibits the endocytosis of nanomaterials, leading to a risk of obtaining a false negative result (Doak et al., 2012). Therefore, if cytochalasin B is used, it is recommended that this agent is added post treatment or a delayed co-treatment protocol is used in order to ensure a period of exposure of the cell culture system to the nanomaterial in the absence of cytochalasin B (see OECD project 4.95, referred to in the next paragraph). Furthermore, depending on the type of nanoform, accumulation in the cytoplasm may interfere with the reading of the test with the risk that micronuclei may be obscured by the nanomaterials (Charles et al., 2018; Jalili et al., 2018).

A project on the adaptation of the *in vitro* mammalian cell micronucleus assay (OECD TG 487) for nanomaterials testing is ongoing in the OECD Working Party on Manufactured Nanomaterials rolling work plan (Project 4.95: Guidance document on the adaptation of *in vitro* mammalian cell-based genotoxicity TGs for testing of manufactured nanomaterials) ⁽⁸⁶⁾. The project focuses on the physico-chemical characterisation of nanomaterials and protocol modifications (selection of cell type with respect to uptake mechanisms, use of cytochalasin B, timing of exposure to nanomaterials, specification of controls, dose ranges and dose metrics). Two intermediary reports studying gold, silica and silver nanoparticles were produced from this project: one in

⁽⁸⁴⁾ [https://www.oecd.org/officialdocuments/publicdisplaydocumentpdf/?cote=env/jm/mono\(2014\)34&doclanguage=en](https://www.oecd.org/officialdocuments/publicdisplaydocumentpdf/?cote=env/jm/mono(2014)34&doclanguage=en)

⁽⁸⁵⁾ https://www.anses.fr/en/system/files/nanogenotox_final_report.pdf.

⁽⁸⁶⁾ [ENV-CBC-MONO\(2022\)15.en.pdf \(oecd.org\)](#).

2018 on the physico-chemical characterisation in water and in serum-containing cell culture media (Mehn et al., 2018) and the other in 2020 on *in vitro* cytotoxicity and cellular uptake in five different cell lines (Bogni et al., 2020).

At this time, it is recommended that genotoxicity tests for nanoforms are conducted in accordance with the available OECD test guidelines as for any other forms of a substance, with the exception of specific adaptations to methods required for nanomaterials, as mentioned here and in the OECD guidance on sample preparation and dosimetry ⁽⁸⁷⁾.

Further information on potential interference of nanoforms with commonly used assays is provided in ECHA *Guidance on Information Requirements and Chemical Safety Assessment – Appendix R7-1 for nanomaterials applicable to Chapter R7a Endpoint specific guidance* ⁽⁸⁸⁾.

7.5.2.5. Toxicokinetic studies for nanoforms

The testing methodology for toxicokinetic studies, described in Section 6.5 of this guidance, should be followed together with the recommendations provided in this section. It is recommended that toxicokinetic studies for nanoforms are conducted in accordance with the principles of OECD TG 417, as for any other forms of a substance, except for specific adaptations required for nanoforms, as mentioned below.

Regarding the route of administration, it is generally accepted that gavage ensures that the correct dose is given. However, since gavage administration is intragastric, the first organs of the digestive tract (mouth and oesophagus) are not directly exposed. To take account of these initial contact areas, the dose can be administered either in drinking water or in feed. However, it is then more difficult to estimate the dose ingested by each animal. Furthermore, the stability of the nanoforms in water or feed needs to be verified.

In oral studies, nanoforms are generally poorly absorbed through the intestinal barrier (only a small percentage by mass of the dose administered) (Shi et al. 2013; Geraets et al. 2014; Jones et al. 2015). However, accumulation in certain systemic organs, in particular the liver, spleen and kidneys, is often reported (Loeschner et al., 2011; Kermanizadeh et al., 2015; Heringa et al., 2018; Brand et al., 2020; Krause et al., 2020). It therefore appears necessary to assess distribution after repeated administration and over a long enough time period (such as 14 days) (see Section 7.5.2.6). The exposure time should take into account the low level of intestinal absorption and any problems with the LOD of the analytical methods used. Similarly, the doses tested should be estimated taking these limitations into account, but doses should not cause toxicity that could affect the kinetics.

Considering that some publications report a gender effect on the distribution, accumulation or toxicity of nanoforms (Kim et al. 2007; Kim et al. 2011; Chen et al. 2013; Mohammadpour et al. 2019; Han et al. 2020; Tassinari et al. 2020), toxicokinetic studies should be performed in both sexes. It is important to note that greater variability in quantification data can be expected for females, since the period of the oestrous cycle at the time of testing can influence gut permeability (Braniste et al. 2009). In order to reduce this variability, it may be necessary to synchronise the cycles of the females before testing.

For the distribution of nanoforms, special attention should be paid to the gastrointestinal tract, brain and reproductive organs (as this information will be essential for initiating further toxicity tests) and to organs where accumulation of nanoforms, such as the spleen, liver and kidneys, has already been regularly observed. For the gastrointestinal tract, in addition to samples from

⁽⁸⁷⁾ [https://one.oecd.org/document/ENV/JM/MONO\(2012\)40/en/pdf](https://one.oecd.org/document/ENV/JM/MONO(2012)40/en/pdf).

⁽⁸⁸⁾ https://echa.europa.eu/documents/10162/17224/appendix_r7a_nanomaterials_en.pdf/1bef8a8a-6ffa-406a-88cd-fd800ab163ae?t=1633348005491.

the small and large intestine, it is important to have samples with lymphoid tissues (Peyer's patches and mesenteric lymph nodes), which are potential passage zones for nanoforms (Geraets et al. 2014; Janer et al. 2014; Bouwmeester et al. 2018).

Negative controls receiving only the vehicle must be included in the toxicokinetic studies. It is also important to check that the doses used did not significantly alter the intestinal epithelium, which could call into question the results obtained for systemic distribution. Collecting histological data on the gastrointestinal tract is therefore recommended. Additional data on intestinal permeability using markers (creatinine, different sizes of dextran, 51Cr-EDTA) can also be generated.

In some cases, external contamination (e.g. via feed) may generate high baseline levels in negative controls, preventing detection of any increase in the systemic organs of treated animals (assuming low levels of passage through the intestinal barrier). Although this is often difficult to achieve, some solutions can be proposed to lower this baseline as much as possible: for example, as silica is an element found in the diet of rodents but also in glass labware, special experimental conditions and also suitable feed with a low silica content are needed (Aureli et al., 2020).

Conventional quantification methods are mainly limited to analyses using ICP-MS or inductively coupled plasma–optical emission spectrometry (ICP-OES). However, with the exception of single particle technology (spICP-MS), these methods are unable to distinguish particulate forms from ionic forms. Therefore, in order to distinguish the different particles in the nanoforms, additional analytical methods are required (e.g. electron microscopy), but these can analyse only a limited number of samples. For some nanoforms that cannot be analysed by ICP and/or in order to facilitate their detection, nanoforms labelled by fluorescence or radioactivity techniques or consisting of metallic cores can be used. In this case, it is important to check beforehand that (1) the labelling does not affect the physico-chemical properties of the nanoforms (it could modify their behaviour) and (2) the labelling is stable and is not lost during the study, which would lead to false conclusions.

Further studies to assess the persistence and excretion of nanomaterials can be performed by adapting the experimental design with recovery periods after repeated treatment.

7.5.2.6. Repeated dose toxicity

The testing methodology for repeated dose toxicity, described in Section 6.6 of this guidance should be followed together with the recommendations provided in this section.

It is recommended that the subchronic (90-day) toxicity study for nanoforms is conducted in accordance with the latest version of the OECD guideline (TG 408, updated 2018), as for any other forms of a substance, with the exception of specific adaptations required for nanoforms as mentioned below.

Investigation of the potential for accumulation. A satellite group should be added to the 90-day oral toxicity study in rodents to investigate whether and to what extent the nanoform accumulates. In the satellite group, the tissue distribution (e.g. in gastrointestinal tract, brain, sexual organs, liver, spleen and kidneys) of the nanomaterial should be determined after a short dosing period (i.e. 14 days) and at the end of the 90-day study. If distribution results are obtained in the 14-day toxicokinetic study, a comparison of the distribution results from the 90-day study could show whether a redistribution effect in certain organs occurs over time. The analytical parameters (extraction yield, LOD and LOQ) of the methods used for the assays should be indicated for each of the organs studied.

The tissue distribution after an elimination period should also be determined (e.g. 30 days), for example in animals that were dosed for the short 14-day period. Note that further guidance on

the duration on the elimination period for toxicokinetic assessment will be developed by the OECD and may depend on the dissolution potential of the material.

Nanoforms are generally quickly, often within minutes, taken up by tissues (Landsiedel et al., 2012) and information on concentration in blood may therefore have a limited value. In such cases, a rough estimate of toxicokinetic plasma parameters ($t_{1/2}$, AUC, bioavailability, C_{max} and T_{max}) would be sufficient to evaluate the potential for accumulation. Because nanoforms are taken up by the mononuclear phagocyte system, they typically distribute to the liver and spleen (Landsiedel et al., 2012); however, they are also detected in other tissues such as kidney, lymph nodes at the organ of entry, bone marrow, lung, brain and heart. For further details, see ECHA *Guidance on Information Requirements and Chemical Safety Assessment – Appendix R7-1 for nanomaterials applicable to Chapter R7a Endpoint specific guidance* ⁽⁸⁹⁾. Nanomaterial retention within the gut wall is also an important determinant, particularly when discriminating between retention in epithelial cells and immune-competent M-cells in Peyer's patches. In the gastrointestinal tract, gut-associated lymphoid tissue, particularly in Peyer's patches and mesenteric lymph nodes, is of importance and is related to potential nanomaterial accumulation and immune responses.

The difference in organ concentrations between days 0, 14 and 90 and after the elimination period (e.g. 4 weeks), as well as the rate of elimination should be used to assess the likelihood of accumulation, given the anticipated exposure pattern in humans. For further details, see EFSA guidance on risk assessment of nanomaterials ⁽⁹⁰⁾.

7.5.2.7. Reproductive toxicity

The testing methodology for reproductive toxicity, described in Section 6.7 of this guidance should be followed. It is recommended that reproductive toxicity tests for nanoforms are conducted in accordance with the available OECD test guidelines, as for any other forms of a substance.

The following test methods to investigate reproductive and developmental toxicity of nanoforms should be considered:

- OECD TG 421: Reproduction/developmental toxicity screening test;
- OECD TG 422: Combined repeated dose toxicity study with the reproduction/developmental toxicity screening test;
- OECD TG 443: Extended one-generation reproduction toxicity;
- OECD TG 414: Prenatal developmental toxicity study (2018).

For specific neurodevelopmental toxicity, OECD TG 426 (Neurodevelopmental toxicity) may also be considered.

ECHA has published a critical review on the reproductive and developmental toxicity of nanomaterials ⁽⁹¹⁾. The review gives an overview of current knowledge on the reproductive and developmental effects of nanomaterials that have been reported in the literature. A scientific discussion of the data provides answers to a series of key questions in relation to kinetic

⁽⁸⁹⁾ https://echa.europa.eu/documents/10162/17224/appendix_r7a_nanomaterials_en.pdf/1bef8a8a-6ffa-406a-88cd-fd800ab163ae?t=1633348005491.

⁽⁹⁰⁾ <https://efsa.onlinelibrary.wiley.com/doi/epdf/10.2903/j.efsa.2021.6768>.

⁽⁹¹⁾ https://euon.echa.europa.eu/documents/2435000/3268573/critical_review_of_studies_on_reproductive_and_developmental_toxicity_of_nanomaterials_en.pdf/c83f78ef-7136-ef4b-268c-c5d9b7bf1fea?t=1586154196963.

properties of the nanomaterials (e.g. penetration into gonads and across the placenta and into the fetus and critical organ systems) and types of adverse effects on fertility and development observed.

Based on the analysis of 111 studies covering mainly *in vivo* testing of manufactured nanomaterials, observation and methodological recommendations were made. Nano-titanium dioxide and -silver were the two most frequently studied nanomaterials and together accounted for almost 50 % of the works listed. Other nanomaterials also studied were zinc oxide, silicon oxide and nanomaterials containing carbon.

8. References and background documents

8.1. General sources

Armitage, J.M., and Gobas FAPC. (2007), 'A terrestrial food-chain bioaccumulation model for POPs', *Environmental Scientific Technology*, Vol. 41, No 7, pp. 4019-25, <http://doi:10.1021/es0700597>.

Arts, J., Hadi, M., Irfan, M., Keene, A., Kreiling, R. et al. (2015), 'A decision-making framework for the grouping and testing of nanomaterials (DF4nanoGrouping)', *Regulatory Toxicology and Pharmacology*, Vol. 71, No 2, pp. S1–S27.

Aureli, F., Ciprotti, M., D'Amato, M., do Nascimento da Silva, E., Nisi, S. et al. (2020), 'Determination of total silicon and SiO₂ particles using an ICP-MS based analytical platform for toxicokinetic studies of synthetic amorphous silica', *Nanomaterials*, Vol. 10, p. 888.

Avramescu, M., Rasmussen, P., Chénier, M. and Gardner, H. D. (2017), 'Influence of pH, particle size and crystal form on dissolution behaviour of engineered nanomaterials', *Environmental Science and Pollution Research*, Vol. 24, pp. 1553–1564.

Baskaran, S., Lei, Y. D. and Wania, F. (2021), 'A database of experimentally derived and estimated octanol–air partition ratios (K_{OA})', *Journal of Physical and Chemical Reference Data*, Vol. 50, 043101, <https://doi.org/10.1063/5.0059652>.

Beekhuijzen, M., de Raaf, M. A., Zmarowski, A., van Otterdijk, F., Peter, B. et al. (2014), 'The underestimated value of OECD 421 and 422 repro screening studies: Putting it in the right perspective', *Reproductive Toxicology*, Vol. 48, pp. 81–87.

Bogni, A., Ponti, J., Drewes, C., Kinsner-Ovaskainen, A. and Bremer-Hoffmann, S. (2020), *In vitro cytotoxicity and cellular uptake evaluation of gold, silica and silver nanoparticles in five different cell lines: Caco-2, A549, CHO, V79 and TK6*, Publications Office of the European Union, Luxembourg.

Borm, P., Klaessig, F., Landry, T., Moudgil, B. and Pauluhn, J. (2006), 'Research strategies for safety evaluation of nanomaterials, part V: Role of dissolution in biological fate and effects of nanoscale particles', *Toxicological Sciences*, Vol. 90, p. 23.

Bouwmeester, H., van der Zande, M. and Jepson, M. A. (2018), 'Effects of food-borne nanomaterials on gastrointestinal tissues and microbiota', *WIREs Nanomedicine and Nanobiotechnology*, Vol. 10, e1481.

Bowen, D. E., Whitwell, J. H., Lillford, L., Henderson, D., Kidd, D. et al. (2011), 'Evaluation of a multi-endpoint assay in rats, combining the bone-marrow micronucleus test, the comet assay and the flow-cytometric peripheral blood micronucleus test', *Mutation Research*, Vol. 722, pp. 7–19.

Brand, W., Ruud, J., Peters, B., Braakhuis, H. M., Maślankiewicz, L. et al. (2020), 'Possible effects of titanium dioxide particles on human liver, intestinal tissue, spleen and kidney after oral exposure', *Nanotoxicology*, Vol. 14, No 7, pp. 985–1007.

Braniste, V., Leveque, M., Buisson-Brenac, C., Bueno, L., Fioramonti, J. et al. (2009), 'Oestradiol decreases colonic permeability through and junctional adhesion molecule-A in epithelial cells', *Journal of Physiology*, Vol. 587, No 13, pp. 3317–3328.

- Charles, S., Jomini, S., Fessard, V., Bigorgne-Vizade, E., Rousselle, C. et al. (2018), 'Assessment of the *in vitro* genotoxicity of TiO₂ nanoparticles in a regulatory context', *Nanotoxicology*, Vol. 12, pp. 357–374.
- Chen, J., Wang, H., Long, W., Shen, X., Wu, D. et al. (2013), 'Sex differences in the toxicity of polyethylene glycol-coated gold nanoparticles in mice', *International Journal of Nanomedicine*, Vol. 8, pp. 2409–2419.
- Cordelli, E., Fresegna, A. M., Leter, G., Eleuteri, P., Spanò, M. et al. (2003), 'Evaluation of DNA damage in different stages of mouse spermatogenesis after testicular X irradiation', *Journal of Radiation Research*, Vol. 160, No 4, pp. 443–451.
- Dekkers, S., Oomen, A., Bleeker, E., Vandebriel, R., Micheletti, C. et al. (2016), 'Towards a nanospecific approach for risk assessment', *Regulatory Toxicology and Pharmacology*, Vol. 80, pp. 46–59.
- Deloid, G. M., Cohen, J. M., Pyrgiotakis, G. and Demokritou, P. (2017), 'Preparation, characterization, and *in vitro* dosimetry of dispersed, engineered nanomaterials', *Nature Protocols*, Vol. 12, No 2, pp. 355–371.
- Doak, S., Manshian, B., Jenkins, G. and Singh, N. (2012), '*In vitro* genotoxicity testing strategy for nanomaterials and the adaptation of current OECD guidelines', *Mutation Research*, Vol. 745, Nos 1–2, pp. 104–111.
- Donahue, N., Acar, H. and Wilhelm, S. (2019), 'Concepts of nanoparticle cellular uptake, intracellular trafficking, and kinetics in nanomedicine', *Advanced Drug Delivery Reviews*, Vol. 143, pp. 68–96.
- Dirven, Y., Eide, D. M., Henriksson, E. W., Hjorth, R., Sharma, A. K. et al. (2023), 'Assessing testicular germ cell DNA damage in the comet assay: Introduction of a proof-of-concept', *Environmental and Molecular Mutagenesis*, Vol. 64, No 2, pp. 88–104, <https://doi.org/10.1002/em.22527>.
- ECHA (2021), 'Categories of materials in SCIP', Substances of concern in articles as such or in complex objects (products) (SCIP) database, https://echa.europa.eu/documents/10162/28213971/material_categories_for_the_scip_database_en.pdf/47142a6a-2634-52ce-ced0-49670928c3c2.
- EFSA Panel on Food Contact Materials, Enzymes and Processing Aids, Lambré, C., Manuel Barat Baviera, J., Bolognesi, C. et al. (2023), 'Scientific opinion on the safety assessment of "waxes, paraffinic, refined, derived from petroleum-based or synthetic hydrocarbon feedstock, low viscosity" for use in food contact materials', *EFSA Journal*, Vol. 21, No 2, e07761, <https://doi.org/10.2903/j.efsa.2023.7761>.
- EFSA Scientific Committee, Hardy, A., Benford, D., Halldorsson, T., Jeger, M. et al. (2017), 'Clarification of some aspects related to genotoxicity assessment', *EFSA Journal*, Vol. 15, No 12, e05113, <https://doi.org/10.2903/j.efsa.2017.5113>.
- EFSA Scientific Committee, More, S., Bampidid, V., Benford, D., Bragard, C. et al. (2021), 'Guidance on technical requirements for regulated food and feed product applications to establish the presence of small particles including nanoparticles', *EFSA Journal*, Vol. 19, No 8, e06769.

Elespuru, R. S., Pfuhler, M., Aardema, T., Chen, S., Doak, A. et al. (2018), 'Genotoxicity assessment of nanomaterials: Recommendations on best practices, assays, and methods', *Toxicological Sciences*, Vol. 164, No 2, pp. 391–416.

Franz, R. and Welle, F. (2017), *Mathematic modelling of migration of nanoparticles from food contact polymers. Nanosubstances in food contact materials: Application, legislation and testing*, DEStech Publications, Inc., Lancaster, Pennsylvania.

Franz, R., Bott, J. and Störmer, A. (2020), 'Considerations for and guidance to testing and evaluating migration/release of nanoparticles from polymer based nanocomposites', *Nanomaterials*, Vol. 10, pp. 1113–1129, <https://doi.org/10.3390/nano10061113>.

Gelbke, H.-P., Fleig, H., Meder, M. and German Chemical Industry Association (2004), 'SIDS reprotoxicity screening test update: Testing strategy and use', *Regulatory Toxicology and Pharmacology*, Vol. 39, pp. 81–86.

Geraets, L., Oomen, A. G., Krystek, P., Jacobsen, N. R., Wallin, H., et al. (2014), 'Tissue distribution and elimination after oral and intravenous administration of different titanium dioxide nanoparticles in rats', *Particle and Fibre Toxicology*, Vol. 11, p. 30.

Giles, H. F., Wagner, J. R. and Mount, E. M. (2005), 'Testing properties', in: *Extrusion – The definitive processing guide and handbook*, William Andrew Publishing, pp. 195–205, <https://doi.org/10.1016/B978-081551473-2.50022-2>.

Gobas, F.A.P.C., de Wolf, W., Burkhard, L.P., Verbruggen, E. and Plotzke, K. (2009) 'Revisiting bioaccumulation criteria for POPs and PBT assessments', *Integrated Environmental Assessment and Management*, Vol. 4, pp. 624–37, https://doi:10.1897/IEAM_2008-089.1.

Goss, K.U., Brown, T.N. and Endo, S. (2013), 'Elimination half-life as a metric for the bioaccumulation potential of chemicals in aquatic and terrestrial food chains'. *Environmental Toxicology and Chemistry*, Vol. 32, pp. 1663–71, <http://doi:10.1002/etc.2229>.

Hagiwara, Y., Kumagai, H., Ouwerkerk, N., et al. (2022), 'A novel in vitro membrane permeability methodology using three-dimensional Caco-2 tubules in a microphysiological system which better mimics in vivo physiological conditions', *Journal of Pharmaceutical Sciences*, Vol. 111, No 1, pp214–224, <https://doi.org/10.1016/j.xphs.2021.11.016>.

Han, H. Y., Cho, J. W., Seong, E., Park, E. J., Lee, G. H. et al. (2020), 'Amorphous silica nanoparticle-induced pulmonary inflammatory response depends on particle size and is sex-specific in rats', *Toxicology and Applied Pharmacology*, Vol. 390, 114890.

Hendriks, G., Atallah, M., Morolli B., Calléja, F., Ras-Verloop, N. et al. (2012), 'The ToxTracker assay: Novel GFP reporter systems that provide mechanistic insight into the genotoxic properties of chemicals', *Toxicological Sciences*, Vol. 125, No 1, pp. 285–298, <https://doi.org/10.1093/toxsci/kfr281>.

Hendriks, G., Derr, R. S., Misovic, B., Morolli, B., Calléja, F. M. G. R. et al. (2016), 'The extended ToxTracker assay discriminates between induction of DNA damage, oxidative stress, and protein misfolding', *Toxicological Sciences*, Vol. 150, No 1, pp. 190–203, <https://doi.org/10.1093/toxsci/kfv323>.

Hendriks, G., Kirkland, D., Vanparys, P., van Benthem, J. and Adriaens, E. (2023), 'The ToxTracker assay: A stem cell-based reporter assay for mechanistic carcinogenicity hazard screening',

Heringa, M. B., Peters, R. J. B., Bleys, R. L. A. W., van der Lee, M. K., Tromp, P. C. et al. (2018), 'Detection of titanium particles in human liver and spleen and possible health implications', *Particle and Fibre Toxicology*, Vol. 15, p. 15.

ISO (2019), *Nanotechnologies – Considerations for Performing Toxicokinetic Studies with Nanomaterials*, ISO, Geneva.

JaCVAM (Japanese Center for the Validation of Alternative Methods) (2020), *IL-2 Luciferase Assay Report of the Peer Review Panel*, https://jacvam.jp/files/list/12/12_01_peer%20review%20report.pdf.

Jalili, P., Gueniche, N., Lancelleur, R., Burel, A., Lavault M. T. et al. (2018), 'Investigation of the *in vitro* genotoxicity of two rutile TiO₂ nanomaterials in human intestinal and hepatic cells and evaluation of their interference with toxicity assays', *NanoImpact*, Vol. 11, pp. 69–81.

Janer, G., Mas del Molino, E., Fernández-Rosas, E., Fernández, A. and Vázquez-Campos, S. (2014), 'Cell uptake and oral absorption of titanium dioxide nanoparticles', *Toxicology Letters*, Vol. 228, No 2, pp. 103–110.

JECFA (Joint FAO/WHO Expert Committee on Food Additives) (2006), *Combined Compendium of Food Additive Specifications – Volume 4: Analytical methods, test procedures and laboratory solutions used by and referenced in the food additive specifications*, JECFA Monographs, Food and Agriculture Organization of the United Nations, Rome.

Jones, K., Morton, J., Smith, I., Jurkschat, K., Harding, A.-H. et al. (2015), 'Human *in vivo* and *in vitro* studies on gastrointestinal absorption of titanium dioxide nanoparticles', *Toxicology Letters*, Vol. 233, No 2, pp. 95–101.

Kelly, B.C., Ikonomidou, M.G., Blair, J.D., Morin, A.E. and Gobas, F.A.P.C. (2007), 'Food web-specific biomagnification of persistent organic pollutants', *Science*, Vol. 317, pp. 236–239.

Kermanizadeh, A., Balharry, D., Wallin, H., Loft, S. and Møller, P. (2015), 'Nanomaterial translocation – The biokinetics, tissue accumulation, toxicity and fate of materials in secondary organs: A review', *Critical Reviews in Toxicology*, Vol. 45, No 10, pp. 837–872.

Kim, J. S., Kuk, E., Yu, K. N., Kim, J. H., Park, S. J. et al. (2007), 'Antimicrobial effects of silver nanoparticles', *Nanomedicine: Nanotechnology, Biology and Medicine*, Vol. 3, pp. 95–101.

Kim, J. S., Sung, J. H., Ji, J. H., Song, K. S., Lee, J. H. et al. (2011), '*In vivo* genotoxicity of silver nanoparticles after 90-day silver nanoparticle inhalation exposure', *Safety and Health at Work*, Vol. 2, pp. 34–38.

Krause, B. C., Kriegel, F. L., Rosenkranz, D., Dreier, N., Tentschert, J. et al. (2020), 'Aluminum and aluminum oxide nanomaterials uptake after oral exposure – A comparative study', *Scientific Reports*, Vol. 10, 2698.

Landsiedel, R., Fabian, E., Ma-Hock, L., van Ravenzwaay, B., Wohlleben, W. et al. (2012), 'Toxicity/biokinetics of nanomaterials', *Archives of Toxicology*, Vol. 86, pp. 1021–1060, <https://doi.org/10.1007/s00204-012-0858-7>.

Loeschner, K., Hadrup, N., Qvortrup, K., Larsen, A., Gao, X. et al. (2011), 'Distribution of silver in rats following 28 days of repeated oral exposure to silver nanoparticles or silver acetate', *Particle and Fibre Toxicology*, Vol. 8, p. 18.

Magdolenova, Z., Collins, A., Kumar, A., Dhawan, A., Stone, V. et al. (2014), 'Mechanisms of genotoxicity – A review of *in vitro* and *in vivo* studies with engineered nanoparticles', *Nanotoxicology*, Vol. 8, No 3, pp. 233–278.

Marucco, A., Prono, M., Beal, D., Alasonati, E., Fisicaro, P. et al. (2020), 'Biotransformation of food-grade and nanometric TiO₂ in the oral-gastro-intestinal tract: Driving forces and effect on the toxicity toward intestinal epithelial cells', *Nanomaterials*, Vol. 10, No 11, 2132.

McLachlan, M.S., Czub, G., Macleod, M. and Arnot, J. (2011), 'Bioaccumulation of organic contaminants in humans: a multimedia perspective and the importance of biotransformation'. *Environmental Science and Technology*, Vol. 45, pp.197–202, <http://doi:10.1021/es101000w>.

Mehn, D., Gioria, S., Ponti, J., Jimenez, O., Drewes, C. C. et al. (2018), *Physicochemical characterisation of gold, silica and silver nanoparticles in water and in serum-containing cell culture media*, Publications Office of the European Union, Luxembourg, <https://doi.org/10.2760/818663>.

Misra, S., Dybowska, A., Berhanu, D., Luoma, S. and Valsami-Jones, E. (2012), 'The complexity of nanoparticle dissolution and its importance in nanotoxicological studies', *Science of the Total Environment*, Vol. 438, pp. 225–232.

Mohammadpour, R., Yazdimamaghani, M., Cheney, D. L., Jedrzkiewicz, J. and Ghandehari, H. (2019), 'Subchronic toxicity of silica nanoparticles as a function of size and porosity', *Journal of Controlled Release*, Vol. 304, pp. 216–232.

OECD (2012), *Guidance document 116 on the conduct and design of chronic toxicity and carcinogenicity studies, supporting test guidelines 451, 452 and 453*, 2nd edition, OECD Publishing, Paris, <https://www.oecd-ilibrary.org/docserver/9789264221475-en.pdf?expires=1695996794&id=id&accname=guest&checksum=76AC479C7DBCBB53072DEE15AFA3973C>.

OECD (2017), 'Overview on genetic toxicology test guidelines', *Series on Testing and Assessment*, No 238, OECD Publishing, Paris, <https://www.oecd.org/publications/overview-on-genetic-toxicology-tgs-9789264274761-en.htm>.

OECD (2018), 'Evaluation of *in vitro* methods for human hazard assessment applied in the OECD testing programme for the safety of manufactured nanomaterials', *Series on the Safety of Manufactured Nanomaterials*, No 85, OECD Publishing, Paris, [https://one.oecd.org/document/ENV/JM/MONO\(2018\)4/en/pdf](https://one.oecd.org/document/ENV/JM/MONO(2018)4/en/pdf).

OECD (2020), 'Guidance document for the testing of dissolution and dispersion stability of nanomaterials and the use of the data for further environmental testing and assessment strategies, No 318', https://www.oecd.org/content/dam/oecd/en/publications/reports/2020/07/guidance-document-for-the-testing-of-dissolution-and-dispersion-stability-of-nanomaterials-and-the-use-of-the-data-for-further-environmental-testing-and-assessment_988b598b/f0539ec5-en.pdf.

Paini, A., Asturiol, D., Fortaner Torrent S, et al. (2020), 'EURL ECVAM *in vitro* hepatocyte clearance and blood plasma protein binding dataset for 77 chemicals', *European Commission, Joint Research Centre (JRC) [Dataset]*, PID: <http://data.europa.eu/89h/a2ff867f-db80-4acf-8e5c-e45502713bee>.

- Panse, N. and Gerk, P.M. (2022), 'The Caco-2 model: Modifications and enhancements to improve efficiency and predictive performance', *International Journal of Pharmacology*, Vol. 624:122004, <https://doi.org/10.1016/j.ijpharm.2022.122004>.
- Peckys, D. and De Jonge, N. (2011), 'Visualizing gold nanoparticle uptake in live cells with liquid scanning transmission electron microscopy', *Nano Letters*, Vol. 11, pp. 1733–1738.
- Peterson, R. J., Montoro Bustos, A. R., Toman, B., Johnson, M. E., Ellefson, M. et al. (2019), 'Determining what really counts: Modeling and measuring nanoparticle number concentrations', *Environmental Science*, Vol. 6, pp. 2876–2896.
- Pfuhler, S., Elespuru, R., Aardema, M., Doak, S., Donner, E. et al. (2013), 'Genotoxicity of nanomaterials: Refining strategies and tests for hazard identification', *Environmental and Molecular Mutagenesis*, Vol. 54, No 4, pp. 229–239.
- Reuter, U., Heinrich-Hirsch, B., Hellwig, J., Holzum, B. and Welsch, F. (2003), 'Evaluation of OECD screening tests 421 (reproduction/developmental toxicity screening test) and 422 (combined repeated-dose toxicity study with the reproduction/developmental toxicity screening test)', *Regulatory Toxicology and Pharmacology*, Vol. 38, pp. 17–26.
- Rotroff, D.M., Wetmore, B.A., Dix, D.J., et al. (2010), 'Incorporating human dosimetry and exposure into high-throughput *in vitro* toxicity screening', *Toxicological Sciences*, Vol. 117, No 2, pp. 348–358, <http://doi:10.1093/toxsci/kfq220>.
- Senaldi, C., Crutzen, H. and Hoekstra, E. (2021), *Review of standards related to materials in contact with drinking water*, JRC technical report, Publications Office of the European Union, Luxembourg.
- SCCS (Scientific Committee on Consumer Safety) (2019), *Guidance on the safety assessment of nanomaterials in cosmetics*, Publications Office of the European Union, Luxembourg.
- SCENIHR (Scientific Committee on Emerging and Newly Identified Health Risks) (2006), *The appropriateness of existing methodologies to assess the potential risks associated with engineered and adventitious products of nanotechnologies*, http://ec.europa.eu/health/ph_risk/committees/04_scenihr/docs/scenihr_o_003b.pdf.
- Shi, H., Magaye, R., Castranova, V. and Zhao, J. (2013), 'Titanium dioxide nanoparticles: A review of current toxicological data', *Particle and Fibre Toxicology*, Vol. 10, p. 15.
- Speit, G., Vasquez, M. and Hartmann, A. (2009), 'The comet assay as an indicator test for germ cell genotoxicity', *Mutation Research*, Vol. 681, pp. 3–12.
- Störmer, A., Bott, J., Kemmer, D. and Franz, R. (2017), 'Critical review of the migration potential of nanoparticles in food contact plastics', *Trends in Food Science and Technology*, Vol. 63, pp 39–50.
- Tantra, R., Bouwmeester, H., Bolea, E.C., Rey-Castro, C. A. et al. (2016), 'Suitability of analytical methods to measure solubility for the purpose of nanoregulation', *Nanotoxicology*, Vol. 10, No. 2, pp. 173–184.
- Tassinari, R., Di Felice, G., Butteroni, C., Barletta, B., Corinti, S. et al. (2020), 'Hazard identification of pyrogenic synthetic amorphous silica (NM-203) after sub-chronic oral exposure in rat: A multitarget approach', *Food and Chemical Toxicology*, Vol. 137, 111168.

Utembe, W., Potgieter, K., Stefaniak, A. and Gulumian, M. (2015), 'Dissolution and biodurability: Important parameters needed for risk assessment of nanomaterials', *Particle and Fibre Toxicology*, Vol. 12, No 11.

Van Der Zwaag, D., Vanparijs, N., Wijnands, S., De Rycke, R., De Geest, B. et al. (2016), 'Super resolution imaging of nanoparticles cellular uptake and trafficking', *ACS Applied Materials and Interfaces*, Vol. 8, pp. 6391–6399.

Warheit, D. B. and Donner, E. (2010), 'Rationale of genotoxicity testing of nanomaterials: Regulatory requirements and appropriateness of available OECD test guidelines', *Nanotoxicology*, Vol. 4, pp. 409–413.

Wetmore, B.A., Wambaugh, J.F., Ferguson, S.S., et al. (2012), 'Integration of dosimetry, exposure, and high-throughput screening data in chemical toxicity assessment', *Toxicological Sciences*, Vol. 125, No 1, pp.157-174, <http://doi:10.1093/toxsci/kfr254>.

Zheng, H. and Olive, P. L. (1997), 'Influence of oxygen on radiation-induced DNA damage in testicular cells of C3H mice', *International Journal of Radiation Biology*, Vol. 71, pp. 275–282.

8.2. Specific sources

8.2.1. Migration testing and analysis of information

4MSI common approach on organic materials in contact with drinking water – Part A: Methodologies for testing and accepting starting substances to be included in the positive list.

4MSI common approach on organic materials in contact with drinking water – Draft Part C: Procedures and methods for testing and accepting products made of organic materials.

4MS Common Approach: Assessment of cementitious products in contact with drinking water.

4MSI Common Approach: Acceptance of enamels and ceramic materials used for products in contact with drinking water – Part C: Procedure and methods for testing and accepting products or components made of enamels or ceramic materials.

4MSI Common Approach: Acceptance of metallic materials used for products in contact with drinking water – Part C: Procedure and methods for testing and accepting final materials as used in a product made from compositions on the positive list.

Andersson, J. O., Helander, T., Höglund, L., Shi P. F. and Sundman, B. (2002), 'Thermo-Calc and DICTRA, computational tools for materials science', *Calphad*, Vol. 26, No 2, pp. 273–312.

Bale, C. W., Bélisle, E., Chartrand, P., Decterov, S. A., Eriksson, G. et al. (2016), 'FactSage thermochemical software and databases – 2010–2016', *Calphad*, Vol. 54, pp. 35–53.

Bratinova, S., Raffael, B. and Simoneau, C. (2009), *Guidelines for performance criteria and validation procedures of analytical methods used in controls of food contact materials*, Publications Office of the European Union, Luxembourg.

Davies, R. H., Dinsdale, A. T., Gisby, J. A., Robinson, J. A. J. and Martin, S. M. (2002), 'DATA – Thermodynamic and phase equilibrium software from the national physical laboratory', *Calphad*, Vol. 26, No 2, pp. 229–271.

ECHA (2019), *Plastic Additives Initiative – Supplementary information on scope and methods*, https://echa.europa.eu/documents/10162/17228/plastic_additives_supplementary_en.pdf/79bea2d6-8e45-f38c-a318-7d7e812890a1.

EFSA (2020), 'Note for guidance for the preparation of an application for the safety assessment of a substance to be used in plastic food contact materials', *EFSA Journal*, Vol. 6, No 7.

Engelsen, C., Kaunisto, T., Rod, O., Kloppenborg, S., Latva, M. et al. (2018), *Material and product innovation through knowledge-based standardization in drinking water sector*, MaiD final report, Nordic Innovation, Oslo.

European standards: EN 15768, EN ISO 17294-2, EN ISO 12846 and EN ISO 17852.

Kaunisto, T., Latva, M., Engelsen, C., Kloppenborg, S., Rod, O. et al. (2017), *Nordic Drinking Water Quality*, MaiD report 1, Nordic Innovation, Oslo.

Mäkinen, R. (ed.) (2008), *Drinking Water Quality and Network Materials in Finland*, summary report, Finnish Institute of Drinking Water / Prizztech Ltd, <https://www.theseus.fi/bitstream/handle/10024/850604/Summary%20report%20ed1.pdf>.

Roine, A. (2021), HSC Chemistry [software], Metso Outotec, Pori, Finland, www.mogroup.com/hsc.

Senaldi, C., Crutzen, H. and Hoekstra, E. (2021), *Review of standards related to materials in contact with drinking water*, JRC technical report, Publications Office of the European Union, Luxembourg.

8.3. Test methods and standards

AFNOR standards can be purchased from the website of the French Institute for Standardisation (AFNOR) (<http://www.afnor.org/en/>).

ASTM standards may be obtained from the American Society of Testing Methods (ASTM), West Conshohocken, Pennsylvania (<http://www.astm.org>).

DIN standards can be purchased from the website of the German Institute for Standardisation (DIN) (<http://www.din.de>).

EPPO guidelines may be obtained from the Secretary of the European and Mediterranean Plant Protection Organisation (EPPO), Paris (<http://www.eppo.int/>).

EU methods are described in the test methods regulation (Regulation (EC) No 440/2008), published in the *Official Journal of the European Union*. They are regularly updated, with new methods introduced as required.

ISO 1382: Rubber – Vocabulary, ISO, 2020.

ISO 8466-1: Water quality – Calibration and evaluation of analytical methods and estimation of performance characteristics, Part 1: Statistical evaluation of the linear calibration function, ISO, 2021.

ISO 8466-2: Water quality – Calibration and evaluation of analytical methods and estimation of performance characteristics, Part 2: Calibration strategy for non-linear second-order calibration functions, ISO, 2001.

ISO 21675: Water quality – Determination of perfluoroalkyl and polyfluoroalkyl substances (PFAS) in water: Method using solid phase extraction and liquid chromatography-tandem mass spectrometry (LC-MS/MS), ISO, 2019.

ISO (2017), *Nanotechnologies – Use and application of acellular in vitro tests and methodologies to assess nanomaterial biodurability*, ISO, Geneva.

OECD test methods can be obtained directly via links on the OECD website (<https://www.oecd.org/en/topics/sub-issues/testing-of-chemicals/test-guidelines.html>).

OECD (2012), *Guidance document 116 on the conduct and design of chronic toxicity and carcinogenicity studies, supporting test guidelines 451, 452 and 453*, 2nd edition, OECD Publishing, Paris, <https://www.oecd-ilibrary.org/docserver/9789264221475-en.pdf?expires=1695996794&id=id&accname=guest&checksum=76AC479C7DBCBB53072DEE15AFA3973C>.

OECD (2023), 'Guidance for the regulatory assessment of (quantitative) structure-activity relationship models, predictions and results based on multiple predictions', *Series on Testing and Assessment*, No 386, OECD Environmental Health and Safety Publications, Paris, <https://www.oecd.org/chemicalsafety/testing/series-testing-assessment-publications-number.htm>.

CEN/TR 16364: Influence of materials on water intended for human consumption. Influence due to migration. Prediction of migration from organic materials using mathematical modelling: Helsinki, SFS:2012.

SFS-EN 1233:en: Water quality. Determination of chromium. Atomic absorption spectrometric methods, Helsinki, SFS:2013.

SFS-EN 12873-1:en: Influence of materials on water intended for human consumption. Influence due to migration. Part 1: Test method for factory-made products made from or incorporating organic or glassy (porcelain/vitreous enamel) materials, Helsinki, SFS:2014.

SFS-EN 14944-3:en: Influence of cementitious products on water intended for human consumption. Test methods. Part 3: Migration of substances from factory-made cementitious products, Helsinki, SFS:2008.

SFS-EN 15664-1: Influence of metallic materials on water intended for human consumption. Dynamic rig test for assessment of metal release. Part 1: Design and operation, Helsinki, SFS:2014.

SFS-EN 15664-2: Influence of metallic materials on water intended for human consumption. Dynamic rig test for assessment of metal release. Part 2: Test waters, Helsinki, SFS:2010.

SFS-EN 15768:en: Influence of materials on water intended for human consumption. GC-MS identification of water leachable organic substances, Helsinki, SFS:2015.

SFS-EN 16056:en: Influence of metallic materials on water intended for human consumption. Method to evaluate the passive behaviour of stainless steels, Helsinki, SFS: 2012.

SFS-EN 16058:en: Influence of metallic materials on water intended for human consumption. Dynamic rig test for assessment of surface coatings with nickel layers. Long-term test method, Helsinki, SFS:2012.

SFS-EN 16618:en: Food analysis. Determination of acrylamide in food by liquid chromatography tandem mass spectrometry (LC-ESI-MS/MS), Helsinki, SFS:2015.

SFS-EN 16691:en: Water quality. Determination of selected polycyclic aromatic hydrocarbons (PAH) in whole water samples. Method using solid phase extraction (SPE) with SPE-disks combined with gas chromatography mass spectrometry (GC-MS), Helsinki, SFS:2015.

SFS-EN ISO 5961: Water quality. Determination of cadmium by atomic absorption spectrometry (ISO 5961:1994), Helsinki, SFS:2001.

SFS-EN ISO 12020: Water quality. Determination of aluminium. Atomic absorption spectrometric methods (ISO 12020:1997), Helsinki, SFS:2001.

SFS-EN ISO 11885: Water quality. Determination of selected elements by inductively coupled plasma optical emission spectrometry (ICP-OES) (ISO 11885:2007), Helsinki, SFS:2009.

SFS-EN ISO 12846: Water quality. Determination of mercury. Method using atomic absorption spectrometry (AAS) with and without enrichment (ISO 12846:2012), Helsinki, SFS:2012.

SFS-EN ISO 14911: Water quality. Determination of dissolved Li⁺, Na⁺, NH₄⁺, K⁺, Mn²⁺, Ca²⁺, Mg²⁺, Sr²⁺ and Ba²⁺ using ion chromatography. Method for water and waste water (ISO 14911:1998), Helsinki, SFS:2012.

SFS-EN ISO 15586: Water quality. Determination of trace elements using atomic absorption spectrometry with graphite furnace (ISO 15586:2003), Helsinki, SFS:2005.

SFS-EN ISO 17294-2:2016:en: Water quality. Application of inductively coupled plasma mass spectrometry (ICP-MS). Part 2: Determination of selected elements including uranium isotopes (ISO 17294-2:2016), Helsinki, SFS:2016.

SFS-EN ISO 17852:en: Water quality. Determination of mercury. Method using atomic fluorescence spectrometry (ISO 17852:2006), Helsinki, SFS:2008.

SFS-EN ISO 17943: Water quality. Determination of volatile organic compounds in water. Method using headspace solid-phase micro-extraction (HS-SPME) followed by gas chromatography-mass spectrometry (GC-MS) (ISO 17943:2016), Helsinki, SFS:2016.

SFS-EN ISO 17993:en: Water quality. Determination of 15 polycyclic aromatic hydrocarbons (PAH) in water by HPLC with fluorescence detection after liquid-liquid extraction (ISO 17993:2002), Helsinki, SFS:2004.

SFS-EN ISO 18412: Water quality. Determination of chromium(VI). Photometric method for weakly contaminated water (ISO 18412:2005), Helsinki, SFS:2007.

SFS-EN ISO 18857-2: Water quality. Determination of selected alkylphenols. Part 2: Gas chromatographic-mass spectrometric determination of alkylphenols, their ethoxylates and bisphenol A in non-filtered samples following solid-phase extraction and derivatisation (ISO 18857-2:2009), Helsinki, SFS:2012.

SFS-EN ISO 18862: Coffee and coffee products. Determination of acrylamide. Methods using HPLC-MS/MS and GC-MS after derivatization (ISO 18862:2016), Helsinki, SFS:2019.

SFS-EN ISO 23631: Water quality. Determination of dalapon, trichloroacetic acid and selected haloacetic acids. Method using gas chromatography (GC-ECD and/or GC-MS detection) after liquid-liquid extraction and derivatization (ISO 23631:2006), Helsinki, SFS:2006.

SFS-EN ISO 23913: Water quality. Determination of chromium(VI). Method using flow analysis (FIA and CFA) and spectrometric detection (ISO 23913:2006), Helsinki, SFS:2009.

SFS-EN ISO/IEC 17025: General requirements for the competence of testing and calibration laboratories (ISO/IEC 17025:2017), Helsinki, SFS:2018.

SFS-EN ISO 21920-2: Geometrical product specifications (GPS). Surface texture: Profile. Part 2: Terms, definitions and surface texture parameters (ISO 21920-2:2021), Helsinki, SFS:2022

SFS-EN 923: Adhesives. Terms and definitions, Helsinki, SFS:2018.

TCIPAC methods may be purchased from the Collaborative International Pesticides Analytical Council (<http://www.cipac.org>).

US EPA test guidelines for pesticides and toxic substances can be obtained from the US EPA website (<https://www.epa.gov/test-guidelines-pesticides-and-toxic-substances>).

VDI guidelines can be obtained from the website of the Association of German Engineers (VDI) (<http://www.vdi.de>).

9. Annex I – Material categories

9.1. Organics

Table 5 lists the options that an applicant can choose when applying for a starting substance to be included in the organics EUPL. The applicant should choose one material category (level 2) and one subcategory (level 3) for each material in which they intend to use the starting substance.

The main material categories are:

- plastics and polymers (excluding elastomers);
- rubber and elastomers;
- silicone;
- lubricants ⁽⁹²⁾;
- coatings;
- adhesives.

In IUCLID when submitting a dossier, plastics and polymers, rubbers and elastomers, silicones and lubricants can be selected as material categories. For any submission, the applicant will have the option to indicate whether a substance is used in/as a surface layer (e.g. an organic coating). If this is chosen, then the details of the coating can be given (see Table 5).

It should be noted that composite resins containing more than one polymer, for example epoxy-urethane hybrid resins or acrylic-urethane hybrid resins may be described using the 'other (specify)' category offered by the IUCLID software, which allows provision of the necessary details. In addition, as there is no specific EUPL for composite materials, polymer-matrix composites (e.g. glass fibre reinforced resins) are included in the organic material categories. These can be specified either by selecting the corresponding material from the list or using the category 'plastic and polymers, other (specify)' and describing the material.

Table 5. Material categories and subcategories for organic materials

Material category (level 2)	Material subcategory (level 3)
Plastics and polymers (excluding elastomers)	• Chlorinated polyethylene (PE-C) • Chlorinated polyvinyl chloride (CPVC) • Copolymers of acrylonitrile-butadiene-styrene (ABS) • Copolymers of styrene-acrylonitrile (SAN) • Epoxide plastic (EP) • Epoxide resins • Laminated glass-epoxy resin • Polytetrafluoroethylene (PTFE); polyvinylidene fluoride (PVDF); other) • Glass fibre-reinforced plastic (GRP) • Other polyesters • Other polyesters, not elsewhere specified

⁽⁹²⁾ Lubricants as the material category means semi-solid materials applied on surfaces to reduce friction. It is not the same as the technical function lubricant in Section 11.1 of Annex III.

Material category (level 2)	Material subcategory (level 3)
	• Polyacetals, including copolymers • Polyamides, including copolymers • Polybutylene (PB) • Polycarbonates (PCs), including copolymers • Polyethylene of raised temperature resistance (PE-RT) • Polyethylene terephthalate (PETP) • Polyethylene, cross-linked (PEX) • Polyethylene, high density (HDPE) • Polyethylene, low density (LDPE) • Polyethylene, medium density (PEM) • Polyphenylene oxide • Polyphenylene sulphide (PPS) • Polyphenylsulphone (PPSU) • Polypropylene (PP) • Polysulphides and polysulphones • Polyurea resin • Polyurethane resin • Polyurethanes • Polyvinylchloride (PVC), hard • Polyvinylchloride (PVC), soft • Polyvinylidene chloride (PVDC) • Reinforced epoxy resin • Plastic and polymers, other (specify)
Rubber and elastomers	• Chloroprene (chlorobutadiene) rubber (CR) • Ethylene-propylene-non-conjugated diene rubber (EPDM) • Fluororubber /fluoropolymer rubber / fluoroelastomer • Halo-isobutene-isoprene rubber (CIIR or BIIR) / chloro butyl rubber • Natural rubber • Nitrile rubber /(acrylo)nitrile-butadiene rubber (NBR) • Styrene-butadiene rubber (SBR) • Urethane rubber / polyurethane (PUR) • Rubber and elastomers, other (specify)
Silicones	• Silicones (or polysiloxanes), including copolymers • Silicone rubber • Silicones, other (specify)
Lubricants	—
Coatings	Specify as needed using, for example, subcategories of 'plastics and polymers
Adhesives	• Acrylate and cyanoacrylate ^(a) • Silicone ^(a) • Epoxy ^(a) • Polyurethane ^(a) • Solvent-based adhesive ^(b) • Other (specify)

^(a) Adhesives obtained by polymerisation.

^(b) Contains, for example, solvent cement also known as PVC glue.

9.2. Metallics

Table 6 lists the material category options that an applicant can choose from when applying for a metallic composition to be included in the EUPL. The applicant should choose one category (level 2) and, where applicable, one subcategory (level 3).

In the case of metallics, 'material categories' matches the 'metallic composition categories' shown in Annex IV to this guidance. In Annex IV, metallic composition categories are defined by a category composition (with its constituents and impurities) and a reference material composition (with its constituents and impurities). Every application for a metallic composition should identify the metallic composition category the composition falls under, and in doing so the metallic material category (level 2) should be declared.

Table 6. Material categories and subcategories for metallic materials

Material category (level 2)	Material subcategory (level 3) (a)
Copper–zinc alloys	• CW501L-DW (CuZn10) • CW506L-DW (CuZn33) • CW507L-DW (CuZn36) • CW508L-DW (CuZn37) • CW509L-DW (CuZn40) • CW510L-DW (CuZn42)
Copper–zinc–aluminium alloys	• CuZn42Al (CC773S)
Copper–zinc–aluminium–tin alloys	• CuZn35Al1.5Sn
Copper–zinc–aluminium–silicon–iron alloys	• CuZn35AlSiFe
Copper–zinc–arsenic alloys	• CW707R (CuZn30As)
Copper–zinc–arsenic–aluminium alloys	• CuZn35Al-C • CW702R (CuZn20Al2As)
Copper–zinc–arsenic–antimony–aluminium alloys	• CC771S (CuZn36AlAsSb-C)
Copper–tin–zinc–phosphorus–sulphur alloys	• CuSn4Zn2PS
Copper–zinc–silicon–phosphorus alloys	• CW724R (CuZn21Si3P) • CC768S (CuZn21Si3P-C)
Copper–silicon–zinc–manganese–phosphorus alloys	• CC245E (CuSi4Zn4MnP-C) • CC246E (CuSi4Zn9MnP-C) • CuSi4Zn4MnP • CuSi4Zn9MnP
Copper–tin–phosphorus alloys	• CW453 K (CuSn8)
Copper–tin–lead–phosphorus alloys	• CuSn10-C
Copper–nickel–manganese–iron alloys	• CW352 (CuNi10FeMn) • CW352H (CuNi10Fe1Mn) • CW353H (CuNi30Fe2Mn2) • CW354H (CuNi30Mn1Fe)

Material category (level 2)	Material subcategory (level 3) ^(a)
Copper alloys not elsewhere categorised	Other copper alloys for product group D (group entry)
Coppers	- CW024A (Cu-DHP) - CW004A (Cu-ETP) - CW008A (Cu-OF) - CW020A (Cu-PHC) - CW021A (Cu-HCP) - CW023A (Cu-DLP)
Tinned copper pipes and tinned copper fittings	CW024A (Cu-DHP) with a tin layer of 1 µm thickness
Aluminium	Aluminium
Steel/iron	- Galvanised steel - Carbon steel according to EN 10025 / EN 10213 / EN 10222 - Cast iron according to EN 1561 / EN 1563 - Stainless steels according to EN 10088 and EN 10283
Platings	- Electroplating of the outer surface (tin plating applied by a galvanic process) - Tin/nickel platings applied by a galvanic process on the external surface - Electrolytic nickel/chromium platings
Passive compositions	- NiCr7030 - Ni55Ti45 - Ti1 according to ISO 23515:2022 ^(b) - Ti2 according to ISO 23515:2022 - Ti3 according to ISO 23515:2022 - Ti4 according to ISO 23515:2022 - Ti-6Al-4 V according to ISO 23515:2022 - Other passive metallic materials for product group D (group entry)
Hard brazing alloys and soft soldering alloys	Hard brazing and soft soldering alloys meeting the maximum concentrations of the specified impurities (group entry)

^(a) This is the metallic composition that is applied for.

^(b) These four passive titanium compositions are considered equivalent to the titanium grades 1, 2, 3 and 4, according to ASTM B265 as shown in the 4MS Initiative positive list of compositions for metallic materials (<https://www.umweltbundesamt.de/en/topics/water/drinking-water/distributing-drinking-water/approval-harmonization-4ms-initiative#publication-of-common-approaches>).

9.3. Cementitious materials

In the case of applications for organic constituents of cementitious materials, material categories are less critical than the technical function of the constituent. Applicants should specify whether the cementitious mixtures for which the applied-for entry is used fall under one or both of the following material categories:

- bulk cementitious material;
- cementitious lining.

Note, however, that this categorisation will not become a condition of use for the organic cementitious constituent in the EUPL.

9.4. Enamels, ceramics and other inorganics

There will be one EUPL for this material type covering the following material categories (material category (level 2)):

- enamels (note that this encompasses glasses);
- ceramics;
- other inorganic materials.

These will contain their respective accepted compositions, including the composition ranges.

Table 7 lists the options that an applicant can choose when applying for an inorganic material to be included in the EUPL. The applicant should choose one material category (level 2) and one material subcategory (level 3). In addition, the applicant will have the option to indicate if the applied-for entry is to be used as an enamel layer.

Table 7. Material categories and subcategories for enamel, ceramic and other inorganic materials

Material category (level 2)	Material subcategory (level 3) ^(a)
Enamels	• Enamel • Soda-lime glasses • Lead crystal and crystal glass cat. 1 • Borosilicate glass • Fused quartz and fused silica glass • Optical glass • Sintered glass • Glass ceramic • Other special glass • Enamels, other (specify)
Ceramics	• Oxide ceramics • Non-oxide ceramics • Inorganic carbon ceramics • Ceramics, other (specify)
Other inorganics	• Inorganic coatings • Other inorganic materials not elsewhere classified

^(a) This is the metallic composition that is applied for.

10. Annex II – Product groups and their conversion factors

10.1. Metallic materials

The intended application of a metallic composition in a product must belong to one of the product groups in Table 8, replicating Table 3 of Annex II to Commission Implementing Decision (EU) 2024/365. The selection of suitable product groups should take into consideration the information given in Table 8 and the paragraphs below.

Table 8. Product groups for metallic compositions

Product group	Examples of product or component	Assumed contact surface 'a'
A	Pipes. This group concerns all pipes, e.g. pipes in domestic distribution system, pipelines in water supply systems.	100 %
B	Fittings, ancillaries in buildings installations (e.g. pump bodies, valve bodies, water meter bodies, water heaters).	10 %
C	- Components of products of product group B (e.g. the spindle of a pump or the moving parts in water meters in building installations). The sum of the surfaces in contact with drinking water of all these components has to be less than 10 % of the total wetted surface of the product. - Fittings, ancillaries in water mains and water treatment works with permanent flow (e.g. pump bodies, valve bodies, water meter bodies used in water supply systems).	1 %
D	Components of fittings and ancillaries in water mains and in water treatment works (C2).	Trivial (< 0.1 %)

- **Product group A: up to 100 % contact surface.** For pipes in a building installation the same metallic composition can be used for all diameters. A single metallic composition can contribute to nearly 100 % of its surface in contact with drinking water, for example copper, galvanised steel or stainless steel. The evaluation of the conditions of use must assume the maximum possible percentage. A metallic composition that is accepted for use in pipes should be considered accepted for use in all product groups (e.g. fittings or components).

This product group also includes uncoated metallic pipelines used in water supply systems and water treatment processes.

- **Product group B: up to 10 % contact surface.** Fittings or ancillaries can be produced from one metallic composition or from slightly different metallic compositions throughout the building installation. Because of their potential to release metals (e.g. lead) into drinking water, there is a need to restrict the total contact surface area of products made from these alloys. For assessments of metallic compositions for these products, a contribution of 10 % water contact surface area is assumed.

This product group also includes the main metallic parts of pumps and valves used in building installations. A metallic composition that is accepted for use in product group B should be considered as also accepted for use in product groups C and D.

- **Product group C: less than 1 % contact surface**

C1: for technical reasons, there might be a need to produce small parts from metallic compositions not accepted for product group B. Other metallic compositions with higher release rates may be accepted in these parts if their use will not significantly increase the total contamination of drinking water (i.e. increase metal releases above the MTC_{tap}). The use of such metallic compositions should be restricted to components whose sum of water contact surface does not exceed 1 % of the total surface in contact with drinking water; for example, the body of a water meter would need to be produced from an accepted metallic composition for product group B, but a moving part may be produced from a metallic composition listed for product group C. The sum of the water contact surfaces of all parts in one product made from product group C compositions should be less than 10 % of the water contact surface of the product.

C2: this group also includes the main metallic parts of fittings and ancillaries used with water mains and in water treatment works. For these products a permanent flow of drinking water must be ensured.

- **Product group D: trivial contact surface**

This product group encompasses components of fittings and ancillaries in water mains and in water treatment works (C2). The sum of the water contact surfaces of all parts in one product made from product group D compositions should be less than 10 % of the water contact surface of the product. For these products a permanent flow of drinking water must be ensured.

10.2. Organic, cementitious, enamel, ceramic and other inorganic materials

Table 9 shows product groups, example products and their respective S/V ratios, CFs and risk groups. This table and the values can be used in describing the product/component, area of use and relevant product group of the intended use (in accordance with Commission Implementing Decision (EU) 2024/365).

The risk groups (Table 10) describe the general risk of product types based on contact surface and time and are based on the CFs presented in Commission Implementing Decision (EU) 2024/368. While they can be useful, they are more relevant for the purpose of accepting products than applying for inclusion in EUPLs.

Table 9. Product groups for organic, cementitious, enamel, ceramic and other inorganic materials

Product group	Examples of product or component	S/V (dm ⁻¹)	CF	Risk group
A1	Pipes and pipe linings ID < 80 mm (domestic installations, buildings)	40	20	1
A2	80 mm ≤ ID < 300 mm (service piping)	5	10	1
A3	ID ≥ 300 mm (mains piping)	1.25	5	1
B1	Fittings, ancillaries ID < 80 mm (domestic installations, buildings)	4	2	2
B2	80 mm ≤ ID < 300 mm (service piping)	0.5	1	2
B3	ID ≥ 300 mm (mains piping)	0.125	0.5	2
B4	Housing of treatment steps and their coatings	0.0125	0.05	3
B5	Water abstraction devices	0.00125	0.005	4
C1	Components of fittings, ancillaries ID < 80 mm (domestic installations, buildings)	0.4	0.2	3
C2	80 mm ≤ ID < 300 mm (service piping)	0.05	0.1	3
C3	ID ≥ 300 mm (mains piping)	0.0125	0.05	3
C4	Housing of treatment steps and their coatings	0.00125	0.005	4
C5	Water abstraction devices	0.000125	0.0005	4
D1	Small components of fittings, ancillaries ID < 80 mm (domestic installations, buildings)	0.04	0.02	4
D2	80 mm ≤ ID < 300 mm (service piping)	0.005	0.01	4
D3	ID ≥ 300 mm (mains piping)	0.00125	0.005	4
D4	Small components of housing of treatment steps and their coatings	0.000125	0.0005	4
D5	Small components of water abstraction devices	0.0000125	0.00005	4
E1	Storage systems (reservoirs) In domestic installations, buildings, water volume < 10 l	4	4	1
E2	In domestic installations, buildings, water volume ≥ 10 l	2	2	2
	In water supply	0.25	1	2
F1	Products for storage systems In domestic installations, buildings			
F1.1	Parts of storage systems (> 10 % of the total surface)	4	4	1
F1.2	Parts of storage systems (1 – 10 % of the total surface)	0.4	0.4	2
F1.3	Parts of storage systems (≤ 1 % of the total surface)	0.04	0.04	3
F2	In water supply			
F2.1	Products covering the total surface or a substantial part of that (e.g. coatings)	0.250.0025 0.00025	1	2
F2.2	Products covering < 1 % of the total surface		0.01	4
F2.3	Elements of raw water basins, covering < 1‰ of the total surface		0.001	4

Table 10. Risk groups for organic, cementitious, enamel, ceramic and other inorganic materials

Risk group	CF (d/dm)
1	≥ 4
2	0.4 to < 4
3	0.04 to < 0.4
4	< 0.04

11. Annex III – Technical functions for starting substances and organic cementitious constituents

Technical functions are of relevance to the EUPLs for organics and cementitious materials only.

11.1. Organic materials

In the case of organic materials, applicants will be required to identify the technical function of starting substances on two levels (see also Table 11).

- **First level.** The application should identify whether the starting substance displays one (or more) of the following technical functions:
 - monomer or other reactant;
 - additive;
 - polymer production aid;
 - aid to polymerisation;
 - other.

The first level technical function will be shown in the organics EUPL and will become a condition of the approval of the starting substance.

- **Second level.** The application should identify whether the starting substance displays one (or more) of the technical functions shown in Table 11. This information will be considered in the assessment of the application but may not be shown in the organics EUPL and will not form a condition of the approval of the starting substance.

Table 11. Technical functions of starting materials applied for inclusion in the organics EUPL

First-level technical function	Second-level technical function
Additives	Adhesion agent
	Adhesion promoter for fillers
	Anti-foaming agent with a function in the final article
	Antioxidant
	Anti-skinning agent
	Anti-static agent
	Blowing agent used in the manufacture of expanded polymers
	Colouring agent
	Corrosion inhibitor

First-level technical function	Second-level technical function
	Desiccant
	Emulsifier (if it has a function in the final material)
	Extrusion aid
	Filler
	Flame retardant
	Hardening agent
	Impact modifier (except for substances that are capable of functioning as the main structural component of the final material)
	Lubricant
	Optical brightener
	Plasticiser
	Protective colloid
	Reinforcement
	Release agent
	Stabiliser of the final material
	Surface treatment for fillers
	UV stabiliser
	Viscosity modifier
	Additives, other (specify)
Aids to polymerisation	Accelerator
	Activator
	Catalyst
	Catalyst deactivator
	Catalyst modifier
	Catalyst support

First-level technical function	Second-level technical function
	Chain scission reagent
	Chain stop reagent
	Chain transfer or extending agent or molecular weight regulator
	Complexing agent
	Desensitising agent
	Initiator and promoter
	Photoinitiator
	Polymerisation inhibitors
	Redox agent
	Vulcanisation agent for elastomers
	Aids to polymerisation, other (specify)
Monomers or other reactants	Agent for post-treatment of polymers
	Base oil
	Comonomer
	Cross-linking agent (that is incorporated into the polymer)
	Hardener (curing agent)
	Macromolecule obtained from microbial fermentation
	Monomer
	Pre-polymer
	Resin
	Surface treatment agent that is incorporated into the polymer
	Monomers or other reactants, other (specify)
Polymer production aids	Anti-caking agent
	Anti-cluster agent

First-level technical function	Second-level technical function
	Anti-crusting agent
	Anti-foam reagent / degassing agent necessary during the manufacturing process
	Anti-scaling agent
	Build-up suppressant
	Coagulant
	Dispersant
	Emulsifier necessary during the manufacturing process
	Flow control agent
	Lubricant
	Nucleating agent
	pH regulator
	Phlegmatising agent
	Solubilising agent
	Solvent
	Stabiliser during manufacturing process
	Surfactants
	Suspending agent
	Viscosity modifier in polymerisation process
	Polymer production aids, other (specify)
Other	In-can preservative

11.2. Cementitious materials

Cementitious materials contain a range of ingredients which, for the purposes of the DWD EUPL, are organised as follows.

- Table 1 of Annex III to Commission Implementing Decision (EU) 2024/367 shows the different generic constituent categories that are the generic ingredients used in cementitious mixtures. These generic constituent categories or generic ingredients show the technical

functionality of the ingredient and may encompass inorganic or organic constituents of cement mixtures. Inorganic constituents do not require to be listed in the EUPL for cementitious materials in accordance with point 3 of Annex V to the DWD. Therefore, applications for the use of, for example, aggregates or mixing water should not be submitted to ECHA. The types of constituents falling into each generic constituent category are shown in Table 12 together with any corresponding technical specifications. It should be noted that the technical specifications are given only as a reference and are not legally mandatory: constituents outside these standards may be used as long as legal obligations are fulfilled.

- Specific organic constituents of cementitious materials may be relevant for inclusion in either the cementitious EUPL or the organic EUPL, or occasionally (in the case of organic additions) either of the EUPLs, as explained in Table 12.

Table 12. List of generic constituents of cementitious materials

Generic constituent category	Types of cement constituents per category	Technical specifications	Types of specific organic cementitious constituents within the scope of the EUPL	EUPL requirements	Conditions of use
Cement	Cement	• EN 197-1 (Common cements) • EN 197-4 (Low early strength blast furnace cements) • EN 197-5 (Portland composite cement CEM II/C-M and composite cement CEM VI) • EN 197-6 (Cement with recycled building materials) • EN 413-1 (Masonry cement) • EN 14216 (Very low heat special cements) • ETA-07/0019 and NF P15-314 (Ciment prompt naturel (natural prompt cement)) • EN 14647 (Calcium alumina cement; for raw water applications only)	Organic grinding aids or other organic substances used for the preparation of cement	EUPL for cementitious materials	
Inorganic additions	• Fly ash for concrete • Silica fume • Blast furnace slag • Limestone • Siliceous additions • Trass • Tempered rock flour • Natural pozzolanic additives	• EN 450-1 (Fly ash for concrete) • EN 13263-1 (Silica fume) • EN 15167-1 or BS 6699 (Ground granulated blast furnace slag) • BS 7979, NF P18-508 or LNEC E 466 (Limestone additions) • NF P18-509 (Siliceous additions) • DIN 51043 (Trass) • NF P18-513 (Metakaolin)	Organic grinding aids or other organic substances used for the preparation of inorganic additions	EUPL for cementitious materials	

Generic constituent category	Types of cement constituents per category	Technical specifications	Types of specific organic cementitious constituents within the scope of the EUPL	EUPL requirements	Conditions of use
Aggregates	- Aggregates for concrete - Aggregates for mortar - Aggregates for grout	- EN 12620 (Aggregates for concrete) - EN 13139 (Aggregates for mortar) - EN 13055-1 (Lightweight aggregates for concrete, mortar and grout)	Not relevant	Not relevant	Recycled concrete aggregates (RCA) derived from concrete that has not been used (e.g. surplus pre-cast units or returned ready mixed concrete), has previously been approved for contact with drinking water, and has not been contaminated in storage, will not require further testing. Other sources of RCA and recycled aggregates will need testing for metals, admixtures and other organic substances.
Mixing water	Mixing water for concrete, including drinking water	EN 1008 (Mixing water for concrete) – Drinking water	Not relevant	Not relevant	
Organic additions	- Polymer dispersions and powders - Auxiliaries for polymer additions		Organic substances	EUPL for organic materials EUPL for cementitious materials	Starting substances that are authorised in the EUPL for organic materials that are authorised for use in the manufacture of coatings may be used to manufacture organic additions for use in cementitious materials.

Generic constituent category	Types of cement constituents per category	Technical specifications	Types of specific organic cementitious constituents within the scope of the EUPL	EUPL requirements	Conditions of use
Admixtures	- Organic constituents of admixtures - Inorganic constituents of admixtures - Auxiliaries for constituents of admixtures (e.g. biocides, defoamers)	- EN 934-1, 2, 3, 4 or 5 (Admixtures for concrete, mortar and grout) - EN 934 may not cover all used admixtures; some of the materials added during the mixing process of a cementitious material in a quantity not more than 5 % by mass of the inorganic binder content of the cementitious material do not comply with a (harmonised) standard	Organic substances	EUPL for cementitious materials	Only biocidal active substances of product type 6 (preservatives for products during storage) in accordance with Regulation (EU) No 528/2012) are permitted to be used as auxiliaries to cementitious admixtures.
Fibres	- Organic fibres for concrete - Inorganic fibres for concrete	- EN 14889-1 (Steel fibres for concrete) - EN 14889-2 (Polymer fibres for concrete)	Organic substances	EUPL for organic materials	Starting substances that are authorised in the EUPL for organic materials may be used to manufacture polymer fibres for use in cementitious materials. NB: Fibres made of metallic compositions approved on the EUPL for metallic materials are also authorised for use in cementitious materials.
Formwork release agents	- Formwork release agents - Auxiliaries for formwork release agents		Organic substances	EUPL for cementitious materials	
Curing compounds	Organic curing compounds	Not applicable to factory-made products. Member State requirements for site-applied products will vary	Organic substances	EUPL for cementitious materials	

In the case of cementitious materials, applicants will be required to identify the technical function of the specific organic cementitious constituent that is applied for on two levels.

- **First level.** The EUPL for cementitious materials includes a column titled 'Generic constituent category'. For this column to be correctly filled in, the application should identify whether the constituent displays one (or more) of the following technical functions:
 - organic grinding aid for cement;
 - organic grinding aid for inorganic additions;
 - organic addition;
 - admixture;
 - formwork release agent (mixture);
 - curing compound (mixture);
 - other.

The first level technical function will become a condition of the approval of the organic cementitious constituent. Note that the list does not include fibres, as the starting substances for those must be listed in the organic materials EUPL (see DWD Guidance Volume III).

- **Second level.** The technical functions identified above should be further elaborated by the applicant by identifying a second level of the technical function, as shown in Table 13. Note that a biocide must always be identified as such under its second-level technical function. Note the absence of fibres from Table 13 as a result of the requirements for the listing of their starting substances in the **organics EUPL** (see Table 12).

Table 13. Technical functions of constituents applied for inclusion in the cementitious EUPL

First-level technical function	Second-level technical function
Organic grinding aid for cement	• Organic grinding aid • Component of a grinding aid • Other
Organic grinding aid for inorganic additions	• Organic grinding aid • Component of a grinding aid • Other
Organic addition	• Polymeric dispersion or powder • Other (including auxiliaries)
Admixture	• Polymeric component of an admixture • Pre-polymer for a polymeric component of an admixture • Biocide (BPR product type 6) • Defoamer • Other component
Formwork release agent (mixture)	• Formwork release agent • Component of a formwork release agent (including auxiliaries)

First-level technical function	Second-level technical function
Curing compound (mixture)	• Curing compound • Component of a curing compound (including auxiliaries)
Other	The user must specify.

12. Annex IV – Existing metallic composition categories (as at January 2024)

Note that the most up-to-date version of Table 14 will be available on the ECHA website.

Table 14. Existing metallic composition categories

Category number	Metallic composition category	Category constituents (% (w/w))	Category impurities (% (w/w))	Reference material constituents (% (w/w))	Reference material impurities (% (w/w))	Elements of relevance in the migration water	Most critical test water
1	Copper–zinc alloys	Cu: ≥ 57.0 % Zn: remainder	Al: ≤ 0.1 % Fe: ≤ 0.5 % Ni: ≤ 0.2 % Pb: ≤ 0.2 % Sn: ≤ 0.5 % Each other impurity: < 0.02 %	Cu: 57.0–59 % Zn: remainder	Al: ≤ 0.05 % Fe: ≤ 0.3 % Ni: 0.15–0.25 % Pb: 0.15–0.25 % Sn: ≤ 0.3 % Each other impurity: < 0.02 %	Cu, Ni, Pb, Zn	Test water 1 according to EN 15664-2
2	Copper–zinc–aluminium alloys	Cu: ≥ 57.0 % Zn: remainder Al: 0.1–0.3 %	Fe: ≤ 0.3 % Pb: ≤ 0.2 % Sn: ≤ 0.3 % Each other impurity: < 0.02 %	Cu: 57.0–59.0 % Zn: remainder Al: 0.1–0.2 %	Fe: ≤ 0.3 % Pb: 0.15–0.25 % Sn: ≤ 0.3 % Each other impurity: < 0.02 %	Al, Cu, Pb, Zn	Test water 1 according to EN 15664-2
3	Copper–zinc–aluminium–tin alloys	Cu: ≥ 62.0 % Zn: remainder Al: 0.5–2.0 % Sn: 0.3–1.0 %	Fe: ≤ 0.5 % Mn: ≤ 0.1 % Ni: ≤ 0.3 % Pb: ≤ 0.2 % Si: ≤ 0.2 % Each other impurity: < 0.02 %	Cu: 64.0–66.0 % Zn: remainder Al: 1.40–1.50 % Sn: 0.66 %–0.70 %	Fe: ≤ 0.10 % Mn: ≤ 0.10 % Ni: 0.06–0.10 % Pb: 0.15–0.25 % Si: ≤ 0.2 % Each other impurity: < 0.02 %	Al, Cu, Ni, Pb, Zn	Test water 1 according to EN 15664-2
4	Copper–zinc–aluminium–silicon–iron alloys	Cu: ≥ 62.0 % Zn: remainder Al: 0.5–0.9 % Si: 0.5–0.9 % Fe: 0.04–0.2 %	Ni: ≤ 0.1 % Pb: ≤ 0.2 % Sn: ≤ 0.1 % Each other impurity: < 0.02 %	Cu: 62.5–64.5 % Zn: remainder Al: 0.5–0.68 % Si: 0.5–0.68 % Fe: 0.12–0.15 %	Ni: 0.05–0.09 % Pb: 0.09–0.15 % Sn: ≤ 0.1 % Each other impurity: < 0.02 %	Al, Cu, Fe, Ni, Pb, Zn	Test water 1 according to EN 15664-2

Category number	Metallic composition category	Category constituents (% (w/w))	Category impurities (% (w/w))	Reference material constituents (% (w/w))	Reference material impurities (% (w/w))	Elements of relevance in the migration water	Most critical test water
5	Copper–zinc–arsenic alloys	Cu: ≥ 61.0 % Zn: remainder As: 0.02–0.15 %	Al: ≤ 0.1 % Fe: ≤ 0.5 % Mn: ≤ 0.1 % Ni: ≤ 0.3 % Pb: ≤ 0.2 % Sn: ≤ 0.5 % Each other impurity: < 0.02 %	Cu: 61.5–63.5 % Zn: remainder As: 0.10–0.15 %	Al: ≤ 0.1 % Fe: ≤ 0.1 % Mn: ≤ 0.1 % Ni: 0.21–0.35 % Pb: 0.15–0.25 % Sn: ≤ 0.1 % Each other impurity: < 0.02 %	As, Cu, Ni, Pb, Zn	Test water 1 according to EN 15664-2
6	Copper–zinc–arsenic–aluminium alloys	Cu: ≥ 61.0 % Zn: remainder As: 0.02–0.15 % Al: 0.2–1.0 %	Fe: ≤ 0.5 % Mn: ≤ 0.1 % Pb: ≤ 0.2 % Sn: ≤ 0.3 % Each other impurity: < 0.02 %	Cu: 63.0–64.5 % Zn: remainder As: 0.11–0.14 % Al: 0.2–0.4 %	Fe: ≤ 0.3 % Mn: ≤ 0.1 % Pb: 0.15–0.25 % Sn: ≤ 0.3 % Each other impurity: < 0.02 %	Al, As, Cu, Pb, Zn	Test water 1 according to EN 15664-2
7	Copper–zinc–arsenic–antimony–aluminium alloys	Cu: ≥ 60.0 % Zn: remainder As: 0.02–0.10 % Sb: 0.02–0.10 % Al: 0.02–1.0 %	Fe: ≤ 0.5 % Mn: ≤ 0.1 % Ni: ≤ 0.2 % Pb: ≤ 0.2 % Sn: ≤ 0.5 % Each other impurity: < 0.02 %	Cu: 62.0–65.0 % Zn: remainder As: 0.03–0.04 % Sb: 0.04–0.05 % Al: 0.45–0.58 %	Fe: ≤ 0.2 % Mn: ≤ 0.1 % Ni: 0.12–0.20 % Pb: 0.15–0.25 % Sn: ≤ 0.3 % Each other impurity: < 0.02 %	Al, As, Cu, Ni, Pb, Sb, Zn	Test water 1 according to EN 15664-2
8	Copper–tin–zinc–phosphorus–sulphur alloys	Cu: remainder Sn: 3.0–9.0 % Zn: 1.0–5.0 % P: 0.01–1.0 % S: 0.2–0.7 %	Fe: ≤ 0.3 % Ni: ≤ 0.3 % Pb: ≤ 0.2 % Sb: ≤ 0.1 % Each other impurity: < 0.02 %	Cu: remainder Sn: 4.6–5.0 % Zn: 1.0–3.0 % P: 0.01–0.06 % S: 0.52–0.65 %	Fe: ≤ 0.3 % Ni: 0.21–0.35 % Pb: 0.15–0.25 % Sb: 0.09–0.15 % Each other impurity: < 0.02 %	Cu, Ni, Pb, Sb, Zn	Test water 1 according to EN 15664-2

Category number	Metallic composition category	Category constituents (% (w/w))	Category impurities (% (w/w))	Reference material constituents (% (w/w))	Reference material impurities (% (w/w))	Elements of relevance in the migration water	Most critical test water
9	Copper–zinc–silicon–phosphorus alloys	Cu: 60.0–80.0 % Zn: remainder Si: 0.5–5.5 % P: 0.01–0.3 %	Al: ≤ 0.1 % Fe: ≤ 0.5 % Mn: ≤ 0.05 % Ni: ≤ 0.2 % Pb: ≤ 0.1 % Sn: ≤ 0.5 % Each other impurity: < 0.02 %	Cu: 75.0–77.0 % Zn: remainder Si: 2.7–3.0 % P: 0.02–0.06 %	Al: ≤ 0.05 % Fe: ≤ 0.3 % Mn: ≤ 0.05 % Ni: 0.15–0.25 % Pb: 0.09–0.15 % Sn: ≤ 0.3 % Each other impurity: < 0.02 %	Cu, Ni, Pb, Zn	Test water 1 according to EN 15664-2
10	Copper–silicon–zinc–manganese–phosphorus alloys	Cu: ≥ 80.0 % Si: 0.5–5.5 % Zn: remainder Mn: 0.01–0.2 % P: 0.01–0.3 %	Al: ≤ 0.3 % Fe: ≤ 0.3 % Ni: ≤ 0.1 % Pb: ≤ 0.1 % Sn: ≤ 0.5 % Each other impurity: < 0.02 %	Cu: remainder Si: 2.5–3.5 % Zn: 8.0–10.0 % Mn: 0.03–0.09 % P: 0.05–0.10 %	Al: ≤ 0.3 % Fe: ≤ 0.3 % Ni: 0.06–0.10 % Pb: 0.06–0.10 % Sn: ≤ 0.3 % Each other impurity: < 0.02 %	Cu, Mn, Ni, Pb, Zn	Test water 1 according to EN 15664-2
11	Copper–tin–phosphorus alloys	Cu: remainder Sn: 1.5–9.0 % P: 0.02–0.4 %	Fe: ≤ 0.1 % Ni: ≤ 0.2 % Zn: ≤ 0.2 % Each other impurity: < 0.02 %	See table note (a)	N/A	Cu, Ni	Test waters 1, 2 and 3 according to EN 15664-2
12	Copper–tin–lead–phosphorus alloys	Cu: remainder Sn: 9.0–13.0 % Pb: 0.2–1.0 % P: 0.01–1.0 %	Fe: ≤ 0.2 % Mn: ≤ 0.2 % Ni: ≤ 0.2 % S: ≤ 0.05 % Sb: ≤ 0.1 % Zn: ≤ 0.5 % Each other impurity: < 0.02 %	Cu: remainder Sn: 10.6–11.0 % Pb: 0.84–1.0 % P: 0.01–0.1 %	Fe: ≤ 0.2 % Mn: ≤ 0.1 % Ni: 0.15–0.25 % S: ≤ 0.05 % Sb: 0.09–0.15 % Zn: ≤ 0.5 % Each other impurity: < 0.02 %	Cu, Ni, Pb, Sb, Zn	Test water 1 according to EN 15664-2
13	Copper–nickel–manganese–iron alloys	Not defined in the first EUPL.					

Category number	Metallic composition category	Category constituents (% (w/w))	Category impurities (% (w/w))	Reference material constituents (% (w/w))	Reference material impurities (% (w/w))	Elements of relevance in the migration water	Most critical test water
14	Coppers	Cu: ≥ 99.9 % P: ≤ 0.04 %	Others total: ≤ 0.1 % Each impurity: < 0.02 %	CW024A (CU-DHP)	CW024A (CU-DHP)	None: no need for comparative testing	—
15	Tinned copper pipes and tinned copper fittings	Sn + Cu: ≥ 99.90 %	Impurities of the tin layer As: ≤ 0.01 % Bi: ≤ 0.01 % Cd: ≤ 0.01 % Cr: ≤ 0.01 % Ni: ≤ 0.01 % Pb: ≤ 0.01 % Sb: ≤ 0.01 % Each other impurity: < 0.02 %	CW024A (Cu-DHP)			
16	Aluminium	Not accompanied by category composition or reference material composition.					
17	Steel/iron	Not accompanied by category composition or reference material composition.					
18	Platings	Not accompanied by category composition or reference material composition.					
19	Passive compositions	Not accompanied by category composition or reference material composition.					
20	Copper alloys not elsewhere categorised	Not accompanied by category composition or reference material composition.					
21	Hard brazing alloys and soft soldering alloys	Not accompanied by category composition or reference material composition.					

^(a) CW453 K (CuSn8) is accepted only for product group C. For the acceptance of this material or another material belonging to category 11 (copper–nickel–manganese–iron alloys) for product group A or B, a complete test according to EN 15664-1 with all three test waters according to EN 15664-2 is required. A reference material will be defined when a material is accepted for product group A or B.